

Validation Evidence for Diagnostic Products

Learn to Make a Bounded Evidence Judgment

BOOK 1C · v7

Contents

Introduction – What Does This Evidence Let Us Say?	2
Chapter 1 – From Clinical Need to Lifecycle Learning	8
Chapter 2 – The Report Is a System Outcome	33
Chapter 3 – One Word, Several Evidence Jobs	54
Chapter 4 – From Claim and Risk to Evidence Plan	84
Chapter 5 – Reference Processes and Comparator Limits	109
Chapter 6 – Specimen Fitness and the Pre-Analytical Chain	139
Chapter 7 – Measurement, Calibration, Trueness, and Uncertainty	160
Chapter 8 – Detection and Reporting Boundaries	184
Chapter 9 – Variation, Challenge, Reagents, Instruments, and Time	206
Chapter 10 – Oncology NGS as an End-to-End Examination System	230
Chapter 11 – From Finding to Meaning	251
Chapter 12 – Clinical-Performance Study Design	276
Chapter 13 – Oncology Claims, Companion Diagnostics, and Bridging	304
Chapter 14 – Digital Pathology: The Image-Production Chain	331
Chapter 15 – Digital Pathology: Readers, Algorithms, and Workflow	353
Chapter 16 – From Performance to Clinical Use	376
Chapter 17 – When Evidence Must Travel	402
Chapter 18 – The Integrated Validation Review	424
Working Terminology and Distinctions	443
Abbreviations	448
Source and Citation Navigation	452

INTRODUCTION

What Does This Evidence Let Us Say?

Begin with an ordinary result

A laboratory professional reviews a draft report about tissue from a person with cancer. The laboratory has examined material from the sample, and software has helped turn measurements into a proposed finding. The professional checks and releases the report. A treating clinician may then consider the finding alongside the patient's other information in a discussion about care.

Nothing has gone wrong, yet the report raises several evidence questions. It may be accurate for one kind of specimen while saying little about another. A calculation may be correct even when the material entering it is unsuitable. A finding may be real while a statement about treatment response exceeds the clinical evidence. And a well-supported report may arrive too late, or in a form that changes how its user understands it.

The book asks **whether the evidence supports this claim for this product, with these materials and people, under these conditions**. An answer should preserve what has been shown and identify what remains uncertain. A reassuring collection of documents becomes useful when we can explain how each record supports the conclusion being considered.

That explanation begins with the care problem and the information the product is meant to provide. It follows a specimen or image through the product to the person who uses the result, then follows the product into use. If a material, component, setting or workflow changes, the old evidence may remain useful. It may need a narrow bridge—a focused comparison showing why its conclusion still applies across that difference—or a new investigation. Which response fits depends on the change and what the earlier evidence represented.

The reader and the decisions

Readers may come from laboratory work, clinical practice, pathology, development, statistics, quality, regulatory review or consulting. Readers new to diagnostics can begin here too. The starting ideas arrive where the questions need them: what a specimen is, how a report comes to exist, what another method or result used for comparison contributes, and why a favorable percentage has a limited meaning. Another volume is not a prerequisite.

The numerical examples use fractions, percentages, averages and simple expressions of uncertainty. They show inputs, units and intermediate reasoning. Often the difficult question comes before the calculation: which observations belong in the numerator and denominator? Statistical experience does not remove the need to check whether the studied people, specimens and conditions match the proposed use. Less experienced readers should still be able to follow the reasoning and recognize when design or interpretation needs a statistician, laboratory specialist, clinician or human-factors specialist.

The aim is a judgment another professional can follow: what the whole product does, which claims it makes, what each study measured and whom it represents, which evidence remains useful, and what work would resolve a consequential gap. An unknown need not become a failure. Developing these skills does not confer authority to release a product, approve a submission, decide conformity or choose a patient's treatment, nor does it replace the specialist methods needed for those decisions.

Book 1C develops the scientific and quantitative evidence reasoning for diagnostic products. Related volumes take particular questions further: Book 1A covers medical-device software, artificial intelligence (AI) and some human-use methods; Book 1B covers cybersecurity; Book 1D follows development choices for AI and machine-learning products; Book 2 applies the European Union's In Vitro Diagnostic Medical Devices Regulation; and Book 3 covers consulting practice. The essential explanations for this book appear here first.

Follow the products without merging their evidence

HelixBridge Dx is a fictional tissue-oncology sequencing product. DNA, or deoxyribonucleic acid, carries genetic information; sequencing reads many pieces of it. Laboratory and computational work helps identify **variants**, differences in that DNA. HelixBridge is described as identifying variants associated with treatment eligibility. Such a finding can inform a clinical question without establishing that a treatment will work for a particular person.

The proposed **HelixBridge Dx Liquid** branch starts with a different material path. Whole blood is drawn, plasma is separated, and cell-free DNA is isolated from that

plasma. Cell-free DNA consists of fragments circulating outside cells; circulating tumor DNA is the tumor-derived part and may be only a fraction of it. These are different material states. Evidence about handling and examining tissue cannot simply be relabeled as evidence for the liquid path. The distinction will matter to specimen fitness, detection, reporting and clinical interpretation.

PathSight Cloud follows an image route. A tissue slide is scanned into a whole-slide image for a pathologist to examine on a display. PathSight prioritizes images in a worklist and highlights suspicious regions; the pathologist makes the diagnosis. An image-processing component, an algorithm score, a reader using the assistance and the complete workflow can each be studied, but those studies answer different questions. Their evidence cannot be read as though the image were a sequencing specimen.

These recurring products are fictional composites, and each chapter supplies only part of their evidence. Missing component identities, denominators, releases and regulatory decisions stay unknown. Separate fictional examples supply other calculations or observed field events. Those facts belong to those examples; they do not become HelixBridge or PathSight history. The chapter source trails identify the basis for general scientific and regulatory statements. Fictional cases have no scientific or legal authority of their own.

How the questions develop

The order in Contents follows the dependencies between the questions. The first four chapters establish a map that later chapters make more precise. Within a case, an initial judgment rests on the facts available at that point. An additional exhibit or changed condition may then justify a revision. Keeping those judgments distinct makes it possible to see which fact changed the reasoning.

The feedback follows the question and explains what the supplied facts support. A defensible conclusion may be narrow: a study supports an analytical result under its stated conditions while the proposed clinical use remains unestablished. Other cases contain adequate bounded evidence. Before a decision to repeat a study, the relevant question is what the existing study already answers and what remains open.

The two professional lenses work with the same facts from different positions. A **Reviewer Lens** considers what the available record establishes now: the results and conditions worth crediting, the unsupported inference and the clarification that could change the judgment. A **Consultant Lens** considers the choices ahead: defining the claim, comparing evidence or design options, ordering dependent work, assigning an owner and specifying closure. The reviewer assesses; the consultant advises and helps plan. The manufacturer, laboratory or other authorized decision-maker owns the product and final action. Neither perspective promises approval, qualification or a business outcome.

Tables and examples keep the reasoning visible: what was counted, what was compared, which conditions stayed fixed, what was excluded and which conclusion follows. Source notes distinguish binding law, nonbinding guidance, voluntary standards and the author's teaching framework. Working Terminology retrieves distinctions already taught; Abbreviations expands short forms; Source and Citation Navigation points to the chapter-local trails. Each aid leads back to the explanation and its limits.

Chapter 18 comes after the earlier chapters as an independent assessment with an unfamiliar diagnostic. Its sequence keeps the complete initial judgment, a later exhibit, the revised judgment and the final judgment ahead of the worked feedback. Its identity, facts, answers and source trail belong within that sequence.

Why each question leads to the next

Part I connects the care problem to an evidence question. Chapter 1 separates a clinical need and hoped-for outcome from the proposed use and claim. It follows the specimen or image to a person and a decision, then places the product's evidence and later information on the same map. Chapter 2 opens the configured-product box. Physical preparation, components, software, data, interfaces and human work combine to produce the report. This reconstruction matters because a study cannot represent an unidentified arrangement merely because the product name matches.

Once the studied object is visible, the words *verified* and *validated* need more detail. Chapter 3 asks which job the record performed: a design requirement, laboratory method, analytical examination and whole clinical use each raise a different question. Chapter 4 turns the claim, risk and configuration into a plan. It asks which observation would answer the question, which reference or comparison fits, and what decision logic belongs in place before results are known. The first part gives each record a place in the reasoning.

Part II asks what the examination can establish. A result agrees with another process—but how was that comparison answer assigned, and how far can it be trusted? Chapter 5 begins there. Chapter 6 then follows the material before examination: how it was collected, transported and prepared, and what was lost before a report became possible. Tissue and the separately proposed liquid path show why a specimen label alone cannot establish fitness or account for missing results.

Those material and reference questions make the numbers interpretable. Chapter 7 asks what is measured and how the result relates to a reference, calibration, variation, systematic difference and uncertainty, using small worked examples. Chapter 8 asks when a signal becomes detectable, when a rule assigns a category and when a result can be reported, including invalid and uncertain outputs. Chapter 9 asks whether comparisons across lots, instruments, operators, challenging specimens and time

actually separate their effects. Mentioning several factors does not establish what a study tested. Chapter 10 joins these questions across the full tissue-oncology sequencing examination, including cases that never become complete reports. Its comparison with the proposed liquid branch shows where the tissue reasoning can and cannot travel.

Part III follows the finding into clinical meaning and use. Chapter 11 begins with a technically identified finding and asks what supports the further assertion about its meaning. The interpretation's evidence and version are part of that question. Chapter 12 puts clinical-performance numbers back inside their study: who was eligible and enrolled, who received an evaluable result, how the reference was determined, what was counted and which population the fraction describes. Those choices determine what sensitivity, specificity and an interval can reasonably say.

The next question is which clinical assertion is being made. Chapter 13 separates disease association, prognosis and differential treatment response, then examines transfer across configuration, population and intended role. Companion-diagnostic terminology keeps its jurisdictional limits. Chapters 14 and 15 take the evidence questions into digital pathology. The first follows tissue through slide preparation, scanning, processing, storage and display so the studied image can be compared with the image used. The second distinguishes an algorithm alone, a pathologist alone, a pathologist with assistance and the worklist, keeping reader, case, reference and workflow units visible. Chapter 16 then asks whether favorable performance supports the proposed use, a narrower claim or a further clinical-utility question. A good percentage cannot silently become demonstrated patient benefit.

Part IV asks whether earlier conclusions still apply. Chapter 17 considers a changed component, specimen path, user or use condition, and information returned from the field. It separates what stayed the same, what changed and what remains unknown before deciding whether existing evidence applies, needs a bounded bridge or calls for new work. Chapter 18 tests the reasoning without familiar case cues. Reconstructing a new product, making a complete initial judgment, revising after additional information and defending a final scope and next question draws on the whole route without naming a decisive chapter in advance.

The same route can be entered through a later question. Uncertain reference results lead back to Chapter 5 and then the relevant material and study chapters. A changed image pipeline leads to Chapters 14–15 and 17. A persuasive clinical percentage leads to Chapter 12, then the claim questions in Chapters 13 or 16. Each return asks which earlier assumption the new problem reaches.

Throughout the book, a useful conclusion stays attached to the claim, product, material, people and conditions the evidence can defend. It gives credit for what is

supported, keeps the remaining question visible and makes the next piece of work specific to the decision.

CHAPTER 1

From Clinical Need to Lifecycle Learning

Begin with a result that makes sense

A laboratory receives tissue from a person with cancer. It prepares material from that tissue and reads many pieces of DNA, the molecule that carries genetic information. Laboratory work and software help identify differences in the DNA, called **variants**. A report presents a finding for professional review and possible clinical use. What would evidence have to establish before that finding could help a treatment discussion?

The fictional **HelixBridge Dx** gives that question a concrete setting. It is a tissue-oncology sequencing product described as identifying variants associated with eligibility for a named therapy. Finding a variant is one contribution to a clinical decision. The finding neither chooses the treatment nor demonstrates that the treatment will work for this person.

A second product begins with tissue too, but follows a different path. A prepared slide is scanned into an image that a pathologist can examine on a display. This is **digital pathology**. The fictional **PathSight Cloud** highlights suspicious regions and prioritizes images in the **worklist**, the queue of cases waiting for review. The pathologist makes the diagnosis. Before that happens, the scanner, image preparation, software and display have helped determine what the pathologist encounters.

Nothing has gone wrong in either story. Yet each already contains several questions. Did the tissue examination reliably find the variant? What supports its treatment-related meaning? Did the highlighted image reach a pathologist in a useful form under the relevant workload? Evidence can answer one of these questions well while leaving another open. Giving a study full credit requires knowing where its answer ends.

That is the chapter's question: **what can we conclude from this evidence, about this product and this use?** A short map will eventually connect the care problem, the product's role, the material and information route, the studied state, and information returned from use. Its purpose is to make the relationships visible before judging them. No earlier volume is needed, and an absent fact can remain “not supplied.” An unidentified release or denominator does not become known because a plausible answer comes to mind.

Decision

For a diagnostic result you can picture, what care problem lies behind it? What change might it help achieve, what role does the product have for a particular person and material, and what assertion would need evidence? How does human material become information that person can use? Which parts of that first picture are known, and which are assumptions?

These initial questions will return after the cases. The change in an answer matters as much as the final wording: it reveals which relationship became clearer.

Reviewer Lens. The tempting first move is to count the documents or read the strongest result. A more useful opening move is to establish what result the file is asking that evidence to support. Two separately identified records can both be valuable while their relationship remains unresolved. The first review question is about that relationship; a large document inventory cannot answer it.

Consultant Lens. Before studies begin, the product and laboratory teams can agree which product arrangement each study will represent and how a later change will be compared with it. That decision is inexpensive while the plan is open and difficult to reconstruct afterward. The consultant makes the dependencies clear; manufacturer and laboratory owners decide the claim, release and evidence plan. If a later difference appears, identifying what changed and which output it could affect comes before choosing a comparison, targeted testing or a narrower claim.

What would the report need to establish?

“Improves cancer care” sounds like a reason to develop a product. It gives a study planner very little to measure. The missing steps lie between the problem that motivates the product and the assertion its evidence must support.

Suppose clinicians need more information when considering treatment for a person with cancer. That is a **clinical need**: a health or care problem worth addressing. It leaves the choice of product open. A team might hope its product will provide better information for a treatment discussion. That hoped-for change is the **intended outcome**. It tells us why the team is interested, but the hoped-for change has not happened merely because the team named it. These are planning labels used in this book.

The next question is what contribution the product itself proposes to make. For PathSight, the known contribution is prioritizing slide images and highlighting suspicious regions for a pathologist who diagnoses. Material enters, information reaches a particular user in a setting, and that person has an action to take. Together those details describe the proposed **intended use or intended purpose**. HelixBridge's

described role concerns tissue variants and treatment eligibility; exact report and professional-use details still need definition before a study can be matched to that role.

Once the role is clear enough, we can ask which assertion needs support. “Identifies variants associated with eligibility for a named therapy” supplies the beginning of a **claim**. It is more assessable than “improves care,” but specimens, variants, population and report role still matter. For PathSight, assistance to a pathologist asks whether the assistance reaches and helps the intended reader. Automated diagnosis would be a different role with a different evidence question. A statement about either role does not establish its truth, authorization, conformity or applicability to every product configuration.

The four questions below help locate what is missing when a proposed study seems disconnected from a clinical goal.

Question	What the answer gives us	What remains to be shown
What care problem motivates the work?	Clinical need	Which product and use could address it
What change is hoped for?	Intended outcome	Whether that change occurs and has evidence
What will this product do, for whom and under which governing source?	Intended use or purpose	Whether it performs that role and produces benefit
What bounded assertion needs support?	Claim	Whether the assertion holds for the proposed state and conditions

The first two rows can easily be filled with plausible stories. In the HelixBridge case, however, the possible need for treatment-relevant information and the wish to improve a discussion are *planning illustrations*. Neither is supplied as a case fact. Its later map therefore keeps both “not supplied.” What is supplied is the tissue-variant role associated with eligibility for a named therapy. PathSight has a different role involving an image and a pathologist. Even after release, an invalid result or a late image only acquires its full significance in relation to what that separate service promises about availability or timing; the event alone cannot define the whole promise.

A legal use statement also needs a source, because the shared teaching question does not collapse different legal meanings. In the United States, 21 CFR 801.4 ties intended uses to the objective intent of those responsible for labeling. Their statements, product design and distribution circumstances can help establish that intent, so one convenient sentence may not be the whole record. In the European Union, Article 2(12) of the In Vitro Diagnostic Medical Devices Regulation ties intended purpose to specified

manufacturer-supplied information, including labeling and performance-evaluation material. The labeling standard ISO 18113-1:2022 pairs “intended use” and “intended purpose” within its voluntary in vitro diagnostic scope. That pairing does not merge the two legal definitions. A legal decision needs the relevant jurisdiction and source beside it.

Reviewer Lens. A precise claim can be a good starting point before any performance conclusion is available. The hard judgment is which missing boundary would change the evidence needed. In HelixBridge, the therapy, population, variant classes and report role cannot be inferred from the word *oncology*. For PathSight, the pathologist's continuing diagnostic role must survive the transition from product description to study appraisal. A request aimed at these boundaries is more useful than asking for a more persuasive version of “improves care.”

Consultant Lens. Separating problem, hoped-for change, product role and claim gives clinical, laboratory, pathology, product and regulatory owners a decision they can actually make before fixing a protocol. If the claim reaches beyond what the proposed workflow can observe, the options become visible: narrow the claim, change the product's role or plan different evidence. The consultant explains those choices and their dependencies. A polished use statement is useful planning work, but it leaves the performance question to the evidence.

Which product produced the result?

A tissue finding cannot enter a useful report merely as “a difference was found.” It needs a location and a description. Following how it acquires them shows why the product studied is more than its application name.

The physical route starts with tissue, preparation and sequencing. The sequencer produces readings of DNA; an analysis program examines those readings to propose candidate findings. To locate and describe a finding, the program uses controlled reference information. The store of that information is a **knowledgebase**. The program does the analysis; the knowledgebase supplies information the analysis uses. Identifying one does not identify the other.

Why might the reference description matter? A **gene** is a region of DNA whose information can be copied into RNA, or ribonucleic acid. The RNA copy is a **transcript**. Different parts can be included, so one gene may have more than one transcript form. To describe a detected DNA difference consistently, the knowledgebase holds a selected sequence description: a **reference-transcript record**. This record is reference information, distinct from biological RNA in this specimen.

Assigning the finding to a reference location is the operation called mapping in the case records. The selected record can affect the finding's stated position or name and which analysis path the program runs. The next step governs what reaches the report: **report**

rules determine which processed findings appear and how they are presented. A professional can then review the report and consider its use. An unexpected behavior at the assignment step is an anomaly; recognizing it requires knowing the ordinary route first.

That route lets us identify the **configured product**: the working arrangement that produced a result or was used in a study. Particular specimen conditions, instruments, software and reference information belong to the study's snapshot. A later arrangement can carry the same product name while differing at one of those points. The configuration record needs the parts relevant to the claim and comparison, not just the analysis program's label.

The image route has its own arrangement. Tissue becomes a slide; a scanner produces the whole-slide image; software processes it and can prioritize or highlight it for the pathologist. The path includes both what happens to the material and what happens to the information afterward. An image-processing result cannot identify the reader's task by itself, just as a sequence analysis cannot identify the professional's treatment-facing use. Chapter 2 will open these physical, computational and human connections in more detail.

Both routes fit into the book's **specimen-to-decision spine**:

clinical need → intended outcome → intended use or purpose → claim → human specimen → collection or preparation → examination by a configured product → information or report → intended user → bounded decision or action

The left side explains why the examination exists; the middle follows what is examined and how; the right side reaches the person who can act. This is a teaching map. The U.S. in vitro diagnostic definition in 21 CFR 809.3(a) expressly covers products intended for collection, preparation and examination of human specimens. The European Union's in vitro diagnostic regulation has its own device, purpose and performance definitions in Article 2. Neither is a legal definition of the entire longer spine, which does not decide ownership, causation, classification or device boundaries.

The same trace matters once a service is in use. Consider two **separate, unnamed fictional services**. They are unrelated to HelixBridge, its proposed liquid branch and PathSight; their observations add no release history to those products.

At the sequencing service, records for **40 consecutive tissue specimen attempts** under an identified released configuration show **two initial invalid statuses before report release**. Case and run identities were retained. No incorrect released variant result is reported. Those identities locate the observations in a specimen-to-report route, although the cause remains unknown.

At the separate digital-pathology service, **one of 25 consecutive supported-image cases** reached the pathologist later than a local worklist target. The image and worklist event were retained, and the pathologist completed review. Here the trace reaches the reader and establishes that review happened. The observation is about timing; it does not measure model accuracy. Comparing either service's field route with a premarket study would require that study's own configuration and conditions.

There are therefore three different boundaries to keep visible. The route drawn for **orientation** includes enough material, information and human work to locate the question. The **evidence boundary** specifies what a record actually covers: initial result availability in the 40 attempts, or worklist timing in the 25 supported-image cases. The **legal product boundary** depends on the particular regulated product and jurisdiction. A scanner, supplier or office can belong on the orientation map without that drawing settling its legal status.

Reviewer Lens. An output is a useful starting point because it forces the review through actual connections. The question is whether the study followed the relevant input, transformations and user path. A reference source may be identified while its relationship to the proposed product remains open. In the field notes, retained case and event identities are positive evidence: they make a focused investigation possible. They supply traceability without supplying a cause.

Consultant Lens. A team can discover a missing study relationship by comparing the proposed use path with the protocol's path before collecting data. Laboratory owners know specimen handling and report release; product owners know software and rules; suppliers know managed reference data; pathology services know the slide and scanner route; clinical users know the decision. If a material step or human action falls between those owners, the company can change the protocol, product or claim while the choice is still explicit. Assuming the missing step is somebody else's problem leaves the eventual result unattached to the full use.

Figure reference: Recurring specimen-to-decision lifecycle. Its three lanes trace tissue NGS, liquid NGS and digital pathology through the shared evidence route.

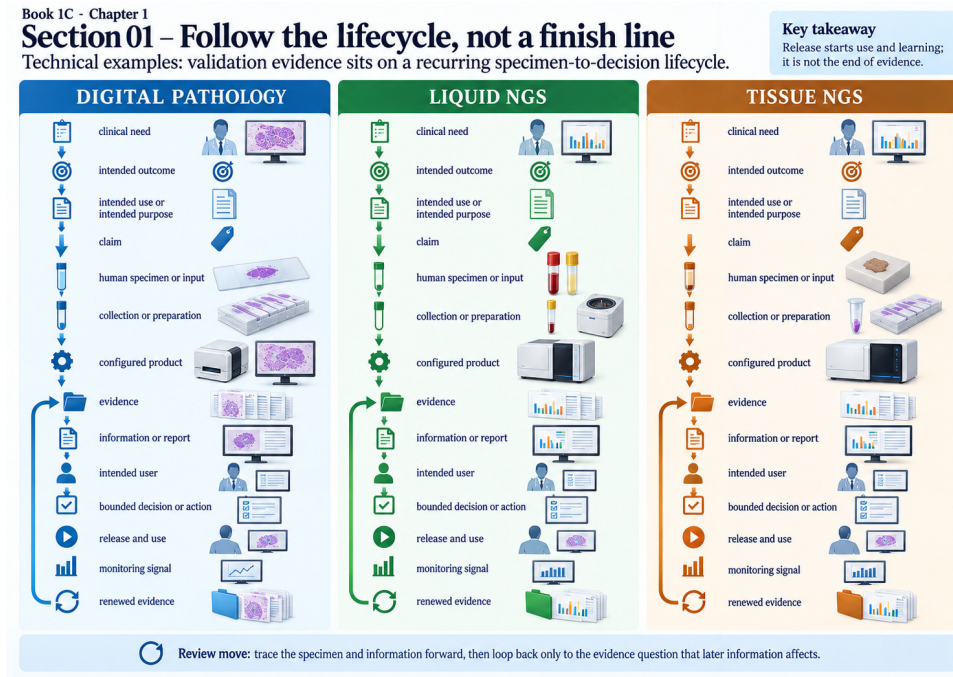


Figure 1.1. Recurring specimen-to-decision lifecycle – Technical review plate.

What does a folder labeled “validation” answer?

Suppose a requirement says, “An invalid specimen status must stop a report from being released.” A test supplies that status and checks the release behavior. If the test succeeds, what can the reviewer conclude about the tissue report as a whole?

The answer starts with what was actually examined. The specific question a record is meant to answer is its **evidence job**. A useful account of that job connects five things in one sentence: **the object; the question about it; the configuration and conditions; the point in the product's life; and the conclusion supported**. These are what allow two records in the same folder to answer different questions without either being defective.

From a written requirement to a usable product

In the hypothetical release-block example, the written requirement is the **design input**. The rule and report screen built to satisfy it are **design outputs**. Testing whether those outputs meet that input is **design verification**. A successful result can establish the required release behavior in the tested state and conditions. It leaves a further question: can the resulting product serve the specified application or intended use in its actual setting? That is the design-and-development question addressed by **design validation**.

The distinction matters because blocking release is only one relationship in the specimen-to-report route. The design clauses of ISO 13485:2016 distinguish verification and validation in this context, particularly clauses 7.3.6 and 7.3.7. Laboratory, software and clinical uses of those words have their own objects and questions. A shared word on the folder is insufficient to make their conclusions interchangeable.

The hypothetical test is also separate from the unnamed sequencing field note. Two observed invalid statuses before release do not establish that this particular hypothetical control exists in that service. One record describes a proposed behavior and its test; the other describes what happened to two attempts. Both can contribute to reasoning about result availability, through different evidence jobs.

From a reliable finding to a justified clinical use

Now suppose the question concerns the examination itself. Can it detect, measure, classify or report what it claims to examine under the represented material and conditions? That calls for **analytical evidence**. An **analytical-validation** study of a tissue assay might challenge known variants in relevant specimens and count detections and invalid results. Its title alone does not identify which findings or conditions were actually tested. A pathology processing study might instead examine whether image details survive specified scanner and preparation settings.

Even a strong result leaves the next link open. Reliably detecting a variant does not establish why it is associated with a clinical state or treatment-related purpose. For an in vitro diagnostic question, that association is the concern of **scientific validity**. Evidence for the association and evidence for the assay's detection ability support different links. An association cannot make an unreliable assay reliable, and perfect detection cannot create an unsupported biological meaning.

Once that relationship has support, there is still the question of what this product's results establish in the population and use being claimed. **Clinical performance** concerns their relationship to the clinical condition or state. The study needs a way to determine the comparison state, people or specimens representing the intended population, and an account of invalid and missing results. A convincing average in one study group cannot stand in for another population or for observed benefit to every patient. The European Union's in vitro diagnostic regulation connects scientific validity, analytical performance and clinical performance within Article 56 and Annex XIII's performance-evaluation framework. This does not assume PathSight qualifies under that regulation.

The person at the end of the route presents another question. Can the intended user notice, understand and act on the information during real work? **Human-use evidence** addresses that link. A PathSight score can be calculated correctly while the display and worklist determine what the pathologist encounters. A technically correct tissue report

still needs professional review. In the unnamed pathology service, the late arrival and completed review establish a concrete workflow observation; they do not establish a diagnostic error.

What if the release-block test was intended to control a risk? Then the same test may also help answer whether a specific unwanted situation was anticipated and whether that control worked. Its success remains attached to that question. A release record and information returned from use answer another: which state entered use, what happened there, and which earlier assumption might need review? A single record may contribute to several conclusions, but the reasoning must connect it to each one.

The following locator helps choose the question to open when a report merely says “validated.”

Question to open	What needs an identity	What a favorable answer still leaves open
Does the proposed role address the stated problem?	Need, outcome, user, input, setting, output and action	Actual performance or benefit
Does the built product meet the design question?	Requirement or specified use, design context and configuration	Questions beyond that design result
Does the examination work for the represented material?	Specimen, output, conditions and product state	The further clinical meaning
What supports the biological or clinical association?	What is examined and its relation to the clinical state or purpose	Whether this product detects it reliably
How do product results relate to the claimed clinical state?	Population, use and comparison state	Benefit to every patient
Can the intended person use the information?	User, display, task and workflow	Questions outside the observed use
Did the relevant control work?	Unwanted situation, control and tested conditions	A verdict on the whole product
Does later information affect an earlier conclusion?	Released state, observation or change, and prior evidence	Automatic failure or automatic continued applicability

The analytical row illustrates how to use the table. A well-supported result for a studied tissue and output can remain positive while the clinical row stays open for a broader treatment assertion. Making both rows “pass” would overstate the evidence; making

both “fail” would discard it. The same separation applies along the pathology route from image production to scoring, display, reader behavior and worklist timing. A scanner comparison does not answer a sequencing detection-limit question. Tissue-assay evidence does not establish performance for the separately proposed liquid branch.

The five-part sentence now gives the field notes precise jobs. In the unnamed sequencing service, the object is the identified released tissue route; the question is initial result availability; the conditions include the observed interval of 40 consecutive attempts; the observation occurs in use; and two attempts had initial invalid statuses. In the separate pathology service, the sentence concerns arrival at the reader and completed review for the 25 supported-image cases. Neither sentence ends with a model-accuracy or variant-detection-sensitivity conclusion.

Reviewer Lens. Reading the five-part sentence before the result prevents a strong number from silently answering a different question. A reader-performance claim calls for evidence of a pathologist using the relevant images, display and workflow; a model score alone cannot identify that experience. Conversely, a narrow record is not deficient because it does not answer an unclaimed question. The reviewer's task is to credit the stated result and locate the unsupported extension, if there is one.

Consultant Lens. The productive moment to choose evidence jobs is while claim and protocol can still change. Clinical specialists establish which meaning question matters; laboratory and pathology specialists define examination conditions; statisticians define the quantity to estimate; human-factors specialists define the user task; product and quality owners identify configuration and controls. Clear input states let those studies be sequenced. A criterion for closing each question, agreed before results arrive, makes it possible to preserve a favorable study while planning only the remaining work.

What do two field fractions tell us?

The two unnamed services have done something useful before either observation is explained: they retained the records needed to count and trace it. What do the resulting fractions say, and which earlier evidence question could they reopen?

For the sequencing service, the numerator is **2 initial attempts with an invalid status**. The denominator is **40 consecutive tissue specimen attempts** under the identified released configuration. Dividing gives $2 \div 40 = 0.05$. Multiplying that proportion by 100 expresses it as **5% of the initial attempts in this interval**. The result describes initial availability of an interpretable result in this observed set. It is not a rate among all patients who might need testing, an estimate of variant detection, or a comparison with an earlier period. The retained case and run identities give the laboratory somewhere to investigate; the cause remains unknown.

The pathology service counted a different event and unit. Its numerator is **1 supported-image case arriving after the local worklist target**; its denominator is **25 consecutive supported-image cases**. Here $1 \div 25 = 0.04$, or **4% of those cases**. The worklist record locates a timing event, and the completed pathologist review establishes that the case was ultimately reviewed. The fraction supplies no measure of model accuracy, no exact delay duration and no clinical consequence. Neither 5% nor 4% is a universal acceptance threshold.

The difference between the fractions points toward different investigations. The sequencing laboratory can examine whether the two invalid attempts share a specimen, run, preparation or product condition and whether earlier evidence represented that result state. The pathology service can examine the routing event and whether the delay affects its promised worklist role. An unknown cause is not itself a demonstrated product defect. Retained trace and completed review deserve credit while the relevant question remains open.

To connect either event to prior evidence, the earlier study or released state must first be identified. Then the later state or information can be compared with it. The book calls this a **changed, unchanged and uncertain comparison**. “Unchanged” needs evidence of sameness at the relevant level. “Changed” needs a documented difference. “Uncertain” preserves what identity, timing, mechanism or applicability has not resolved. A missing release date belongs in that last category; it is not another observed event.

The comparison is useful because a return to earlier evidence can have different outcomes. If the observed route was correctly bounded and the exception did not affect a claimed condition, a documented explanation may preserve the prior conclusion. If an affected condition was not represented, a focused comparison or new evidence may be needed. Risk reassessment, a change to the claim or another response may also be appropriate. The mechanism and claim determine the choice. A later signal neither erases all previous evidence nor grants permission to carry every earlier result to a different state.

This return from use or change to a particular evidence question is a **lifecycle return**. It is the book's teaching method, not a regulator's required worksheet. Chapter 17 examines what makes a bridge—a focused comparison supporting continued applicability—convincing. Here the task is to locate the relationship that would need that support.

Different sources help with different parts of the return. For control of design changes and feedback within a quality system, ISO 13485:2016 addresses relevant activities in clauses 7.3.9, 8.2.1, 8.4 and 8.5.1. For production and later information affecting medical-device risk, ISO 14971:2019 addresses the lifecycle in clauses 4.1, 9 and 10.1–10.4. Both retain their voluntary-standard scope; the risk standard does not

prescribe the same response to every signal. The U.S. Food and Drug Administration (FDA) describes total-product-lifecycle oversight as an agency approach. Under the European Union's in vitro diagnostic regulation, Article 56 and Annex XIII connect performance evaluation with continuing information for applicable devices. None of those sources turns the book's map into a universal required development order.

For a U.S. quality-system decision, the binding context must also be identified. FDA's **Quality Management System Regulation** took effect on February 2, 2026. Current 21 CFR Part 820 incorporates ISO 13485:2016 by reference with U.S.-specific provisions; applicable U.S. law and more specific requirements control conflicts. That locates the quality-system question without deciding either fictional product's status. Regulatory and quality specialists apply the relevant requirements to the actual product.

Reviewer Lens. The event record comes before the explanation. In one service there were two initial invalid attempts with retained identities; in the other there was one locally delayed case and completed review. Those are useful, bounded observations. The next pass asks which promise or prior assumption they could affect. A targeted comparison is warranted when its timing, scope or mechanism could change the conclusion. Relabeling either fraction as “clinical failure” would lose the question the record can actually answer.

Consultant Lens. A useful monitoring plan decides in advance which observations and denominators sites will retain and who can investigate them. After an event, laboratory, pathology, product and risk owners can compare the released state with the studied state and explain their respective parts. A bounded, explained exception with no claim effect can support closure. An affected condition calls for work that tests that relationship, with a criterion and an owner, before a broader claim or release decision is advised. The consultant helps make those alternatives concrete without inventing an event outcome.

Figure reference: Changed, unchanged, uncertain and reopened evidence. It shows how a later difference sends a question back to the affected evidence.

Book 1C - Chapter 1

Section 02 – Classify the later input

Technical examples: a signal opens a bounded evidence question; it is not a verdict.

Key takeaway
Changed does not mean everything fails; unchanged does not mean everything transfers.

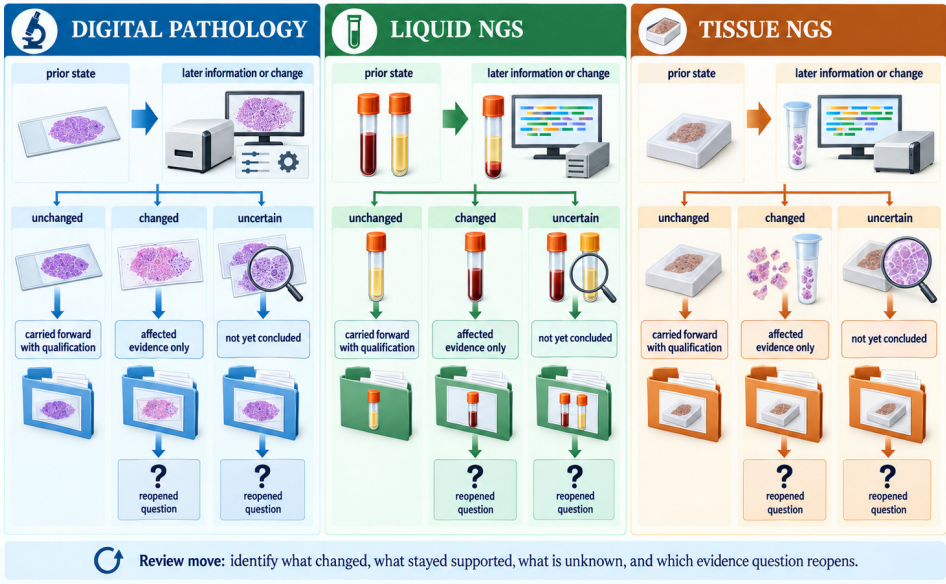


Figure 1.2. Changed, unchanged, uncertain, and reopened evidence – Technical review plate.

One page another person can understand

A reviewer can recognize every term we have used and still lose the thread in a large file. A fact about a component, an unanswered release question and a favorable study conclusion can appear beside one another until they seem to describe the same state. A **lifecycle and evidence-job map** keeps those relationships visible.

The page has one claim and one focal state. Different claims or versions needing different conclusions need separate views. At its beginning are the four questions from the opening: clinical need, intended outcome, intended use or purpose with its jurisdiction and source, and the bounded claim. Unknown wording remains unknown. These entries tell another person what the rest of the page is trying to establish.

The middle follows the specimen or image to the decision. At each stop it distinguishes a supplied fact and its source, an explicit unknown or boundary, and what the evidence establishes there. The table below shows the stops and the question each brings to the record.

Stop on the route	What needs to be known there?
Human specimen and population	Which material and people does the record represent?
Collection or preparation	What happened before examination?

Stop on the route	What needs to be known there?
Configured examination product	Which working arrangement produced the evidence?
Information or report	What output reached the next step?
Intended user and setting	Who encounters the output, and under which conditions?
Decision or action	What bounded use can that person make of it?

For the configured-product row, a named analysis program is a fact about the arrangement. It becomes evidence for performance only when a study supplies the relevant question and result. Its identity, where that identity was recorded, what else remains unknown and the supported conclusion are separate entries. This separation is just as important for the user row: identifying a pathologist does not establish successful use of a particular display.

The end of the page returns to a prior conclusion when later information is available. Its questions are sequential: which release or use state is involved; what information, signal or change arrived; what is established as unchanged; what changed; what remains uncertain; and which evidence job reopens, with which owner responsible for the next fact or decision. Without a supplied release, that first entry stays open. A blank cannot authorize a release chronology.

Two kinds of open entry matter. **Not supplied** means the fact is absent. **Not yet decided** means the known facts pose a question whose judgment is unresolved. Neither means an adverse result. The page can be adequate for orientation while the evidence-sufficiency decision remains open.

Reviewer Lens. The map earns its space when it exposes the smallest missing relationship that could change a judgment. “More validation” gives the next reader little direction; identifying a configuration comparison or an affected analytical scope gives them a question. A clearly applicable positive result belongs on the page too. For PathSight, keeping algorithm, display and reader routes distinct allows a strong model result to remain strong without becoming a workflow conclusion by proximity.

Consultant Lens. Before evidence collection, the same page gives the proposed study a named product state and a reason to exist. A consequential unknown creates choices: identify the configuration more precisely, revise the claim, alter the workflow, plan a targeted study or justify a limitation. Laboratory, clinical, statistical, product and regulatory specialists supply their methods; the company or site decides the course. The consultant coordinates the relationships, including what each choice leaves unresolved.

The initial HelixBridge view

Here is the page's first application. Every major document is described as present. One record is labeled analytical validation and identifies a particular application, report-rule set and knowledgebase. Before any later information is added, what does that let the page say?

The exact labels below identify the supplied record once. In the rest of the chapter, they become the validation application, validation report rules and knowledgebase used in analytical validation.

Part of the first map	Supplied entry	What remains open
Clinical need	Not supplied	An oncology modality does not identify a specific unmet need.
Intended outcome	Not supplied	Eligibility information does not show that care changed.
Intended use or purpose	Identifies variants associated with eligibility for a named therapy	Exact U.S. or EU wording, therapy, population, variant classes, treatment-facing role, confirmatory language and limitations
Claim	Identify variants associated with eligibility for a named therapy	Support for the unidentified commercial configuration; authorization and conformity outcomes
Specimen and product	Tissue-oncology sequencing; application 7.0 and report rules R19 in the analytical-validation view	Exact laboratory preparation, instrument, reagent, service, report and commercial component identities
Evidence state	All major documents described as present; analytical-validation record identifies application 7.0, report rules R19 and knowledgebase DB-18	Coherence, applicability and adequacy; counts, classes, criteria, failures and strata
Release and use	Not supplied	Release date, deployed sites, laboratory and manufacturer decisions, and regulatory outcomes
Later information	Reserved for supplied use, monitoring or change information	Any comparison or applicability judgment based on that later information

The evidence-state row supplies something real: an identifiable object for the analytical-validation record. It supplies no result details that would justify “analytically sufficient.” The commercial arrangement is also unresolved. Describing a knowledgebase as “current” would not give it a reproducible identity. U.S. submission preparation and European in vitro diagnostic strategy work are under way, but no clearance, approval, certification or conformity outcome is supplied.

This is why the columns stay separate. A later technical report belongs beside the state it examined, not inside this study cell as though both necessarily used the same arrangement. For PathSight, the corresponding page would follow image, scanner, preparation and display to a pathologist. Any actual deployed observation would need its own released configuration and event record. The two unnamed field notes belong to their separate services' release-and-use rows; they cannot fill these named products' empty cells.

Reviewer Lens. This first page is useful precisely because it knows its limits. It preserves the identified analytical-validation view and exposes the unidentified commercial state without manufacturing a study pass. That is a positive orientation judgment. The remaining task is to decide which missing relationship would affect the claim, rather than treating every open entry as a demand for a complete new study.

Consultant Lens. The first ownership questions are who controls the exact claim and who can identify the commercial arrangement. Their answers determine whether the next work is a traceable comparison, a bounded bridge, a claim revision or a study. Speculative details would make the page look fuller while making that choice less reliable. The planning value lies in knowing which decision depends on which fact.

Two records, then more information

The first HelixBridge map supplies an initial evidence state. PathSight now supplies another. Each will receive further information after the initial questions. The comparison concerns possible changes to named products; it does not presume that either changed arrangement was commercially released.

The sequencing record

The initial HelixBridge view is the one just mapped: tissue-oncology sequencing, the validation application and report rules, and the knowledgebase used in analytical validation. The broad role concerns variants associated with therapy eligibility. Study counts, relevant classes, acceptance logic and commercial identity remain unresolved.

Decision

What claim and product state can this evidence be attached to? What can be concluded now, and which unknown matters most before the evidence could support the product being considered?

The pathology record

PathSight prioritizes whole-slide images and highlights suspicious regions; the pathologist diagnoses. To read its record, it helps to follow how a score is produced and presented. During development, software learns patterns from images by adjusting numerical settings, the **model weights**. In use, it applies those settings to new images to produce scores.

The images have already passed through other operations. A scanner's **sharpening** changes edge contrast and apparent fine detail before PathSight analyzes the image. **Normalization** adjusts incoming images toward a consistent appearance before scoring. After scoring, a **low-confidence display rule** governs how cases treated as low-confidence appear in the worklist. A collapsed display shows less detail by default. These descriptions locate operations; the next records establish which states are actually supplied.

The case states support for two scanner families, labeled Scanner A and Scanner B in the record. We will call them the first and second scanner families. It identifies the validation model as M3.2 and states that commercial model weights are the same as those evaluated weights. The validation set contains mostly images from the first scanner family.

Decision

What product role, image route and model identity are known? What is the strongest conclusion available now, and what remains unknown about the evidence's scope?

Further information for the two records**The knowledgebase in the separate technical report**

A separate HelixBridge technical report concerns knowledgebase DB-19 and describes **strong overall agreement**. This knowledgebase added a reference-transcript record that makes a previously rare mapping anomaly execute more often. The validation application and report rules are recorded as unchanged statements, as are the tissue modality and broad proposed purpose. The record does not prove that every other component matched between the two evidence configurations.

A supplier-managed publication job can publish a new knowledgebase without changing the validation application's version label. This describes a capability, not a dated observed release. The exact commercial configuration, affected variant classes and failure denominator are not supplied.

Decision

Which entries are now unchanged, changed or uncertain? Which evidence jobs reopen? What is the strongest conclusion about carrying the earlier evidence to the knowledgebase in the separate technical report or to an unidentified commercial state? Which new fact changes the initial judgment, and which conclusion remains undecided?

The later image and display view

After validation, PathSight's normalization was upgraded to improve throughput. Its interface now collapses low-confidence cases by default. The second scanner family contributes proprietary sharpening, while the validation images were dominated by the first family.

A high-volume **simulation** recorded **two positive, low-score slides exceeding expected turnaround time**. The denominator, exact time and clinical consequence are not supplied. The changed configuration's commercial release is not established.

Decision

Where do the differences fall between slide and pathologist? Are unchanged weights sufficient to carry the earlier conclusion forward? Which evidence questions reopen, and which conclusions remain unknown? What can the simulation establish separately from a commercial release?

The two judgments before feedback

For each record, a complete judgment has a claim and known product states, a material route and an information-and-user route, and evidence attached to the state it represented. Which relationships are unchanged, changed or uncertain after the additional information? For each reopened job, can the object, question, configuration and conditions, lifecycle position and possible conclusion be distinguished? What is supported now, and what still cannot be concluded? These are revised judgments; the initial ones remain the comparison that shows what the added facts did.

How a reviewer reasons through the two comparisons

Worked reasoning for the two records

HelixBridge: locate the affected relationship

The initial judgment had a useful but limited positive conclusion: the analytical-validation record named an evidence object. The further information supplies another object, the knowledgebase in the separate technical report, and a favorable reported overall comparison. It also supplies a specific difference: a new reference-transcript record makes a previously rare mapping anomaly execute more often. The application, report rules, tissue modality and broad purpose are recorded as unchanged statements. They do not establish identity of every non-knowledgebase component.

That difference matters because the reference record participates in assigning sequence information to a reported variant or interpretation. The anomaly question therefore points to particular outputs and variant classes. “Strong overall agreement” remains a favorable result reported for its own comparison. Without affected-class detail, however, it cannot settle the question raised by the new transcript record. The earlier analytical-validation record also retains its original identity; a later difference does not erase it.

The unresolved relationships are exact commercial configuration; whether the non-knowledgebase components matched across the evidence states; affected classes and mappings; failure denominator; report consequences; release timing and sites; and regulatory outcome. The supplier's publication capability explains how a knowledgebase could change while the application label stays fixed. It establishes neither use of that route nor the date of a commercial release.

A useful next request connects the exact configurations and update control to the potentially affected analytical output, clinical interpretation and report claim. If a state was released, labeling, field traceability and supplier governance could matter too; release is still unknown here. Software specialists may need to determine whether the supplier service appears in the relevant software bill of materials, while security specialists may need to compare the assessed service scope. Those are questions about the evidence boundary. They add no security event to the case and supply no reason to repeat every study.

The revised conclusion is bounded: **the earlier knowledgebase's evidence cannot yet be carried to the knowledgebase in the separate technical report or to an unidentified commercial state without assessing the transcript change, affected scope and relationship between configurations.** This concerns applicability. It establishes no analytical failure, new clinical outcome, blanket invalidation or

regulatory decision. A later comparison might preserve evidence for unaffected outputs and assumptions; an affected claim-relevant class might require targeted work. Neither result has been supplied.

PathSight: follow the image into the reader's work

The initial view identifies the assistive role, pathologist's diagnostic responsibility, validation model and unchanged commercial weights. Those are positive facts. The later view adds differences before and after scoring: normalization changes image processing; the second scanner family's proprietary sharpening raises a comparison with the mostly-first-family validation images; and the collapsed low-confidence display changes what the pathologist sees by default.

The two delayed positive, low-score slides add an observation in a high-volume simulation. No delay fraction can be calculated without the number of simulated cases. Without exact timing, event path, comparison and clinical context, no cause or harm can be assigned. The observation directs attention to score-based worklist action and human use. It does not demonstrate a model-performance defect or a postmarket event.

The comparison now needs to ask whether changed image preparation produces inputs comparable to those studied; whether second-family scanner conditions and difficult tissue strata were adequately represented; what the collapsed display does to attention and turnaround; whether reader-study conclusions apply to that interface; and what an appropriate fallback permits. Product and risk owners also have relevant software-change, security and monitoring questions to consider. Exact commercial configuration and release, generalizability to the second scanner family, and reader-workflow applicability remain unknown.

The revised conclusion follows the changed route: **unchanged weights identify one unchanged component, but do not establish applicability of the validation evidence to the changed preparation, scanner conditions, interface and high-volume workflow.** Different questions may close differently. A documented comparison might settle a narrow preparation question; a claim about pathologist performance with a new display might require direct human-use evidence. The content of the comparison determines the bridge, and that content is not supplied.

The two products share a reasoning method. Their mechanisms remain distinct: reference information and tissue-variant reporting in one, image and interface changes reaching a pathologist in the other. A conclusion about either product is not evidence for the other.

Reviewer Lens. The difficult part is giving positive evidence its proper weight without letting it answer the whole question. HelixBridge has an identified earlier record and a favorable reported later overall comparison. PathSight has unchanged weights and actual observations from its simulation. Each deserves acknowledgement. A complete

configuration comparison showing no relevant impact could close a narrow question; repeating a product version label cannot. The request should seek the relationship capable of changing the judgment.

Consultant Lens. For HelixBridge, knowledgebase and assay teams can identify transcript-dependent outputs, while laboratory and clinical owners determine which reported classes and interpretations matter to the claim. Only then can the product owner choose among bridging, retesting, limiting a claim or holding a release. For PathSight, image, model, workflow and pathology owners can compare studied and changed routes and define the observation that would close each inference. The consultant orders those dependencies and explains alternatives; the manufacturer or site makes the authorized decision after assessment.

Laboratory and clinical view. The actual specimen, result, report and professional action are what make those owner decisions meaningful. A laboratory can explain relevant tissue conditions and variant interpretations without asserting that the present HelixBridge snapshot has established them. Pathology staff can compare the studied display with what a pathologist would see and do. If a later field event arises, preserving its result state, denominator and workflow context keeps a technical discrepancy from turning into an assumed patient outcome.

Quality and regulatory view. Quality owners identify the controlled state and process behind a record. Regulatory owners identify the jurisdiction, applicable rule and decision. Design validation under an ISO standard, a U.S. quality-system duty, a European in vitro diagnostic performance-evaluation duty and this book's map can concern one product while doing different jobs. These two comparisons concern scientific evidence and product states. Neither assigns clearance, approval, certification or conformity.

A report from an unnamed product

The next product's modality is deliberately unspecified. This is a separate fictional case, unrelated to HelixBridge, its liquid branch, PathSight or Chapter 18. The table gives the complete supplied fact set so the judgment can be made from the same record throughout.

Part of the record	Supplied fact
Product role	The product examines a human specimen and returns a report to an ordering clinician for consideration in a patient-management decision.

Part of the record	Supplied fact
Recorded route	Collection at a referring site, transport to a central laboratory, preparation, examination in a controlled product configuration, report generation and return to the ordering clinician.
Earlier evidence	A prior evidence package is tied to a controlled configuration, but component identifiers and evidence results are not supplied.
Information from use	During routine use, a monitoring channel receives reports of unexpected invalid outputs.
Supplier information	Separately, a supplier discloses a material-process change for one supplied component; its timing, affected units, mechanism and relationship to the invalid outputs are not established.

The route row is read from left to right as a single specimen's journey: collection, transport, preparation, examination, report and clinician. It makes the physical and information paths visible without naming a modality. The other rows describe the evidence and later information available about that route.

Many clinical details remain unknown: modality, analyte—the thing examined—disease, specimen type, intended population, exact use or claim, and desired outcome. Component and release identities, particular users and sites, report count and denominator, prior invalid rate, and earlier study results and criteria are also absent. Cause, risk-control effect, jurisdiction and final owner decision are unresolved. Familiar facts from the named products cannot fill these fields.

Decision

Where do the five facts belong on the clinical-need, specimen-to-decision and lifecycle-return parts of the map? Which fields must remain unknown, and which evidence jobs may need review? What does receipt of the invalid-output reports establish? What does the supplier disclosure establish? What bounded next decision follows, without supplying an unstated modality, threshold, count, cause or field action?

The judgment before feedback rests on those five facts. Its most useful feature is the relationship it says can, or cannot yet, support the next decision.

How a reviewer handles the two later inputs

Transfer reasoning

The known route runs from human specimen through referring-site collection, central-laboratory transport and preparation, controlled but unidentified examination, report, ordering clinician and consideration in patient management. Clinical need, intended outcome, exact use, claim and many conditions remain unknown. The earlier package belongs with its own controlled configuration. Without identifiers, methods or results, its applicability to the current state cannot be decided.

The monitoring channel establishes receipt of reports of unexpected invalid outputs during routine use. The supplier disclosure separately establishes a material-process change. No chronology, affected-unit match or mechanism links those facts. Drawing a causal arrow would supply information the record lacks; ignoring either input would discard potentially relevant information.

The next questions concern the invalid-output definition and denominator; affected specimens, units and configurations; the component and its change date; whether changed units could have contributed to the observed outputs; and the implications for claim, performance, risk and earlier conclusions. These questions form a review path, not a universal study list. The bounded next decision is to **preserve the inputs separately, establish their scope and possible relationship, and assess affected prior conclusions before choosing further evidence or action**. Causation, performance, safety, compliance, authorization, conformity and field action remain undecided.

Reviewer Lens. Event traces and component traces can distinguish outcomes that a broad request for validation would blur. A response might show that changed material never entered the affected units. That could close the proposed causal route while the invalid-output signal is assessed on its own. A matching trace could instead require focused investigation. The supplied fact sheet establishes neither result, so the request should be designed to distinguish them.

Consultant Lens. Site and supplier owners can preserve their separate records and agree which units and time windows are comparable. The consultant helps define what finding would close or escalate each question. The laboratory owner handles current specimens and reports under its procedures; product and quality owners decide change and risk actions. Advice from these five facts can make those decisions ready for evidence, but cannot justify an invented field action.

Return to the report

The opening tissue report now carries a more precise question: which claim, product state and conditions does its evidence actually support? In the HelixBridge file, an identified analytical-validation view and a separate technical report with favorable overall agreement remain useful records. The reference-transcript difference, anomaly question, supplier update capability and unresolved commercial state have distinct places. Treating them separately preserves what is known while exposing the relationship still needed for transfer.

PathSight's unchanged model weights also retain their value, while image, interface and reader questions remain distinct. The two unnamed deployed services contribute a different kind of knowledge: observations with actual denominators and bounded result states. They add no release history to either named product. Across these examples the useful reasoning runs from **observation** □ **possible mechanism** □ **affected assumption** □ **discriminating evidence** □ **bounded conclusion**. Each link says more than “validated” or “needs more validation.”

The initial picture can now be reconsidered. Has the care problem stayed separate from the desired outcome? Does the proposed role identify the material, user and action? Is the claim an assertion that evidence can address? Which earlier assumptions have become explicit unknowns, and which supplied facts now support a conclusion? Revision makes the map more dependable without turning it into a performance verdict.

Chapter 2 takes the next step inside the configured product: exactly which physical, computational, data, interface, environmental and human components produced the report or assistive display? Later evidence can revise the map. Its value is that the relationship needing revision will be visible.

Questions that test the map

Could these questions now be answered in ordinary words, without relying on a folder title?

- Why are clinical need, intended outcome, use and claim different?
- What material path and information path lead from a specimen or image to a decision?
- Which five elements make an evidence-job statement useful?
- How can a changed component leave some earlier evidence intact while reopening another conclusion?
- Which configuration field remains unknown when only a product name is supplied?
- How does a monitoring signal differ from a finding about cause or adequacy?

The professional perspective adds one more distinction: what can a reviewer conclude from the record now, and what can a consultant help the responsible owners decide to do next?

Source trail and scope

The following locators support the general technical and regulatory propositions above. The case observations and teaching numbers are fictional; they do not establish technical or legal rules. Standards are paraphrased rather than reproduced.

- 21 CFR 801.4, first four sentences: U.S. intended-uses proposition. The eCFR is authoritative but unofficial; the provision is not an EU definition or complete evidence method.
- 21 CFR 809.3(a): U.S. IVD definition and its collection, preparation and examination span. It does not define this book's whole orientation spine.
- 21 CFR Part 820, §§820.1(b), 820.3(b), 820.7(b) and 820.10, with FDA's QMSR implementation page: current U.S. regulation, incorporation by reference and February 2, 2026 effective date. FDA's page is explanatory; the regulation controls. Checked September 24, 2026.
- Regulation (EU) 2017/746, Article 2(2), (12) and (37)–(44), Article 56, Articles 78–79, Annex I §3 and Annex XIII Parts A–B: EU IVD purpose, performance and continuing evaluation context. The 2025-01-10 consolidation is a navigation aid without independent legal effect; the base act and applicable amendments govern. No PathSight qualification or product conformity is inferred. Checked September 24, 2026.
- ISO 13485:2016(E), Edition 3, Introduction 0.1 and clauses 1, 7.1, 7.3.2, 7.3.6–7.3.9, 8.2.1, 8.4 and 8.5.1: bounded design, quality and feedback propositions. ISO edition record. Voluntary international standard; legal effect depends on jurisdictional incorporation or adoption.
- ISO 14971:2019(E), Edition 3, Introduction and clauses 1, 4.1, 9 and 10.1–10.4: bounded risk-management lifecycle and later-information propositions. ISO edition record. Voluntary international standard; it does not determine a product's performance or prescribe one response to every signal.
- ISO 18113-1:2022(E), Edition 2, clauses 1 and 3.1.37: paired use/purpose terminology within this voluntary IVD-labeling standard. ISO edition record. Its terms do not replace U.S. or EU law.
- FDA, Total Product Life Cycle for Medical Devices, “Total Product Life Cycle Approach”: nonbinding U.S. agency oversight description, not a required universal development order or this book's map template. Checked September 24, 2026.

CHAPTER 2

The Report Is a System Outcome

Start with what reaches a person

A reviewer is shown a report bearing the fictional HelixBridge Dx name. An initial product-package summary sits beside it. The report looks finished. Which study describes the arrangement that produced it? Before judging the findings, the reviewer needs a connection between this visible result and the material, programs, reference information and people behind it.

Chapter 1 gave that question a purpose: evidence supports a particular claim for a particular product, material, user and set of conditions. In an ordinary tissue sequencing route, a laboratory receives tissue, prepares it, extracts DNA—the material carrying genetic information—and reads many pieces with a sequencer. Software proposes variants, or DNA differences. Rules and reference information help determine what appears in a report; a professional reviews it before it reaches a clinician. That route explains why a study of one program might answer only part of the question about the report.

A digital-pathology route begins differently. Tissue is prepared on a glass slide, a scanner makes an image, and software may organize or mark it for a pathologist, who diagnoses. Here the visible result may include both a highlighted region and the case's place in the waiting queue. These are possible ordinary routes, not complete descriptions of either named product. Their value is to show what the report or screen leaves out.

A useful reconstruction follows the parts and exchanges that could matter to the selected output and claim. It identifies what is known and leaves missing relationships visible. A study can then attach to the arrangement it actually examined; a release or field record can attach to its own state. The resulting inventory is a working aid for evidence reasoning, not a regulator-issued form or an adequacy verdict.

Chapter 1's route ran from clinical need and intended outcome through proposed use and claim, specimen or image, configured examination, output, user and decision, with information returning from use.

Decision

Where does a study belong on that route when the exact configuration is unknown? Where would a specimen-handling condition belong? Which earlier conclusion might a reference-data or scanner change reopen?

How the map locates a missing relationship

Specimen handling belongs before measurement. A study belongs beside the state it examined, even if that state's detailed identity is still missing from the file. A later change sends a question back to conclusions that depended on the changed part. Locating the blank is already useful: it says which relationship needs evidence without inventing the missing configuration.

We can now open Chapter 1's configured-product box. “Validated software” will be easier to interpret once we can tell which software, material, conditions and output were examined.

Trace backward before judging forward

What would explain this visible result?

A report states a finding. A display presents a slide, an overlay highlights a region, and a worklist prioritizes cases. Each verb tells us something to follow backward. The same report logo can accompany a different reference record or reporting rule; a familiar image can reach a pathologist through a different scanner, display setting or queue rule. The useful starting point is the particular statement, presentation, highlight or order that needs explaining.

How far should that backward walk extend? Far enough to identify the material, equipment, software, data, active choices, exchanges, people and conditions that could affect the result or its use. This makes Chapter 1's **configured product** more precise: the working arrangement is bounded by the output and claim being examined. A nearby service belongs in this inquiry when its contribution could matter to the output; physical proximity alone gives no reason to include it. The resulting boundary helps reconstruct evidence. Whether something belongs legally to a regulated product, and whether its evidence or controls are adequate, require their own judgments.

That distinction gives a study a definite home. Before release, its observations describe the instrument, software, material and users it actually represented. A later released arrangement needs an identifiable relationship to that study. Chapter 1's two separate deployed services show why this remains useful in practice. In the unnamed tissue sequencing service, **two of 40 consecutive specimen attempts** had initial invalid

statuses before report release; case and run identities were retained, and no incorrect released variant result was reported. In the unrelated digital-pathology service, **one of 25 consecutive supported-image cases** arrived after a local worklist target; the image and event were retained, and the pathologist completed review. These records locate result-availability and reader-timing questions in their own released services. They supply no HelixBridge or PathSight configuration, cause or clinical consequence.

Reviewer Lens. The first useful record connects an output to the relevant state and time. If it does that clearly, the identity question can close even while a performance question remains open. Asking for every component in a building would obscure the relationship being judged. For an image, however, stopping at the model may stop too early: the scanner supplied its input and the display delivered its assistance. Enough detail means another reviewer can follow the consequential route, not that every available technical label has appeared.

Consultant Lens. Before a study, the product and site owners can agree which changes could alter one selected output and which controlled records would reveal them. That conversation gives the study a reproducible object. If it happens only after results arrive, the team may discover that an important state was never recorded. The consultant makes the options and closure condition clear; the responsible owners choose the product and study boundary. A successful reconstruction is one another person can repeat from the records.

What made the software's input?

The backward walk reaches a file, but the file began as physical work. In a possible tissue sequencing route, the laboratory checks the received specimen and its handling, extracts DNA, prepares fragments and uses an instrument to generate readings. The materials used in those steps are **reagents**. Included materials or conditions that help reveal whether a step behaved as expected are **controls**. If preparation leaves too little suitable material, an attempt may end with an invalid status instead of a variant report. A software version cannot identify the specimen requirements, reagent lot, instrument or handling conditions behind that attempt.

For a pathology image, the corresponding questions reach through the scanner to a thin tissue section, its stain and glass slide. The scanner's optics and settings affect what is captured; later processing and the display affect what the pathologist sees. A **whole-slide image** represents the slide digitally. It is not the physical slide. Even when a highlight is the feature under review, the image supplied to the software remains part of the explanation.

The retained records in the deployed notes make these questions practical. For an invalid sequencing attempt, the case and run identities offer a route back through receipt, preparation, instrument and software result states to where a report ceased to be available. For the late pathology case, the retained image and event offer a route

through arrival, queueing and presentation. These are ways to investigate; neither note has identified a faulty step. The completed pathology review remains a favorable observation while its arrival time is examined.

Which source can help a reviewer recognize the physical scope without supplying missing product facts? The U.S. Food and Drug Administration (FDA) describes a specified tissue-based sequencing tumor-profiling device type in 21 CFR 866.6080. Its device-description and report provisions span specimen handling, extraction, components, instruments, software, methods, thresholds, controls, external-information links and the report. This binding provision concerns its specified device type. It neither classifies HelixBridge nor defines every sequencing route.

FDA's final, nonbinding 2018 germline sequencing guidance offers a process-length comparison from collection through variant assessment and reporting. Its scope expressly excludes tumor sequencing and cell-free DNA testing. The comparison therefore cannot supply authority for HelixBridge's tissue or proposed liquid claims.

For the slide route, FDA's final, nonbinding 2016 whole-slide-imaging guidance connects tissue preparation, image acquisition and display within its specified surgical-pathology technical context. It helps identify which steps merit inspection. The actual PathSight chain and reader performance still need PathSight records.

Reviewer Lens. A precise software record can draw attention away from an unidentified input. Following the specimen and run, or slide and scanner, checks that the data being interpreted belong to the material and conditions under consideration. A documented material trace can settle that identity question. Its success should be credited without asking it to establish clinical interpretation; an incomplete trace should lead to a focused request for the missing connection.

Consultant Lens. Laboratory and pathology owners can decide early how specimen, slide, instrument and case identities will remain connected through a study and later use. The consultant helps select records that permit the trace, then brings the representativeness question to the relevant specialist. If a study assumes a material step is fixed, leaving that step unidentified until after the study creates an avoidable applicability question. Settling its identity first lets the team judge whether to hold conditions fixed, study variation or narrow the proposed claim.

Why can the same program produce a different result?

An ordinary transaction in a separate fictional metal-fittings factory makes the question visible. The warehouse dispatches a batch against an open order and sends a dispatch record to the accounting office. The invoicing program receives shipment and order identifiers and the shipped quantity. This arriving record is **software input**. No price or quantity value is supplied in the example; the point is what the program needs to receive.

The received record cannot explain the invoice by itself. The program already holds the open order and active invoice settings for this transaction. That retained or selected context is **software state**. It is different from the newly arriving dispatch message. To calculate the charge, the program also consults an approved price schedule maintained by the sales team. This is **external reference data**: information maintained separately and consulted during the work. “External” describes that relationship; the sales team can be inside the same company.

With those pieces available, the program matches the shipment to the order and calculates a charge. Those operations are **software transformations**. Yet a calculated amount does not automatically become an invoice the clerk may send. When the records agree, a rule offers a draft to the clerk; otherwise it routes the transaction for checking. This **criteria-based selection** determines what happens to the calculated result. Calculation and permission to advance have different jobs.

The draft presented to the clerk is **software output**. In this successful transaction, the clerk reviews and sends it, the office records it in a ledger, and an accountant later reconciles the invoice, shipment and ledger entry. Those human actions complete a route that the program's draft alone cannot establish. The office performs real calculation and selection; people retain release and reconciliation. The same program version could calculate differently if the consulted price schedule or active rule changed. That answers the opening question without assuming anything went wrong here.

These six software checkpoints are an instructional aid, not a mandatory execution order or a regulator's taxonomy. A correctly prepared invoice supplies no evidence about diagnostic computation, clinical interpretation or patient consequences. What transfers to a diagnostic product is the set of questions, each applied to that product's actual route.

For a generic tissue sequencing execution, the incoming data are readings with specimen and run identifiers. Active case settings and selected report options supply context. A separately maintained reference sequence or knowledgebase supplies comparison information; it is neither the specimen's own readings nor the application version. Software places short readings at likely positions on the reference, compares the positioned readings to propose DNA differences, and attaches descriptions to those candidates. The result at this point is an intermediate candidate list. These operations have done useful work, but there is no released clinical report yet.

What allows a candidate to advance? In a deliberately simple teaching rule, a candidate within the declared portion of examined DNA permitted for reporting goes to a laboratory professional for draft-report review; one outside that portion is held out. Applying the condition filters candidates. A rule may also use a specified decision point, such as a minimum quality value. These choices operate on already proposed

candidates: they cannot make an unmeasured specimen adequate or establish what a finding means clinically. Professional review and release occur later in the whole laboratory route. This rule is an illustration; it supplies no content or behavior for HelixBridge's package report rules.

A generic pathology execution asks the same questions of different material. A scanner-produced image and case identifier arrive. Active model, image-processing and worklist settings provide context. The software may divide the image into small regions, adjust their appearance, score each region for a learned pattern and combine those scores into a case-level value. The appearance adjustment is the normalization introduced in Chapter 1. A preset cutoff or ranking criterion can then select an earlier worklist position or a highlighted region. The score is intermediate; the priority and highlight reach the pathologist, who diagnoses. These image operations and their clinical meaning differ from mapping DNA readings. A particular image system may or may not consult external reference data; the category supplies no PathSight database or training-set fact.

The question at the end of either route is now sharper: which inputs, held settings, references, operations and choices accompanied the output the study actually examined? After release, the retained identities in the separate deployed notes permit the corresponding investigation. The sequencing attempts can be traced to where their route stopped before a report; the pathology event can connect image arrival, active queue choice and presentation time to the local target. Retained identities make those questions answerable in principle. They do not identify a cause or turn the late completed review into a diagnostic error.

Reviewer Lens. The persuasive software result is a connected trace: specimen-linked readings or an identified image, the state used, consulted references, intermediate result, choice rule and what reached the professional. A trace can establish that an output came from the studied state while leaving performance or meaning for another evidence job. The common overreach is to let a precise program label stand for all the other inputs. The factory's price schedule makes the gap easy to see; the diagnostic record must demonstrate its own equivalent relationships.

Consultant Lens. Before collecting evidence, laboratory, software and pathology owners can compare three practical approaches: freeze the output-relevant state, record the states that vary, or narrow the claim to the route that can be observed. Each has a different evidentiary reach. The consultant helps make that tradeoff explicit and defines closure as a repeatable output-to-state trace plus the relevant claim comparison. When a deployed signal later appears, that trace lets the service owners test a possible mechanism before choosing corrective work.

How did the result reach the right person?

A file can exist before the next program receives it, and an image can exist before the pathologist encounters it. The crossing between two parts of the route is an **interface**. For this inventory, a crossing matters when its content, direction, state, timing or reliability could affect the selected output or use. The useful question therefore continues beyond “which program?” to “what crossed, for which case, in which direction, and under what conditions?”

In a generic sequencing route, a prepared specimen becomes instrument readings, an analysis program receives a file, a professional receives a draft and a clinician receives a released report. A recognizable case identifier and a successfully delivered file address different relationships from the meaning of the finding. For digital pathology, slide-to-scanner and image-to-software exchanges precede software-to-worklist and worklist-to-pathologist exchanges. File format, identifying information, processing choices, display and queue behavior can each matter to what arrives. “Cloud” names a location or service arrangement; it does not describe the exchange.

The whole-slide-imaging guidance's technical model permits acquisition and display to be distinct and physically separated. That makes the file and presentation path worth tracing across locations; it does not prove remote combinations equivalent. The separate deployed pathology note shows why time belongs on this map. Review was completed, but arrival was after the local target. A record can locate that event without measuring diagnostic accuracy.

Someone performs each step in a setting. The people and organizations are the actors; the relevant equipment and workflow conditions make up the environment. These are concrete questions about who could send, receive, see or act on the information. When a supplier or remote service contributes to the result, its exchange remains relevant even if it operates outside the product team's offices.

An exclusion from the focal configuration therefore needs both a reason and an account of any dependency that remains. If another function supplies information the product relies on, excluding that function from one study does not explain away the input it supplies. FDA's final, nonbinding 2023 device-software submission guidance addresses interactions between reviewed functions and other functions in its U.S. premarket context. Its final, nonbinding 2017 interoperability guidance helps frame purpose, exchanged information, users and environment for electronic interfaces. Those questions orient this reconstruction; they do not turn it into a generic software-development or cybersecurity checklist or decide legal inclusion.

The entries also need their **provenance**: which package, study, release record or field observation supplies each fact. A study's exact configuration and a package's stated role can both be useful without becoming the same kind of evidence. “Unknown” means the

relevant identity or relationship has not been supplied. “Not applicable” needs a reason tied to the product and question. Neither is a substitute for an observed adverse result.

Configuration sheet: what arrangement is represented?

A **configuration sheet** brings the output-producing route together. For each position below, it holds the exact supplied entry, the kind of state and source it comes from, and why it could affect the output or what remains unresolved.

Position on the route	What the entry needs to distinguish
Visible output and use	What reaches someone, and the stated job it serves.
Specimen, tissue or slide	Material identity, preparation and handling conditions.
Instrument, scanner and equipment	The equipment and relevant operating state.
Software and active settings	A named version versus what was active for this output.
Data and outside references	The identity of separately maintained information used.
Processing and choice rules	What was calculated and what could advance or appear.
People and setting	Who participated, with relevant equipment and workflow conditions.
Exclusions and dependencies	What is outside this inquiry, why, and what still crosses the boundary.
Release, timing and applicability	Which study, use or release state is represented and how it relates to the proposed conclusion.

The factory's price schedule illustrates the reference row. The known entry is an approved schedule maintained by sales; its source is the teaching transaction. It matters because the program consults it to calculate the charge. An actual configuration record would need enough identity to retrieve the schedule used for the selected invoice. Nothing in the factory example supplies that identifier for any diagnostic product. The same columns can faithfully hold a precise fact or a precise statement of what is unknown.

Interface sheet: how did the parts participate?

The **interface sheet** asks about relationships. It is useful when the configuration sheet names two things but has not shown how they interact.

Question about the crossing	What the record distinguishes
Which endpoints?	A demonstrated exchange versus a relationship still to investigate.
Why and in which direction?	The purpose, sender and recipient.
What crossed?	Actual content and identifying information, with relevant format.
Under what conditions?	Active state, time, reliability and other context that matters to the question.
Who and where?	People or organizations and their operating or use environment.
What remains dependent or unknown?	Missing facts and assumptions retained across an exclusion.

For the factory's dispatch-to-accounting crossing, the example supplies direction and content: the warehouse sends shipment and order identifiers and quantity to the office. It does not supply transmission timing or reliability measurements. That row can be partly known without being a failed exchange. Similarly, a study that precisely identifies one product configuration earns a positive entry; an unidentified handoff calls for its own targeted question.

Reviewer Lens. Matched sending and receiving records for one case can settle delivery and identity while leaving interpretation open. The temptation is to ask a well-populated configuration list to carry those missing relationships. A narrow request for the actual crossing is often clearer: what left, what arrived, which case it described, and which timing or setting matters? If a service is excluded from the focal study, its output-relevant dependency still needs to be visible. The record format is secondary to whether the connection can be followed.

Consultant Lens. Owners can choose identifiers and event records before a study so a representative transaction remains traceable end to end. Comparing a direct handoff with an external-service route then becomes a practical design discussion: what can each owner observe, and what happens if the exchange fails? The consultant connects those owners and the evidence needed for closure. Legal inclusion and performance adequacy retain their own decisions; completing the two sheets cannot silently decide either.

A route instruction makes the six jobs visible

A route-guidance service receives an estimated location and current conditions, while retaining a destination and active route. It consults map and closure information, calculates candidate routes, chooses among them by configured criteria and communicates an instruction. A change in conditions can therefore change what the person hears without the arriving location being the only relevant information.

The technical names are **software input**, **software state**, **external reference data**, **software transformation**, **criteria-based selection** and **software output**. The navigation comparison explains these information roles only. Recalculation after a road-condition change does not model a software update or a validation decision, and it supplies no conclusion about specimen fitness, variant truth, image quality, performance or clinical use. Those questions return to the actual material, software and professional route.

Figure references: Route-guidance mapping of the six software checkpoints and technical trace of visible outputs through the bounded system. The first is limited to information flow; the second returns to material, software, interface and human routes before any case answer.

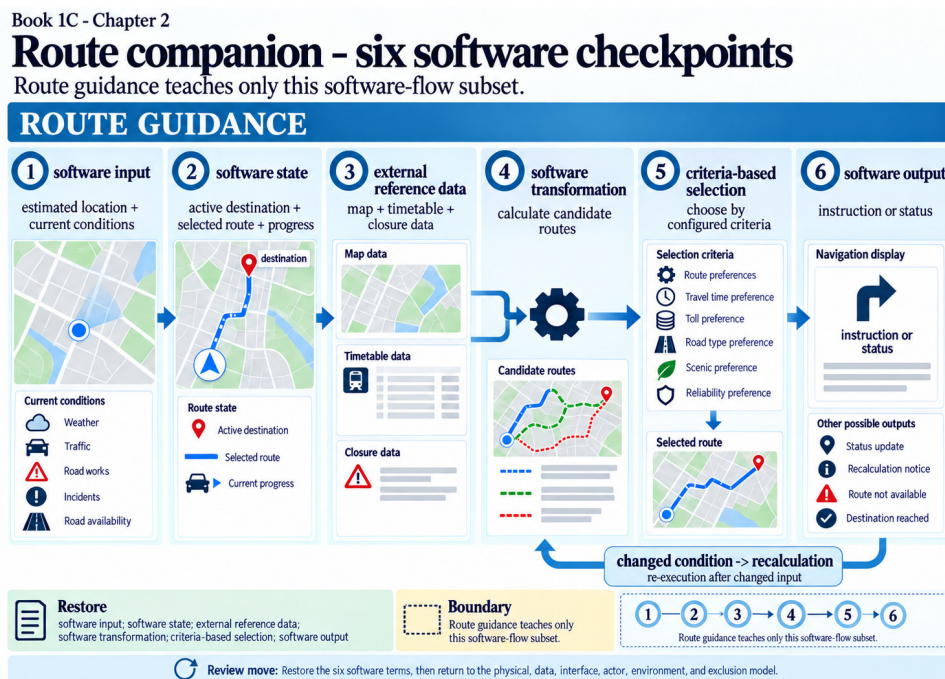


Figure 2.1. Trace the visible output back through the bounded system – Route-guidance transfer.

Book 1C - Chapter 2

Section 02 – Trace the output back to the system

The visible output is not the whole configuration.

Key takeaway

A visible diagnostic output depends on a bounded physical, computational, data, interface, and human configuration.

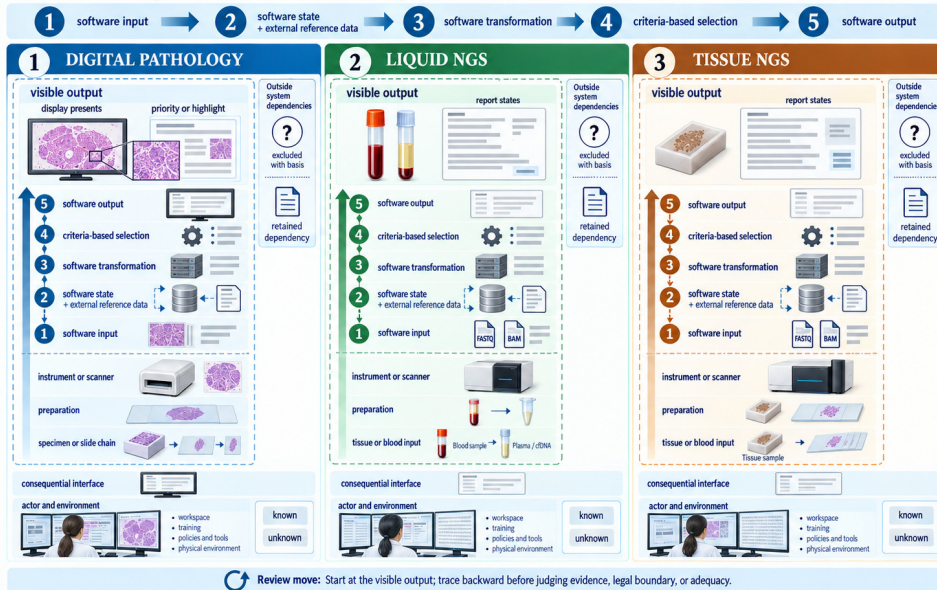


Figure 2.2. Trace the visible output back through the bounded system – Technical review plate.

Two package records to reconstruct

The generic routes have shown what kinds of relationships to look for. These two initial packages supply their own, more limited facts. The questions concern those snapshots alone; Chapter 1's study and change records are outside this exercise. A missing fact stays missing even if a familiar workflow makes one answer seem likely.

The package beside the HelixBridge report

The table contains the six supplied package statements. It gives the starting evidence for deciding what can be connected to the report shown in the opening.

Entry	Supplied statement
Source and kind of product	Initial-package view; HelixBridge Dx is a tissue-based oncology next-generation sequencing assay.
Stated purpose	Identify variants associated with eligibility for a named therapy.
Laboratory assay	Described as stable.
Analysis program	Bioinformatics application 7.0.
Reporting	Report rules R19.

Entry	Supplied statement
Outside reference	Cloud knowledgebase described as current.

For example, the first two rows locate the stated tissue-oncology role and treatment-eligibility purpose. The last four are entries about the laboratory assay, application, rules and reference information in that package. The report itself supplies no variant, therapy or component details in this exercise. The commercial target remains unidentified.

Two decision turns

Decision

Which of *stable*, the application version, the reporting label and *current* provide reproducible identifiers, and which provide descriptive wording? What relationship to the displayed report does each establish? What is the strongest conclusion about the unidentified commercial target? Does the reporting label supply a threshold or behavior?

PathSight Cloud's initial package says it assists pathologists by prioritizing whole-slide images and highlighting suspicious regions. The pathologist makes the diagnosis. It names model M3.2 and claims support for Scanner A and Scanner B, the first and second scanner families.

Decision

What is the strongest statement the package supports about the second family? What separate statement can be made about the scanner and complete state behind a particular priority or highlight? Which interface facts are directly supplied, and which need another record?

What would the two sheets contain?

For each case, what belongs in a configuration record, and what belongs in a relationship still to investigate? In HelixBridge's record, how do the six package statements differ from facts about the displayed report? In PathSight's record, which human role can already be credited? Which missing identities or relationships warrant “unknown,” and is there a reason for any “not applicable” entry? Where direction and content are absent, would an endpoint pair be a demonstrated exchange or a question? Finally, what maximum applicability conclusion could each inventory support, and which decisions would it leave open?

How a reviewer reconstructs both initial packages

The worked reconstruction

HelixBridge: preserve the package's value and its limit

The tissue modality, purpose, application version and reporting label are useful facts about this package. “Stable” and “current” faithfully reproduce its wording but cannot identify a particular laboratory setup or retrievable knowledgebase version. The other problem is relational: even an exact package label has not been connected to the displayed report or commercial target. These gaps concern identity and applicability; the exhibit has supplied no adverse performance result.

This configuration sheet separates a supplied entry from its source and from the relationship still needed. That separation makes the package usable without filling its blanks from the generic sequencing route.

Part of the route	Supplied entry and source	What remains unresolved
Visible output and use	A HelixBridge-branded report is shown. The initial package states tissue-variant identification associated with eligibility for a named therapy.	Report content, population, variant classes, treatment-facing role and the package-to-report link.
Specimen and preparation	The initial package states tissue-oncology sequencing and calls the laboratory assay “stable.”	Exact laboratory setup, preparation, conditions and their relation to this report. The adjective supplies no component identity.
Instruments and equipment	None identified in the initial package.	Which instrument and state supplied the readings for this report.
Program and active settings	The initial package names analysis application version 7.0.	Settings, use in this report, and inclusion in the commercial target.
Data and references	The package describes a cloud knowledgebase as “current.”	Retrievable reference identity, access state and time for this execution.
Processing and reporting choices	The package names a report-rule set; its content is not supplied.	Thresholds, filters, reporting behavior and any link to this report.

Part of the route	Supplied entry and source	What remains unresolved
People and setting	No laboratory actor, site or environment supplied.	Who performed, reviewed or received the result, and under what conditions.
Excluded records and dependencies	This exercise contains no later study, change event or supplier role.	Exclusion from this snapshot does not establish absence from the product. Later records need their own state.
Release and applicability	An initial-package view and an unidentified commercial target.	The report-producing and commercial configurations, release timing and connection from package to output.

The program row contains an exact label, so a reviewer can distinguish that package entry from an unspecified application. What is still needed is evidence that it was active for this report. The reference row has an earlier problem: another reviewer cannot retrieve a specific knowledgebase from “current.” One gap needs a link to execution; the other also needs a reproducible identity. Treating both as “more validation needed” would hide this difference.

The interface sheet goes one step further. Its pairs of endpoints identify questions about possible exchanges. They are not asserted arrows in HelixBridge's architecture. Each row retains the direction, content, state, timing, reliability, people and setting questions that the package cannot answer.

Relationship to investigate	What the package establishes	Missing exchange facts and retained question
Laboratory assay □ package application	Both are listed in one summary.	Whether the exchange exists, its purpose and direction, readings or file content and case identity, execution/report time, state and reliability. Laboratory people and environment are unknown. What input and conditions reached the target application?
Package application □ knowledgebase called “current”	Both entries appear in the package.	No query or copied reference is established. Direction, content, reference identity, access state, time and reliability are unknown, as are service and laboratory actors. How was reference information used or updated?

Relationship to investigate	What the package establishes	Missing exchange facts and retained question
Package application/report rules □ displayed report	The package labels and displayed report are available.	No relationship, direction, rule content, intermediate result or report payload is supplied. Active state, time, reliability, reviewer and laboratory environment remain unknown. Which components and rules, if any, governed this report?

The middle row is especially easy to overdraw. A cloud knowledgebase beside an application label may suggest a query, but the package has supplied neither a query nor a copied reference. The honest row preserves both endpoints and the question about their relationship. A later report-linked record could turn particular unknowns into known entries while retaining the current package evidence.

The strongest conclusion is therefore a useful partial reconstruction: the application and reporting entries are exact package labels; “stable” and “current” are descriptions; none of the four is established as governing the displayed report or unidentified commercial target. The package's value survives that limit.

PathSight: a known role with an unresolved image route

Here one human role is explicit. The pathologist receives assistance and retains diagnosis. The package also supplies a model label and support claim for two scanner families. That gives the inventory a useful starting point without identifying a particular scanner, processing state or display, or providing a performance result for the second family.

The configuration sheet uses the same distinction between a supplied fact and the scope of that fact. A stated role belongs in the record even though no study of that role has been supplied.

Part of the route	Supplied entry and source	What remains unresolved
Visible output and use	Initial package: whole-slide priorities and suspicious-region highlights assist a pathologist; the software does not diagnose.	A particular displayed instance, reader response or worklist effect.
Tissue, slide and image	Whole-slide-image context in the initial package.	Particular tissue, stain, slide preparation, handling and image-production conditions.

Part of the route	Supplied entry and source	What remains unresolved
Scanner and equipment	The package claims support for the first and second scanner families.	The actual scanner, its internal software and color settings, and evidence applicability to the second family. A support claim is not observed performance.
Software and active settings	The package names a model.	Division into image regions, appearance adjustment, settings, interface, queue rules and exact output-producing state.
Data and references	No external reference is identified in this snapshot.	Whether one is used and its identity. No database or training set may be inferred from a generic image-system description.
Processing and choices	Prioritization and highlighting are stated functions.	Actual scoring, cutoffs, ranking, combining processed scores, highlighting and display criteria.
People and setting	Pathologist is the stated user and diagnosis owner.	Site, scanner operator, display and worklist environment.
Excluded records and dependencies	No study, later changed state or release event belongs to this exercise.	Later facts cannot turn this package's support claim into an observed performance result here.
Release and applicability	One initial-package view.	Exact commercial or output-producing configuration, release state, and generalizability to the second scanner family.

The people row can already say something positive: the product supports a pathologist who makes the diagnosis. The scanner row answers a different question. Its support statement identifies a claimed family, not the equipment and conditions behind one image. Knowing the first relationship does not fill the second, and the uncertainty in the second does not erase the first.

That distinction gives the interface sheet one known direction at the level of the claimed role. It still leaves the actual image's display content, arrival time and reader action unknown.

Relationship	What is supplied	Missing exchange facts and retained question
Claimed scanner support □ PathSight processing	The two-family support claim identifies a relationship to investigate.	Purpose and direction, image format, identifying details and processing handoff. Scanner internal software, color settings, state, time, reliability, operator, site and technical environment are unknown. Which route produced this image?
PathSight assistance □ pathologist	Direction and role-level content: priority and suspicious-region highlight for the user who diagnoses.	Exact display content, interface, queue, timing, reliability, case state and site. Reader behavior and workflow performance need their own evidence.
Package model □ priority/highlight function	Model label and functions appear in the package.	The implemented relationship, intermediate score, cutoff or other content, active processing and selection choices, time and reliability. Software/service environment is unidentified. Which function used this model, and how did its result reach the display?

Reading the middle row shows why the whole sheet should not be called unknown. The direction toward the pathologist and the claimed kind of assistance are supplied. What remains missing is the case-specific exchange and what happened in use. By contrast, the first row cannot establish that the second scanner family produced a particular image or performed successfully in a study.

The strongest sentence about that family is: **the initial package claims support for the second scanner family**. Its generalizability remains an evidence question. FDA's surgical-pathology technical model can suggest component types to inspect, but cannot populate a PathSight-specific cell.

What would resolve the next decision?

Reviewer Lens

The worked records deserve credit for what they identify: HelixBridge's exact package labels and PathSight's assistive role, user, model and scanner-support claim. The next request follows the particular missing link. A selected report or image needs a trace through material, active components, settings, references, exchanges and time, followed by comparison with the state represented by the evidence. An empty field matters when the claim depends on it. The two separate deployed notes would be judged against their own released services; their retained records cannot fill these named-package blanks. That separation allows a narrow adequate entry and an unresolved applicability entry to coexist.

Consultant Lens

The company's next options depend on why the relationship is missing. If the intended state is settled but poorly described, product and laboratory owners can establish a reproducible baseline and link it to a report or image. If the state is still changing, they can freeze the study configuration, narrow the claim to the observed route or plan a comparison with the proposed state. An external-service owner can document an exchange and active version while quality and regulatory owners determine their respective control and legal questions. Closure means a traceable state and justified evidence applicability; it is more than filling every cell.

For PathSight, image and pathology owners can determine whether the second scanner family's conditions are represented in planned work or call for a separate comparison. The support claim cannot choose between those options. For HelixBridge, knowledgebase and report owners can identify which reference was “current” at report time and whether the package report rules were active, then compare that arrangement with the studied one. These are prospective choices with owners and dependencies. The supplied records promise no particular comparison will suffice.

Laboratory and clinical view

The people handling material and reviewing outputs can expose handoffs a software diagram misses. The sequencing laboratory can connect specimen and run identities through preparation, analysis, review and release. The pathology service can connect an image to the reader and the reader's action. This is why the separate deployed notes retain value: an invalid attempt can be investigated without alleging an incorrect released variant result, and a late image can be investigated while completed review remains credited. Human participation completes the route; it does not make every component or performance claim known.

Transfer to an external analysis service

A separate, unnamed report-producing product sends an input package to an analysis service operated by another organization. The service returns a result payload used to compose the visible report. The record does not identify the input content, the service's active software or configuration, any external reference data, transformations or selection criteria, or the exact returned payload.

Decision

Which supplied fact makes the service relevant to the output? What would an interface record need to establish about input, active state, references, processing, choice rules, return, people, environment, timing and reliability? What is known about the fallback if no usable result returns? What do external ownership and remote location establish about legal product inclusion, responsibility, adequacy or security? What bounded conclusion can a reviewer support now?

How a reviewer follows the service's contribution

The return payload supplies the connection

The service belongs in this report's evidentiary trace because its returned payload helps form the visible result. Unlike the uncertain endpoint pairs in the package exercises, this record supplies two directions: product to service for the input, and service to product for the return. Actual content and format, active state, timing, reliability, people and environment still require records. The service's application, settings, references, transformations and choice rules matter because they can affect the returned content.

Availability introduces another question: if the service fails or cannot return a usable result, whether and when the report can be composed may change. The actual fallback is not supplied. External ownership and remote location neither remove the service's contribution nor settle its place within a regulated product or the parties' legal duties. The bounded finding is that its interface and dependencies belong in the evidence trace of this report. Legal inclusion and control requirements need the product- and jurisdiction-specific basis; tracing the returned payload does not make the whole external organization part of the device or establish security or performance adequacy.

Reviewer Lens. A report-linked transaction record can show what was sent, which service state was used and what came back. If it connects a known service configuration to the payload used in the selected report, the trace earns credit. The next question is whether the relevant study applies to that state. A missing vendor detail merits a

request when the conclusion depends on it; it does not supply an unsupported failure or legal finding.

Consultant Lens. Product and service owners can agree before use on exchanged data, version and reference identity, correct specimen association, and behavior when results are late or unavailable. A fixed route can then be observed in a study. Where it remains unsettled, owners can defer the claim, constrain the service route or define a targeted comparison after specifying the interface. The consultant coordinates these dependencies and the evidence closure condition. Quality, regulatory and security specialists determine the duties within their own remit.

Return to the visible result

The report in the opening still bears the same name. The useful question is now specific: what connects it to the material route, active program, reference information, reporting choices and people that produced and used it? The initial HelixBridge package supplies some exact labels and some descriptive wording, while the report-producing and commercial relationships remain unresolved. PathSight's initial package identifies assistance and its intended professional, a model and a scanner-support claim, while its particular image-to-display state and second-family performance remain unknown. Neither reconstruction requires an invented failure.

The two sheets place a study, release record or field observation beside the arrangement it actually describes. Material becomes data, programs transform it and choose what advances, interfaces carry it, and people receive and use it. Chapter 3 asks what someone means by calling one of those objects *verified* or *validated*.

Retrieval check

What makes a configuration bounded? How do the six software checkpoints differ, and why do the material and human routes still matter? Why is an application version different from a reference-data identity? What can a support claim for the second scanner family establish without an image trace or study? How could an excluded external service still affect a report? Where would one consequential interface belong on Chapter 1's lifecycle route, and what known facts and unknown relationships would its row contain?

Source trail and scope

This chapter's two-sheet inventory and six software checkpoints are instructional syntheses, not prescribed templates or normative taxonomies. The fictional package records and deployed notes supply teaching facts only. The sources below support the bounded general propositions indicated; each keeps its own authority and scope.

- 21 CFR 866.6080, paragraphs (a), (b)(1)(ii)(B), (b)(1)(iii)(B)–(F) and (b)(2): binding U.S. provisions for the **specified tissue-based NGS tumor-profiling device type**. They illustrate the material, software, reference, control and report span; they do not classify HelixBridge. The eCFR is authoritative but unofficial and displayed Title 21 current through September 22, 2026 when checked September 24, 2026.
- FDA, *Considerations for Design, Development, and Analytical Validation of NGS-Based IVDs Intended to Aid in the Diagnosis of Suspected Germline Diseases*, final nonbinding U.S. guidance, April 13, 2018; Scope, printed pp. 4–5, and Elements, printed pp. 6–7. It is used only to contrast a possible NGS chain. It expressly excludes tumor sequencing and cell-free DNA testing, so it is **not authority for the HelixBridge tissue or liquid product**.
- FDA, *Technical Performance Assessment of Digital Pathology Whole Slide Imaging Devices*, final nonbinding U.S. guidance, April 2016; Sections II–IV(A), printed pp. 2–3. Its surgical-pathology technical model supports tracing slide, acquisition and display, including separated subsystems. It supplies no PathSight configuration or clinical-performance result.
- International Medical Device Regulators Forum (IMDRF), *Software as a Medical Device (SaMD): Clinical Evaluation*, final nonbinding international regulator-forum document, September 21, 2017; Section 6 and Figure 9. It is one constituent example of input, reference information, algorithms and output in software; it does not define this chapter's six-part teaching framework or create an in vitro diagnostic rule.
- FDA, *Content of Premarket Submissions for Device Software Functions*, final nonbinding U.S. guidance, June 2023; Sections VI.D–VI.E, printed pp. 18–22. Its software inputs, outputs, requirements and architecture relationships inform the inventory and retained-dependency questions. This chapter does not apply its software-engineering recommendations as a generic diagnostic-product checklist.
- FDA, *Design Considerations and Pre-market Submission Recommendations for Interoperable Medical Devices*, final nonbinding U.S. guidance, September 2017; Sections IV–V, especially V.A–V.C, printed pp. 4–10. It informs purpose, exchange, timing, metadata, users and environment questions for electronic interfaces, without deciding the legal boundary or evidence adequacy of a particular product.

CHAPTER 3

One Word, Several Evidence Jobs

The decision hidden inside one word

A pathologist receives a prioritized slide image with suspicious regions highlighted. PathSight Cloud supplies that assistance; the pathologist examines the tissue and makes the diagnosis. Now imagine reviewing two short study accounts about this fictional product. What would each account let you say about the work in front of that person?

The first asks whether the model identified in validation met a main performance goal chosen before images from several sites were analyzed. The source reports that it did. A separate reader study reports improved average sensitivity without unacceptable loss of specificity. These are two favorable reports, each answering a different question.

The reader result needs a little unpacking. **Sensitivity** concerns recognizing findings the study's reference process designated positive; **specificity** concerns avoiding flags for those it designated negative. The exact units, reference process, workflow and calculations are not supplied. We know the direction of the reported reader result, but cannot yet inspect how it was calculated or judged.

Suppose a cover note compresses both accounts into “**PathSight was validated.**” A colleague might read that as a software conclusion, another as evidence that pathologists can use the interface safely, and another as support for every claimed scanner. The same sentence would then authorize three different inferences without showing the relationship behind any of them. Discarding the studies because the sentence is vague would lose their favorable reports. Extending those reports to every interpretation would add evidence that has not been supplied.

The useful next step is to recover the question behind the word. What was examined, what would count as success, and what actually happened? Whose framework gives the term its meaning, and which use could the result support? Those questions let a positive result stay positive at an honest scope. They do not supply PathSight's European legal classification, second-scanner support, regulatory outcome or relationship to an exact commercial configuration.

A compact **evidence-job glossary** can hold the repaired statements: up to eight entries, small enough for two pages, each connecting a question to its source and limit. Its value is practical. A colleague should be able to see which decision an entry informs and which decision remains open, without mistaking the glossary for a universal dictionary or a specialist test protocol.

Retrieve before you rename

Chapter 2 left us with configurations and exchanges behind a visible result. What distinguishes a product name from that bounded configuration? Which state did a study examine, and what would connect it to a proposed commercial state? If an adjacent service is excluded from the focal product description, which output-relevant exchange remains visible? Why can a titled document be **present** without its evidence being **applicable**? How did Chapter 1's five-part evidence statement keep those relationships together?

How the configuration becomes part of the evidence question

A name can identify a product while leaving its active material, software, data and user conditions unknown. A study attaches first to the state it examined. Its relationship to another state needs evidence, including any consequential exchange with an excluded service. A study title cannot establish that relationship.

Chapter 1's form was **object + question + configuration and conditions + lifecycle position + supported conclusion**. The two additions here are the basis for judging the observations and the discipline or jurisdiction supplying the terminology. A Chapter 2 inventory entry therefore provides the object, source and unresolved applicability for an evidence-job entry; it does not become a ninth job merely because it has its own row.

The configuration tells us what could have produced a result. The next question is what that result can establish. The reasoning runs in this order:

claim → configured object → qualified question → planned criterion → observation → bounded decision → use or reopened question

A development check can establish a specified behavior in one build. A study can examine performance for a use. A laboratory can investigate its own implementation, and a released service can supply an observation that reopens a question. The position in that sequence changes what is being decided. Detailed software, security and conformity methods remain with their specialist sources; here we need enough precision to choose the right question.

Start with the relationship, then check its source

Consider a generic digital-pathology worklist, separate from PathSight. Its requirement says that an image marked urgent by an authorized rule must appear in a designated high-priority queue. In a staged test, the rule places the image there. That observation answers a useful question: did the implemented behavior meet the prior requirement? It leaves another question for actual users, interface and workflow: can pathologists act as intended with the resulting system? A successful placement deserves credit even while that use question remains open.

The first relationship is **verification**: evidence that specified requirements were fulfilled. The second is **validation**: evidence that requirements for a specific intended use or application were fulfilled. The International Organization for Standardization supplies these general quality-management relationships in ISO 9000:2026. They help us orient the work; the words alone identify no requirement, condition, observation, criterion or configuration. A tissue-sequencing filter can likewise implement its declared rule correctly while the larger assay still needs evidence about report meaning and use.

Which edition supplies those words matters in the same way that a named software version matters. The current general vocabulary is the 2026 edition; the 2015 edition has been superseded. Yet the U.S. **Quality Management System Regulation**, effective February 2, 2026, incorporates ISO 13485:2016 and **Clause 3 of ISO 9000:2015** through 21 CFR 820.7. The incorporated medical-device standard also has its own dated reference to that 2015 vocabulary. A dated incorporation stays attached to its named edition when a newer general edition appears. Substituting the newer clause number would obscure the source being applied; identifying the correct edition still leaves product evidence sufficiency to be judged.

Reviewer Lens

A reviewer gains more by asking what the staged worklist check closed than by debating whether its cover sheet chose the best word. Its positive result can stand at the tested scope. The intended-use relationship remains a separate inquiry. When a writer invokes a standard or U.S. rule, the edition belongs beside that inquiry, because apparently conflicting vocabulary may actually come from different dated sources.

Consultant Lens

The inexpensive decision is the one made before a team collects evidence: which product decision is pending, and which source supplies its criteria? Renaming a completed report later cannot recover an unstated intended use. The specialist owns the framework and criteria; the consultant helps preserve a requirement-to-result trace that another team can understand without guessing the object, edition or intended use.

Make the seven fields one reasoning sequence

A separate, deliberately invented tissue-sequencing teaching record says, “**The release-block function passed.**” The team wanted an initial invalid result to offer no report-release action. What would make “passed” more than a status on a screen? Following this one check will give the glossary its seven fields.

The received statement is only “The release-block function passed.” Keeping that wording visible prevents an unnoticed enlargement to “the assay passed validation.” Before interpreting it, we need to find what the team actually tested.

The record identifies a release-block function in a teaching build, tested with a staged initial invalid-result condition. That fixes the **evidence object**: one function under those conditions. Chapter 2’s configuration trace would supply the corresponding version and active state in a real file. It would not make the specimen, whole assay, released report and deployed service interchangeable objects.

With that object fixed, the question becomes answerable: when the staged result is invalid, does this implemented function withhold the report-release action as specified? Whether every intended user can safely resolve an invalid result would require a different inquiry.

The plan supplies a **criterion**: the release action must be unavailable for the staged invalid result. The test then supplies an **observation**: the screen withholds the action and records the event. Comparing that observation with the prior criterion supports the team’s **decision** that the specified behavior was met in this test. This plan–observation–decision relationship is the **acceptance logic**.

The event record is part of what was observed; the stated acceptance criterion concerns withholding release. Keeping the two descriptions separate prevents the report from silently inventing another planned criterion. If the screen had instead offered release, the stated criterion would not have been met. Rewriting the criterion afterward would not change what happened. Other invalid paths, anomalies and the full configuration would still need inspection for a broader conclusion.

Who gives the word “passed” its authority here? This is an internal, illustrative product-software requirement check. No jurisdiction, standard or guidance is stipulated as controlling it. A real record would identify the actor and distinguish local wording, nonbinding guidance, a voluntarily adopted standard and a binding rule. The same observed behavior could inform several records, but it would not make their authority identical.

The check also belongs at a particular time: development of the identified build. A changed rule, interface or build would reopen the relationship to that result. A later invalid attempt in use would instead be a new observation to investigate; it could not retroactively tell us whether this earlier staged check passed.

We can now repair the statement: *In the identified teaching build, the release-block function met its specified criterion for the staged invalid-result condition observed; applicability to other invalid paths, the full assay and deployed use is not established by this check.* It is a positive conclusion with a visible stopping point. The example supplies neither a HelixBridge rule nor an explanation for the two initial invalid attempts in the separate deployed sequencing note. It says nothing about how often invalid attempts occur in use.

The seven fields have emerged from following the check: received statement, object, question, acceptance logic, discipline and jurisdiction, lifecycle position, and scope with the nearby conclusions that do not follow. “Not supplied” and “applicability not established” remain useful entries when a real record cannot complete that chain. They expose the next relationship to resolve without converting an unknown into a result.

Reviewer Lens

The hard part of reading “passed” is often resisting a request that is too broad. For this check, the planned release rule and observed screen/event can resolve the stated software question. A demand for whole-assay validation would change the question. An experienced reviewer can close the local check and identify separate owners for the remaining invalid routes or intended-use evidence.

Consultant Lens

Before testing, the software owner can make the intended invalid condition and no-release criterion explicit. Afterward, keeping the observation separate from the team's judgment makes either a pass or a disagreement inspectable. The consultant's useful handoff connects all three to the same build, then sends broader assay and user questions to their owners. One green status would hide that division of work.

Follow the image into the next question

The release-block check answered one question about a function. To see where the other jobs enter, follow a generic image toward a pathologist. The examples that follow explain possible evidence relationships; they add no studies or controls to PathSight or HelixBridge.

Does the built route meet its requirement?

An accepted image needs to keep its case identity as it enters the worklist. Suppose the manufacturer made that a prior requirement. As in Chapter 1, that prior requirement is a **design input**; the implemented image-to-case link and worklist display specification are corresponding **design outputs**. A representative link and display can then be checked against the requirement. This is the design-and-development verification relationship in the medical-device quality standard, ISO 13485:2016 Clauses 7.3.6–7.3.7.

A correct link still leaves the manufacturer's proposed use to examine. Can a representative resulting system serve that use with its intended images, users and workflow before release? That is the separate design-and-development validation relationship. User and performance evidence may remain necessary after the link check passes. The standard's adoption or incorporation determines its legal effect; its vocabulary does not decide a fictional product's status.

A report-field requirement in the ordinary sequencing route works the same way: a rule produces fields that can be compared with the requirement, while the use of the resulting report also depends on the specimen path, meaning and users. The design owner needs input, output, criterion, result and representative configuration together. If a later worklist case arrives late, that trace helps locate the reopened question: timing requirement, interface behavior or intended workflow. The separate anonymous pathology note records such an event without telling us which relationship caused it.

What did the software itself establish?

Following the link inside the product brings us to a software-stage question. Did the model pass the scored image and case identity to the intended worklist? Did a sequencing filter apply its declared rule? In the U.S. Food and Drug Administration's (FDA's) nonbinding 2023 device-software submission guidance, comparing one development-stage output with that stage's input is software verification.

The same guidance gives software validation a wider use relationship: specifications must conform to user needs and intended uses and be fulfillable in actual or simulated use, including integration in the final device when appropriate. A build identifier and count of completed tests cannot show that relationship alone. The useful record connects requirements, expected and observed results, anomalies, integration state and environment.

Our invented release-block observation fits the narrower software question: one implemented requirement was met. Nothing in that result establishes the analytical correctness or clinical meaning of a variant. Similarly, a worklist interface can meet its software input requirement while the usefulness of its priority and its effect on pathologist work remain to be examined.

Would the route resist an unauthorized action?

A normal route working as specified raises a further question: what happens when an action is attempted without permission? Return briefly to an accounting office. A normal invoice test might show that an authorized clerk can send the invoice. A security test instead tries the action with an account lacking that permission. Its planned control would refuse the send and record the attempt; the tester still has to observe whether that happens.

The diagnostic counterpart could attempt to substitute an altered report-rule file or use an unauthorized account at an image service. These are illustrative challenges, with no claim about the named products' controls. Their purpose is to challenge a **security control** against a specified threat, including possible effects on access, integrity or availability. A valid-input test would leave that adversarial question unanswered.

The record therefore needs the system boundary and version, threat assumption, expected control behavior, challenged condition, observed result and remaining risk. FDA's nonbinding February 2026 cybersecurity guidance connects this risk and test work with design evidence while preserving its particular question. Security evidence can continue from design through end of support. The invoice comparison explains a test mechanism; invoice correctness establishes no diagnostic computation or patient-safety result.

Reviewer Lens – design, product software and security

The design, software and security records meet at the same route, which makes it easy for their conclusions to blur. A case-identity requirement, an implemented software handoff and a control challenged by an unauthorized action each need their own expected-to-observed comparison. The reviewer can credit a passed requirement while leaving representative-product use or residual risk open. The anonymous sequencing note's retained case and run identities offer a starting point for investigating a release path; they establish no named-product software or security failure.

Consultant Lens – design, product software and security

Shared build and interface identities make these records easier to connect before the studies begin. The design owner chooses representative use, while software and security owners set their specific conditions and criteria. If a worklist event later arrives late, the trace helps the consultant route it after inspection to the relevant timing, software or use owner. Routing by the word “validation” alone could send the investigation to a team whose evidence cannot answer it.

Can the person use what arrives?

Even an intact, correctly linked image leaves the pathologist's interaction to examine. Can the person see which image has priority and understand why? In the sequencing route, can the reviewer distinguish an invalid result from a draft that may be released? A technically correct interface can still invite a consequential use error.

Human-factors validation addresses this interaction through intended users, use environments, critical tasks and use-related hazards. It observes what people do and whether the relevant risk controls work. FDA's nonbinding August 2026 human-factors guidance places validation testing near the end of development in that use-risk context. The resulting task conclusion can inform design validation while keeping its own criteria and boundary.

The fact that a study includes readers does not tell us that it observed these tasks. PathSight's supplied reader summary concerns average sensitivity and specificity; it supplies no critical tasks, user groups, use conditions, use errors, hazards or residual-risk analysis. A successful task observation would have the reverse limit: it would not establish diagnostic performance by itself.

The separate deployed pathology note illustrates why interaction and timing remain live after introduction. One of **25** supported-image cases reached its pathologist after a local worklist target; the pathologist completed review. That event identifies a timing question for its own service. It establishes neither cause nor clinical error nor a PathSight human-factors result.

Reviewer Lens

A reader study's title can make the user question feel answered. The more useful first comparison is between what the study actually observed and the task claimed to be controlled. Intended users performing specified critical tasks under relevant conditions, judged against planned criteria, can support a bounded use conclusion. A timing record alone needs a further relationship to the interface or task before a reviewer can assign cause.

Consultant Lens

The intended users and consequential tasks are worth deciding while the interface can still change. User-research, clinical and risk owners then have a common basis for observations and closure. The consultant coordinates that plan with software and workflow records; the owners decide whether a use risk closed, needs redesign or needs further observation. An interface or workflow change sends the question back to whichever conclusion depended on it.

What makes the result meaningful for its clinical purpose?

A user reaching and understanding an output brings us to its intended clinical purpose. For a tissue-sequencing test, reading a variant correctly leaves open why that marker matters and how the device's results relate to the target clinical setting. Where the European Union's In Vitro Diagnostic Medical Devices Regulation (IVDR) applies, these relationships are brought together in **performance evaluation**: planned assessment and updating of the relevant data throughout the lifecycle.

The material or marker being examined is the **analyte**. Starting with that concrete object makes the regulation's three pillars easier to distinguish.

First, what connects the marker with the clinical condition or physiological state? That association is the question of **scientific validity**. A correct measurement alone cannot establish it.

With that proposition identified, can the device correctly detect or measure the analyte? This is **analytical performance**. It concerns the device and the material it examines, so its conditions and criterion need to remain attached to the result.

Then, where applicable, how do the device's results relate to a clinical condition or physiological or pathological state in the target population and for the intended user? That is **clinical performance**. Moving from a correct variant reading to this conclusion requires the relationship to clinical use to be shown.

One dataset may contribute to several pillars if separate criteria and conclusions connect it to each question. The integrating evaluation still reaches beyond one study titled “validation,” and it receives postmarket information as well as premarket reports. Any claim that a clinical-performance element is inapplicable depends on the regulation's actual conditions. “Software only” cannot supply that analysis.

Software also requires a prior qualification decision. The nonbinding Medical Device Coordination Group guidance MDCG 2020-1 describes a valid clinical association or scientific validity, followed by **technical performance** for software under the EU Medical Devices Regulation or analytical performance for software under the in vitro diagnostic regulation, and clinical performance. Intended purpose and legal qualification determine which branch applies. PathSight's name and assistive image role do not resolve that qualification. Its two favorable study summaries therefore cannot be silently recast as a completed conclusion under either regulation.

Reviewer Lens

The reviewer first needs the qualified intended purpose because it determines which performance question the record is answering. A credible analytical finding remains credible for its analyte and studied conditions, even if association or clinical performance remains open. For a pathology summary, a focused question about the relevant branch is more useful than forcing the study into a familiar regulatory heading. The supplied PathSight facts establish neither that branch nor support for the second scanner family.

Consultant Lens

Regulatory and clinical owners determine qualification and intended purpose; analytical and clinical specialists choose the pillar questions and criteria. If that work happens early, a consultant can show where one study may support several propositions and where another source is needed. A living evidence map retains favorable findings, unfavorable findings, gaps and update owners. Closing one supported proposition does not require pretending every other proposition closed with it.

Why can another source call software processing “analytical validation”?

Suppose the immediate object is a qualified medical-device software program receiving image inputs and producing scores. The question “does it reliably produce the expected scores?” is about processing. Another question asks whether those scores serve the proposed clinical role for intended users. In the International Medical Device Regulators Forum's N41 framework, **Software as a Medical Device (SaMD)** uses *analytical validation* for correct, reliable processing of inputs into accurate, repeatable and reproducible outputs, and *clinical validation* for outputs achieving the intended purpose in the target health-care situation.

The corresponding hypothetical sequencing-software example follows identified sequence inputs to expected variant candidates, then asks separately whether those outputs serve the medical purpose for relevant specimens and users. Neither example classifies PathSight or HelixBridge's assay components as that kind of software. A processing check also leaves the professional's use of the output to be established.

Why not translate this directly into the previous section's analytical-performance entry? The object changed. There, the question concerned detection or measurement of an analyte; here, it concerns software input-to-output processing. The words sound close enough to invite a substitution while the evidence relationship differs.

The source boundary matters too. N41 remains an available, nonbinding international framework with jurisdiction-dependent adoption. FDA withdrew its own 2017 **adoption guidance** on January 6, 2026; it did not withdraw the international document. Presenting N41 as current FDA policy would therefore misstate the

authority. Product qualification and jurisdictional adoption have to travel with the term.

Reviewer Lens

“Analytical validation” on a cover sheet is a reason to open the question, not select a framework. Image-to-score reliability, analyte measurement and a locally titled assay study can lead to different criteria. The reviewer looks for the identified inputs and outputs, intended purpose, criterion and applicable qualification. The HelixBridge study label later in this chapter supplies too little to make that selection.

Consultant Lens

Source-native wording helps software and regulatory owners preserve what their study actually addressed. The consultant's translation explains that meaning to the wider team. After a reference-state or image-processing change, the owners may need to reconsider both input-to-output representation and the intended-purpose conclusion. Agreement on the word “analytical” cannot settle either relationship; a separate processing answer and, where claimed, clinical-purpose answer can.

What if the object is the way the product is made?

So far we have followed an output toward its user. A different question begins before there is a finished product to examine. Suppose a manufacturer mixes or fills assay reagent lots, and inspection of the finished containers cannot reveal every defect in that step. A few acceptable containers cannot establish that the step will consistently produce acceptable material.

Process validation addresses that consistency question. ISO 13485:2016 Clause 7.5.6 addresses a manufacturer's production, servicing or installation process where output cannot be, or is not, verified later and deficiencies may become apparent only after use or delivery. The process and its ability to achieve planned results under defined conditions are the object. Equipment, personnel, methods, criteria and change controls may therefore matter to the conclusion.

Scanner servicing by a manufacturer could raise the corresponding question if those conditions are met. An end-user laboratory preparing specimens or implementing an examination method does not become a manufacturer production process simply because its work also occurs around diagnostic material.

Now suppose a program sets or records the reagent-filling parameters. A capable process still leaves that application's role to examine. The same clause separately addresses software used in production or service provision: evidence concerns its assigned production or service role before initial use and, as appropriate, after change, proportionate to risk. A scanner manufacturer's service application similarly has its own support role.

These applications differ from the product software analyzing a patient's specimen or image. Process consistency does not establish the supporting application's performance, and neither result supplies diagnostic performance. Legacy Global Harmonization Task Force process guidance can help explain the process question, but it excludes software validation and is explanatory guidance predating current rules, not current compliance instruction.

Reviewer Lens

The finished-container example gives the reviewer a useful starting question: what could escape later verification, and how did the process evidence address it? Defined conditions and achieved planned results can close that manufacturing question. If the record also relies on an application, its assigned use, criterion and change decision need their own trace. A single clause containing both topics makes conflation easy; it does not make one result evidence for both.

Consultant Lens

Process and application owners can share settings and risk information while keeping separate closure records. That separation is inexpensive when the filling or service step is being planned. If a later complaint concerns a reagent lot or servicing condition, it allows the consultant to compare the affected process state and software settings with those studied before advising on action. A local examination-method issue goes to the laboratory owner, whose decision has a different object.

Can the laboratory achieve the performance here?

An established method reaches an end-user medical laboratory with developer performance claims. The laboratory still has to decide whether its people, equipment and conditions can achieve the required performance before it reports patient results. That question concerns local implementation. It cannot be answered merely by relabeling the manufacturer's design or software record.

Which local job applies depends partly on the method's status. Under the U.S. Clinical Laboratory Improvement Amendments (CLIA), 42 CFR 493.1253 distinguishes an unmodified FDA-cleared or approved test system, for which specified performance is verified locally, from a modified or other listed system, for which applicable performance specifications must be established. Jurisdiction and laboratory facts must first establish applicability. The cover-sheet word “verified” omits the status that selects the branch.

The voluntary medical-laboratory standard ISO 15189:2022 gives its own qualified distinction: verification of required performance before introducing a method, and validation for intended use of a laboratory-developed, out-of-scope or modified method. The Clinical and Laboratory Standards Institute's EP19-Ed3 describes a

developer who establishes and validates performance and an end-user laboratory who verifies established claims during implementation. Those conventions clarify actor and stage; neither replaces applicable binding law.

What might the local work examine? Repeated results can show how closely measurements agree under stated conditions—**laboratory-method precision**. Comparing results with a defined comparison method addresses **method comparison**. Knowing these names supplies neither their required values nor their observed results. Both still need the criterion-to-observation relationship we followed on the release screen.

For a tissue-sequencing method, local specimens and instrument conditions could be compared with the established performance the laboratory intends to rely on. Whether the laboratory verifies or establishes specifications depends on method status and the governing framework. A pathology service could similarly ask whether its own scanner, display and readers achieve required local performance; that example selects no CLIA category and makes no in vitro diagnostic classification for PathSight.

After introduction, the separate deployed sequencing note supplies a fresh question. **Two initial invalid results among 40 attempts**, with retained case and run identities, allow an investigation of whether specimen, run or result handling departed from the conditions supporting usable-result performance. The record supplies no answer or cause. It remains an observation about its own service.

Reviewer Lens

The first useful fact may be who is deciding. Developer and end-user laboratory can measure the same characteristic while deciding different things. Method status, local configuration, prospective requirements and observations then let a reviewer assess the particular implementation decision. A satisfactory local result may support that bounded decision under the applicable framework. It need not carry a manufacturer design-validation or patient-outcome conclusion to be valuable.

Consultant Lens

Before patient-result reporting, the laboratory director and method specialist can choose the applicable branch and criteria. A consultant helps keep developer claims, local plan, observations and introduction decision distinct, with a responsible owner for unknown method status or jurisdiction. Borrowing the developer's successful label would be easy, but it would leave the laboratory's own achievement question unanswered.

One artifact can inform more than one job

Suppose an intended-user study records that a pathologist saw an urgent worklist entry and acted on it. Must the team create another copy of that observation for an interface requirement? The observation can inform both records. For the interface, the question is whether specified display behavior occurred. For the use-related control, it is whether the intended person could carry out the task under relevant conditions. Criteria, denominators and conclusions may differ even when the observed event is shared.

A tissue-sequencing dataset can similarly inform a software requirement and an analytical-performance analysis through separate traces. Reuse becomes defensible when each question has a method and result fit for it; a shared green status would hide those relationships.

Planning the questions separately makes later investigation more precise. The anonymous deployed notes—two initial invalid sequencing results among 40 attempts and one late pathology case among 25 supported images—could reopen one or several relationships in their respective services. Their identities and timing help locate the investigation. They add no release or event to HelixBridge or PathSight.

Reviewer Lens

When an artifact appears in two places, duplication itself is not the concern. Its method, conditions and result need to fit each question. A display observation may close the specified-behavior question while leaving intended-user capability unresolved. A later delay calls for its event and local-target trace before a reviewer can know which earlier conclusion, if any, to reopen.

Consultant Lens

The study owner can preserve one artifact while design, clinical and use-risk owners maintain their own criteria and decisions. That avoids both duplicated evidence and merged accountability. The consultant connects the records and closure states; a post-use investigator can then identify the particular link under examination and return the finding to its owner.

Which question belongs behind each label?

Decision

For each of the eight jobs just encountered, what would repair its received label into an object, question, acceptance logic, source qualifier, lifecycle position and boundary? Where a framework contains two jobs, what keeps them distinct? For one Chapter 2 configuration entry, which of those fields is already fixed and which question or criterion remains unknown? A missing result can stay missing while the wording becomes clearer.

A compact reference for the eight repaired entries

Reference for comparison

The table helps locate the relationship a received label may be trying to describe. Each row belongs inside the seven-field record, beside the source's actual wording and identified configuration. It supplies a planned basis for judgment; it supplies no observation or case-specific decision. Its purpose is to make an eight-entry glossary usable without turning it into eight imagined study results.

Received label and source	Object and question	Basis, timing and boundary
Design verified or validated; ISO 13485:2016, with adoption and dated U.S. incorporation identified	Outputs meeting design inputs; separately, representative product meeting specified-use requirements	Planned methods and criteria tied to inputs, outputs and observations. Before release and after applicable change. Other performance, security, usability and laboratory conclusions need their own support.
Product software verified or validated; nonbinding U.S. FDA software guidance	A build and integration state meeting development inputs; separately, specifications meeting user needs and intended use	Expected and observed results, anomalies and use environment. Development through relevant change. This is neither production software nor diagnostic performance.
Cybersecurity validated; nonbinding U.S. FDA cybersecurity guidance	A named control in a system boundary, challenged against its threat	Threat assumptions, expected control behavior, test observations and remaining risk. Design through end of support. General correctness and diagnostic performance remain separate.

Received label and source	Object and question	Basis, timing and boundary
Production or service validation complete; ISO 13485:2016 Clause 7.5.6, adoption qualified	Process consistency; separately, a supporting application's production or service role	Requirements, risks, planned criteria, results and change decisions. Before use and after relevant change. Process, supporting application and product software are separate objects.
Laboratory method verified or validated; applicable U.S. law, ISO 15189 and CLSI conventions distinguished	An examination method or test system achieving required established performance locally, or needing performance and fitness established	Method status, claims, local plan, criteria and observations. Before patient reporting and after applicable change. No manufacturer-design, regulatory performance-evaluation or patient-outcome conclusion follows automatically.
European performance established; applicable binding in vitro diagnostic regulation and qualified nonbinding software guidance	Intended-purpose proposition and its scientific, analytical and applicable clinical questions; technical and clinical questions for the other software branch	Intended purpose, legal branch, pillar criteria, data, gaps and updates. Planning through continuous evaluation. An evaluation is more than one study; technical and analytical performance retain different branches.
Software as a Medical Device analytically or clinically validated; nonbinding IMDRF N41 with adoption qualified	Reliable input-to-output processing; separately, intended purpose in the target health-care situation	Identified inputs, outputs, reliability, use, population and clinical propositions. Evaluation and updates. These are not interchangeable with analyte-centered European terms; FDA's adoption guidance was withdrawn.
Human-factors validation complete; nonbinding U.S. FDA human-factors guidance	Intended users carrying out critical interface tasks and effective use-related controls	Users, environment, tasks, hazards, observations and remaining risk. Near development's end and after relevant change. May inform design validation; supplies no automatic software, analytical, clinical or security conclusion.

The product-software row can now be read with the release-block example. The label is only the starting phrase. The identified teaching build and invalid condition fix the object; the no-release rule fixes the question and planned criterion. The withheld action and recorded event are the observations. “Met in this staged test” is the decision. The FDA source qualifier explains a software vocabulary without making the invented example a legally classified product. A changed build could reopen applicability, and the boundary prevents this local result from filling a whole-assay validation entry.

The selected Chapter 3 technical figure: one record, separate evidence questions depicts three generic lanes—digital pathology, liquid NGS and tissue NGS—in which one artifact branches toward separately qualified questions. Read each branch from object through question and acceptance logic to a bounded conclusion. It is a conceptual map, not evidence for either named case and not permission to merge the branches. A text-only reading is: one record may travel to several owners, but each owner must preserve the configuration, criterion, observation, decision and unresolved scope for that owner's question.

Book 1C - Chapter 3

Section 03 – One record, several evidence jobs

Technical examples: reuse evidence only for the questions it actually answers.

Key takeaway

A shared artifact can support more than one question, the conclusions stay separate.

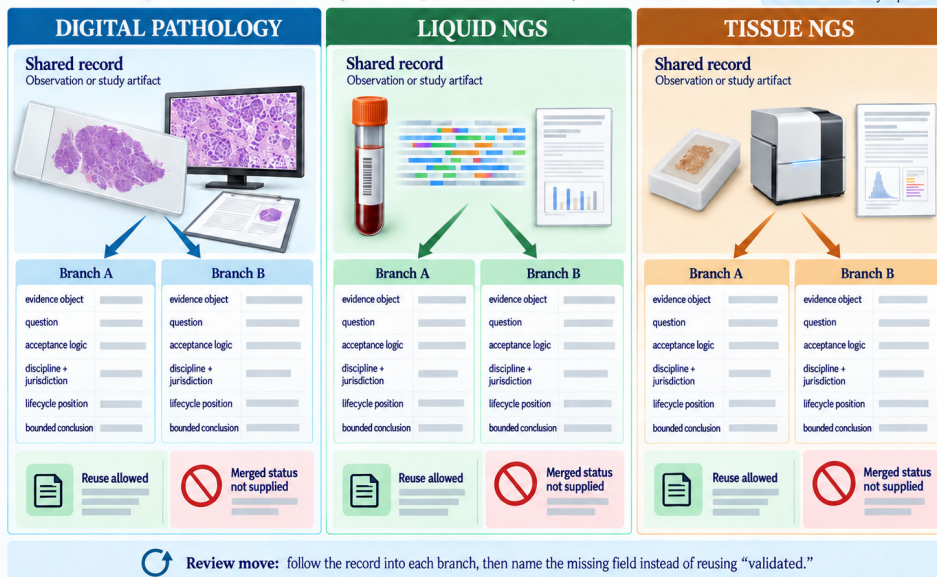


Figure 3.1. One record can inform several separately qualified questions – Technical review plate.

What can a study title establish for HelixBridge?

The distinction becomes useful when a record offers less than a test result. HelixBridge Dx is the fictional tissue-oncology sequencing assay proposed to identify variants associated with eligibility for a named therapy. This chapter's controlled occurrence identifies a study called **analytical validation** and a partly identified validation setup: the validation application is 7.0, the validation report rules are R19, and the knowledgebase used in analytical validation is DB-18.

Those exact identities belong to this study record. It supplies no exact wet-lab identifier, instruments, reagents or other computational dependencies. Study design, sample count, variant classes, criteria, failures, strata, observations and results are also absent. The exact commercial configuration and release state, intended population, treatment-facing language and regulatory outcome remain unresolved. The validation report rules are not identified as implementing the invented release-block behavior.

What is positive here? **Provenance**—the record's origin and the state to which it is attached—gives us an identified study label and part of its validation configuration. There is no supplied criterion-achievement result to grade. The word “analytical” could sound like the international software-processing job, European analyte performance or a locally defined assay study. The title alone selects none of them. Keeping the study in the evidence inventory while asking for its actual question and result preserves what is known.

The seven-field view below helps distinguish an identified study from an interpreted result.

Field	What this occurrence supplies
Received wording	A study called analytical validation.
Object	Part of the tissue-sequencing validation setup: validation application, report rules and knowledgebase identified above.
Question	Not supplied beyond the study label.
Acceptance logic	Requirements, prospective criteria, observations and result basis not supplied.
Discipline and jurisdiction	A controlled case label; technical framework and jurisdiction not supplied.
Lifecycle position	A validation view; relationship to the exact commercial state not established.

Field	What this occurrence supplies
Boundary	Study and configuration provenance. Software, security, usability, laboratory, process, European performance-pillar, commercial-applicability and regulatory conclusions remain unestablished.

The object row contains usable identity while the acceptance row contains an explicit unknown. Neither row cancels the other. The identified validation application tells an owner which study configuration to retrieve; it cannot tell the reviewer whether that study met its criteria.

Reviewer Lens

A reviewer can acknowledge the study's presence without endorsing an unreported outcome. The narrow next request is its protocol, criteria, results, failure handling and comparison with the configuration under review. “Analytical validation unclear” would obscure which of those relationships is absent; a precise note preserves the provenance and leaves result and applicability unjudged.

Consultant Lens

Assay and regulatory owners need to identify the proposition the study is meant to support, its governing framework and the owner of the commercial-state comparison. The consultant can organize that relationship without changing the source's title. Closure then means a criterion-to-observation conclusion for the selected object and evidence connecting any different target state. A more confident cover sheet cannot supply either.

Guided micro-cases

A. One result, three possible jobs

PathSight's multi-site study reportedly met its prespecified overall endpoint; the separate reader study reportedly improved average sensitivity without unacceptable loss of specificity.

Decision

If the pending decision concerns software behavior, intended-user task performance or support for both scanner families, would the same next request answer all three? Which relationship belongs to each, and which favorable facts remain?

How the pending decision changes the request

A software conclusion needs requirements, build, test level, expected results and anomalies. An intended-user conclusion needs users, critical tasks, conditions, hazards, observed interactions and controls. A claim spanning both scanners needs the applicable framework, scanner strata, study configurations, endpoint, reference process, denominators and estimate precision. The summaries supply neither those relationships nor a reason to erase the two reported positive results. The useful request selects the missing relationship for the actual decision.

B. Two jobs in one clause

A manufacturer process cannot be fully checked through each finished output, and an application records its settings. Both records cite the production-and-service clause, ISO 13485:2016 Clause 7.5.6.

Decision

What would let one of these records close while the other remains open?

How two objects lead to two closure decisions

The process record needs evidence of consistently achieving planned results under its conditions. The application's record needs evidence that it performs its assigned support role before use and after relevant change. Shared settings and risk analysis can link the records; they cannot make one conclusion stand for the other. Neither result establishes diagnostic product-software or performance conclusions.

C. The actor changes the job

A developer and an end-user laboratory both examine precision.

Decision

What additional facts would tell us whether the two teams are deciding the same thing?

How the actor changes the meaning of the measurement

The developer asks what performance the method can claim; the laboratory asks whether it can achieve required performance locally before patient reporting. Actor, method status, adopted framework, criteria and lifecycle point distinguish those decisions even if both use repeated measurements. Those facts are needed before selecting verification or validation terminology.

Reviewer Lens

Across the three cases, the reviewer is selecting a relationship to inspect. For PathSight it may be the study-to-second-scanner claim. For the manufacturer it may be the process or supporting application. For the laboratory it begins with actor and method status. Enough evidence to close one of those questions may leave the others open; the finding should say which.

Consultant Lens

The consultant can make each request actionable by connecting it to its owner: product and study owners for scanner transfer, process and application owners for manufacturing, and the end-user laboratory for local introduction. Closure requires an owner able to connect a configured object and planned rule to observations and a decision. “More validation” names no such piece of work.

Application exercise: repair before you infer

Return now to PathSight with the complete controlled snapshot for this exercise. The two study reports are familiar; the scanner distribution below adds a relationship that the opening did not supply.

- PathSight assists pathologists by prioritizing whole-slide images and highlighting suspicious regions; the pathologist retains diagnosis.
- The initial package names the model **M3.2** and claims support for **Scanner A and Scanner B**, the first and second scanner families.
- The validation configuration's images came mostly from the first scanner family.
- A multi-site performance study reports that the model identified in validation met its prespecified overall endpoint.
- A separate reader study reports improved average sensitivity without unacceptable loss of specificity.
- Endpoint definition, sites, cases, scanner strata, failures, precision, reader population, case set, reference process, estimand, denominators and workflow are not supplied.

- Exact end-to-end commercial configuration, evidence for the second scanner family, reader-study applicability, regulatory outcome and European qualification remain unresolved.

To interpret the reader result, we would first need to know how the study assigned the positive or negative status against which performance was judged. That is its **reference process**. We would then need the precise quantity it intended to estimate, its **estimand**. A description such as “average sensitivity” leaves those details unspecified.

The calculation also needs a **denominator**, the quantity used as its divisor. For an illustrative case-level sensitivity calculation, this would be the reference-positive cases actually included, not every potentially eligible case. Finally, how uncertain is the estimated quantity? **Precision of a study estimate** addresses that uncertainty, commonly through an interval. It differs from the earlier laboratory-method precision, which concerns how closely repeated measurements agree. None of these details is supplied for the PathSight studies.

Decision

How could “PathSight was validated” become the strongest supported statement? What is the maximum identified object, which questions were reported, and which parts of the acceptance logic are present or missing? How do framework, lifecycle and configuration limit the conclusion? Which five or more nearby conclusions would overreach, and what one focused request would address the second-scanner claim? Every other case fact remains unknown.

How a reviewer preserves both positive reports

Worked reasoning

The largest identified object is a partly specified validation configuration containing the model named in the record, with images mostly from the first scanner family, in an assistive pathologist workflow. Exact scanner states, image preparation, threshold, processing after scoring, queue, interface, cloud service, fallback and commercial configuration remain unresolved. The two reported questions concern a multi-site overall endpoint and a separate reader sensitivity/specificity comparison; neither is fully defined.

The first report gives us one piece of planned acceptance logic: the overall endpoint was **prespecified**. It also reports the judgment that the endpoint was met. We lack the definition, criterion, observations, strata, failures, precision and analysis needed to examine that judgment independently.

The reader report is favorable too, but “without unacceptable loss” supplies no threshold. The estimand, denominator, reference process, precision and workflow are also absent. Missing details limit the conclusion we can independently draw; they do not turn either reported positive into a failed study.

No jurisdiction-specific classification is supplied, and no relationship to an exact commercial state has been established. A bounded repair can retain the observations as reported:

Note

For the partly specified PathSight validation configuration containing the model identified in validation, with images mostly from the first scanner family, the source reports that a multi-site study met its prespecified overall endpoint and that a separate reader study improved average sensitivity without unacceptable loss of specificity. Endpoint, criteria, methods, strata, denominators, precision and workflow are not supplied; applicability to the second scanner family or an exact commercial state is not established.

This statement preserves two favorable reports. It establishes no software-requirement coverage, product-software validation, security-control effectiveness, human-factors validation, second-scanner generalizability, qualified European performance conclusion, regulatory authorization or conformity. Nor does it transfer diagnosis from the pathologist to the model.

The focused request is the study-to-claim comparison for the second scanner family. It would identify counts and states for each scanner family, image preparation and end-to-end configurations, endpoint and acceptance criteria, target and difficult-tissue strata, failures, denominators, precision and the analysis supporting transfer to the claimed use. These are parts of one relationship to resolve. While it stays open, the two source-reported study results remain in the record.

Professional artifacts

Reviewer artifact: evidence-statement repair note

“Validation unclear” gives the study owner little help. The useful reviewer note connects the received statement with the decision that needs support. The table below shows that short trace. Its planning row states what would count; its next row keeps what occurred distinct from the judgment about it. The closing row allows bounded acceptance, a focused request or unresolved applicability. Unknown information does not automatically establish a deficiency, nonconformance or unsuccessful study.

Part of the note	What another reader needs to see
Received statement	Original wording, source and date.
Decision at issue	The decision that could change with the statement's meaning.
Object and configuration	Identified product, process, software, method, interface, control or proposition, with state and conditions.
Qualified question	Actual question, actor, discipline, framework and jurisdiction.
Planned acceptance logic	Prospective requirement or criterion and method.
Observations and decision	What happened; separately, whether it met the plan and with what limits.
Applicability	Study lifecycle position and configuration compared with the state under review.
Supported conclusion	The narrow positive or negative conclusion justified by the record.
Conclusions that do not follow	The nearby jobs this result could mistakenly be taken to answer.
Request or disposition	Missing relationship to resolve, bounded scope accepted or applicability left open.

For PathSight, the “decision at issue” row could identify second-scanner support. The observations row retains the reported overall endpoint and reader results in the validation view, mostly using the first scanner family. The last row then seeks the scanner-specific analysis and study-to-claim relationship, or limits the conclusion to the studied scope. This gives the owner a route to closure and keeps a favorable result visible while its wider applicability is unresolved.

Consultant artifact: cross-framework translation register

The consultant faces another problem: different teams may mean different things by the same status word. A translation register keeps each team's original wording beside its source and intended decision. The blank row below is intentional. It becomes meaningful only for a selected decision with actual evidence; it is not an invitation to give every term in the organization its own row.

Team's words	Source and authority	Object and question	Owner and decision time	Shared meaning	Limit and next owner
--------------	----------------------	---------------------	-------------------------	----------------	----------------------

Consider one possible row from the invented release-block check. The team's words are “the function passed,” and its source is an internal software-requirement check. The object is the release-block function in the identified teaching build. The development lead owns the pre-release question: does the function withhold release in the staged invalid state? The shared meaning is that the observed withheld action met the planned no-release criterion in that test.

The final column carries the boundary onward: other invalid routes remain with the software test owner, and intended assay use remains with the assay and use owners. The row can hold a positive local result and a visible handoff together, without becoming a regulator's product-wide verdict.

The source column distinguishes law, guidance, an adopted voluntary standard, consensus language and local wording. The object column exposes where another team might infer a different job; the owner column identifies who decides now and which specialist owns the remaining depth. A laboratory's “method verified” can consequently translate into local implementation without changing the manufacturer's design conclusion. Even a “validation green” dashboard becomes useful if it opens to the question, criterion, configuration, observations and unresolved decisions. The register coordinates accountability; it does not transfer it.

Operational translation

A software owner can keep requirement-to-result and intended-use traces separate while connecting security and human-factors findings to design. Manufacturing and service owners distinguish process capability from the applications supporting it. A laboratory starts with its method status and local conditions; a team crossing jurisdictions retains the source and edition. The handoff works when the next owner can see both the supported conclusion and the specific relationship still open.

Changed-case transfer

An unfamiliar cover note offers a useful final test. This unnamed mass-spectrometry example is separate from HelixBridge, its liquid branch, PathSight and both deployed teaching notes. Its eight supplied facts are:

1. The laboratory's cover note says only that the method was validated.
2. An end-user medical laboratory plans to implement an established quantitative mass-spectrometry examination method.
3. The laboratory must determine whether the method's required precision and method-comparison performance can be achieved in that laboratory before use.

4. The prospective plan requires **both** the developer-specified precision criterion and the developer-specified method-comparison criterion to be met; their values and observed results are not supplied.
 5. The identified setup is analyzer **MS-2**, preparation protocol **P-4** and analysis software **version 3.2**, with no stated modification to the established method. These identifiers belong only to this separate method.
 6. The end-user medical laboratory performs and reviews the evidence work.
 7. The decision comes before the laboratory introduces the method for reporting patient results.
 8. The question concerns medical-laboratory implementation; no jurisdiction, accreditation adoption or regulator-specific legal effect is supplied.
-

Decision

Which evidence job do these facts describe, with what source qualification? How would the cover note change if it identified object, question, both criteria, configuration, actor, discipline and jurisdiction, lifecycle point and scope? What still cannot be concluded? Every other fact remains unknown; an unknown is neither a success, a failure, an absence nor a finding of inapplicability.

How the local method decision remains qualified

Transfer reasoning

The object is the established examination method in the identified analyzer, preparation protocol and software 3.2 setup. The end-user laboratory is asking whether it can achieve both required precision **and** method-comparison performance locally before patient-result reporting. Both developer-specified criteria must be met. Meeting just one would not satisfy that prospective rule.

No criterion values, study design, samples, replicates, comparator, observations or results have been supplied. Consequently, neither criterion achievement nor approval to introduce the method can be inferred. The supplied record identifies a plan and its decision structure, rather than a study outcome.

A qualified replacement for the cover note could therefore read:

Note

For the established quantitative mass-spectrometry method in the identified analyzer and preparation setup with analysis software 3.2, the end-user laboratory plans to determine before patient-result reporting whether it can achieve both developer-specified precision and method-comparison criteria. Their values and observations are not supplied, so neither criterion achievement nor an introduction decision is established. In a qualified medical-laboratory framework this is an **implementation verification-type** question; the controlling term and legal or accreditation effect require the applicable jurisdiction and adoption facts.

The cautious “verification-type” wording preserves the local-achievement concept described by ISO 15189 and CLSI without inventing their adoption here. CLIA applicability is also unknown. The source's word “validated” remains the received wording, while the repaired account identifies the actual planned job.

There is no positive or negative study result to grade, and no conclusion about manufacturer design, product or production software, security, usability, a specific regulatory analytical or clinical-performance framework, patient outcomes, authorization, conformity or commercial state. The method is unfamiliar; the seven-field reasoning still works.

From a repaired statement to a plan

PathSight still has its two favorable reported study results. HelixBridge still has an identified study and part of its validation setup. The unfamiliar laboratory has a prospective local question with two criteria and no supplied outcome. Their conclusions differ because the records differ. In each, the repaired wording connects what is known to the decision it can actually inform.

Chapter 4 takes that question into an evidence plan: which work addresses the claim and risk, what result would count, and how remaining uncertainty will be handled. Naming the job makes that plan answerable; it does not supply the plan.

Reasoning checklist and retrieval

A useful final check follows the same reasoning as the release-block test. Is the original statement visible? Can the object, configuration, conditions and actor be identified? Does an explicit question connect the planned criterion to a separate observation and decision? Is the source qualified by authority, jurisdiction and edition where relevant, and is the evidence located before release, at local implementation, in use or after change?

The conclusion then needs two further comparisons: how far does the observed result apply to the decision state, and which nearby inference would exceed it? A shared artifact can support several answers through separate traces. An unknown stays unknown until the responsible owner resolves it.

Decision

What are the seven fields? How do general verification and validation differ, and why is that only a starting point? Why do the 2026 and 2015 ISO vocabulary editions both matter in the U.S. quality-system discussion? Which two design questions are distinct? How can one artifact support several jobs? How does product software differ from production software? What are the three European in vitro diagnostic performance pillars? Why is international software analytical validation a different relationship? Which actor and method facts select a laboratory implementation question? Finally, how could “the system was validated” be repaired in one line without inventing an outcome?

The relationships behind the ten answers

The seven fields are received statement, object, question, acceptance logic, discipline and jurisdiction, lifecycle position, and scope with conclusions that do not follow. General verification concerns specified requirements; validation concerns requirements for intended use or application. Neither supplies the actual object, criterion or observation. The 2026 edition gives current general vocabulary, while the U.S. rule's dated incorporation still identifies the 2015 vocabulary clause.

The design questions compare outputs with inputs and the representative resulting product with its specified use. One artifact can inform both or other jobs when separate methods, conditions, criteria and conclusions make each relationship visible. Product software performs a device function; production or service software supports making or servicing the product. A result for either must stay attached to its own role.

The European in vitro diagnostic pillars concern scientific validity, analytical performance and applicable clinical performance. Analyte detection or measurement differs from the international software framework's reliable input-to-output processing. For a laboratory, developer versus end-user actor, established versus modified or other method status, local conditions, criteria, lifecycle point and applicable authority govern the qualified question.

A one-line repair therefore needs the identified object and state, actual question, criterion and source, what was observed if anything, decision time and supported scope. If a fact is absent, “not supplied” is more accurate than a guessed result. Book 1D

develops reference assignment, integrated verification and human-use evaluation in greater depth; those methods retain the same source-qualified boundaries.

Source trail

These sources support the chapter's distinctions at their named authority, scope and date; they do not establish fictional-case performance or regulatory status. Licensed standards are paraphrased, not reproduced. Status-sensitive records were checked on September 24, 2026; recheck before publication.

- ISO 9000:2026 official record, licensed Clauses 3.11.12–3.11.14 (printed p. 22), supplies current general verification, test and validation vocabulary. The official 2015 public terms listing, document p. 39, supplies the former Clauses 3.8.12–3.8.13 for dated-reference context. Neither edition is a universal diagnostic taxonomy. The FDA QMSR page gives the February 2, 2026 effective date; 21 CFR 820.7(a)–(b) fixes the incorporated editions. The underlying CFR provision is binding U.S. law; eCFR is its authoritative but unofficial presentation.
- ISO 13485:2016 official record, licensed Clauses 2–3, 7.3.6–7.3.7, 7.3.9 and 7.5.6, supports the distinct design, process and production/service software jobs. Adoption or incorporation determines legal effect. Legacy GHTF process-validation guidance, printed pp. 3–8, is explanatory only, predates current rules and excludes software validation.
- FDA device-software submission guidance, final 2023, Sections IV and VI.H, supports the bounded product-software distinctions; it is nonbinding U.S. guidance and not a diagnostic performance method. FDA cybersecurity guidance, final February 2026, Sections V.A and V.C, supports security risk/control and test distinctions; it does not supply general software or clinical conclusions. FDA human-factors guidance, final revised August 2026, Sections 2–5 and 8, supports the intended-user/critical-task/use-risk framing. Each is nonbinding guidance, not product-specific evidence.
- 42 CFR 493.1253(b)–(c) is the underlying binding U.S. laboratory provision for establishment and verification of performance specifications; application depends on method and laboratory facts. ISO 15189:2022 official record, licensed Clauses 3.31–3.32 and 7.3.1–7.3.3, and CLSI EP19-Ed3, report pp. 3–6, 15–22 and 35, supply their separately qualified voluntary laboratory conventions. Neither displaces applicable law.
- Regulation (EU) 2017/746, Article 2(38)–(44), Article 56 and Annex XIII Part A 1.1–1.3, is the binding IVDR basis for the performance pillars and continuous evaluation; the consolidated text is a navigation aid and applicable amendments control. Nonbinding MDCG 2022-2, Sections 3–4 and 6, and MDCG 2020-1,

definitions and Sections 4.1–4.4, explain the IVD and qualified software branches. They do not classify PathSight or supply EU conformity conclusions.

- IMDRF N41:2017, Sections 5.2–5.4 and Figures 6–8, is a nonbinding international SaMD terminology source with jurisdiction-dependent adoption. The FDA withdrawn-guidance register records withdrawal of **FDA's adoption guidance** on January 6, 2026, not withdrawal of IMDRF's document.

HelixBridge and PathSight are controlled fictional composites. The separate deployed observations and unnamed mass-spectrometry transfer are teaching records only. None is scientific, clinical, legal or regulatory authority for a real product.

CHAPTER 4

From Claim and Risk to Evidence Plan

What would this study let us decide?

A diagnostic-product team has put its proposed studies in a folder. A colleague covers their titles and asks what each activity will let the team decide. Which promise will it examine? In which product state? What would the team do with a favorable result, and what would change if the result were unfavorable? A list that looked reassuring a moment ago may leave all four questions unanswered.

The missing connection begins with the person who will rely on the result. A laboratory professional needs a report that can be interpreted for its proposed use. A pathologist needs the promised assistance in the workflow where tissue is examined. The team needs to decide what to learn about those uses, how to observe it and what the observation would mean. That reasoning must exist before data make a preferred answer tempting. A study can answer one question well while leaving another open.

The book calls the connected account a **claim-to-study evidence matrix**. Each row follows one claim condition, risk or uncertainty through a question, the product state examined, a method and a prospective decision rule. It also distinguishes evidence that exists from work still to come. This is an author-created planning aid. It does not replace a protocol, statistical analysis plan, risk-management file or the European Union's regulated performance evaluation plan, and it is not a regulator-issued form. Its value is that another person can follow why an observation would change a decision.

What the earlier questions contribute

Decision

Which links from Chapter 1 connect a diagnostic claim to the later clinical use of a report? Why might the software version from Chapter 2 fail to identify the product that produced it—what two other kinds of state could matter? Which questions from Chapter 3 would repair “validation is complete”?

How the earlier questions make a plan answerable

A claim concerns a result used for a purpose. The specimen or image, physical and computational processing, interpretation, report and human action connect the purpose to the output. The software version identifies only part of that route. The reference-data state and report-rule state, for example, may also affect what emerges. Evidence therefore belongs to an identified configuration and conditions.

Chapter 3's evidence job asks what was examined, which question was asked, what acceptance logic applied, which framework gives the question meaning, when the evidence was generated and how far its conclusion travels. One activity can inform several jobs only through separate answers. A configuration entry and a related evidence-job entry now give a prospective question somewhere to attach. Their original source, known state and unresolved applicability remain intact; another case cannot supply their missing facts.

The distinction is easiest to see in time. Before an observation, the plan says what should be investigated. The activity then shows what was actually investigated. Appraisal asks what that evidence supports now. A convincing first answer cannot manufacture the other two. An event during later use may reopen the question or reveal a narrower range of studied conditions, but it cannot rewrite what the team planned earlier.

Reviewer Lens. A study list becomes readable when one pending claim decision supplies its purpose. The useful next question is which activity could change that decision and why. A missing connection may require an existing protocol or configuration comparison; it need not mean every document is inadequate. Equally, a favorable result deserves credit only for the question and state it covers. The narrow request is for the missing relationship, with a clear account of what would settle it.

Consultant Lens. The claim owner knows the proposed use, the configuration owner knows the product state, and the evidence owner knows the question being examined. Their descriptions can each be accurate and still fail to connect. Bringing them together early prevents a study being commissioned before anyone can explain the decision it serves. Later, the same conversation may locate sufficient existing evidence. The consultant makes the connection and closure condition clear; the responsible owners make their respective decisions.

Start with the promise a person will rely on

A tissue-sequencing program can produce every specified report field while the meaning of the reported finding remains a separate question. Follow the ordinary route: the laboratory receives tissue, reads its genetic material, and uses software and rules to select findings for a report that a professional reviews. A software check can establish that required fields appeared. A promise that a variant finding supports a proposed treatment-facing use also depends on the specimen, analytical and clinical relationships. Which of those promises is the team asking the study to support?

That question determines which conditions matter. The intended patients, specimen, user, setting and workflow belong in the plan when changing them could change the decision. The same reasoning applies to digital pathology. An image is made from a glass slide; assistance may prioritize the case or highlight a region; a pathologist examines the image and diagnoses. Seeing a priority label on screen answers a display question. Establishing what prioritization contributes to the intended workflow reaches further. Naming those consequential conditions makes the latter question specific without turning every descriptive detail into a separate study.

There is a labeling reason to preserve that connection too. In its applicable U.S. in vitro diagnostic context, 21 CFR 809.10, a provision of the Code of Federal Regulations, addresses intended use, procedure type, limitations, expected values and performance-characteristic information. The provision helps identify information relevant to the proposed use. It neither prescribes this matrix nor makes a statement on the label evidence that the performance was achieved.

Which unknown could change the decision?

Suppose the study does not identify the software state it used. That is an **uncertainty**: something inadequately known. It can prevent a reviewer applying the study to the proposed product without establishing any particular harm. A **risk question** goes further into a possible harmful consequence and the information needed to understand or control it. A route by which a misleading report could reach care can matter even when its frequency is unknown. Neither an unknown nor a high-risk label selects a method or numerical cutoff by itself.

The useful question connects the two ends: *what must be observed in this product state, under these conditions, to make this decision?* For an initial invalid sequencing result, the question might be whether the release action is blocked. For a valid report it might concern correct identification of variants. In pathology, a late worklist entry might prompt an inquiry into the image-to-reader route without establishing a missed diagnosis. An answer earns its place in the plan when it could support a bounded claim, limit that claim, change a control, call for further evidence or leave a defined gap open.

Chapter 1's two separate deployed teaching notes show how carefully that question must preserve the observed event. In the unnamed released tissue-sequencing service, **two initial invalid statuses occurred among 40 consecutive specimen attempts**, before report release. Dividing two by 40 gives 0.05; expressed as a percentage, $2 \div 40 = 5\%$ of attempts in that interval. The divisor counts attempts. It does not count final failed reports or patients harmed.

In the separate deployed pathology service, **one of 25 consecutive supported-image cases reached the pathologist after a local worklist target**. One divided by 25 is 0.04, so $1 \div 25 = 4\%$ of those cases were late by that target. The pathologist completed review. This is a timing observation; it is neither a clinical-error rate nor a sensitivity calculation. Neither note supplies a cause, clinical consequence or a performance result for HelixBridge or PathSight. Both retain identities that let the responsible service follow the actual attempt or case into a focused inquiry.

Reviewer Lens. Percentages can make a thin record look like a completed judgment. The first reading should recover what the numerator describes and what entered the denominator. Here an initial invalid attempt and a late supported-image case are visible; an acceptance failure or harm estimate is not. The next request can concern the event trace and its relation to the evaluated conditions. In a prospective study, the corresponding discipline is to check that the planned population, material, users and workflow are the conditions in which evidence will actually be generated.

Consultant Lens. The clinical and claim owners decide which condition could change the promised use. After deployment, the service owner can use the retained case or event identity to ask a specific question about what happened. The consultant helps translate that question into the next decision: compare configurations, investigate a route, narrow a claim or choose a suitable method. A risk color or an unexplained percentage makes none of those options concrete. Closure depends on resolving the selected relationship.

What did the release block actually establish?

Chapter 3's invented teaching check introduced an initial invalid sequencing result. In the identified teaching build, the screen withheld the report-release action and recorded the event. The specified behavior was met in that staged condition. That positive finding is useful. How far can it take us toward saying the reporting route is safe, or that the assay's variant findings support their claimed use? This mechanism is separate from HelixBridge's validation report rules.

The first question starts before the block. How could an invalid state arise, reach a user as a misleading report and affect care? Information about the frequency and consequences of that sequence contributes to **risk estimation**. An invalid state alone is not the entire sequence. The other service's two invalid attempts among 40 show that

initial invalids occurred there; they do not estimate release or harm in this teaching system. No harm observation or probability is supplied here, and frequency information alone does not decide acceptability.

Once a control is chosen, there is a more immediate question: does the identified build contain it and behave as specified? This is **control implementation**. The withheld release action and recorded event support the selected block's presence and behavior in the staged test. The conclusion can be positive without claiming that every invalid route was examined.

Presence leaves the next link open. Does withholding that action actually reduce the route from an invalid state to a misleading released report under relevant conditions? That is **control effectiveness**. The same observation may contribute if the connection to prevented release is established. Other release routes, overrides and use conditions may need attention. The missing causal connection calls for evidence; a different label alone calls for neither a second report nor automatic repetition of the test.

Even an effective block leaves a further decision. What risk remains from this particular harmful sequence, and does it meet the responsible team's applicable criteria? That is the **individual residual risk** judgment. “Residual” means remaining after the control has been taken into account. The observed staged behavior does not establish the amount of remaining risk or its acceptability. This example supplies no acceptable-risk level.

The device also has remaining risks beyond this one sequence. Evaluating them together, in relation to the benefits of the intended use and by the planned method and criteria, is the **overall residual risk** judgment. A well-controlled release route contributes to it. It cannot stand in for the evaluation of the remaining risks together.

Now return to a report that is valid and releasable. Does it identify the claimed variants for the specified tissue, people and use conditions, and support the treatment-facing proposition? That is **diagnostic-performance claim support**. Preventing an invalid report from leaving the system protects one route. It does not establish the correctness or clinical meaning of the valid findings that do leave. This sixth judgment keeps the performance promise visible beside the five risk judgments.

The first five distinctions follow work described by the International Organization for Standardization's medical-device risk standard ISO 14971:2019, particularly Clauses 4–8. The six-part teaching arrangement is the book's synthesis, not a table from that standard or a complete risk-management method. The standard calls for appropriate risk-acceptability criteria without supplying a universal acceptable level. Its legal effect depends on adoption or incorporation in the applicable context.

A separately imagined pathology worklist alert follows the same chain. Its appearance could establish implementation. Evidence that it changes the risk-relevant route to the pathologist would address effectiveness. The remaining delay risk and overall device

risk would still need their own judgments. The other service's one late case among 25 can motivate a timing inquiry but establishes none of those alert conclusions. No such alert is asserted for PathSight, and an effective alert would not itself demonstrate diagnostic performance.

Reviewer Lens. The difficult move is often stopping at an adequate local conclusion. The staged block worked as specified in the observed condition; that should remain clear even when its broader use is unresolved. The next request follows the unfinished link: other release paths for effectiveness, remaining-risk information and criteria for acceptability, or variant evidence for the performance claim. An invalid attempt during use does not automatically contradict the implementation result. Its route has to be understood before deciding which earlier conclusion it touches.

Consultant Lens. Six judgments need distinct ownership and reasoning, but they need not generate six studies. The risk owner holds the harmful-sequence and residual-risk questions, the design or software owner holds implementation, and analytical and clinical owners hold performance support. A well-fitting observation can travel to more than one owner with a separate criterion and conclusion. The consultant helps identify whether the unresolved work is another observation, a missing configuration or causal connection, or an explicitly open judgment.

How a plan carries that reasoning

Imagine returning to the moment before the staged block check. The team has a question and a proposed way to answer it, but no observation yet. What makes this a useful plan? Another reader must be able to follow why this behavior matters, what will be examined and how the result will affect the decision. The matrix preserves that path in thirteen fields, arranged below in three parts so the relationships remain readable. Related questions can point to the same configuration or study record without copying it.

A. What is the decision about?

The first table is useful before selecting a method: it tests whether there is a specific question for a specific product state.

Field	What another reader needs to understand
1. Claim and consequential condition	The particular promise and the use condition that matters to it.
2. Decision	What could be supported or changed by the answer.
3. Risk or uncertainty	The possible harm or unresolved issue behind the question.

Field	What another reader needs to understand
4. Evidence question	What must be observed or established to answer it.
5. Configuration and time	The exact product state examined and when it is examined.
6. Conditions	The relevant population, specimen or image, user, setting and workflow, including what is unknown.

For the block, the limited promise is that an initial invalid state is not releasable. The question concerns the release action in the specified staged condition, because a misleading report reaching a user is the possible route being controlled. “Sequencing software” is too broad for Field 5: the build and rule state are needed. Field 6 identifies the relevant conditions so a later reader can tell whether the observation answers this question. For a pathology-priority claim, the image route and pathologist workflow may do that work. These entries establish where a result would apply; they contain no result themselves.

B. How could the observation answer it?

The second table helps judge the proposed method before it is carried out.

Field	What connects observation to judgment
7. Method and any reference	What the activity observes, why it answers the question and what a comparison contributes when needed.
8. Judgment served	Which of the six judgments it addresses; why each additional role is justified.
9. Prospective acceptance logic	The interpretation rule, its technical reason and when it was fixed.

A staged invalid input and inspection of the release action can answer the block's specified-behavior question. That gives Field 7 its method and Field 8 an implementation role. Effectiveness would require the further relationship to prevented release. Field 9's **prospective acceptance logic** is the rule chosen before outcome information can shape the judgment: in the specified staged condition, the release action must be unavailable. Its reason is the intended block on that release route. A criterion copied from a favorable observation afterward has a different history, even if its words match.

C. What has happened, and what can follow?

The final table becomes especially useful when a plan, result and decision are all present in the same file.

Field	What must remain distinguishable
10. Result state	Not generated; available but unappraised; applicable; a candidate for bridging; nonresponsive; or unknown. The observation remains separate from its appraisal.
11. Decision rule or disposition	Why to support, limit, bridge, generate evidence or leave unresolved; whether this is conditional or an executed decision.
12. Source authority and scope	Identity, date or edition, jurisdiction, exact passage, supported proposition and limits.
13. Remaining gap and closure	What is open, what would settle it and who must decide next.

Before the block check, Field 10 can truthfully say **not generated**. The plan may still be ready to execute. Later, Chapter 3's teaching observation supplies the withheld action and recorded event. Compared with the prior rule, that supports “met for this staged condition.” The observation is retained when its **result state** is appraised; appraisal does not replace the underlying record. Field 11 then distinguishes an anticipated decision route from the decision actually made. None of this fills a whole-assay performance row.

Fields 12 and 13 prevent another quiet leap. A source must support the proposition for which it is cited, and the next owner must be someone able to resolve the remaining question. If no person is identified, the role remains an assignment to make. A complete-looking row cannot create performance evidence, source authority or an owner.

Decision

For one configuration entry from Chapter 2 and its related Chapter 3 evidence job, what bounded promise, decision, uncertainty, question, state and conditions would occupy the first six fields? Which evidence judgment belongs beside them? Which parts are known facts, and which are still proposals or unknowns awaiting a method, criterion, result and closure decision?

The distinction in that last question protects the planning exercise: a sensible proposed question remains a proposal. Its usefulness does not turn it into a newly discovered fact about the case.

Reviewer Lens. Reading the fields as one argument reveals more than counting filled cells. A well-bounded question may have an unsuitable method; a suitable method may have no interpretable result yet; an interpretable result may belong to another configuration. Conversely, a prospective plan with “not generated” in its result field

may be ready for execution. Enough for that decision is different from enough for a claim. A local observation that meets its prior criterion can close its own question while a broader use question remains open.

Consultant Lens. Separate owners often already hold the necessary records. The useful early work is to connect claim, configuration, study, risk and clinical decisions so another reader can follow the exact state, planned rule, observation and bounded disposition. Repeating all the files inside a new table makes maintenance harder without strengthening the inference. Where an existing study answers the question, the closure record can point to it; where it does not, the owner and missing link become specific.

What can this method actually observe?

The release-block question can be answered by introducing the specified staged invalid state and inspecting the action offered on screen. It does not need a diagnostic reference method to establish whether that action was available. Now change the question: did the reported sequencing variants correctly identify the target? A screen observation is no longer enough. The study needs a defensible way to establish the target's status and interpret the comparison.

That is where a **comparator**, the method or result used for comparison, needs an explained role. Agreement tells us that two results matched. If neither method independently establishes the target status, matching alone cannot establish correctness. The **reference process** is how the status used for judgment was assigned. A respected name on the comparison method cannot substitute for showing what that process establishes for this question. Chapter 5 follows that relationship in detail.

A pathology timing log makes the same limit easy to see. It can show when a supported image entered and left a worklist. That is useful timing evidence. It cannot establish whether suspicious tissue was correctly recognized or what a pathologist would have done without assistance. Answering those different questions may require different observations, users or a reference process. The word “assistive” does not select a reader-study design or endpoint.

The source used to justify the method also has boundaries. The U.S. Food and Drug Administration's 2007 diagnostic-statistics guidance distinguishes reference-standard comparisons from other-method comparisons for qualitative binary diagnostic results in its premarket-notification, or 510(k), and premarket-approval contexts. It is nonbinding guidance. It does not provide a universal tissue-sequencing or pathology reference method, cover quantitative or ordinal outcomes, or serve as a general manual for multiple samples from one patient.

A clinical-study standard leaves a different boundary. ISO 20916:2019 addresses regulatory clinical-performance studies of in vitro diagnostic devices using human

specimens and expressly excludes analytical-performance studies. Its planning clauses can help with clinical-study work within that scope. They cannot be treated as the analytical-validation protocol for HelixBridge merely because both activities use specimens.

Reviewer Lens. The useful first reading of a method is what it actually observes. For the block's specified behavior, the staged input and screen observation may be sufficient. For a correctness claim, the missing relationship may be target assignment; for a reader claim, it may be the intended-user comparison. Preserving those boundaries lets a timing result remain a good timing result. A source mismatch belongs beside the method it affects, where the reviewer can explain why that citation fails to answer this question.

Consultant Lens. The method specialist appraises the procedure while the clinical or claim owner explains the inference needed. Their agreement is the substantive work behind the method field. A comparator may be necessary for one question and unnecessary for another; either choice needs a reason. If a source covers a neighboring job, the consultant can route the unresolved question to the appropriate specialist. Expanding the meaning of the citation would hide the gap until the result has already been collected.

What would count before the result is known?

“We will review the data” leaves the block's decision unfinished. Three statements complete it. The **planned criterion** is that the release action must be unavailable in the specified staged initial-invalid condition. The **observation** from Chapter 3 is that the action was unavailable and an event was recorded. The **disposition**, the decision drawn from that comparison, is that the specified behavior met its criterion in the staged build and condition. The plan made the comparison possible; the observation supplied the answer. Neither establishes that all invalid routes were blocked.

Before interpreting the answer, the plan also needs to say what enters it. The deployed sequencing note counted 40 consecutive attempts, two initially invalid. A later repeat of an invalid attempt, if one occurred, would raise a handling question; it would not justify silently replacing the attempt denominator with a count of final reports. The pathology note counted 25 supported-image cases against a local worklist target. It supplied neither a sensitivity denominator nor a count of clinical errors. The **analysis set**—the observations included in the calculation or judgment—has to fit the selected question, with the unit, exclusions, invalids, repeats and missing observations handled explicitly where relevant.

A confirmatory plan, one intended to test a stated proposition, therefore fixes its objective, observation or estimate, analysis set, criterion and consequences before outcome information can shape those choices. The rule should explain what would

support the bounded claim, require a limitation or further work, or leave the decision unresolved. Its reason must fit the technical and clinical question. A numerical threshold is only one possibility: a categorical result, a structural behavior such as the blocked action, or an escalation rule can be appropriate. No general source in this chapter supplies one diagnostic pass percentage for all products. The absence of such a percentage makes justified criteria necessary, rather than optional.

When was the rule fixed, for which configuration, and who could see outcomes then? These questions establish whether the rule was genuinely prospective. The nonbinding FDA 2013 pivotal-device-study guidance recommends prespecified analysis and cautions about post-hoc choices in its U.S. pivotal clinical-investigation scope. ISO 20916:2019 Clauses 5.5.1 and 5.5.3 distinguish a clinical-performance protocol and justified amendments from the later report. Neither source supplies a criterion for the illustrative block or HelixBridge's unknown analytical study.

What changes when the plan changes?

Suppose an illustrative pathology team discovers, before viewing outcomes, that it cannot assemble the image set its plan specified. A dated amendment can explain why the question or method changed and what that change means. Compare that with a team viewing favorable results, removing a difficult image group from the analysis and calling the revised rule “prespecified.” Both histories need to remain visible. Their timing and opportunity for outcome-driven selection differ, so they cannot carry the same interpretation merely because the final table looks alike.

A justified later analysis may still be informative. It can support a narrower conclusion or suggest a follow-up question without acquiring the status of the original confirmatory test. Likewise, a matrix assembled after a study can locate the old plan, observations and gaps. It cannot make an after-the-fact criterion prospective. The original plan, amendments, dates, reasons, access to results and later reconstruction each contribute to appraisal. Transparency permits a bounded judgment; it does not require automatic rejection of every old observation.

The selected Chapter 4 technical figure: criterion and amendment history traces a generic question, evaluated configuration and criterion fixed before outcomes to later observation and bounded appraisal. A separate lane preserves amendment or reconstruction date, reason and outcome access. Its teaching point is temporal: the same words written before and after seeing results do not have the same evidentiary history. The figure supplies no case cutoff, result, completed matrix row or exercise answer.

Book 1C - Chapter 04

Preserve criterion history

Fix the rule before outcome information; preserve every later change.

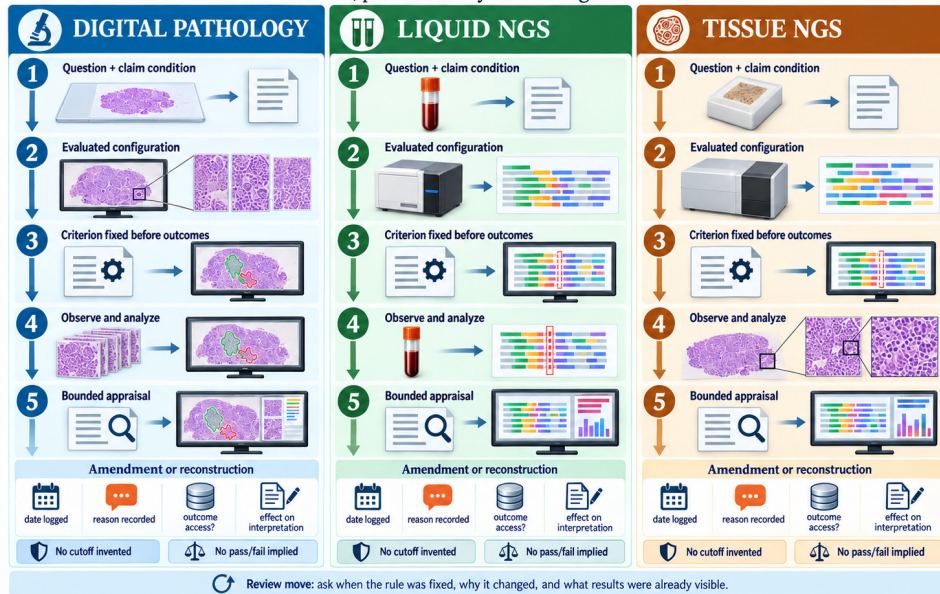


Figure 4.1. Preserve criterion and amendment history – Technical review plate.

Reviewer Lens. A dated rule and a favorable result are only the beginning of the comparison. The reviewer needs to know which observations entered the judgment and whether the rule changed after outcomes were visible. This is especially easy to lose when initial invalid attempts become final reports in a summary or late cases become an undefined “error rate.” A transparent later analysis can retain value with its limitations stated. The focused concern is a claim to prospectivity or breadth that the actual history cannot support.

Consultant Lens. The study owner and statistician or method specialist can settle the analysis set and acceptance logic before confirmatory outcomes are available, when competing results have not yet made one rule attractive. When circumstances require an amendment, the consultant helps make the reason, date, outcome access and decision consequences intelligible. Closure means a justified plan state followed by appraisal of the evidence actually generated. It cannot mean arranging a favorable result into a retroactive pass.

Which question does the source answer?

A regulation, a study-planning standard and an actual study report may sit next to each other in the folder. They do different work. The regulation can establish an applicable obligation; the standard can describe a kind of study; the report can show what happened for an identified product state. The first two do not generate the observations in the third, and a study does not confer a regulatory outcome on itself. This is why source authority and scientific fitness remain separate questions in the plan.

For a U.S. in vitro diagnostic labeling issue, 21 CFR 809.10 addresses intended-use and performance information within its applicability provisions. 21 CFR 860.7 supplies safety, effectiveness and valid-scientific-evidence context for its specified classification, special-control and premarket-approval decisions. These provisions do not mandate the book's matrix or select a particular assay study. A nonbinding FDA guidance may help with a technical question in its own scope without becoming a legal requirement.

For an applicable European Union in vitro diagnostic device, Article 56 and Annex XIII of Regulation (EU) 2017/746 establish the performance-evaluation framework. Available evidence relevant to the intended purpose is identified and appraised; additional data address gaps when needed. That order gives the missing information a precise meaning: a consequential question remains unanswered after appraisal. A missing document title alone does not establish such a gap, and suitable existing evidence need not be replaced by a new study. Nonbinding Medical Device Coordination Group guidance MDCG 2022-2 explains these planning relationships without replacing the Regulation. The regulated performance evaluation plan has its own requirements; this introductory matrix is neither that plan nor a completeness test for it. Detailed European application belongs to Book 2.

Technical sources need the same restraint. The 2019 editions of the risk-management and clinical-performance-study standards cited here are current in the source record, but publication status does not establish jurisdictional adoption. ISO 14971:2019 does not choose a universal acceptable risk or validate a product by citation. ISO 20916:2019 does not turn into an analytical-performance protocol. A binding obligation can matter without proving a particular observation sufficient; a useful study can inform a technical judgment without determining the legal one. A single ranking of sources cannot answer both.

Reviewer Lens. The precise reading follows the citation to its proposition: which date or edition, jurisdiction, authority and passage support this statement? A second reading then follows the actual evidence to the population, material, users, state and claim. Closing the source-status question is useful, but it cannot close an unexamined study. Conversely, an applicable study deserves its bounded conclusion even if its title differs from the expected one. A request becomes more useful when it identifies which of these two relationships is unresolved.

Consultant Lens. Regulatory owners select the applicable U.S. or European route; technical owners assess the method and evidence. Keeping their decisions adjacent helps each see what the other needs while retaining accountability. The practical choices may be a justified use of existing evidence, a narrower claim, a question about bridging a difference, or a defined need for new data. The consultant makes the source and product-specific rationale clear enough to compare those choices. A current international standard cannot supply an authorization or conformity outcome.

What the HelixBridge file lets us plan

HelixBridge Dx, the fictional tissue-oncology sequencing product, has a package described as containing all major documents. Its broad purpose is to identify variants associated with eligibility for a named therapy. In this Chapter 4 snapshot, the therapy, intended population, variant classes and exact treatment-facing wording are not supplied. A U.S. marketing submission and a separate European in vitro diagnostic regulation strategy are being prepared. Neither preparation is an authorization or conformity outcome.

The record also identifies a study called **analytical validation**. Its partly specified validation configuration contains the validation application, the validation report rules and the knowledgebase used in analytical validation. Their source identifiers are preserved together in the table below. The exact commercial configuration remains unresolved. The wet-laboratory, instrument, reagent and other dependency identifiers needed to reconstruct the full evaluated system are not supplied. Neither are the method, sample count, criteria, failures, strata or results. This is a limited planning snapshot; it supplies no release or change event.

What is already positive?

The study is identified, and the package is described as containing the major documents. Those are useful inventory facts. They give the inquiry a starting point without showing which claim element the study addressed or whether its data apply to the proposed commercial state. A matrix reconstructed now can locate the original records and criteria. Only their actual dated history could show that those criteria preceded the original study outcomes.

Where does the argument still need a connection?

The following table applies all thirteen fields to this snapshot. It is useful for distinguishing a supplied fact from a proposed question or conditional next step; its planning questions are author reconstruction, not additional case facts.

Field	What this snapshot supports
1. Claim and condition	Tissue-based variant identification associated with therapy eligibility. Exact therapy, population, variant classes and wording remain undefined here.
2. Decision	Whether identified analytical evidence supports the relevant claim element for the proposed commercial state. This is not the marketing or conformity decision.
3. Uncertainty	Claim detail, complete studied state, study basis and commercial applicability remain unresolved.
4. Evidence question	Which claim elements and conditions did the analytical work examine, and how do they relate to the state proposed for use?
5. Configuration and time	Validation application 7.0; validation report rules R19; knowledgebase used in analytical validation DB-18. Physical and other dependencies are missing here. The commercial state is separate and unresolved.
6. Conditions	Tissue modality is known. Consequential specimen-chain, population, user, setting and workflow details are not established by the excerpt.
7. Method and reference	Not supplied. The actual method and the role of any comparison need appraisal.
8. Judgment served	Candidate analytical-performance claim support. The study title establishes no control-implementation, effectiveness or residual-risk conclusion.
9. Acceptance logic	Unknown. The original criteria, rationale, timing and claim relationship need to be recovered.
10. Result state	Study named; appraisal information not supplied. No pass or fail established.
11. Decision route	Appraise the bounded claim and configuration relationship; retain a limit, bridge question or gap where it cannot be justified.
12. Source authority	The analytical question and the separate U.S. submission and European strategy each need their own source fit. Neither regulatory outcome is established.
13. Closure and owner	Exact claim and configuration records plus the study basis could permit appraisal. The appropriate decision owner needs assignment; no individual is supplied.

Consider Fields 5, 9 and 10 together. The configuration identifiers let a reviewer locate particular analytical work. They do not tell the reviewer which rule judged its results or what those results were. The narrow next request therefore connects the evaluated-state record to the proposed commercial-state record and recovers the exact claim element, original method, criterion timing, observations and results. Each item repairs a distinct break in the argument. “Not supplied in this excerpt” says what this reading can establish; it does not prove that the original study lacked a method or failed.

What could the next record change?

If the retrieved records show a suitable question and method, defensible original criteria, interpretable findings and applicability to the bounded claim and state, they could support that relationship. A consequential difference could instead permit only a narrower condition, require a supported bridge or call for new evidence. Where the relationship remains unjustified, that specific gap remains open. These are possible appraisal outcomes. The supported positive conclusion now is study identity and partial configuration provenance; performance adequacy, commercial applicability and regulatory outcomes remain undetermined.

Reviewer Lens. A demand for “more analytical validation” can obscure the most useful work: connecting the study already named to the claim and proposed state. The response might close an analytical subquestion while leaving treatment-facing use or a jurisdictional decision open. The reviewer can make that possible by distinguishing study presence, an appraised result and claim support, and requesting the smallest record chain that could establish the missing relationship. There is no need to discard the positive inventory facts.

Consultant Lens. The claim owner resolves therapy-facing wording, the configuration and study owners recover the original state and evidence, and regulatory owners assess the separate U.S. and European implications. The consultant helps each see what counts as closure and where the next handoff occurs. Assigning an owner role is legitimate here; inventing an individual is not. A complete record may justify reuse, a limitation or further work. The consultant cannot select the conclusion before those records exist.

What question comes before choosing the pathology study?

This exercise selects a narrower PathSight Cloud record than Chapter 3 did. The initial package describes an assistive whole-slide-image product that prioritizes images and highlights suspicious regions for a pathologist, who retains responsibility for diagnosis. No study, result, scanner state or changed configuration is selected for this occurrence. The earlier records still exist; the exercise deliberately limits the evidence in hand.

Decision

What human-workflow decision needs an answer before relying on this assistance, and what does “assistive” leave unknown? What can be asked without borrowing a method, endpoint or outcome from another record?

How the clinical role leads to a question

The pathologist retains diagnosis, while prioritization and highlighting contribute to the work that leads there. A planning question can therefore ask under what conditions one of those functions affects the intended workflow. “Assistive” establishes the division of roles; it leaves actual reliance, queue effects and safe work under particular conditions unresolved. It also supplies no method, criterion, observation or result.

The tissue-sequencing example contributes its question-to-decision reasoning. Its assay method, reference and study type do not transfer with that reasoning. A pathology method needs the facts and authority appropriate to its own question.

Reviewer Lens. The stated role and pathologist's ownership deserve credit. The focused request is which workflow decision is pending and which product state and use conditions would make an observation applicable to it. Requesting a particular reader-study title from this limited record would choose the method before understanding the question. Importing Chapter 3's favorable study summaries would also violate the selected evidence boundary.

Consultant Lens. Pathology and product owners can decide whether the concern is queue timing, highlighting, user interpretation or another consequential relationship. Qualified specialists can then select a method that observes it. The consultant connects the condition, question and owner before a test is specified. Closure requires a supported answer for that selected question; the word “assistive” cannot provide it.

Can one activity answer two questions?

An unnamed evidence activity is proposed for two of this chapter's six judgments. No product, method, result or pass is supplied.

Decision

For two judgments that are easy to confuse, what extra relationship would make one observation relevant to both? When could the original activity suffice, and when would another activity be needed?

How shared evidence keeps separate conclusions

Implementation asks whether the control is present; effectiveness asks whether it reduces the relevant route to harm. One observation can contribute to both when its method and the control-to-risk connection justify both readings. If that connection is absent, two labels cannot create it. Another activity may be needed, but that follows from the missing inference, not from the number of labels.

The distinction changes again for individual and overall residual risk. Evidence about one remaining risk can contribute to an evaluation of the remaining risks together without constituting that overall evaluation. For control effectiveness and diagnostic-performance support, the unfinished connection is especially consequential: preventing a misleading output does not show that valid outputs meet the diagnostic promise. In each pair, the appropriate question is what the existing observation and its conditions actually establish.

One shared record can therefore preserve several separately justified conclusions. The next record gives much less information and asks whether even a prospective decision has been made answerable.

What is missing from this proposed plan?

The following table is the entire anonymous record available for this exercise. Its three entries show exactly what the reader may use.

Item supplied	Entry
Claim condition	A bounded intended-use condition has been identified for examination.
Uncertainty	One uncertainty associated with that condition has been identified.
Proposed activity	An evidence activity has been named.

For example, the last row supplies the fact that an activity has been named. It supplies no disease, instrument, method detail, threshold or result. The other two entries have the same limited status. Every proposed repair must respect that boundary.

Decision

Across the thirteen fields, which missing relationships prevent this record from supporting a prospective decision, and why does each matter? Keeping the three supplied entries intact, what must be fixed before outcome data can be seen? How do question, method, acceptance logic, result state, source fit and closure differ here? What one focused request could make the plan answerable, and what conclusions would still remain open?

How a reviewer repairs the argument without inventing the study

Worked reasoning

The three entries preserve a useful inventory: a condition, an uncertainty and an activity have been identified. They do not say which decision connects them. The first repair is the relationship among Fields 1–4: what needs deciding about the condition, why the uncertainty matters to that decision and what answerable question follows. This requires the plan owner's reasoning, not invented clinical detail.

The word “bounded” cannot fill the next two fields. Field 5 needs the actual state to be evaluated. Field 6 needs the consequential population, specimen or image, users, setting and workflow, including their unknowns. Without them, a later result could be applied to a different state or use from the one examined. These are information to recover, not evidence that those conditions were absent from the original work.

The activity then needs its reason. Field 7 asks what the method observes and why it can answer the question, including the role of a comparator or reference when the inference needs one. A comparator is not obligatory merely because the field mentions it. Field 8 identifies whether the evidence serves risk estimation, control implementation, control effectiveness, individual residual risk, overall residual risk or performance support. If more than one role is claimed, each needs a separate explanation.

For confirmatory work, Field 9 needs a justified rule fixed before outcomes can shape it. The observation and analysis set must be defined, with relevant handling of invalids and repeats, and with an account of which result would support, limit or leave the decision open. The supplied record cannot yield a numerical criterion. Field 10 is **result not supplied**: naming the activity establishes neither execution nor a usable outcome. Field 11 may describe a conditional decision route, but no final disposition can be made from these facts.

Field 12 needs a source whose authority and technical scope fit the selected question. Field 13 then identifies the remaining gap, what would close it and the next owner role

to assign. A useful focused request is for the plan owner to connect the stated condition and uncertainty to the pending decision, evaluated configuration, reason for the activity and prospective interpretation rule. A satisfactory response could make the **plan** ready to execute or appraise. It would still not establish that the activity happened, passed, applied to a released state or supported the product claim.

This is why “add a pass/fail percentage” fails as a repair: it invents the missing basis. “The whole product is invalid” makes an equally unsupported leap from a thin excerpt. If existing records already supply the missing relationships adequately, they can receive a bounded positive judgment. New work is not required merely to replace their titles.

The initial judgment should retain the three inventory facts. The revised judgment changes when their relationships become answerable. Comparing those two judgments reveals whether the reasoning improved: which fact remains, which inference was repaired and which question still awaits evidence? A tidier set of blanks would not produce that change.

Who can settle the open relationship?

The missing relationship in a plan often crosses several owners. HelixBridge's snapshot makes the division concrete: the proposed promise, the state examined and the original study record may each need clarification before anyone can appraise support.

The reviewer identifies the decision at stake

The useful request asks how the analytical work using the knowledgebase from validation relates to the bounded claim and proposed commercial state, including the original method and rule. That trace could support a particular analytical element while leaving treatment-facing use, the complete commercial state, U.S. authorization and European conformity unresolved. The reviewer needs the missing relationship explained; a second document title alone would not answer it.

The manufacturer connects the records and their owners

The claim owner defines the exact promise, the configuration owner identifies studied and proposed states, and the study owner recovers the original plan and result. Each decision needs its actual responsible owner when known and a clear closure condition. “Method not supplied in this excerpt” is an honest description of available information. It does not prove the original study omitted a method. A plan can likewise be ready for execution while its result correctly remains “not generated.”

The clinical setting tests whether the conditions fit

The laboratory or clinical team can compare the planned material or image, users and workflow with the setting in which the result will be produced and used. A mismatch matters when it changes the meaning of a report or the person expected to act on it. Identifying that relationship does not require inventing a clinical result or an operational change. Qualified specialists retain the method, study and care decisions beyond this planning account. The consultant can connect the handoffs without taking over those decisions.

What reopens when only the intended population changes?

In an unnamed, nonrecurring hypothetical, the assay configuration, specimen type and claimed output remain the same. The intended population changes from **the population represented in the original evidence** to **a population outside that evidence**. These are Population A and Population B in the source record. This is separate from HelixBridge and PathSight.

No clinical characteristics, prevalence, jurisdiction, method or study results are supplied for either population. The fixed assay, specimen and output and the changed intended population are the complete facts available.

Decision

Which evidence questions reopen, and why? What remains unchanged or unknown? For each question, what would justify direct use of existing evidence, a bridge, new evidence or a remaining gap? What can be concluded now, and which owner role could settle the next decision?

How applicability changes without inventing a new assay

Transfer reasoning

The only stated change is the intended population. An explanation based on changed hardware, assay configuration, specimen type or output would contradict the exercise. Yet holding those elements fixed does not show that the original evidence applies to the newly intended population. The original method, results and limits are absent, as are the new population's characteristics, prevalence, spectrum of conditions, exclusions, risks, reference process, users, settings and workflow. These are possible questions to investigate, not differences already established.

The comparison begins question by question: what did the original evidence represent, and why might population matter to that inference? Evidence for a purely specified component behavior may remain relevant under the same configuration, but these facts

do not prove that all relevant conditions are unaffected. A clinical or use-related proposition may depend on characteristics not supplied. Being outside the original evidence population makes an applicability argument necessary without erasing earlier observations.

Four conditional routes follow from that comparison. **Direct use** can be justified for a particular question when the relevant conditions and inference remain applicable and the existing evidence adequately answers it. An **evidence bridge** connects a defined difference through evidence-backed reasoning to the proposed conditions; calling reuse a bridge does not create the connection. **New evidence** becomes necessary when a consequential question cannot be answered or bridged from existing work. A **remaining gap** is the honest state when the facts needed to choose among those routes are unavailable. The fact sheet selects no route for the product as a whole.

The risk questions remain separate too. A different intended population alone establishes neither a new hazard, an effective control nor acceptable individual or overall residual risk. The useful next question identifies which risk relationship could change and what information its responsible owner would need. The possible evidence route stays attached to that question and the relevant owner role; no individual owner's name can be supplied from this record.

The initial judgment can therefore preserve the unchanged elements and explicitly unresolved applicability. A defensible revised judgment would change only when the missing comparison is supported. Three tests help distinguish that reasoning from an automatic choice: has an unstated population difference slipped in, has a conditional route for adequate existing evidence remained possible, and is there a specific consequential question that would require new evidence if it could not be answered or bridged? Both unrestricted reuse and repetition of everything skip that question-by-question appraisal.

Back to the covered study titles

The colleague can now cover the titles without hiding the argument. Each activity has a question tied to a claim or risk, a state and conditions to which the observation belongs, a reason for the method and a rule for the decision. Planning quality, execution, an interpretable result, applicability and bounded support remain distinguishable. A supported answer can close one relationship while an honest gap directs the next owner to another.

Chapter 2's configuration and Chapter 3's qualified evidence job now have a prospective question to connect them. The anonymous application and changed-population exercise show what happens when that connection is missing or challenged. Chapter 5 follows one especially consequential part of the method field:

what the reference or comparator can establish before agreement is interpreted as correctness.

Questions that keep the inference visible

“Supported,” “repair needed” and “unresolved” are states of a particular inference, each justified by its record. They are not scores for completed cells. The following questions return to the places where a plan can appear complete while its argument remains unfinished.

- Which bounded claim condition and decision make the evidence question consequential?
- Does the activity observe the right object in the relevant state and conditions?
- Which of the six judgments does it serve, and what justifies any additional role?
- Does the method, including a reference or comparator when needed, answer this question?
- Was the acceptance logic justified and fixed when claimed, with the unit, exclusions, invalids and repeats accounted for?
- Can the original plan, amendment, later reconstruction, observation and decision still be distinguished?
- Does each source support its exact proposition at the stated authority, date, jurisdiction and scope?
- Are a complete plan, execution, an interpretable result, applicability and claim support still separate states?
- Can adequate existing evidence receive a bounded positive conclusion while each remaining gap has an owner and closure condition?

The reasoning without the field names

Decision

How does the reasoning move from a consequential claim condition through question, method, planned rule, observation and appraisal to a decision? Why can one activity serve two judgments without merging them? What separates a justified amendment from a rule chosen after results? Why does a current authoritative citation not establish study sufficiency? When the population changes but the assay does not, what must be reconsidered before deciding to repeat work?

The next chapter's examination of reference assignment can change a specific inference in the method field. The original question and prospective rule still matter when that happens. Even a retrospective account of older evidence needs to retain when each plan, criterion and result came into being.

Source trail

These sources support bounded planning distinctions, not the fictional cases' performance or regulatory outcomes. The thirteen-field matrix, six-judgment teaching arrangement, examples and conditional conclusions are author synthesis. Licensed standards are paraphrased. Status-sensitive sources were checked September 24, 2026; recheck before publication.

Source and exact locator	Role in this chapter and boundary
21 CFR 809.10, (a)(2), (b)(2), (10)–(12)	U.S. IVD labeling anchors for intended use and bounded performance information; not a universal study method or this matrix. The eCFR is an authoritative, unofficial presentation of underlying law.
21 CFR 860.7, (b)–(g)	U.S. safety/effectiveness and valid-evidence context for the provision's stated classification, special-control and premarket-approval decisions; no fictional-case authorization or universal pathway rule follows.
FDA diagnostic-statistics guidance, 2007, Sections II, V–VI	Nonbinding reference-standard versus other-comparator distinction for the guidance's qualitative binary diagnostic-test setting; it does not cover quantitative, ordinal or multiple-samples-per-patient designs as a general manual.
FDA pivotal-device-study guidance, 2013, Sections 2.2, 6, 8–10	Nonbinding intended-use, study-design and prespecification context in its pivotal clinical-investigation scope; not a comprehensive analytical-validation method.
Regulation (EU) 2017/746, Article 56; Annex XIII Part A 1–1.3	Binding EU IVD performance-evaluation planning, existing-data appraisal and gap-to-additional-data framework where applicable. The consolidated view is a navigation aid; the base act and applicable amendments control. It does not make this matrix an EU performance evaluation plan.

Source and exact locator	Role in this chapter and boundary
MDCG 2022-2, Sections 6.1–6.3 and 7	Nonbinding EU explanation of intended purpose, risk and performance-evaluation planning; not an IVDR substitute or product-specific conformity finding.
ISO 14971:2019, Clauses 1, 4.1–4.5, 5–8, 9–10.4	Current licensed international standard, reviewed at printed pp. 1, 7–16; bounded source for risk estimation, control implementation/effectiveness, residual-risk and post-use review distinctions. It neither sets universal acceptable risk nor supplies this chapter's matrix or jurisdiction-specific legal effect.
ISO 20916:2019, Clauses 1, 5.1–5.3, 5.5.1–5.5.3, 8.2	Current licensed international standard, reviewed at printed pp. 1, 12–19 and 25–27; clinical-performance-study planning, amendment and report distinctions within its stated scope. Analytical-performance studies are excluded.

HelixBridge and PathSight are controlled fictional composites; the deployed teaching notes and unnamed changed-population exercise are separate fictional records. They supply only their stated facts, not scientific or legal authority.

CHAPTER 5

Reference Processes and Comparator Limits

What stands on the other side of “agreement”?

A tissue specimen reaches a laboratory. Sequencing and software help identify a variant, and the finding reaches a report. A study says the result agrees with something else. Chapter 4 gave that study a bounded question, a product configuration and a decision to inform. We now need to follow the *other* answer: how was the condition or property used for comparison assigned?

The fictional HelixBridge Dx identifies tissue variants associated with eligibility for a named therapy. Its selected records contain a favorable phrase, “**strong overall agreement.**” The therapy, exact population and variant classes remain unspecified. The following table locates that phrase so that it stays attached to the record that contains it.

Table 5.1 – Two evidence states in the HelixBridge record

Part of the record	What is supplied
Analytical-validation study	Application 7.0, report rules R19 and knowledgebase DB-18
Separate final technical report	Knowledgebase DB-19; “strong overall agreement”
Knowledgebase difference	The separate report's knowledgebase added a transcript that makes a previously rare mapping anomaly execute more often

The first row identifies the validation application, validation report rules and knowledgebase used in analytical validation. The second identifies the knowledgebase in the separate technical report. These are two evidence states within one shared case snapshot, not two replications. The table does not establish that all other components matched or identify what result each state produced. It supplies no comparator, reference material or procedure, variant-assignment rule or reference result. Counts, strata, failure denominator, acceptance criterion and the basis for “strong” are also absent.

Decision

What can the agreement line support now? Which missing reference-process record would most directly change that judgment?

The initial judgment will return after the comparison has a clearer meaning. Its possible conclusions include a defensible assignment for a particular condition, useful agreement with another method, and a case whose status cannot be resolved. An imperfect reference may still do an important job within limits. A five-part **reference-process adequacy memo** will hold that reasoning together. This book's two-page aid is neither an ISO form, a regulatory checklist nor a score. Its value lies in explaining what the process supports and where that support stops.

Which earlier question needs this comparison?

Three questions connect the new work to the earlier chapters. What distinguishes a product name from the configuration represented by a study? What must be added to the word *validated* to identify its evidence job? What relationships make a claim-to-study row answerable?

For a comparison-bearing row, the relevant starting record contains the claim condition, pending decision, risk or uncertainty, evidence question, configuration and time, use conditions, method, evidence judgment and prospective acceptance logic. Its result state, source authority and residual gap remain visible. The matching configuration record from Chapter 2 and evidence-job qualifier from Chapter 3 give those entries their meaning. Which part of that row would change if the reference could assign only a narrower property or population?

How the earlier plan locates the reference question

The relationship is bounded question + configured object + justified method + prospective judgment → decision with a residual gap. A reference belongs in the method when the inference needs a comparison. A staged check of whether an invalid report can be released may be judged against the specified release behavior; it need not assign a patient's tumor-variant status. The comparison-bearing row for this chapter instead asks something for which a second assignment matters. Expanding its reference reasoning leaves the rest of the plan intact and supplies no observed result.

One specimen, two routes to an answer

Follow the tissue specimen again. Collection and preparation precede sequencing; software examines the readings and produces a candidate variant result for reporting. To judge whether it identifies the relevant variant, the study needs a separately described route that can assign that variant status. Material, another procedure and review rules may all contribute. Their output may also be “cannot assign.” Before the two answers meet, specimen or case identity, time, property and conditions must match.

The same need becomes especially visible on a slide image. A pathologist can diagnose the whole slide, while an assistive product can highlight one region or give the image a worklist priority. A whole-slide condition and the location of a lesion require different assignments. A faster reader or a visible highlight supplies neither assignment merely by being observed.

The candidate and reference chains meet at one question, not at a word such as *truth*. The first chain tells us what the configured product and user produced. The second tells us how a target condition or property was assigned—or why it remained unresolved. Their comparison can then support a bounded conclusion. Figure 5.1: Candidate chain and reference-assignment chain meet at one question keeps both identities and their meeting question visible.

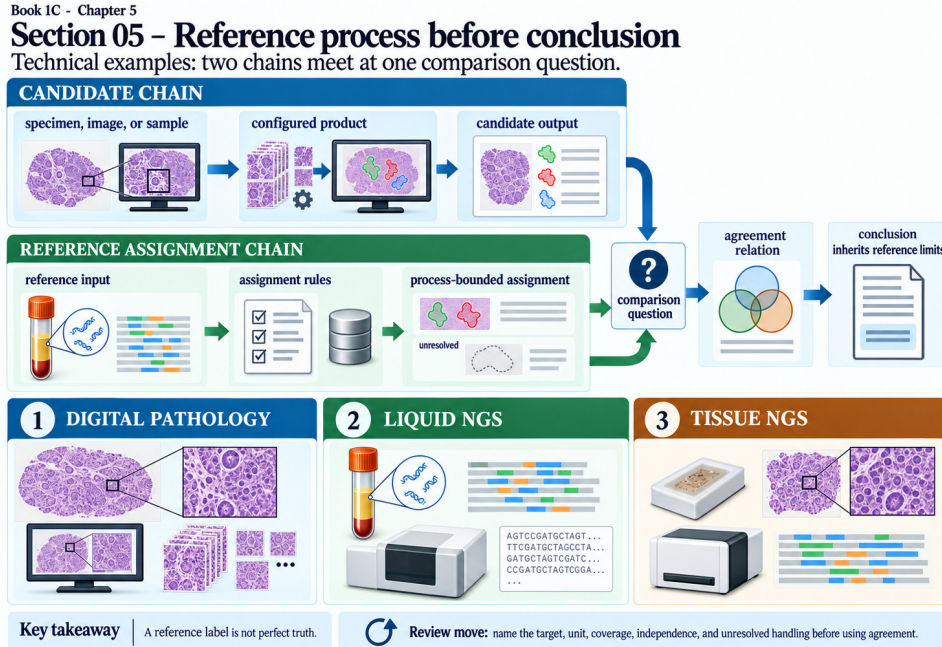


Figure 5.1. Candidate chain and reference-assignment chain meet at one question – Technical review plate.

The second route is chosen for the conclusion the team intends to draw. Choosing it prospectively means its rule exists before favorable results or disagreements can influence its construction. A later candidate configuration, reference release, specimen path or intended conclusion may require the relationship to be appraised again. Chapter 7 develops quantitative calibration and uncertainty; Chapter 12 develops study design and analysis. Here the narrower task is to understand what role the reference can serve.

Reviewer Lens. Following one result is often more informative than reading a list of reference names. A tissue-variant comparison needs the same unit and property on both sides; a pathology comparison first needs a choice among diagnosis, location and reader action. A complete, matched process deserves credit for that question. Where a link is absent, the useful request names the missing record instead of turning the gap into a product-failure verdict.

Consultant Lens. The claim owner fixes the decision, the product team identifies the candidate state, and the scientific or clinical methods owner proposes how the comparison status will be assigned. Agreeing on the unit and stop states before collection prevents later ambiguity about which answers belong together. After a change, a documented crosswalk may be sufficient to preserve applicability. The owner must establish that relationship; a version label cannot do it alone, and a new study is not automatic.

What kind of answer can this reference provide?

Imagine the question concerns a concentration measured on two instruments. A characterized material gives the instruments something with specified properties to examine. A measurement procedure gives a way to assign a value to a specified quantity. Those are useful contributions to comparing numerical results. They leave a further question if the proposed conclusion is that a patient has a clinical condition.

A sequence-change category or a pathologist's diagnosis asks for a different assignment. A clinical process may combine observations to determine the target condition. Another assay may instead provide only a second result with which the candidate can agree or disagree. The table helps distinguish these roles when a dossier uses *reference* for all of them.

Table 5.2 – What the reference supplies

Role	Contribution to the comparison	Boundary of that contribution
Reference material	Material characterized as sufficiently uniform and stable for specified properties and use	Another property, specimen path or clinical conclusion needs its own qualification

Role	Contribution to the comparison	Boundary of that contribution
Reference measurement procedure	A procedure accepted for a defined measurement role: assessing trueness, calibration or characterizing material	Its title does not establish clinical diagnosis or categorical truth
Clinical reference standard	In the cited FDA context, the best available process for assigning the target condition without the investigational output	It may misclassify a subject and may not fit another context
Comparator	The method, process or result placed opposite the candidate	Agreement with it does not establish that it is correct

In the concentration example, **reference material** and a **reference measurement procedure** have complementary jobs. Calibration relates instrument response to assigned quantity values. Trueness concerns how close an average result is systematically to a suitable reference value. Characterization documents the material's specified property. These are measurement jobs; none by itself assigns a disease.

The clinical row concerns a **clinical reference standard** in the FDA diagnostic-study context. “Best available” describes a justified role, with possible error. The last row makes a smaller promise: a **comparator** supplies the other output. When two outputs match, that relation can be useful even before either method's correctness is established. The measurement vocabulary comes from JCGM; FDA uses its clinical-reference wording in a particular diagnostic context, while CLSI EP49 uses assessment of the target condition and a comparator in its clinical-validation framework. These source-specific meanings remain distinct. (JCGM VIM terms 2.7 and 5.13; FDA diagnostic-statistics guidance, Introduction; CLSI EP49 §1.4.1.)

A useful test of any proposed source is whether this sentence can be completed: **the process must assign this property or condition, for this population and material, at this time, to support this conclusion.** The unit then makes the sentence concrete: a variant at a defined genomic position, a region on an image, a whole-slide diagnosis or a person's condition. A gene name or slide label cannot settle all those fields.

What does “unresolved” refer to?

Suppose a suitable process has been identified. It might be available already, need prospective construction from documented components, or be unavailable for the proposed use. Even an available process may be impractical to apply to everyone. That calls for a justified design and analysis, without declaring a more convenient method equivalent. These are study-level possibilities described in the FDA guidance's binary diagnostic context; neither named product has been assigned one of those states here. (FDA diagnostic-statistics guidance, §4.1.)

Availability still does not tell us what happened for one specimen. The process may assign a supported positive or negative status, or leave the case unresolved because a required input is missing or conflicting. And a third uncertainty belongs to the reviewer: the dossier may not explain the method at all. Missing information about a process does not prove that the study had no adequate process; an unresolved individual result does not prove that no suitable process exists.

A prespecified, documented and fit assignment can therefore be relied on within its stated conditions while retaining error and unresolved cases. We will call that a **process-bounded assignment**. This book also uses *operational truth* as an author-defined teaching phrase. Neither phrase promises perfect truth or attributes a new definition to FDA, ISO, CLSI, JCGM or NIST.

Reviewer Lens. An impressive source title is easy to credit beyond its role. The first distinction is what the source actually assigns; the next is what the dossier establishes about its fitness. “Process not described,” “case unresolved” and “no suitable process available” call for different responses. A documented narrower assignment deserves a positive conclusion at that boundary.

Consultant Lens. The target and unit are inexpensive to settle while the protocol is open. A methods owner can then qualify the source, protect a path for unresolved cases and identify the version to which the decision applies. After a reference update, a version and applicability crosswalk may answer the question. Rewording a certificate or comparator name cannot replace the missing clinical qualification.

What do the twenty pairs actually say?

Consider a separate fictional comparison with **20 paired, valid, two-category results**. These teaching counts belong to neither HelixBridge nor PathSight. The comparator's ability to assign the condition correctly has not been established. It calls eight specimens “condition-indicated” and twelve “condition-not-indicated.” How much agreement did the candidate show?

Start with the eight comparator-indicated specimens. The candidate also says indicated for **6**, and says not indicated for **2**. In the twelve comparator-not-indicated specimens,

the candidate says indicated for **1**, and also says not indicated for **11**. The four cells account for every valid pair: $6 + 2 + 1 + 11 = 20$. The matching cells contain $6 + 11 = 17$ pairs; the other cells contain $2 + 1 = 3$ disagreements.

For overall agreement, the **denominator**—the divisor actually used—is all twenty valid pairs. Thus $17 \div 20 = 0.85$, or **85% overall agreement**. That single percentage cannot show which direction the disagreements took.

To see the direction, keep each comparator group intact. **Positive percent agreement** asks how many of the comparator-indicated specimens also had an indicated candidate result: $6 \div 8 = 0.75$, or **75%**. **Negative percent agreement** asks the corresponding question among comparator-not-indicated specimens: $11 \div 12 \approx 0.9167$, or **about 91.7%**. The familiar abbreviations are PPA and NPA. Each fraction answers a different question because it uses a different group. Changing the mix of those groups can change overall agreement even if the two within-group fractions stay the same.

What would let the first fraction be called sensitivity? Its eight specimens would have to belong to a justified target-condition-positive group for the specified design, rather than simply a group called indicated by another method. **Sensitivity** counts candidate results within that assigned positive group; **specificity** concerns the assigned negative group. A fit but imperfect process carries its own limits into those interpretations. Two methods may agree and both be wrong, and disagreement alone cannot identify which is wrong. Independence from the candidate and a masked operator can reduce particular biases without establishing the comparator's relationship to the condition.

The twenty pairs also leave the starting set unknown. There might have been missing or invalid results before these pairs became available; no starting count is supplied. Nor are sampling, precision around the fractions or an acceptance rule given. These are observed agreement calculations, without a claim of passing a criterion. The binary framework itself needs checking before use with several variant categories, a graded pathology result or a numerical result. (CLSI EP12, §§4.2–4.2.5; CLSI EP49 §3.5.6; FDA diagnostic-statistics guidance, §4.1 and Appendix 7.2.)

That keeps the number attached to the intended evidence job. A sequencing team may want agreement with a specified method for particular variant classes, or correctness relative to a fit assignment. A pathology team may want a comparison of reader diagnosis, region location or worklist action. Those choices precede the calculation.

The earlier deployed teaching notes show how easily a different fraction could be substituted. In the separate unnamed tissue-sequencing service, **2 initial invalid statuses among 40 consecutive specimen attempts** gives $2 \div 40 = 5\%$ initial unavailability in that interval. In the separate unnamed pathology service, **1 of 25 consecutive supported-image cases** arrived after its local worklist target, giving $1 \div 25 = 4\%$; the pathologist completed review. These are availability and timing

observations. They identify neither reference-positive groups nor diagnostic disagreements and add no HelixBridge or PathSight event.

Reviewer Lens. Reconstructing the four cells exposes more than the headline percentage. It identifies the positive and negative divisors and leaves a visible question about missing, invalid and excluded pairs. A sound agreement analysis can receive full credit for agreement. Stronger language depends on the target-assignment basis, while precision and study design still need their own appraisal.

Consultant Lens. Before collection, the evidence owner and statistician can agree on the pair unit, output categories, quantity to estimate and unresolved-result handling. The reference owner supplies the assignment basis. That gives the eventual report a short trace from starting cases to each analyzed denominator. An operational signal later needs its own event investigation; reusing its percentage as diagnostic performance would answer the wrong question.

How does the second answer get made?

In a separate illustrative slide assessment, the plan calls for a clinical record and an independent slide review. It identifies when each is obtained and what happens if they conflict. One reviewer cannot inspect the slide. What status should the process produce now?

Its prespecified rule can say “unresolved pending required input.” That is a meaningful output. Filling the gap from the candidate's answer would change the route by which the supposedly separate status was assigned. This example describes a process shape; it is neither a PathSight protocol nor proof of fitness for a particular disease.

Which observations become one assignment?

When several specified observations feed one assignment, the process is a **composite reference process**. A separate sequencing illustration might combine another method's result with planned review of variant identity and region coverage. That raises immediate questions: must both inputs be present, what counts as the same written variant, and what happens when a method cannot examine the region? The rule must be able to stop without an assignment. Otherwise the last reader effectively chooses the rule after seeing which answer would be convenient.

A reproducible process therefore identifies input versions and eligibility, order and timing, the decision algorithm, repeats, stopping rules and unresolved states. FDA's pivotal-study guidance §8.2 and CLSI EP49 §3.3.1 discuss combinations of methods, clinical information and follow-up in their covered clinical contexts. The combination needs advance description and protection from selection or change by the candidate result under evaluation. A fixed rule makes a composite appraisable; it does not make its inputs or decisions infallible. (FDA pivotal-study guidance.)

Who chooses the evidence, and who sees it?

Now imagine a candidate sequencing result is positive and its initial comparative result negative. The study runs an extra reference method only after such disagreements. The extra method's operator cannot see the candidate label. Has the candidate been kept out of constructing the reference?

The operator is protected from one influence by **masking**, also called blinding: controlling which information is visible at a decision. Yet the candidate still selected which specimens received extra evidence. **Construction independence** concerns that earlier link—whether candidate output chooses inputs, further testing or assignments. An operator can be masked while the reference is constructed in a candidate-dependent way.

The reverse can happen too. Every eligible specimen might receive a fixed reference plan, but a reader who sees the candidate result may interpret an ambiguous reference result in its light. Reading order can influence a later interpretation even when the same named methods are used. And cases used to select or tune the candidate configuration create another problem if they are later used to evaluate it. Separating development from evaluation addresses that relationship, rather than either operator exposure or extra-test selection.

For pathology, these controls must respect the setting of the question. A clinician making a care decision may properly see information withheld from a study reference reader. Study masking is not a universal instruction to withhold useful clinical information. The endpoint and design determine the controls, and the record needs to distinguish them. FDA pivotal-study guidance §§8.6 and 8.8 and ISO 20916:2019 §5.5.3.8 and Annex B.8.1–B.8.2 inform these questions within their stated study scopes; the latter's analytical-performance exclusion and Annex B boundary remain applicable.

Reviewer Lens. A useful appraisal follows one ordinary case and one discordant case through the rule. Can another reader reproduce either assignment, including a stop state? Did the candidate cause added testing, and who saw each output before interpreting the next? Reciprocal masking deserves credit where documented, but supplies no answer about reference construction or reuse of development cases.

Consultant Lens. Methods owners decide what evidence will exist; study owners decide who may see it and when; data owners preserve versions and pair identities. Making these choices explicit early is easier than repairing a candidate-triggered process after results have been interpreted. A prospective rule with auditable exceptions can close a description gap. A new masked title cannot close the selection defect.

What may an investigation change?

Return to the twenty valid pairs. Two candidate-not-indicated results sit opposite comparator-indicated results, and one candidate-indicated result sits opposite a comparator-not-indicated result. These three **discordances** are observations that the paired results differ. They do not identify the correct side or establish that both methods addressed the same property under comparable conditions.

An investigation may examine specimen identity, a software path, comparator coverage or a new method. Its conclusion can explain a mismatch or add evidence. A defined later judgment is **adjudication**, and it needs its own endpoint. Deciding whether two pathology diagnoses differ in a clinically consequential way may be different from assigning the original slide diagnosis. A variant review may explain equivalent written representations without proving which assay was analytically correct.

That later record must remain linked to the original pair. If only the three disagreements were checked and their original entries replaced, the revised table could look as though the other seventeen matches had also been independently verified. They were not. Both methods could have made the same error in an unreviewed pair. A justified partial-verification design requires prospective assumptions and suitable statistical treatment; selective repair of the table is not such a design. (CLSI EP12 §4.2.5.2; CLSI EP49 §3.6.3.)

Where does the case go when no assignment is possible?

The slide whose review could not be completed remains a case. So does a specimen with conflicting reference inputs, a missing required record, an invalid test or an attempted procedure that failed. These causes need separate reasons and stages in the record. None is automatically positive, negative or zero cases. The later exercises supply no counts; “not supplied” remains the count, rather than an invented zero.

This also explains why twenty valid pairs are not necessarily twenty starting cases. A justified analysis may use a narrower set, but the route from the starting set through missing, failed or excluded results to the actual divisor must be traceable. The sequencing field note's invalid candidate result concerns candidate availability, not an indeterminate reference assignment. The pathology note's late case concerns workflow, not a reference diagnosis. Their retained identities may support investigation without forcing either event into a reference-positive or reference-negative cell. (CLSI EP49 §§3.4.1.2, 3.6.1 and 3.7.)

Figure 5.2: Preserve original pairings through discordance review shows the record relationship: a later review adds a linked observation while the original candidate–reference pair and unresolved states remain visible.

Book 1C - Chapter 5

Section 05 - Preserve the original pair

Technical examples: investigate discordance without rewriting the comparison.

**Key takeaway**

Discordance is not a verdict.

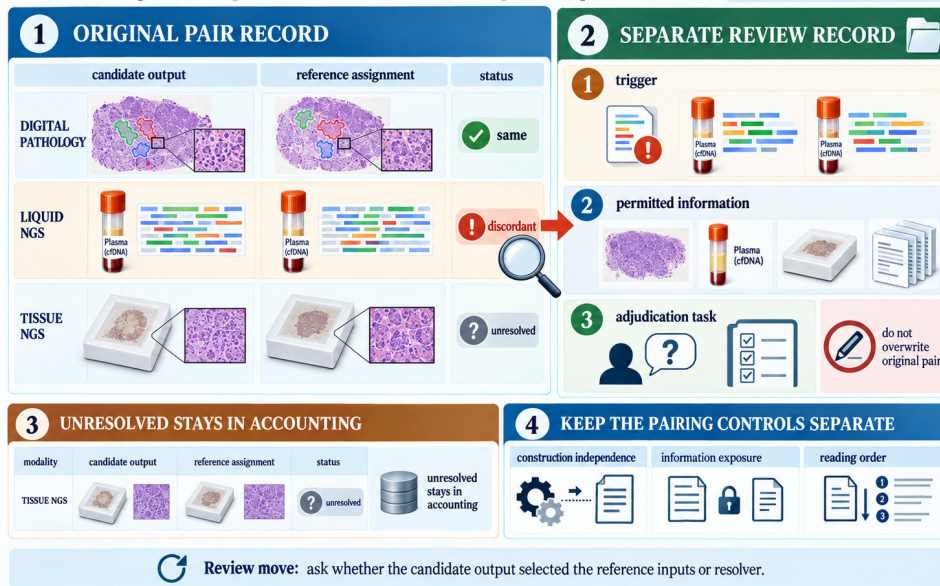


Figure 5.2. Preserve original pairings through discordance review – Technical review plate.

Reviewer Lens. The original paired data, the linked investigation log and the analysis population answer three different questions: what first happened, what was learned later and which cases enter this calculation. A coherent trace can justify a narrower comparison while keeping excluded or unresolved cases visible. The later judgment deserves credit for its own endpoint without silently recoding the first comparison.

Consultant Lens. Data owners can preserve an immutable original record and a separate investigation record from the outset. Clinical or laboratory owners specify when investigation is warranted and what state remains if it cannot decide; a statistician addresses any partial-verification analysis. The closure may be an explanation of a specimen mix-up, a revised future rule or a retained unresolved case. After use, reference changes reopen relevant conclusions without rewriting what originally happened.

Does the reference cover this finding?

A separate illustrative sequencing result reports an insertion at a numbered position. The benchmark writes what appears to be a different position. Before calling the candidate wrong, what would have to match?

Follow the insertion to its qualified region

The position belongs to a **reference assembly**, a versioned map of the genome. A coordinate on one assembly cannot be compared directly with a coordinate on another as if the maps were identical. Once the assembly matches, the written **representation** needs examination: the reference letters and inserted letters used to describe the finding. In repeated letters, equivalent descriptions of an insertion can put it at different nearby coordinates. A text mismatch can therefore raise an equivalence question before it establishes an analytical error.

Even matching the descriptions leaves the question of coverage. A **benchmark call** records a characterized variant for a particular material and release. A **benchmark region** identifies where that release supports the specified comparison. Within a qualified region, assembly, representation and variant-class interpretation still need to match. Outside it, absence of a listed call supplies no reliable absence claim for that question. “Nothing listed” cannot become a verified negative specimen.

Tumor tissue, low variant level or a difficult region can introduce further limits. NIST's Genome in a Bottle resources illustrate material, release, call and region boundaries. Its germline resources do not establish HelixBridge tumor performance or provide a universal patient-like truth source. (NIST Genome in a Bottle FAQ, items 1 and 8–10; CLSI MM23 §§4.1.1.1 and 4.2.1; CLSI EP39 §7.7.)

The named sequencing occurrence here is tissue based. The separate proposed liquid branch begins with whole blood, then plasma, then isolated cell-free DNA; circulating tumor DNA can be a tumor-derived fraction within that cell-free DNA. Changing the specimen name would not carry tissue reference coverage through those steps. No liquid configuration, reference result or bridge is supplied. Chapter 6 follows the material route itself.

Would a qualified material work with another procedure?

The same coverage question appears when two procedures measure a concentration in patient serum. A processed control material may behave differently with one procedure because its surrounding components, the **matrix**, interact with that procedure. **Commutability** asks whether the relationship between the procedures' results on the material is sufficiently like their relationship on the specified patient samples for the material's intended use.

That question ties together the quantity, material, patient-sample type, particular procedure pair and purpose. A material qualified for one pair does not arrive qualified for another. A matrix effect identifies a material–procedure interaction; it does not prove either procedure gives wrong patient results. This is why the qualification needs to accompany the material into its proposed use. (JCGM VIM term 5.15; ISO

17511:2020 §3.10 and §§4.5.1–4.5.5; CLSI EP14 §§1.1, 1.4.2 and 2.1.1; EP30 §§1.1, 1.5.1 and 2.2–2.5.)

For a qualitative sequencing check, an engineered-variant training material may be useful for a defined detection step while bypassing features of a fixed tissue section: extraction, fragmentation, tumor fraction and unrepresented variant classes. The appropriate appraisal is comparability for the intended use—what analyte, matrix, preparation, concentration or level, process and study question were qualified? CLSI EP39 supports that use-specific surrogate appraisal without making every surrogate equivalent to patient material. A useful downstream check remains useful; a later specimen-handling change needs an applicability argument for the earlier qualification.

What is being assigned: a quantity or a category?

In the serum example, an instrument observes a signal and converts it to concentration, an amount per unit volume. **Calibration** supplies the established relation to assigned values. A reference measurement procedure can help assess whether results tend systematically high or low—the question of **trueness**. A certified reference material can supply a documented assigned property value, uncertainty and intended use. Together these objects support the measurement chain, while the separate question of disease assignment remains open. Chapter 7 develops the calibration and uncertainty calculations.

The insertion's identity has a different kind of output. “This sequence change is present” and “this whole slide has the condition” are **nominal categories**: names without amounts. “Mild” and “severe” are **ordinal categories** when their order is known but numerical distances are not. A printed yes/no result can also be derived from a measured quantity and cutoff. The words on the report therefore cannot by themselves select the reference framework; the underlying magnitude, count or cutoff relationship matters.

Measurement standards preserve these boundaries. ISO 17511:2020 covers specified numerical, counting and cutoff-derived results, excluding nominal or ordinal properties without magnitude. ISO 15193:2026 concerns reference measurement procedures; ISO 15194:2026 concerns specified certified reference materials. Their informative annexes discuss nominal and ordinal properties without providing a complete categorical-assignment method. ISO 15195:2018 addresses calibration laboratories using reference measurement procedures, excludes medical laboratories and nominal/ordinal examinations, and supplements rather than replaces ISO/IEC 17025 for that competence scope. Citing any of them alone cannot certify a diagnosis, a sequence truth set or a laboratory's full competence. (The measurement standards' §1 scopes and informative Annex A in the procedure and material standards are located in the Source trail.)

Which answer belongs to the image endpoint?

A pathology study brings the unit question back into view. A whole-slide diagnosis, a lesion-location annotation, a reader's final diagnosis and worklist timing are four different objects. The whole-slide label cannot locate the lesion. A location annotation cannot establish how assistance changes a reader's diagnosis. A clock cannot assign lesion status. A reference process can be well documented and still answer the wrong one of these questions. Before a study, the chosen endpoint determines the assignment inputs and permitted information; after a scanner, display or workflow change, it also determines which earlier comparisons need an applicability review.

Public FDA records provide concrete examples of these different choices. The original P230011 decision summary describes product-specific reference choices, including a composite for RNA fusion findings where breadth and detection capability differed. DEN200080 separates slide diagnosis from localization and describes masked annotation. K232879 distinguishes the original signed-out diagnosis from later classification of disagreement. These frozen records supply illustrations, without giving either fictional product a method, result, acceptance limit or universal rule. (P230011 summary, p. 27; DEN200080 summary, pp. 8–10; K232879 summary, pp. 13–15.)

Reviewer Lens. The boundary is easiest to see next to a particular finding: this insertion, this material on this procedure pair, or this slide-level endpoint. The benchmark's unqualified region stays outside the negative assignments; the material's original qualification remains positive for its stated use. Keeping both conclusions prevents a narrow gap from erasing useful evidence or a useful result from spreading beyond its scope.

Consultant Lens. A scientific owner can choose a source for the exact job and document where its support ends. If only part of a proposed claim is covered, the options are a narrower conclusion or specialist work on the remaining relationship. After a material, reference-release or image-route change, closure may be a justified unchanged-applicability explanation, a targeted bridge or a new study. The gap determines which option is needed.

What conclusion can another reader reconstruct?

The twenty-pair example can now support a precise reference-role judgment. That judgment needs somewhere to retain its basis when it returns to the Chapter 4 evidence row. The five sections of the reference-process adequacy memo follow the inference in order; two working pages can be enough. “Not supplied” identifies an absent fact. “Not applicable, because...” explains why a dimension does not belong to this question. Either entry still needs appraisal for its effect on the conclusion.

1. Requested decision and object. What conclusion is needed, about which condition or property, unit and candidate output? The exact configuration and evidence time connect it to the Chapter 4 row and Chapter 3 evidence job. A question about a display remains visibly different from diagnostic correctness.

2. Assignment process. How do the inputs produce the other answer? Each material, measurement procedure, clinical record, panel, reference process or comparator has a specified role. Inputs and versions, algorithm, order, timing, repeats and stopping rule explain the path from observation to target conclusion. A list of method names cannot show that path.

3. Coverage and information controls. Where does the path apply, and what could influence it? Population, specimen or image, sites, users, workflow, output classes, regions, levels and difficult cases define the relevant coverage. Reference construction, information exposure, reading order and separation of development from evaluation have separate entries because each protects a different link.

4. Disagreement, unresolved states and limits. What remains visible when an answer cannot be assigned or the two answers differ? Original pairings stay linked to later investigation or adjudication. Invalid, equivocal, missing, failed and unresolved states retain reasons and, where supplied, starting counts and analysis denominators. Reference error, uncertainty, version and applicability limits travel with the result.

5. Disposition and next action. What does all that support now: target-condition correctness language, a defined agreement relation or a narrower statement? The residual gap, record that could close it and next decision owner make the judgment usable. If no owner is identified, the needed role remains to be assigned; the record cannot invent a person.

The following four dispositions help express that last judgment. They classify the reference process *for this question*.

Table 5.3 – What the reference judgment permits

Disposition	Supported meaning
Adequate	The reviewed process supports the exact requested conclusion within its stated bounds; it may still err
Adequate with limits	A narrower conclusion is supported, with explicit restrictions or uncertainty
Comparator-only	A defined agreement relation is supported, without direct correctness language

Disposition	Supported meaning
Unresolved	The facts do not establish the role or adequacy for the requested conclusion

The fictional twenty-pair comparison illustrates **comparator-only**. Its identified other method supports the agreement relation used in the calculations, while its target-assignment fitness is unestablished. If even the other method's identity and rule were absent, its role would instead be unresolved. For the quantitative material example, a request spanning the original qualified procedure pair and a new pair could earn **adequate with limits**: credit the first pair and leave fitness for the second unresolved.

These are author-defined reference dispositions, rather than candidate-performance or regulatory decisions. Adequate does not mean infallible, and a known defect must be named as a defect rather than softened into missing paperwork. A completed memo cannot repair an unsuitable process by completeness alone. The four labels are Book 1C constructs, not FDA, European Union, ISO or CLSI outcomes.

Reviewer Lens. The dependency between sections matters more than filled cells. The first unit must be what the second process assigns; the third section must cover the use; the fourth must account for cases the process cannot decide. A comparator-only memo can be entirely useful for an agreement question. Its refusal to claim accuracy protects the supported conclusion.

Consultant Lens. The memo gives owners distinct decisions: the claim owner chooses the inference, methods and data owners supply process and accounting records, and clinical or statistical specialists appraise fit and analysis. The consultant identifies the smallest decision-changing gap. A consequential unsupported subgroup may need its own reference work while evidence for the supported group remains intact.

Return to the agreement line

The opening judgment concerned a favorable phrase attached to the knowledgebase in the separate technical report. With the five-part reasoning now available, does that phrase support a defined agreement relation, target-condition correctness, or a still-unresolved reference role? Which part of the initial judgment survives, and which needs revision? The two evidence states and the transcript change remain the supplied facts.

How a reviewer limits the HelixBridge conclusion

The strongest reference judgment is **unresolved reference fitness**. “Strong overall agreement” remains a favorable reported description, but the excerpt does not identify what HelixBridge agreed with, how that status was assigned or which variant and specimen conditions it covered. Even comparator-only needs an identified comparative process and defined relation. The absence of those facts here does not establish that the actual study lacked an adequate reference.

The study with the knowledgebase used in analytical validation and the final technical report with the changed knowledgebase add an applicability question. The validation application and report rules are identified package components; the record does not prove that every non-knowledgebase component or the reference processes matched across the two comparisons. The added transcript makes a previously rare mapping anomaly execute more often. It supplies neither affected variant classes nor a new comparison result. Earlier evidence can remain useful while affected scope and cross-state comparability need a trace.

The focused next record would identify and version each comparison's reference or comparator, its target and coverage, assignment and information-control rules, original result accounting and the configuration crosswalk between the two evidence states. The affected variant and mapping scope would then locate the change's relevance. Classes, regions, representations, specimen conditions, population, independence, masking, discordance and no-call handling, counts, strata, criteria and uncertainty remain unknown until supplied. The excerpt establishes neither acceptable performance nor a regulatory outcome.

Reviewer Lens. The favorable reported description and the identified transcript change are distinct positive facts. Recovering the reference identity and original pairs comes before interpreting a defined agreement measure; the configuration crosswalk comes before transferring evidence between knowledgebases. A fit narrower region or class may eventually support a bounded conclusion. A missing record today is not demonstrated assay failure.

Consultant Lens. The configuration owner can provide the evidence states and change trace while the scientific methods owner supplies the reference process and coverage. Linking both to the treatment-facing question exposes which mappings need attention. Unaffected scope may be supported by an applicability explanation; affected scope may need targeted evidence. Neither option permits inventing the comparator used in the original work.

A comparison with both operators masked

The anonymous cases that follow are fictional teaching stipulations. They name no real disease, specimen or assay and provide no empirical performance, sample count, panel size or acceptance threshold. Their local labels identify separate fact sets; familiar letters never join one exercise to another. Unknown counts stay unknown.

The first assay and its comparison method

In the first fact set, the local record labels are Assay A, configuration A-1, Condition A, Population A, Specimen A and Method C. In ordinary language, the request concerns whether **the first assay** correctly classifies the condition as present or absent when its specimen is collected, for diagnostic assessment in the population represented by the exercise. Its configuration is locked, specimen handling is fixed, and its only final outputs are condition-indicated and condition-not-indicated.

The established comparison method uses the same output labels for that population and specimen. The record does not establish it as the best available condition assessment or otherwise demonstrate correctness. Difficult-case coverage, error modes and its relationship to the target condition are unspecified.

The comparison method does not use the candidate result, and the candidate does not trigger added comparator testing. Each operator records a result without seeing the other's. Separation of development from evaluation data is not supplied. Original pairs and discordances remain visible. No resolver is supplied, so disagreement remains unresolved as to the target condition. There are no observations, denominators, precision or acceptance rule.

Decision

What belongs in each of the five memo sections? Which disposition is supported, which relation could a later comparison estimate, and which correctness, accuracy, performance or sufficiency conclusions remain unavailable?

How the masked comparison earns a useful narrower role

The five-part judgment

Decision and object. The question concerns binary correctness at specimen collection for the represented population and locked first-assay configuration. No numerical quantity or observed performance result is involved.

Assignment process. The comparison method supplies the second categories. Its exact procedure and ability to assign the condition correctly are unestablished. Shared labels make a comparison possible while leaving their correctness open.

Coverage and controls. The population and specimen are identified; difficult-case coverage is not. Candidate-independent result generation and reciprocal masking are documented strengths. Development/evaluation separation remains unknown. Each control receives credit for the link it actually protects.

Disagreement and limits. The original pairs remain available. With no resolver or fit target-condition process established, a disagreement cannot identify the correct side. The method's unknown error and coverage are limits; no empirical denominator, result or precision can be reported.

Disposition and next action. The supported role is **comparator-only**. An appropriately conducted later study could estimate a defined agreement relation with the comparison method. This fact set contains no such estimate yet, no direct correctness conclusion, no candidate-performance result and no regulatory-sufficiency judgment. A correctness request needs the method owner's evidence for a fit target-condition assessment and its coverage. The study owner also needs to establish development/evaluation independence. These are owner roles to identify; no individual is supplied.

Masking tells us what the operators could see. It cannot tell us whether the comparison method identifies the condition. The positive controls nevertheless make a narrower comparison useful when its pairings and limits remain attached.

Reviewer Lens. Reciprocal masking and preserved pairs are genuine strengths. The missing relation is between the comparison method and the condition. A future agreement analysis may be accepted as agreement without stating that the candidate met a performance criterion when neither results nor a criterion have been provided.

Consultant Lens. The method owner can decide whether the claim needs correctness assessment or whether a narrower agreement question is useful. The study owner protects the locked candidate and original-pair plan. A proposed correctness claim requires a qualified process, coverage and statistical plan before data interpretation; an agreement-only deliverable can remain worthwhile meanwhile.

What changes when the qualification changes?

The next two fact sets keep the same five questions available. Each supplies its own target, process and controls. Their conclusions depend on those facts rather than on the familiarity of a method title.

The adult-and-pediatric request

The local record labels are Assay B, configuration B-4, Condition B, Population B, Serum B and Reference Process B. **The adult-and-pediatric assay** is locked and uses serum under fixed handling instructions for diagnostic assessment. Its only final outputs are condition-indicated and condition-not-indicated. The request concerns correct classification of the condition as present or absent at serum collection, for both adult and pediatric members of the population.

The adult reference process is stipulated to be the documented best available target-condition assessment for adults in that population using that serum. Its exercise qualification covers adults, the serum, both categories and the stated time. No pediatric qualification or coverage is supplied.

The process is fixed before evaluation. The candidate cannot select its inputs, repeats or assignments. Reference and assay operators record their results without seeing the other's output, and evaluation data are separate from those used to establish the locked candidate configuration. Original discordances remain visible; a prespecified investigation may classify them when warranted without rewriting results. An indeterminate or missing reference input remains unresolved. The reference can still err within adult use. No candidate result, count, threshold, precision or acceptance criterion is supplied.

Decision

Which disposition fits the adult-and-pediatric request? What can the reference support for adults, what remains unresolved for pediatric use, and which candidate-performance conclusion must still be withheld?

How the adult qualification retains its value

For the combined request, the disposition is **adequate with limits**. The stipulated process supports adult target-condition assessment for the serum at the stated time. Pediatric fitness remains unresolved. There is no candidate-performance observation in this fact set.

That is a positive adult reference finding. The broader request does not erase it, and the missing pediatric qualification does not prove failure in children. Equally, adults cannot confer pediatric qualification by sharing the assay name. The adult process remains capable of error within its qualified use.

Reviewer Lens. The population attached to a future result must match the population for which assignment is justified. Keeping the adult reference role explicit allows it to be credited without producing a pediatric correctness estimate from adult-only evidence.

Consultant Lens. The claim owner can retain an adult comparison question while the clinical methods owner determines the pediatric assignment evidence needed. Treating them as separate decisions avoids repeating adult work solely to complete a broader sentence.

The proposed review of disagreeing pairs

This independent fact set uses Assay C, locked configuration C-2, Condition C, Population C, Specimen C and Method D as local labels. **The discordance-review assay** is intended for diagnostic assessment of the condition at specimen collection in the stated population, using fixed handling. Its final outputs are exactly condition-indicated and condition-not-indicated; classification correctness is requested.

A proposed composite combines **the initial comparison method** with clinical review. The review occurs only when candidate and comparison method disagree and uses both results. The comparison-method operator is masked to the candidate, but the reviewer is not. Population, specimen and both output labels are nominally included, while difficult-case coverage and component error are unspecified. Development/evaluation separation is also not supplied.

The proposal would replace the original comparator entries after review. It defines no unresolved state for missing inputs or nonconvergence. No empirical result, count, vote, threshold, precision or acceptance criterion is provided.

Decision

What disposition fits the correctness request? What happens separately to construction independence, information exposure, selective resolution, original results and unresolved states? What useful relation remains preservable?

How the original comparison survives a defective resolution plan

The correctness reference remains **unresolved**, with known defects in the proposal. The candidate selects which specimens receive added clinical review and then enters the reviewer's information. Masking the original method's operator protects only that operator's exposure. It cannot remove candidate-dependent construction. Concordant pairs receive no independent check; rewriting the original entries would hide the first comparison; and missing inputs or nonconvergence have no defined state.

The original candidate–comparison-method relation can still be preserved as an agreement record with limits. A selectively revised table cannot be called corrected accuracy. Stronger inference would need an appropriate prespecified process, candidate-independent selection, separate information controls, unresolved handling

and analysis, with clinical and statistical input. Adding another reviewer to the same candidate-informed rule would retain the defect.

Reviewer Lens. The masked operator receives credit for the stated exposure control. The candidate-triggered clinical review is already a known construction flaw, rather than a missing document title. Preserving the first pairs protects the immediately supportable comparison while the proposed assignment and accounting plan needs repair.

Consultant Lens. Clinical methods and statistical owners design any new assignment or valid partial-verification approach before its results are used. Data owners link later investigations to retained originals. That makes two deliverables visible: the preservable agreement record and a still-open correctness question. Redesigning a process cannot promise a favorable future result.

What did the reader study compare?

PathSight Cloud prioritizes whole-slide images and highlights suspicious regions for pathologists. The pathologist remains the diagnostic user and decision owner. This selected validation view identifies the model used in validation, labeled M3.2, and a study distribution mostly from the first scanner family, labeled Scanner A. The source describes a reader study with improved average sensitivity without unacceptable loss of specificity. The full basis of that favorable summary is not supplied.

The missing fields include the exact target condition, assessment unit, endpoint, reference inputs, panel, information exposure, assignment rule and unresolved handling. Reader population, case set, denominators, precision and workflow are unknown. Exact scanner, firmware, preprocessing, threshold, interface and commercial applicability also remain unresolved. The initial-package assistive-prioritization role identifies a question to examine; it does not prove that the claim applies to this validation or a commercial state. This occurrence introduces no later change event.

Decision

How would the five memo sections distinguish a priority or highlight from the reader's diagnosis? What can be said from this selected record, and which endpoint-matched reference information would change the judgment? Which details remain unknown rather than borrowed from the public product examples?

How a reviewer preserves the favorable summary while locating its basis

The endpoint comes before the reference judgment

Reference fitness is **unresolved from the selected record**. The assistive workflow is identified, and the favorable reader-study summary remains an asserted result. Neither establishes how target status was assigned for that result. It cannot be translated into device-alone sensitivity or lesion-localization performance. The unknown method also supplies no basis for declaring the actual study invalid or its reference inadequate.

The next record must connect endpoint and unit to reference inputs, assignment and unresolved rules, and the information each participant could see at each decision. If a panel was used, its qualifications and instructions matter; no panel is stipulated here. Development/evaluation separation, original paired and missing-result accounting, and the evaluated configuration complete the comparison basis. An original signed-out diagnosis, annotation agreement or adjudication could be a kind of record to inspect. None has been supplied for PathSight.

Mostly-first-family images raise an applicability question, without supplying a numerical result for the second family. With no local change event, the questions concern initial adequacy and applicability; they cannot be presented as evidence reopened by an invented release.

Reviewer Lens. A study can concern readers with assistance, readers alone or another object. The endpoint and assignment record identify which result the favorable summary describes. If those records establish a fit comparison for a defined case set, the conclusion can receive bounded credit. An incomplete excerpt is a reason to recover its basis, without deciding that the study itself is defective.

Consultant Lens. The pathology study owner supplies the endpoint, reader workflow and assignment record; the configuration owner supplies the studied scanner-to-display route. Linking them to the assistive claim leaves commercial applicability open until it is demonstrated. Recovering an existing protocol and case accounting may close the question. Whether another reader study is needed depends on what that recovery shows.

A request that can change the judgment

For HelixBridge, the missing relationship begins with the comparison process. For PathSight, it begins with the endpoint and its assignment basis. Those focused requests are more useful than “more validation” because a response can change a specific conclusion. In planning, the same questions establish the rules before results make selective choices attractive. Laboratory and clinical teams document which observations enter assignment, who sees them and what happens when assignment fails. The information appropriate to clinical care may differ from a study reader's permitted information.

A responsible handoff identifies the evidence owner, the scientific or clinical methods owner and the decision their record will permit. Advice can coordinate that work and support a narrower claim. It does not transfer the laboratory's result authority, the pathologist's diagnostic responsibility or the regulator's sufficiency decision to the consultant. The unrelated deployed sequencing invalids and pathology delay likewise remain with their own service owners: one trace follows candidate availability, the other follows timing and completed review. Neither event assigns a target condition. Any new comparison proposed after use needs prospective reference reasoning of its own.

Reviewer Lens. A comparator specification, endpoint-specific assignment rule, case-accounting trace or configuration crosswalk can be the smallest record that changes the inference. When an existing qualified process is adequate for the narrower question, that conclusion should be explicit and unnecessary reference work avoided. Clinical and jurisdiction-specific decisions stay with the appropriate practitioner or authority.

Consultant Lens. Claim definition leads to reference selection, accounting and interpretation before performance is communicated. After a product or reference change, owners compare the changed state with the evaluated state and close affected links. An unexplained result can justify a focused investigation while neither establishing clinical harm nor requiring a complete repeat of validation.

Change only the reference process

The final independent fact set keeps the candidate fixed. Its local labels are Assay U, locked configuration U-1, Condition U, Specimen U and Method R. **The unchanged assay** classifies the condition as present or absent when the specimen is collected, for diagnostic assessment in the symptomatic adult outpatients represented by this exercise. Collection and handling instructions are fixed. Its only final outputs are condition-indicated and condition-not-indicated; there is no third, ordinal or quantitative final output.

The earlier comparison used one non-reference method. It could support only agreement, and no empirical result was supplied. Only that comparison process is now replaced. Candidate, condition, intended use, population, specimen and output categories remain the same.

The proposed replacement has the following complete stipulations. Their order follows inputs into the initial assignments, then the panel decision and accounting.

1. The inputs are a result from **the qualified reference method**, prespecified relevant clinical observations and prespecified follow-up information available by assignment time. The exercise qualifies that method for the condition in the stated population and specimen.
2. Each participant has documented qualifications for the assigned input and condition. All use one locked definition, input set, evidence hierarchy, assignment form and unresolved rule. No participant count or particular credential is supplied.
3. At initial assignment, each participant sees only the specified inputs. The candidate output and other participants' initial assignments are hidden.
4. The candidate cannot select inputs, trigger another method, alter the hierarchy, enter discussion or change an assignment.
5. Matching independent assignments produce the panel assignment. Differing or missing assignments remain unresolved; they are neither counted as votes nor overridden. There is no majority threshold, casting vote or other tie breaker.
6. Original candidate–panel discordances remain visible. Prespecified investigation may classify a discordance when warranted without rewriting either result. Nonconvergence, missing required input and uninterpretable input stay unresolved and remain in accounting.
7. Qualification extends only to this condition, adult outpatient population, specimen, locked candidate configuration, two final outputs and fixed information set. No other population, specimen, configuration, output, difficult-case stratum or time point is covered.
8. Participants and inputs can still be wrong. No empirical performance, sample or panel count, threshold, acceptance criterion, precision or regulatory conclusion is supplied.

The assignment rule in the fifth item is particularly important to read literally: the record describes matching independent assignments and an unresolved outcome otherwise. It provides no numerical voting rule.

Decision

Does the process remain comparator-only, become adequate, become adequate with limits, or remain unresolved for the stated conclusion? How do target and process role, coverage, independence, information exposure, decision rule, unresolved handling and limits support that choice? What does each of the five memo sections now contain, which inferences remain unavailable, and which evidence questions reopened even though the candidate stayed fixed?

The initial transfer judgment concerns this fact set alone. A later revision should make clear which stipulation changed the comparison's role and which conclusion still lacks results.

How a new reference earns a bounded assignment role

What changed, and what the change establishes

Under these teaching stipulations, the new process is **adequate with limits**. It can assign the condition for the stated population, specimen, locked configuration, output categories, assignment time and information set. The positive judgment rests on the qualified reference-method input, relevant clinical information, qualified participants and locked process. A panel's existence or size would not supply that qualification, and the fictional stipulations establish no process for a real assay.

The earlier comparator-only role has therefore changed to a constructed target-condition assessment. Inputs and hierarchy are fixed. Participants initially assign without the candidate output or another participant's answer; the candidate cannot select extra inputs or alter the decision. Matching assignments produce the panel assignment. Missing or differing ones remain unresolved, without negotiated votes. Investigation adds a linked explanation while original candidate–panel pairs remain available.

The limits are part of that positive conclusion: no other population, specimen, candidate state, output, difficult-case stratum, time or information set is qualified. Participants and inputs may err, and unresolved assignments remain in the accounting. There are **no candidate-assay accuracy or clinical-performance results**. A fit basis for comparison does not create observed candidate pairs, denominators, precision or acceptance results. Regulatory sufficiency also remains undecided.

The change reopens reference-input fitness, participant qualifications, exposure controls, construction independence, assignment and unresolved rules, coverage and reference error. The stable candidate configuration supplies valuable continuity while answering none of those new reference questions. A final judgment can therefore change the comparison's supported role while preserving the unchanged candidate and withholding its performance verdict.

Reviewer Lens. The decision-changing facts are the qualified inputs, exclusion of the candidate, locked rule and unresolved handling. Together they support assignment within the stipulated scope. A performance estimate still needs actual candidate pairs, denominators and design; better reference planning cannot create an unobserved result.

Consultant Lens. The methods owner can complete the reference-role change while the product owner keeps the candidate fixed. Data and statistical owners then plan capture and analysis of future pairs and unresolved cases. The consultant can present a completed, bounded reference-planning decision while leaving candidate performance and regulatory application to later evidence and the responsible owners.

Back to the comparison-bearing row

The opening agreement line now points to a specific missing relationship: what assigned the comparison result for HelixBridge, for which variant object and configuration, under which conditions, with what original-pair, failure and unresolved accounting? PathSight's favorable reader-study summary needs a different connection, between its endpoint and reference assignment. Both source descriptions remain on the record without becoming unsupported performance conclusions.

The anonymous cases show the positive side of the same judgment. The first comparison method can support a future agreement question. The adult reference supports its stipulated adult assessment while pediatric fitness stays open. The changed panel process supports a bounded condition assignment while candidate performance remains unobserved. The conclusions differ because the supplied relationships differ.

Returning that reasoning to the Chapter 4 row may change its method description, comparison vocabulary, residual gap or next owner. The planned criterion, actual observation and later disposition remain separate entries. The initial HelixBridge judgment and the final reference judgment can now be compared without erasing either. Chapter 6 asks whether specimens and handling cover the route to which this evidence would apply. Even a well-described reference cannot compensate for an unrepresented material path.

Questions that test the judgment

A memo becomes reconstructable when another reader can answer these questions from its evidence:

- What conclusion, condition or property, unit and decision time are being addressed?
- Which candidate output and exact configuration are distinct from the reference process?

- What role does each reference object actually serve?
- Is the uncertainty about process availability, an individual assignment, or missing process information?
- Which population, specimen, output and consequential boundary cases are covered?
- How are construction independence, information exposure, reading order and development/evaluation separation distinguished?
- Where are the original pairs, later investigations and adjudications retained?
- What reasons and denominators preserve invalid, missing, failed and unresolved states?
- For which material, use and procedure pair does commutability or comparability apply?
- What is the strongest supported conclusion, including useful agreement or a narrower adequate use?
- Which record and next decision owner could close each remaining gap?

A further retrieval test is whether the reasoning holds without the memo headings. How do its five sections connect? Why are comparator and clinical reference standard different roles? Why can independently generated results still fail to estimate correctness? How does masking differ from construction independence? What is lost by checking only discordant pairs and replacing the original table? Why can adequate with limits be a stronger judgment than blanket acceptance or rejection? How does an unresolved reference assignment differ from missing process information? What one-sentence conclusion fits a qualified reference process when no candidate-comparison results are supplied?

Those are eight distinct tests of understanding. Comparing an initial answer with the chapter should reveal which part can stand and which relationship needs repair. The final sentence should retain both the useful reference role and the still-unobserved candidate result.

Source trail

The Chapter 5 source packet records claim-level locators and limits. The technical sources below support bounded paraphrases; the anonymous exercises are teaching stipulations, not empirical or regulatory evidence. The packet's licensed-text review is dated September 4, 2026; the directly relevant EP49 and ISO 17511 clauses and the official edition/status pages for ISO 15193:2026, ISO 15194:2026, ISO 17511:2020, CLSI EP49 and NIST Genome in a Bottle were checked again on September 24, 2026. No licensed method, table, formula or numerical acceptance rule is reproduced.

- **FDA, Statistical Guidance on Reporting Results from Studies Evaluating Diagnostic Tests (2007).** Nonbinding U.S. guidance; Introduction, §4.1 and Appendix 7.2 support reference role, availability, agreement and overall-agreement limits in its stated binary-result context, not a universal method for ordinal, quantitative or repeated-patient data. Official final guidance.
- **FDA, Design Considerations for Pivotal Clinical Investigations for Medical Devices (2013).** Nonbinding U.S. guidance; §8.2 and §§8.6, 8.8 support composite-reference, information-exposure and evaluation-data distinctions. Its exclusions include in vitro diagnostic (IVD) analytical-validation and companion-diagnostic clinical-validity studies; it is not HelixBridge's protocol. Official final guidance.
- **CLSI EP49, first edition (2026), and EP12, third edition (2023).** Voluntary laboratory consensus guidance. EP49 §§1.4.1, 3.1–3.3.1, 3.5.6–3.7 supports target, comparator, original-result and accounting boundaries within its noninterventional clinical-validation framework, not a universal protocol or analytical-validation method. EP12 §§1.1–1.2 and 4.2–4.2.5.2 supports two-final-output agreement language and discrepant-resolution limits; its correction concerns an uncited table. EP49 issuer record; EP12 edition record.
- **ISO 20916:2019 and JCGM 200:2012 (VIM, third edition).** ISO 20916 §1, §5.5.3.8 and Annex B.8.1–B.8.2 inform comparator and conditional bias-control questions only within covered regulatory clinical-performance studies using human specimens; analytical-performance studies are excluded and Annex B has its own study-category boundary. JCGM terms 2.7, 5.13 and 5.15 supply measurement vocabulary, not a clinical truth or acceptance rule. ISO edition record; JCGM record.
- **ISO 17511:2020; ISO 15193:2026; ISO 15194:2026; ISO 15195:2018.** Voluntary international measurement standards, used at their stated scopes. ISO 17511 §1, §3.10 and §4.5 distinguish applicable quantities and cutoff-derived results from nominal/ordinal properties without magnitude and make material commutability procedure- and use-specific. ISO 15193 and 15194 §1 and informative Annex A concern reference measurement procedures and specified certified materials, not a complete categorical-truth process. ISO 15195 §1 concerns calibration laboratories, supplements ISO/IEC 17025, and excludes medical laboratories and nominal/ordinal examinations; it does not independently establish competence. Detailed calibration hierarchies and uncertainty calculations belong to Chapter 7. 17511; 15193; 15194; 15195.
- **CLSI EP14, fourth edition; EP30, second edition; EP39, first-edition guideline; MM23, second edition (2025).** Voluntary, scope-bound material and tumor-molecular guidance. EP14 §§1.1, 1.4.2 and 2.1.1 and EP30 §§1.1, 1.5.1 and 2.2–2.5 address quantitative processed/reference materials, not qualitative yes/no truth; EP30's corrected text was used only at unaffected locators, not for a

companion-datasheet-dependent method. EP39 §§7.1–7.2.2 and 7.7 supports use-specific qualitative surrogate comparability, not specimen equivalence. MM23 §§4.1.1.1 and 4.2.1 supports solid-tumor reference-coverage questions without making an orthogonal method infallible or supplying HelixBridge's comparator. EP14; EP30; EP39; MM23.

- **NIST Genome in a Bottle.** Official benchmark resource; FAQ items 1 and 8–10 explain material and benchmark-region boundaries. Neither its germline resources nor the tumor/normal resources described on its current project page are HelixBridge evidence or a universal patient-like reference. Official FAQ; project page.
- **Frozen FDA product decision summaries.** The original P230011 summary (2024), DEN200080 (2021) and K232879 (2024) illustrate product-specific reference choices, endpoint distinctions and later disagreement classification, respectively. They are not current-labeling substitutes, general rules or facts about the fictional cases. P230011; DEN200080; K232879.

The named HelixBridge and PathSight occurrences retain their selected configurations, chronology and locked unknowns. The anonymous Assay A/B/C/U exercises remain source-neutral and supply no empirical performance, count, threshold or regulatory conclusion. The process-bounded terminology and memo dispositions are Book 1C constructions. Applicable product- and jurisdiction-specific decisions require their own evidence.

CHAPTER 6

Specimen Fitness and the Pre-Analytical Chain

Would tomorrow's tissue follow the studied path?

The specimen row in an evidence plan says **tissue; wet lab stable**. Tomorrow, a laboratory may receive another piece of tissue. What would let a reviewer decide whether it follows the path represented by the evidence?

The row concerns HelixBridge Dx, the fictional tissue-oncology sequencing assay that identifies variants associated with eligibility for a named therapy. The selected record does not identify that therapy. It calls the wet-lab assay *stable*, but supplies no anatomical tissue source, preparation, transport or storage history, examination input, or count of specimens that failed to reach a result.

Decision

What are the first three facts needed to make this specimen row useful? What does the word *stable* leave unidentified? How much can the row establish about the specimen arriving tomorrow?

The initial judgment will return when a second specimen route is supplied. For now, the question is whether study material followed a route that can support the proposed specimen claim. The chapter's **specimen and preanalytics coverage matrix** gives that comparison two connected parts: the claimed and studied material states, and the evidence identifying their configuration, failures, counts and supported conclusion. Its tissue and proposed liquid pages are a reasoning aid devised for this book. A completed page has no official acceptance status; an empty entry may mean only that the record has not been supplied.

What does the earlier evidence row already tell us?

A specimen question belongs beside a claim and a reference-process judgment. Why did Chapter 2 require more than a product name to identify the studied configuration? How did Chapter 4 connect a claim to an evidence question and a prospective decision? Why could Chapter 5's well-characterized reference still be insufficient for the requested conclusion? For one such evidence row, which assumption about the specimen would the study need to establish, without supplying an unstated anatomical site, preservation method or population?

The relationship carried forward

Evidence belongs to a particular configured object and conditions. The plan connects a requested conclusion to a method and decision rule; the reference memo asks whether the comparison can support that conclusion. The next link is physical: did the study material and its route into the examination represent the specimen conditions in the claim? A good reference assignment cannot supply a missing collection history.

Where did the study begin?

An examination receives prepared material and produces a result. The laboratory can identify the input without yet knowing everything that happened before it arrived. To see why that matters, follow ordinary operation back toward the patient.

Someone was sampled for a reason and at a particular time. Material was collected, identified, moved, perhaps stored, and prepared. The examination acted on an input from that preparation; software and a professional could then turn its output into a report. For tissue sequencing, the preparation leads to material containing DNA, the genetic material whose sequence can contain variants of interest. A successful examination of that input tells us something about the examined material. How far backward does that observation reach?

That depends on where the study started. A study receiving already isolated DNA may observe sequencing and interpretation while leaving collection and earlier preparation outside its observed route. Work before a specified examination is called **pre-examination or preanalytical work**. The specified boundary matters: isolation can precede sequencing and still be part of the complete assay being evaluated. Its position earlier in the route does not remove its possible effect on a result.

Digital pathology makes the same boundary visible in another form. Tissue becomes a prepared slide; the scanner makes a whole-slide image; a pathologist examines the image, perhaps with assistive software. An evaluation that starts with acquired image files can study analysis or reading. Opening those files correctly cannot document the

tissue and slide production that came before them. Later delivery to a worklist—the queue awaiting review—adds a timing question distinct from specimen suitability.

Molecular specimen guidance in CLSI MM13 and sequencing-workflow guidance in CLSI MM09 help locate specimen processes and acceptance points. They do not supply the missing product-specific route or an authorization. The useful question remains concrete: what reached this examination, and which earlier handoffs did this study actually observe? (MM13 Chapters 2–4; MM09 §§3.3.2.3–3.3.3.)

Reviewer Lens. Following one study input backward reveals the evidence boundary more clearly than the study title. A documented comparison beginning with isolated DNA may deserve full credit for its downstream question. An image-file study can likewise support its actual analysis or reader question. The request for earlier history should follow the proposed conclusion: it matters when that conclusion reaches into collection, preparation or acquisition.

Consultant Lens. The specimen owner and assay or imaging owner can settle where the study enters the route before fixing the protocol. That decision determines which handoffs need evidence and who can supply it. If an unavailable report or late worklist item appears after use, the stage and its owner give the investigation a starting point. The event alone does not assign the cause to collection, scanning or software.

What exactly reached the examination?

Follow one blood-to-DNA route

Blood is drawn from a vein into a collection device. That venous whole blood is the **specimen**, the material collected from the patient. The patient's reason for testing and sampling time belong alongside it: the person supplies a clinical context as well as material.

Processing separates **plasma**, the liquid portion used for the next step in this route. Plasma is now a **derived sample** because processing has produced it from the collected specimen. A study beginning with plasma therefore starts at a different point from a study beginning with whole blood.

Isolation takes the next step, obtaining DNA fragments present outside cells from that plasma. These are **cell-free DNA (cfDNA)**. The examination uses an input prepared from those fragments. The material sequence is now visible:

patient and sampling context □ **venous whole blood** □ **derived plasma** □ **isolated cfDNA** □ **examination input and result.**

Where is the tumor-derived material in that sequence? **Circulating tumor DNA (ctDNA)** is a possible tumor-derived contribution within the circulating cfDNA. It is not another specimen collected after plasma. How much, if any, is present can depend on

the patient's biology and sampling context. No tumor quantity or meaning of a negative result for the proposed HelixBridge liquid branch is supplied here.

The feature or component the examination addresses is its **analyte**. A numerical question will also need the quantity and unit, developed in Chapter 7. Even when the analyte remains DNA, its surroundings change: plasma and the isolated-DNA preparation contain different surrounding material. That environment is the **matrix**. Sharing an analyte name therefore does not make two inputs the same. A successful isolated-input result alone cannot establish how well blood collection and plasma separation worked.

ISO 20186-3:2019 addresses the pre-examination path from venous blood through plasma to circulating cell-free DNA. Its scope helps identify these handoffs; it supplies none of the proposed branch's missing product conditions. (ISO 20186-3 §1 and §§4–6; MM09 §4.4.4.3.1.)

Which corresponding tissue steps are known?

For HelixBridge tissue, the path still contains blanks: **patient/context unknown** □ **anatomical tissue source and preparation unknown** □ **processed and isolated input unknown** □ **exact configured examination unresolved**. The oncology purpose gives no license to fill them with a familiar preparation.

For example, **formalin-fixed, paraffin-embedded tissue** has undergone chemical preservation with formalin and support in paraffin wax so thin sections can be cut. That physical history can affect the DNA recovered later. Evidence for it needs an applicability judgment before supporting another preparation. ISO 20166-3:2018 addresses this specific tissue-to-isolated-DNA path; it does not assign that preparation to HelixBridge or cover every tissue preparation. (ISO 20166-3 §1.)

In pathology the corresponding route is **patient and tissue** □ **prepared slide** □ **acquired image** □ **configured display and assistive reading** □ **pathologist's diagnosis**. The image is a later object with a production history. PathSight's assistive role and support statement for the first and second scanner families do not identify the preparation and acquisition histories represented in a particular evaluation. The question behind both routes is now sharper: which of these actual states entered the study, and which histories are evidenced?

Why might a liquid signal be small?

Suppose a liquid test finds little tumor-derived DNA. The route gives two different places to look. Biology concerns how much tumor material entered circulation at that patient's sampling time. Handling concerns what collection, holding, plasma separation and isolation preserved or changed. A small signal does not choose between those explanations.

For example, DNA released from blood cells during handling could change the background against which a small tumor-derived contribution is sought. This is a possible mechanism, not a reported HelixBridge event. The sampling context and handling history therefore need separate entries. A favorable handling study can leave patient-context coverage open; a biological explanation cannot make a documented handling deviation disappear.

FDA's final 2024 ctDNA guidance discusses tumor and sampling factors beside controllable preanalytical conditions in **early-stage, curative-intent solid-tumor drug development**. That scope supplies a way to distinguish the questions, not general liquid-biopsy device guidance or a HelixBridge result. (FDA ctDNA guidance §§IV.B–C.)

Figure 6.1: Name every material state and handoff places generic DP, liquid-NGS and unresolved tissue-NGS routes side by side. Read an arrow as a handoff whose conditions may matter, not as evidence that every handoff has been studied.

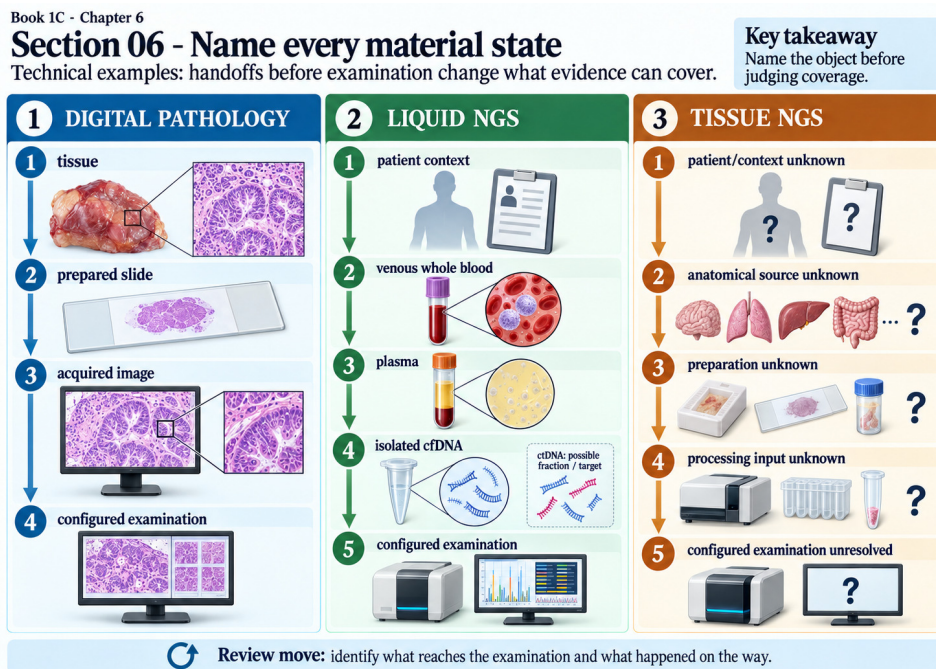


Figure 6.1. Name every material state and handoff – Technical review plate.

Reviewer Lens. A small liquid signal calls for records that distinguish sampling biology from the route taken by the material. A poor pathology image similarly leaves tissue, slide, scan and file/display origins to distinguish. An observed defect at the end cannot identify its origin by appearance alone. A stage that was adequately studied retains that positive judgment while another stage remains unresolved.

Consultant Lens. Clinical owners know why and when sampling occurs; technical owners know the blood or tissue handoffs and image-production states. Joining their

records before study selection makes each planned start and stop explicit. After a signal or image concern, an event-level trace and documented disposition can close the relevant question—or preserve its unresolved cause—more usefully than calling the whole assay or scanner stable.

Which material was waiting?

The possible handling explanation raises a practical question: when did the clock start? Blood may wait at the collection site, travel, and wait again in the laboratory before plasma separation. A courier's transit record covers only part of that interval. Once separated, plasma has its own holding and storage history; isolated DNA has another.

A time claim therefore needs **the material state, exposure interval and conditions, and the endpoint used to judge suitability**. A favorable storage result for isolated DNA can support that result without saying how unprocessed blood behaved earlier. ISO 20186-3:2019 distinguishes the venous-blood and derived-plasma stages; its conditional procedures are not universal time or temperature limits supplied by this chapter. (§§5.1.4–6.7.)

Even before reading a duration, the word *stable* needs an object. This table locates the question when that word appears in a record.

What is called <i>stable</i> ?	What question does that raise?
The wet-lab configuration	Which controlled arrangement is meant, and has it changed?
A reagent or collection-device substance	Does the product retain its specified properties under its claimed storage or use conditions?
A patient specimen or derived sample	Does the material remain suitable for its intended examination along the specified history?

The reagent row shows why the objects matter. A product can remain fit on a shelf while blood collected with it changes before processing. **Specimen stability** concerns the material's suitability over its defined history. The configuration row concerns whether the working arrangement changed. The opening judgment can now ask which of these questions its short phrase actually addresses; the worked comparison will return to the supplied tissue record.

Source scope follows the same separation. CLSI EP25 excludes patient specimens from its reagent-stability scope. ISO 23640:2011 covers specified reagent and collection-device substances while excluding the sample under examination. CLSI PRE04's routine-blood scope excludes molecular pathology and nucleic-acid diagnostic testing. Those exclusions prevent these sources from becoming positive stability

evidence for the molecular specimen route. The relevant molecular questions belong with MM13, MM09 and the venous-blood-to-cell-free-DNA standard. (EP25 §1.1; ISO 23640 §1; PRE04 §1.1.)

A further question concerns what was introduced during preparation. **Interference** asks whether something changes the examination result. **Stability** asks whether material remains suitable over a specified history. The questions may interact, but success on a time question does not establish the interference answer. The European Union's in vitro diagnostic regulation asks for applicable specimen, storage, transport and time-critical documentation in Annex II §6.1.1, and relevant interference, including preparation substances, in §6.1.2.3. These are documentation provisions within that regulation's scope, not a universal study method or acceptance value.

Reviewer Lens. A clock's start and stop can be more consequential than the largest duration in a report. The review follows one material state through the exposure and endpoint; a favorable isolated-DNA result receives credit there. Moving that conclusion to transported whole blood would require a different link. A vague stability phrase first needs its object identified so the request reaches the right record.

Consultant Lens. A timed route lets specimen and assay owners connect each proposed limit to the material state and performance question it protects. It also identifies who will judge a changed interval, storage practice or collection device. Settling these links before choosing study conditions avoids discovering afterward that a useful result describes a different clock from the instruction the site needs.

What was permitted, what was studied, and what was supported?

Suppose a claim permits two collection routes and the study represents only one. Both may be called *blood* or *tissue*. How can the evidence plan show the difference without dismissing the study's useful result?

The first record tells a site which conditions are allowed. The second establishes which conditions were actually evaluated. The third connects observations and decision logic to a conclusion. The table gives the next question when any one of these statements appears in a file.

Statement in the file	Record needed to understand it
The claim permits this condition	Controlled specification defining the allowed condition
A study included this condition	Actual material, handling and configuration records
Evidence supports this condition	Result, prospective criterion, uncertainty and applicability reasoning

In the two-route example, a site instruction can establish permission for both routes while a specimen log establishes study participation for only one. Even that second entry leaves the result and criterion to inspect. If the represented route meets its justified criterion, its favorable conclusion remains valuable. The question is whether the other permitted route also needs support for the proposed claim. Visual completeness across three rows cannot supply the missing relationship.

This claim-to-study-to-conclusion comparison is the book's reasoning aid. The device-type-specific provision at 21 CFR 866.6080 connects intended tissue specimens with handling, extraction, input and failure information for its defined device type; it does not classify HelixBridge or impose one universal sequencing checklist. The original FDA decision summary for P230011 separately describes requirements and interference, extraction and stability studies in that product's formalin-fixed, paraffin-embedded setting. Its conditions and numbers do not become conditions or results for this fictional assay.

Reviewer Lens. The narrow request depends on the break in the chain. Permission alone calls for the linked study entry; an observed condition without its criterion and result calls for those records. When the chain is complete for one condition, that part can be credited. The review need not turn a limitation in coverage into a finding that the represented condition failed.

Consultant Lens. Claim owners define permission, study owners document actual conditions, and decision owners judge what the result supports. Linking those versions and decisions early makes a later route comparison tractable. After a change, the comparison is between old and new permitted conditions and the studied set; that is what tells the owners whether the existing conclusion applies or a bridge is needed.

How did twelve specimens produce eleven attempts?

Twelve specimens arrive in a separate, wholly fictional teaching example. Two are rejected before any examination. Ten can therefore make a first attempt: $12 - 2 = 10$. Among those ten attempts, two return an invalid state and eight produce initial reports: $10 - 2 = 8$.

One of the two invalid specimens is examined again and this time yields a report. The other still has none. The successful repeat adds an attempt, not another arriving specimen. There are now $10 + 1 = 11$ **total attempts** and $8 + 1 = 9$ **final reports**, from the original **12 specimens**. Which denominator belongs under nine depends on the question.

For the burden at first examination, the numerator is two initial invalid attempts and the denominator is ten first attempts. $2 \div 10 = 0.20$, or **20% of first attempts**. The successful repeat does not erase that specimen's initial invalid state.

For eventual report availability from arrival, the numerator is nine specimens with a final report and the denominator is twelve arrivals. $9 \div 12 = 0.75$, or **75% of arriving specimens**. That denominator retains the two specimens rejected before examination and the examined specimen that still has no report.

For the fraction of all attempts that produced reports, the numerator is nine report-producing attempts and the denominator is eleven attempts. $9 \div 11 \approx 0.818$, or **about 81.8% of attempts**. This measures an attempt-level relationship. It cannot replace the 75% specimen-level availability result merely because the numerator happens to be nine in both. None of these invented counts is a target, HelixBridge observation or product-performance claim.

The history that makes all three fractions reconstructable is a **disposition record**. It identifies what happened, where and why. In a real route, a specimen might be rejected at receipt, fail preparation, return invalid or no-call after examination, yield a partial result, or be repeated. Each state needs its product's definition. Preserving excluded-material accounting means preserving its record; it does not necessarily mean retaining the physical specimen. Molecular specimen and workflow guidance and FDA's qualitative diagnostic-statistics guidance support that original accounting within their respective scopes. (MM13 §§3.1, 7.2–7.2.1; MM09 §3.3.3; FDA diagnostic-statistics guidance §5.)

The separate deployed notes from Chapter 1 show how much less can be calculated from a shorter history. The unnamed tissue-sequencing service recorded **2 initial invalid statuses among 40 consecutive specimen attempts**, so $2 \div 40 = 5\%$ for that interval. Its receipt count, later repeats, final availability and cause are absent. Neither nine nor twelve from the example can fill those gaps.

At the unrelated digital-pathology service, **1 of 25 consecutive supported-image cases** arrived after the local worklist target, and the pathologist completed review. $1 \div 25 = 4\%$ concerns delivery timing. It gives no specimen-adequacy, diagnosis or sequencing denominator. Image-acquisition rejection, software no-output and late worklist delivery would occupy different stages in a pathology study; this note supplies no counts for the first two and no established delay cause. These are separate fictional deployed services, with no new HelixBridge or PathSight chronology or inferred clinical consequence.

A study may legitimately analyze only valid outputs for a defined comparison. That can support a good conditional performance result while leaving availability across the whole route unanswered. The analyst needs the trace from starting set to chosen denominator and the reasons exclusions fit that question. When counts are absent, their state is **not supplied**, never zero. Chapter 8 develops report states; Chapter 12 develops analysis sets and the particular quantity a study aims to estimate.

Reviewer Lens. A headline percentage is easiest to appraise after reconstructing receipt, preparation, first attempt, repeat and report. The divisor might count specimens, attempts, valid reports or a selected comparison set. A valid-result comparison deserves its stated credit, while an unanswered availability question stays separate. For a field event, the trace is also the first step toward investigating cause and consequence; the fraction itself proves neither.

Consultant Lens. Before collection, study and statistical owners can define the units and preserve each repeat's first state. Operational owners define invalid, no-call, partial and delayed states for the actual service. Connecting those definitions to analysis and monitoring prevents a recovered report from cleaning its earlier failure out of the evidence. The later disposition should say what the event history supports, including when a cause remains unresolved.

Can a study start with material already on the shelf?

What can the archive's history establish?

An archive may offer patient material for a planned study. Its origin is useful: an **archived patient-derived specimen** was collected from a patient and retained for later use. But what condition does it represent now?

That depends on how it was selected, collected, prepared and stored, and how its source population relates to the intended use. An archive enriched for easy-to-interpret cases may leave difficult inputs and failures unrepresented. A documented archive may instead be well suited to a bounded analytical challenge. Freshness alone cannot resolve the question; the history must represent the condition the study needs.

FDA's qualitative diagnostic-performance guidance discusses appropriately representative archives for sensitivity or specificity claims, including disclosure of selection and target-status assignment. ISO 20916:2019 addresses selection, handling, storage and accountability within its clinical-performance-study scope. That standard excludes analytical-performance studies, and storage under a study record does not itself establish specimen stability. (FDA diagnostic-statistics guidance §5; ISO 20916 §1 and §§5.3, 5.5.3.9.)

What if the study needs a controlled challenge?

A prepared input containing a known target may help answer whether the downstream sequencing system detects that target. The material has a role because of the question it can challenge. A **surrogate sample** stands in for patient material for a specified purpose; **contrived material** is deliberately constructed or altered. Neither label decides whether the relevant properties are adequately qualified.

If that input bypasses collection and plasma preparation, its target content cannot establish those earlier stages. An image patch can similarly challenge an algorithm's response to content while bypassing slide preparation and scanning. A good downstream result can be useful within that boundary. Its extension depends on qualification of the properties that matter to the further question.

CLSI EP39 treats surrogate qualification as study-specific. MM09 distinguishes patient, reference and contrived material in sequencing validation; those roles and their denominators remain separate. CLSI EP35 addresses analytical equivalence of candidate specimen types for a specified examination, including matrix questions, while excluding the broader preanalytical route. Thus a favorable comparison between inputs can leave collection-route suitability open. No recipe or acceptance number is supplied here. (EP39 §§1.1, 7.1–7.3; MM09 §§3.3.2.2–3.3.2.4, 4.4.4.3.1; EP35, second edition, §1.1 and §2.1–2.2.)

Figure 6.2: A downstream challenge begins late and covers only its bounded path locates generic late-entry studies against DP, tissue-NGS and liquid-NGS paths. The useful reading question is where a study began and which earlier stages it did not observe.

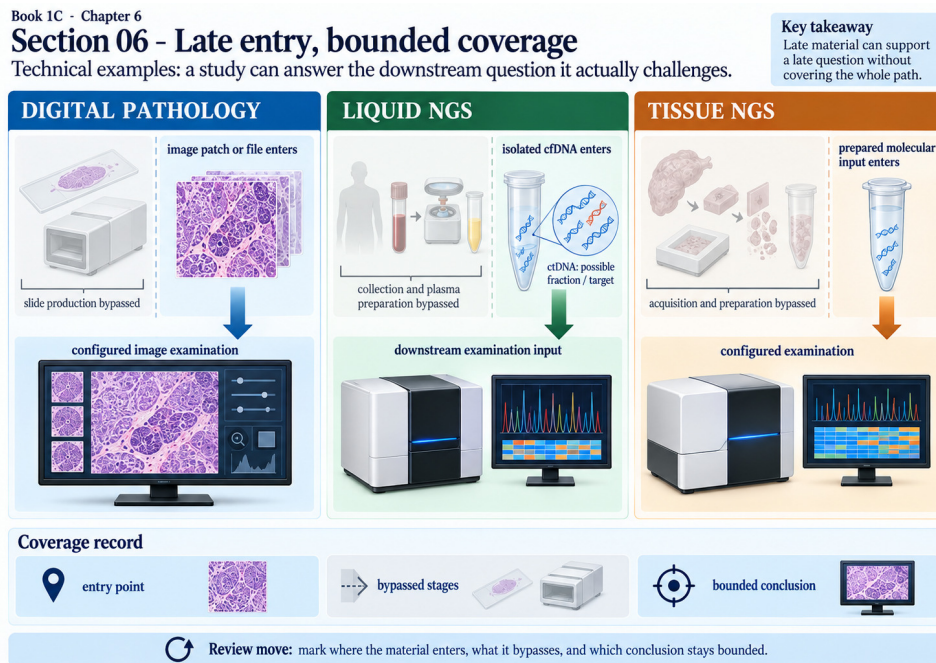


Figure 6.2. A downstream challenge begins late and covers only its bounded path – Technical review plate.

Reviewer Lens. Patient origin is one fact about an archive; selection and handling supply the applicability argument. For a surrogate, the corresponding argument is qualification against the exact challenge, matrix, preparation and input. These can

support positive bounded judgments. The same reading separates an image-content challenge from an observed slide-to-image route and keeps unobserved stages open.

Consultant Lens. Choosing inventory after naming the study question makes it possible to defend why patient, archived, reference or surrogate material is appropriate. A qualification owner can document the properties that matter. Keeping evidence indexed by role and start point then allows a downstream question to close while native-path evidence is planned only for the consequential remaining gap.

What can the two specimen pages say?

The opening specimen row can now be expanded without filling it with guesses. Each page follows patient and sampling context, collected specimen, preparation and derived material, isolated input, and the configured examination. Beside each state are the claimed or proposed condition and the condition actually studied. In the liquid route, a separate tumor-contribution row prevents a biological fraction from being mistaken for another collected specimen.

The crosswalk needs its linked evidence record to reach a judgment. Exact configuration and study identities connect a result to the right object. Exclusions, failures and repeats explain how the counts were reached. The conclusion states the supported condition, remaining difference and next consequential request. A change entry identifies an actual new difference; an absent baseline fact remains a different kind of gap. The Chapter 4 plan row and Chapter 5 reference-process limit stay attached, because specimen coverage cannot settle a different weakness in the comparison.

Reviewer Lens. Reading down the material route and across the claimed and studied states locates the missing relationship. The linked result and accounting then determine its importance. “Not supplied” can call for an existing record; it is not an automatic demand for a new experiment. A represented path meeting its prospective criterion can earn a positive judgment while a broader extension stays open.

Consultant Lens. Claim and configuration owners supply what is proposed, and study owners supply what was actually examined. The consultant helps them address the most consequential gaps first, with an owner and a record capable of closing each question. A later change deserves its own comparison, so neither the baseline nor its original evidence is silently rewritten.

The second route is now supplied

The tissue facts remain those of the opening. The additional record describes the separately controlled **proposed HelixBridge Dx Liquid branch**. It retains the broad variant-identification purpose and proposes venous whole blood, derived plasma and isolated cfDNA, with ctDNA a possible tumor-derived contribution. Its configuration is distinct from the tissue baseline. These are parallel teaching views, with no new dated development or deployment event.

The proposal supplies no detailed population, therapy, clinical role, collection device, handling conditions, plasma preparation, isolation method, input limit, study result or regulatory outcome. Its register identifies preanalytical, end-to-end analytical, clinical-performance and tissue-applicability evidence jobs. No completed result is supplied for those jobs.

Decision

Which claimed or proposed states can the tissue and liquid pages name, and which studied states remain unknown? Does the tissue record establish coverage of the liquid path? Which existing or prospective record would most improve each page? What is the revised answer to the opening request for three facts and a meaning for *stable*?

How the two coverage records support different requests

What the tissue page establishes

The tissue crosswalk answers which patient-to-input states the selected record can actually populate.

Material or context	Claimed condition supplied	Studied condition supplied
Patient and sampling context	Broad oncology variant-identification purpose; full context unknown	Not supplied
Collected tissue	Tissue category; anatomical source unknown	Not supplied
Preparation and derived material	Not supplied; formalin-fixed, paraffin-embedded preparation cannot be assumed	Not supplied
Isolated material entering examination	Exact preparation and input requirements unknown	Not supplied

The preparation row is the crucial temptation: familiarity with oncology tissue could make it easy to insert a standard-looking preparation. The source supplies none. The first row supplies a broad purpose rather than a sampling context, and the tissue row supplies no anatomical source. Every studied cell is *not supplied*. These are absent records, not observed failures or zero specimens.

The selected initial package and unresolved commercial target do not identify an exact specimen configuration. “Wet lab stable” establishes no specimen-stability endpoint or linked study. Exclusions, failures and denominators are also absent. This occurrence contains no new tissue change event: the gap is an unresolved baseline. A focused request seeks the controlled tissue specification and its linked specimen-coverage and disposition records. Specimen fitness is not established by this selection, but the selection does not prove the actual assay lacks adequate evidence.

What the liquid page adds

The liquid crosswalk has more proposed object names. The table asks whether those names are accompanied by studied conditions.

Material or context	Proposed condition supplied	Studied condition supplied
Patient and sampling context	Broad purpose retained; exact population and sampling context unknown	Not supplied
Collected specimen	Venous whole blood; device and handling unknown	Not supplied
Derived sample	Plasma; preparation and storage unknown	Not supplied
Isolated input	cfDNA; isolation, quantity and quality conditions unknown	Not supplied
Tumor contribution	ctDNA may contribute within cfDNA; no quantity or negative-result meaning supplied	Not supplied

The plasma row identifies what is proposed after whole-blood processing. It supplies no method, storage interval or result. The cfDNA row is a further material state with its own absent conditions. The last row describes a possible relation within the DNA; it adds neither a fifth tube nor a known tumor fraction. No studied material or result is supplied in any row.

The exact governing design label is `cfg-hbl-liquid-branch-proposed`, the proposed liquid configuration. Its evidence jobs have no supplied results. Exclusions, failures, repeats and their denominator effects remain unknown. Retaining a broad purpose while changing the specimen and analyte route reopens collection,

preparation, input, reference, performance, reporting and earlier-evidence applicability questions. The next request is an exact proposed configuration and claim-linked evidence plan, with material roles and failure accounting distinguished.

The revised conclusion is that **the tissue record does not establish the proposed liquid path's coverage**. A tissue study, input limit, software state, reference process or report conclusion cannot pass to that path through the family name. No equivalence, performance, release or regulatory verdict follows. Neither product has thereby been shown to fail.

Returning to the first judgment, three useful requests would identify the tissue source, its preparation and handling history, and the actual input and conditions represented by the linked study. Their records must include the relevant stability endpoint and disposition accounting. The opening phrase named a specimen category and a stated assay condition; it did not establish specimen stability. The two pages therefore lead to an existing-record request for tissue and a separate prospective evidence plan for liquid.

What changes when the material record changes?

Archived tissue with no linked history

An unnamed product's proposed claim identifies tissue and refers to a separately controlled collection and preparation path. Its study inventory says only **archived patient-derived tissue**. Anatomical source, preparation, collection device, preservation, transport, storage and their match to the intended path are unspecified. No result, acceptance decision, excluded count or denominator is supplied.

Decision

What can the tissue coverage entry establish, and what narrow conclusion follows? Which missing record would test applicability more directly than asking simply for newer specimens?

What the archive label earns

The studied-material entry can say archived patient-derived tissue. Its match to the claimed path remains unresolved, and no result or count can be entered. Linked selection, collection, preparation and storage history would allow a comparison with controlled intended-use conditions. The archive's age label neither qualifies nor disqualifies it.

The reviewer needs that provenance to judge applicability. A consultant would first ask whether it can be recovered: an immediate fresh collection could discard useful

evidence, while accepting patient origin as representativeness would overstate the existing record. The next decision concerns the archive's actual history and the question it needs to answer.

Two materials in one evidence index

A different unnamed exercise uses the whole-blood □ plasma □ cfDNA path, with ctDNA a possible tumor-derived contribution. Its index names patient-derived plasma for one activity and contrived material for a separate challenge. The contrived entry establishes no whole-blood collection, plasma preparation, background, fragment, matrix or sampling-context coverage. Methods, quantities, results, equivalence and a pooled denominator are absent.

Decision

What can each of the two studied-material rows say? What needs checking before either activity could support a conclusion about the whole path, and what accounting can the index actually provide?

Why the two material roles stay separate

The index establishes two roles, with no adequacy judgment or result for either. Patient-derived plasma leaves preceding whole-blood history undocumented. Target-containing contrived material does not, through that content alone, represent the native route. Separate rows preserve the useful identities without creating a pooled denominator.

The next records need each activity's question, start and end boundary, qualification, configuration, observations and disposition accounting. A reviewer can credit the stated roles while withholding broader coverage. The consultant's first question to the study owners is which distinct gap each material was intended to close; their answers determine whether existing qualification can resolve the gap or additional work is needed.

A qualified challenge that meets its criterion

In a separate fictional stipulation, a surrogate is qualified for a specified **downstream analytical challenge**. Its qualification matches the challenge's analyte, matrix, preparation and input conditions and supports comparability for that purpose. Under a locked downstream configuration, the challenge meets its prespecified, justified criterion. Collection and plasma preparation were bypassed and are expressly outside the requested conclusion. No numerical result or wider product claim is supplied.

Decision

What is the strongest conclusion this record permits? Where does that support end?

The positive conclusion the challenge earns

The qualified surrogate and successful result support the specified downstream challenge under its locked conditions. This is useful positive evidence. Its qualification and criterion are precisely the links missing from a mere inventory entry.

Collection, plasma preparation, whole-product analytical performance and clinical use remain outside that conclusion. A reviewer can accept the downstream answer within its boundary. A consultant can close that question in the plan and assign separate owners to upstream questions only where the claim requires them. Qualification and success are fictional exercise facts; they establish nothing about an actual material or HelixBridge.

Who can close the remaining question?

A manufacturer or developer needs a proposed specimen specification connected to named evidence questions. Each consequential gap has a material state, possible mechanism, performance or availability claim, prospective criterion and residual limit. A laboratory needs instructions and a disposition process usable for its exact implemented examination. Chapter 3's evidence-job distinction still applies: laboratory verification alone does not validate a manufacturer's product, and this matrix supplies no laboratory procedure.

Reviewer Lens. The useful request connects the permitted collection-to-input path with actual evaluated conditions, results and original accounting, including stages a study bypassed. For PathSight's assistive role, that means distinguishing slide preparation and scanner/image states represented by an algorithm or reader study, with the pathologist still responsible for diagnosis. The independent 1/25 late-case note can motivate a route question, but it supplies no PathSight deployment event or cause.

Consultant Lens. Claim, specimen, study and clinical owners can close different parts of that request. A clinical collaborator can explain when sampling occurs and what an unavailable result would mean for the proposed use; the generic matrix is no individual patient-management instruction. In pathology, image acquisition, software and worklist owners remain distinct, including for after-use delivery monitoring. Ordering those handoffs makes the next evidence decision specific enough to act on.

A collection device and additive change

This unnamed final exercise is separate from HelixBridge. Two configurations share the whole-blood → plasma → cfDNA object path. The patient/sampling context, downstream examination configuration and claim wording are stated unchanged. Applicability of earlier evidence is not stated.

The prior configuration uses **local-device-A** with **local-additive-A**; the proposal uses **local-device-B** with **local-additive-B**. These labels identify the prior and proposed collection device and additive once. Their commercial and chemical identities are absent. Three earlier records are identified: collection/handling, the preanalytical path, and end-to-end analytical performance. Their designs, conditions, denominators, results and conclusions are not supplied. There is no next-configuration result or release decision.

Decision

Which object changes directly, and which downstream objects need an evidence-applicability assessment? How do the three earlier records enter that assessment? Which questions reopen, and what request would let the responsible owner determine the further work? What equivalence, acceptability or repetition judgment can this fact sheet support?

How the collection change reaches earlier evidence

Start where the material first differs

Device and additive identities change at whole-blood collection. No new procedure is declared for plasma preparation, cfDNA isolation or downstream examination. Yet the input reaching those procedures might differ. Unchanged procedure identity therefore cannot establish unchanged input behavior.

The collection/handling record can show which device, additive, conditions and disposition rules its earlier conclusion represented. The preanalytical-path record can show preparation, recovery, background, input distribution and compatibility. The end-to-end analytical record can show which performance and reporting questions were evaluated under those inputs. Each title locates a record to inspect; none supplies its conclusions.

Actual identities and mechanism determine which further questions matter: matrix or dilution effects, relevant interference, DNA released from blood cells before plasma preparation, target-fragment integrity and recovery, extraction compatibility, input and adequacy, rejection, invalids and repeats. Each possible effect needs a connection to the

claim and an earlier evidence object. These are assessment questions, not findings that the proposed device is defective.

The decision needs both specifications, the device and additive identities, the change rationale and complete earlier evidence. With molecular and study leads, the change owner can justify a bridge or new study where needed, choosing criteria and materials before results. The supplied record supports reopening applicability. It does not support *no impact, equivalent, minor, acceptable* or *repeat everything*. Chapter 17 extends that reasoning across other kinds of change.

The specimen row now has a route

The opening tissue row now connects a person and sampling context, material states and history, configured examination, identified evidence, disposition accounting and a bounded conclusion. Its missing facts remain visible. The separate proposed liquid page contains more object names but no studied result. Keeping the pages separate prevents tissue evidence from becoming liquid evidence through an unexamined assumption.

The material judgment belongs beside the Chapter 4 plan row and Chapter 5 reference-process limit. A documented archive can contribute evidence, a qualified surrogate can close a downstream question, and a changed collection component can reopen only those relationships it could affect. The conclusion follows provenance, qualification and applicability. Chapter 7 asks the next question: what does a numerical measurement on that material mean? Neither a specimen name nor a calibration certificate answers it alone.

Questions that test the specimen judgment

Can these distinctions now be explained without relying on the case's labels?

- How do whole blood, plasma and isolated cfDNA occupy different material states, and where does ctDNA belong?
- Which tissue-path entries remain unknown, including its preparation, and why can another product's conditions not fill them?
- How do permitted conditions, studied conditions and supported conclusions connect?
- Which object, exposure conditions and endpoint make a stability statement assessable?
- How do rejected material, first unsuccessful attempts, repeats and final reports stay visible in the accounting?
- How do patient-derived, archived, reference and contrived materials retain their actual study roles?

- How can a successful limited challenge support a positive conclusion while leaving bypassed stages open?
- How does an actual change reopen a named evidence question without determining impact or workload?

Returning to the two material paths raises a final connected question. Why can isolated material support useful evidence without representing collection? What does *wet lab stable* fail to establish? How does a missing baseline record differ from evidence reopened by change? How would the collection-device judgment follow from claimed, studied and supported conditions without borrowing the earlier answer's wording?

The relationships behind the final judgment

A study starting with isolated material can establish its downstream answer while leaving collection and preparation outside the observed boundary. A stated stable assay condition supplies no specimen-stability endpoint. Missing baseline information may be recoverable in an existing record; an actual difference requires judging whether an earlier conclusion still applies.

The link runs from a permitted claim condition to the observed study condition and then to the supported conclusion. In the transfer, collection device and additive identities changed directly. Downstream evidence consequently needs an assessment of the input it represented. That is a reason to examine applicability, with neither automatic inheritance nor automatic disposal of the previous work.

Source trail and limits

This chapter's crosswalk is author-defined, not an official acceptance scheme. The source packet's thirteen bounded claims and exact licensed-text locators govern the technical propositions; no procedure, storage time, temperature, input limit, tumor fraction, study count or universal acceptance value is donated to a fictional case. The local Chapter 6 selections and exercise fact sheets govern all case statements.

- **Molecular specimen path:** CLSI MM13, second edition, §§1.1, 3.1, 3.3, 4.2.1–4.2.1.3 and 7.2–7.2.1; CLSI MM09, third edition, §§3.3.2.2–3.3.4.2 and 4.4.4.3.1. These consensus sources guide molecular specimen and workflow reasoning, not HelixBridge facts or authorization.
- **Different material states:** ISO 20166-3:2018, §1 and §§4–6, addresses FFPE tissue to isolated DNA; ISO 20186-3:2019, §1 and §§4–6, addresses venous blood, plasma and isolated circulating cell-free DNA. The official ISO record identifies the latter edition as current after 2024 confirmation. Neither makes the fictional tissue path FFPE or supplies a liquid product condition.

- **Material roles:** CLSI EP39, first edition, §§1.1 and 7.1–7.3, supports study-specific surrogate qualification; CLSI EP35, second edition, §1.1 and §§2.1–2.2, concerns candidate specimen-type analytical comparisons and excludes the broader preanalytical route. No licensed study recipe or numerical condition is reproduced.
- **Clinical-study specimens and counts:** ISO 20916:2019, §1 and §§5.3, 5.5.3.9, is confined to its clinical-performance-study role and excludes analytical-performance studies. FDA's qualitative diagnostic-statistics guidance, §§4–5, informs conditions of use, original accounting and appropriately representative archived collections within its stated context.
- **Scoped public sources:** FDA's final November 2024 ctDNA guidance, §§IV.B–C, addresses early-stage curative-intent solid-tumor drug development, not general device clearance. 21 CFR 866.6080 is device-type specific. P230011's original FDA summary is a frozen, different-product illustration. Regulation (EU) 2017/746, Annex II §§6.1.1 and 6.1.2.3, is cited through a dated consolidation for EU specimen and interference documentation; applicability and amendments need checking for actual use.
- **Exclusions:** CLSI PRE04 §1.1 excludes molecular pathology and nucleic-acid diagnostics. CLSI EP25 §1.1 excludes patient specimens; ISO 23640:2011 §1 excludes the sample under examination. They mark scope boundaries, not positive specimen-stability authority.

The two unnamed micro-cases, unnamed device/additive transfer and bounded surrogate-success exercise are separate fictional teaching records, not developments in HelixBridge, PathSight or their regulatory histories. Their answers follow their prompts; none supplies a real product's empirical result.

CHAPTER 7

Measurement, Calibration, Trueness, and Uncertainty

A number waiting for a decision

Prepared plasma enters an instrument. The instrument produces an optical response. A report will eventually express how much of fictional Analyte Q is present per volume of that plasma, in **milligrams per litre (mg/L)**: mass divided by volume. Between the response and the report, the procedure needs a documented relationship that turns the signal into a concentration. Afterward, a separate rule may use the concentration to assign a category. Each step has a different job.

We will call this fictional examination **the plasma procedure**; its record identifier is **QMP-A**. It is separate from HelixBridge Dx and PathSight Cloud. Its summary says *high precision* and *strong correlation with the comparison procedure*, then proposes that its results are sufficiently trustworthy for decisions near a specified concentration. What does that summary actually let someone decide?

Decision

Which statement in the summary could be retained, and which conclusion would still need support? If “high precision” or “strong correlation” is too vague to interpret, which record would make it meaningful? What is known about repetition conditions, the comparison procedure and range, the values reasonably attributable to one new result, and the basis of the decision rule? No reference value or clinical acceptance limit has yet been supplied.

A useful measurement record follows this whole route. One page can identify the quantity, signal-to-value relationship, reference and material roles, observed spread and comparison difference, and included uncertainty sources. A second can show the exact decision and its limits. This is the book's reasoning aid, not an official form. Its value lies in connecting evidence to a conclusion another person can reconstruct.

The calculation will expose every observation and explain the average, squares and square root as they become useful. No statistics prerequisite or mental-arithmetic skill is assumed. All certificates, conditions, measurements and rules in the plasma example are teaching fiction.

The questions a number still has to answer

Chapters 5 and 6 asked whether the reference was fit for its question and whether the studied material represented the claimed specimen path. A numerical result needs both relationships too.

Decision

What makes a reference suitable for a conclusion rather than merely available? How could a study input pass through the examination while bypassing the patient's collection and preparation? What needs reconsideration when the studied configuration differs from the claimed one? In a claim's reference memo and specimen record, which exact output is involved: signal, concentration, variant identity or clinical interpretation?

How those earlier limits follow the number

The reference needs a justified role and coverage for the proposed inference. The material's entry point determines which part of the specimen path was actually challenged. A configuration difference needs an argument about applicability. Calling the output “the test result” hides these distinctions; identifying the signal, concentration or interpretation makes the relevant relationship visible.

How does an optical response acquire mg/L?

“Analyte Q” names the component of interest, but it does not say what is being measured about it. The plasma procedure seeks **mass concentration of Analyte Q in human plasma, expressed in mg/L**. The material system, quantity and unit turn the component's name into a defined **measurand**, the quantity intended to be measured. The procedure and relevant specimen conditions specify how that quantity is realized here. “Analyte Q positive” would describe a different output. (JCGM metrology vocabulary, term 2.3; ISO 17511:2020 §4.2.)

The instrument initially supplies only an **indication**, its optical response. To assign mg/L to a new response, the procedure needs the relationship between responses and assigned concentration values under specified conditions. Establishing that relationship and using it to obtain a measured value is **calibration**. It explains why the number has this value on this scale; a label saying “calibrated” leaves the relationship to be inspected.

Changing a response setting does a different job. That is **adjustment**, an alteration of the instrument; a later check may show whether a requirement is met. Neither the alteration nor a passing check supplies the whole calibration relationship. Software

teams also use “calibration” for the relationship between predicted probabilities and observed outcomes. That statistical sense concerns predictions and does not establish a plasma concentration value chain. (JCGM calibration, term 2.39.)

A converted number still needs information about how to interpret it, ordinarily including its uncertainty. That fuller account is the **measurement result**. File and version records can trace which optical response and software produced it. To ask how its value relates to a stated reference is to ask for **metrological traceability**: a documented calibration chain, with uncertainty at every link. An intact chain may remain too uncertain for a particular use. It also leaves separate questions about mistakes and whether the reference material behaves appropriately relative to patient material. (JCGM traceability, term 2.41; ISO 17511:2020 §§4.1–4.4.)

Reviewer Lens. Reading backward from one reported value is revealing: what converted the response, what assigned the calibrator's value, and what uncertainty traveled with each link? A well-documented chain deserves a bounded traceability conclusion. The difficult next judgment is whether that chain, specimen and width serve the proposed decision. Requesting the missing use relationship is more useful than asking for another certificate with the same unexplained label.

Consultant Lens. The measurement owner can freeze the conversion procedure and calibration records while the specimen owner fixes the material boundary and the claim owner identifies the decision region. Connecting those versions before collecting data prevents a later argument over which chain the study represented. After a calibrator or procedure change, the owner needs a revised value-transfer and applicability record before carrying the earlier conclusion forward.

What would make that chain credible?

A material can carry an assigned quantity value and uncertainty. A reference measurement procedure can assign or compare values for its defined use. A laboratory using that procedure needs competence to perform it. These are the distinct jobs recalled from Chapter 5. A material certificate cannot certify laboratory competence; a competent laboratory cannot make an unsuitable material behave like patient plasma by issuing a certificate.

The source for the procedure's job is **ISO 15193:2026**; for higher-order certified reference materials and their documentation, **ISO 15194:2026**; and for calibration-laboratory competence using reference procedures, **ISO 15195:2018**. The last is not the general medical-laboratory standard and excludes nominal or ordinal results. The applicable in vitro diagnostic value-traceability relationship is the subject of ISO 17511:2020. These scopes identify where to look, without qualifying any fictional object. (Procedure, material, laboratory.)

Reviewer Lens. The first question is which property the chain would support. A colour reference could support a defined image quantity; it could not by itself establish successful reader prioritization. An identity-assignment reference could support a reported variant's name without supplying mg/L traceability. A missing role remains visible until the record identifies who or what actually performs it.

Consultant Lens. The assay or imaging owner defines the property, the reference owner documents its value or identity assignment, and the claim owner identifies the intended decision. The consultant brings those records together at one controlled configuration. Borrowing a familiar certificate from an unrelated measurement example would conceal the work still needed.

Which part of a report is a quantity?

The plasma procedure's optical response becomes a concentration, then possibly a category. A sequencing report can instead identify a DNA variant. Software may have counted reads and calculated scores to reach that name, but the name answers “which variant?” The output being claimed determines which part of the route needs a measurement model.

The table helps locate that output before choosing a calculation.

Reported output	What its values mean	What must first be defined?
Concentration	A magnitude in a unit	Quantity, specimen system and procedure
Count	Number of specified entities	Entity, counting rule and sampling conditions
Ordered category	Rank, possibly derived from quantified intervals	Whether an underlying quantitative relationship exists
Nominal identity	Which entity or class	How identity and the reference classification are assigned
Category derived from a quantity	A rule applied to a measurement	Both the measurement model and the classification rule

The concentration row fits the plasma procedure: mass per volume has a stated meaning and unit. For the count row, a count of reads is not automatically a count of tumor molecules; the entity and counting rule matter. “Low, medium, high” is an **ordered category** without a promise of equal distances. A variant name is a **nominal identity**; replacing its letters with a code number would not give it magnitude. The last

row contains two evidence jobs because a reliable quantity and a justified categorizing rule must both hold.

This is also a scope boundary in ISO 17511:2020 §1. It covers defined numerical and counting results and some categories produced from specified quantitative relationships, while excluding nominal properties and ordinal scales without magnitude. Scope identifies the applicable model; it supplies no product conclusion. (ISO edition record.)

PathSight makes the distinction concrete. It prioritizes whole-slide images and highlights suspicious regions for the pathologist, who diagnoses. Its selected bench record compares average colour values but does not assess weakly stained tissue near low-confidence boundaries. No exact colour quantity, unit, measurement procedure, reference or uncertainty is supplied. A named scanner calibration profile cannot fill those fields. A later, properly defined colour measurement could answer a bounded image-production question, while prioritization, highlighting and reader workflow would still need their own evidence.

Reviewer Lens. A favourable bench comparison can retain its recorded scope while its relationship to the assistive claim remains open. The narrow request is for the colour quantity or scale, acquisition procedure, fit reference assignment and relevant image material, followed by the link to the claimed behaviour. For a reported variant, the corresponding request concerns identity assignment and categorical performance. The same demand for a concentration interval would miss that difference.

Consultant Lens. The report and decision belong with the clinical claim owner; their underlying numerical and categorical layers belong with assay or imaging specialists. Defining those layers early routes each to an appropriate evidence plan. Timing and invalid-result observations later belong to their event owners and denominators, rather than being relabeled as measurement spread.

Give the plasma calculation an actual record

A specified concentration allows a quantitative model. It does not yet supply the model's inputs. The following record gives them explicitly, so the eventual calculation can be checked without treating the example as a real assay, clinical protocol or licensed-standard recipe.

Which procedure, material and assigned values?

The plasma procedure converts an optical indication into mass concentration of Analyte Q in human plasma using an end-user calibrator. Its exact record is **QMP-A, standard operating procedure (SOP) 1.2; QMP-A-CONFIG-A; Analyzer Q, serial AQ-017; software 1.2; one fictional laboratory**. We will call this the recorded configuration. A transfer record from another analyzer or software state does not establish this one's relationship.

The exercise stipulates plasma separation within **two hours** of collection and measurement within **twenty-four hours at 2–8 °C**, or storage **at or below –70 °C with no more than one freeze–thaw cycle**. These are invented case conditions, never patient-handling advice or universal stability limits. The repeated-measurement material enters as already prepared plasma, so the twenty observations below do not evaluate collection or separation.

The model is confined to **central results of 9.50–10.50 mg/L**. Within that narrow interval, the same *absolute* uncertainty width is assumed. Neither the observations nor a general metrology rule establishes that assumption for a real product or a wider interval.

The next table follows how an assigned value reaches the routine procedure. Exact record identifiers appear beside plain role names so the documents remain distinguishable.

Object in the record	Job in the fictional chain	Supplied value or limit
Reference procedure — Q-RMP-1	Assign the reference material's value	Role stipulated; no real procedure qualification
Reference material — Q-CRM-1	Highest available material reference in this example	10.000 mg/L; standard uncertainty 0.050 mg/L
End-user calibrator — Q-CAL-A	Carry an assigned value into the routine procedure	10.000 mg/L; combined standard uncertainty 0.080 mg/L for an individual lot
Plasma-procedure result — QMP-A	Convert an indication into concentration	Procedure and recorded configuration identified above

Following the middle rows makes the relationship visible. The reference material carries a concentration assigned by the reference procedure. The end-user calibrator carries an assigned value into the routine examination, which converts the optical response to mg/L. A procedure is not another vial or merely a routine comparator. This

proposed arrangement establishes the fictional roles only: it proves no real unbroken chain, International System of Units (SI) endpoint, or patient-material fitness.

The reference-material certificate, **Q-CRM-1-CERT-v1**, states **10.000 mg/L** with standard uncertainty $u_{\text{ref}} = 0.050 \text{ mg/L}$. Standard uncertainty expresses uncertainty on the scale of a standard deviation. The certificate also gives expanded uncertainty **0.100 mg/L**, using factor **2** and approximately **95% coverage** under a bell-shaped, or Gaussian, model. These are different expressions of the same assigned-value uncertainty.

The end-user certificate, **Q-CAL-A-CERT-v1**, gives $u_{\text{cal}} = 0.080 \text{ mg/L}$ for an individual calibrator lot. That value **already includes the 0.050 mg/L reference contribution**, along with later value-transfer uncertainty. Counting 0.050 again would count one source twice. The 0.080 excludes lot-to-lot shifts: this example assigns those to the longer-term routine spread. Actual certificates and studies would need to establish those inclusion boundaries.

The patient-material relationship remains a separate question. **Commutability** asks here whether the reference material shows the appropriate relationship between the reference and routine procedures relative to relevant patient samples for this quantity. Its assigned value and certificate cannot establish that relationship. (JCGM term 5.15; ISO 17511:2020 §§4.5.3–4.5.7.)

Reviewer Lens. The four rows are four claims requiring supporting records. The certificate's inclusion statement is consequential: a precise-looking budget can be wrong before any arithmetic if the reference contribution appears twice. The stipulated chain is sufficient for this teaching calculation. Real traceability and material fitness would require the documents behind each role.

Consultant Lens. Reference owners can document the procedure and material certificate, the calibrator owner can state which sources its width includes, and the laboratory owner can identify the routine configuration. One shared account of those boundaries prevents omission and duplication. A later calibrator lot needs its own value and behaviour trace before the result owner reuses the model.

What do twenty different results tell us?

The next material is **the frozen plasma pool**, recorded as **Q-POOL-A**: fresh-frozen pooled human plasma divided into portions on case day zero, stored at or below **−70 °C**, with **one thaw per portion**. A *separate fictional record* stipulates stability for **sixty days**. The twenty results are not that stability study. For this calculation, the pool is assumed to have precision behaviour similar to typical intended plasma near **10 mg/L**; that assumption does not establish commutability between the reference and routine procedures.

Each row below is one run, one repeated measurement event. There are twenty runs, rather than twenty patients or twenty reference assignments. Two operators alternate. The first reagent and calibrator pair is **R-A/C-A**; the second is **R-B/C-B**. Operators **O-A/O-B** are the first and second operators. The exact identities stay in the data table, whose question is what changed while the results varied. Ambient temperature was 20–24 °C.

Run	Case day	Operator	Reagent / calibrator lot	Result
1	1	O-A	R-A / C-A	9.82
2	4	O-B	R-A / C-A	10.04
3	7	O-A	R-A / C-A	9.88
4	10	O-B	R-A / C-A	10.08
5	13	O-A	R-A / C-A	9.92
6	16	O-B	R-A / C-A	10.12
7	19	O-A	R-A / C-A	9.95
8	22	O-B	R-A / C-A	10.16
9	25	O-A	R-A / C-A	9.98
10	28	O-B	R-A / C-A	10.20
11	31	O-A	R-B / C-B	10.02
12	34	O-B	R-B / C-B	9.96
13	37	O-A	R-B / C-B	10.05
14	40	O-B	R-B / C-B	9.92
15	43	O-A	R-B / C-B	10.08
16	46	O-B	R-B / C-B	9.88
17	49	O-A	R-B / C-B	10.12
18	52	O-B	R-B / C-B	9.84
19	55	O-A	R-B / C-B	10.18
20	58	O-B	R-B / C-B	9.80

Run 1 means that on case day 1 the first operator used the first lot pair and obtained **9.82 mg/L**. Run 11 moves to the second pair on day 31. Reading just the result column would miss that reagent and calibrator lots changed together at the same point in time.

The pattern also deserves attention. In runs **1–10**, the first operator's results rise from **9.82 to 9.98**, and the second operator's from **10.04 to 10.20 mg/L**. In runs **11–20**, the first operator's rise from **10.02 to 10.18**, while the second operator's fall from **9.96 to 9.80 mg/L**. These are observations to investigate. The paired lot change and time cannot be separated into individual causes using this design. One summary of spread will not establish independent random variation, absence of drift, or coverage of all routine conditions. This is original teaching data, not a CLSI EP05 protocol or a universally adequate sample size.

First describe how closely the results repeat

How close repeated values lie to one another under specified conditions is **precision**. A tight cluster obtained on one day with one lot speaks to more constrained conditions, often called **repeatability**. Specified changes within a site give **intermediate precision** a broader question; **reproducibility** concerns a broader set of stated conditions again. The factors actually varied are what make those words informative. This meaning of precision differs from the classification statistic that counts how many positive predictions agree with a reference-positive category. (JCGM precision, term 2.15; CLSI EP05, fourth edition, §1.1.)

For these runs, the first step is to locate the centre of the cluster. Adding the twenty measurements gives **200.00 mg/L**. Dividing by **20 observations** gives their average. The bar over x means average; x_i means the result in row i , with i running from 1 to 20.

$$\bar{x} = (200.00 \text{ mg/L}) / (20) = 10.000 \text{ mg/L}.$$

To measure departures from that centre, subtract **10.000 mg/L** from each result. For row 1, $9.82 - 10.00 = -0.18 \text{ mg/L}$. Simply adding departures would let positive and negative values cancel. Squaring each one keeps its contribution: row 1 gives $0.0324 (\text{mg/L})^2$. The summation sign means to add those squared departures over all twenty rows.

$$\sum_{i=1}^{20} (x_i - \bar{x})^2 = 0.2882 (\text{mg/L})^2.$$

The mean came from the same twenty observations. Once it is fixed, their departures must sum to zero, leaving **19** freely varying departures. The ordinary sample-variance divisor is therefore $n-1 = 20-1 = 19$. The **sample variance** is $0.2882/19 = 0.0151684211 (\text{mg/L})^2$. Taking its square root returns to mg/L and gives the **standard deviation** of the observed results. Here it serves as the estimated longer-term within-laboratory precision contribution, named u_{Rw} :

$$u_{\text{Rw}} = \sqrt{((0.2882 (\text{mg/L})^2) / (19))} = 0.1231601 \text{ mg/L}.$$

The resulting **0.1231601 mg/L** describes pooled variation across days, two operators and paired reagent/calibrator lots in one laboratory. It is more than short-condition repeatability, but it cannot isolate every factor. Independence and bell-shaped behaviour remain assumptions for later arithmetic, not findings established by one standard deviation. Chapter 9 develops designs that separate conditions.

Reviewer Lens. The sum, one squared departure and the divisor are inspectable. So are the day and lot columns, which prevent the numeric result from being mistaken for a complete routine-use estimate. The observed spread deserves credit for this pool and these conditions under the stipulated behaviour assumption. A question about the paired lots or patterns concerns its interpretation and transfer; it does not undo the correct descriptive arithmetic.

Consultant Lens. Laboratory and statistical owners can decide which days, operators, instruments, lots and materials must vary before the study begins. The pool's late entry also belongs in that plan: no favourable spread closes collection or separation. After a lot change or monitored shift, the owner compares the new conditions and pattern with the earlier design before deciding whether its estimate still applies.

Is the cluster in the right place?

A tight cluster could lie consistently away from a suitable reference value. Its spread would still be small. **Trueness** concerns the systematic closeness of the long-run average to a reference quantity value; in metrology it is not itself a numerical score. A mean difference, or **bias** estimate under an appropriate reference comparison, addresses that systematic question. **Measurement accuracy** relates to both systematic closeness and spread and is qualitative in this vocabulary. A classification “accuracy percentage” is a separately defined statistic. (JCGM trueness, term 2.14, and bias, term 2.18.)

The plasma record supplies a second dataset that lets us inspect a difference. This time the routine procedure measured the **reference material ten times**, rather than the frozen pool over twenty runs. Its assigned concentration is **10.000 mg/L**. The third column subtracts that assigned value from each measured result, so positive values indicate results above it on this material.

Replicate	QMP-A result, mg/L	Difference from assigned value, mg/L
1	10.01	+0.01
2	10.03	+0.03
3	10.04	+0.04
4	10.05	+0.05

Replicate	QMP-A result, mg/L	Difference from assigned value, mg/L
5	10.06	+0.06
6	10.04	+0.04
7	10.07	+0.07
8	10.08	+0.08
9	10.06	+0.06
10	10.06	+0.06

The first row gives $10.01 - 10.000 = +0.01$ mg/L; the last gives $10.06 - 10.000 = +0.06$ mg/L. The ten results total **100.50 mg/L**. Their mean is $100.50/10 = 10.050$ mg/L. Subtracting **10.000 mg/L** yields a **+0.050 mg/L mean difference for the reference material**. That is a useful, material-specific observation. It is neither a numerical trueness score nor established patient-sample bias.

The distinction matters before anyone corrects a patient result. The material's assigned value has uncertainty, and its commutability for the reference and routine procedures relative to relevant patient plasma remains unresolved. It could interact differently with the two procedures than patient samples do. These ten repeats also lack the patient-sample comparison design addressed by CLSI EP09c. There is therefore no established basis here for subtracting **0.050 mg/L** from patient results or quietly adding that difference as another uncertainty component. A real correction would need an applicable basis and its own uncertainty.

What if the comparison used a second routine procedure? Such a comparator can be useful without being a higher-order reference. A difference would describe the relationship between the procedures, without deciding which is closer to an appropriate reference quantity value. A patient-sample comparison needs relevant patients, measurement range, paired observations and a justified comparator role. (CLSI EP09c, third edition, §1.1.)

This returns to the opening phrase “strong correlation.” **Correlation** describes whether two sets rise and fall together. If every comparator value equals the candidate value plus the same offset, their correlation can be perfect while every pair differs. **Agreement** asks about the size and pattern of paired differences in the relevant range and whether they meet a prespecified use-specific criterion. A wide range can yield strong correlation while concealing consequential differences near the decision boundary. The comparison guidance cautions against correlation alone as an acceptance criterion; this chapter supplies no regression or limits-of-agreement recipe. Its quantitative comparison also differs from categorical agreement, whose correctness

interpretation still needs Chapter 5's fit reference assignment. (EP09c, §6.2.1; FDA qualitative diagnostic-statistics guidance, §§3, 6.1.)

The same question separates potential imaging studies. Repeated imaging of a defined colour feature under specified scanner and preparation conditions could estimate feature spread; comparison with a suitable colour reference could examine a systematic difference. Neither establishes that a pathologist receives the right image at the right time. For a reported variant identity, repeated agreement with another call set could address categorical consistency. Correctness still depends on assigned reference identities and which attempted specimens reached valid reports. These are possible study choices, not supplied successes for PathSight or HelixBridge.

Reviewer Lens. “+0.050 mg/L on the reference material” preserves exactly what was observed. The next request concerns whether the material relationship and patient comparison support transfer. In a broad summary, the misleading move is to let “high precision” imply trueness, or “strong correlation” imply acceptable paired differences. Enough evidence names the repetition conditions, comparator role and differences in the decision region, and ties the conclusion to them.

Consultant Lens. The statistician and clinical owner can choose a relevant range and difference criterion prospectively, while measurement and material owners choose a design capable of addressing it. Precision, reference comparison and categorical performance need distinct evidence jobs. For the observed material difference, the consultant can frame the possible closure outcomes—supported correction, a narrower claim or no correction—without choosing among them from ten repeats alone.

What belongs around one new result?

The frozen pool has supplied a spread, and the calibrator certificate has supplied an assigned-value uncertainty. Neither source replaces the other: repeated measurements do not erase uncertainty in the calibrator's assigned value, and a certificate does not measure routine spread. Together they can inform the values reasonably attributable to one new specimen's concentration. That is the question of **measurement uncertainty**.

Before combining anything, the table separates this question from two others that also make a conclusion uncertain.

Question	Kind of uncertainty	What would answer it?
Which values can reasonably be attributed to this defined measurand?	Measurement uncertainty	Measurement model, inputs, components and coverage statement

Question	Kind of uncertainty	What would answer it?
How well do finite observations estimate an average, rate or model relationship?	Sampling or model uncertainty	Statistical assessment of estimates and assumptions
Does this material, configuration or reference support the intended inference?	Unresolved applicability or evidence	The missing evidence or a narrower conclusion

The first row concerns a single defined measurement result. **Standard uncertainty** is expressed like a standard deviation in its quantity's unit. A wider interval can be expressed by multiplying a combined standard uncertainty by a coverage factor. The second row concerns how well a study has estimated something: a confidence interval for an estimated mean is a different object from a coverage interval attributed to one specimen result. Data and assumptions matter to both. The third row cannot be repaired by inventing an extra numerical allowance for unstudied material behaviour. Clinical uncertainty adds questions about disease, sampling and care that a mg/L interval also cannot settle. (JCGM measurement uncertainty.)

Chapter 1's separate deployed services show why the second row matters. The unnamed tissue-sequencing service recorded **two initial invalid attempts among 40**, an observed initial-invalid-attempt fraction; final report availability is not supplied. The unrelated pathology service recorded **one late worklist arrival among 25**, with review completed, an observed timing fraction. Finite counts leave uncertainty about an underlying event rate. Neither fraction is measurement uncertainty of a variant identity or colour value, establishes cause, or becomes a HelixBridge or PathSight event. Their respective owners need attempt or case traces and prospective monitoring criteria. A separately defined numeric submeasure could receive its own budget only with an applicable reference and repeated-measurement design.

For the plasma concentration, medical-laboratory uncertainty guidance in **ISO/TS 20914:2019** informs the routine-spread and calibrator components. **ISO 17511:2020** makes allowable uncertainty use-specific for applicable in vitro diagnostic quantity results. Neither supplies this example's acceptance rule. (Uncertainty guidance, §§5.2–5.4, 6.2–6.6; traceability standard §4.3.)

Reviewer Lens. An interval needs a target as well as endpoints: one result, an estimated mean or a clinical probability. Its unit and coverage basis expose which inference it serves. A narrow, well-supported measurement interval can remain useful while patient-material or clinical applicability is unresolved. The missing relationship is evidence to obtain, rather than a reason to relabel every uncertainty as the same thing.

Consultant Lens. Measurement owners establish component boundaries, statisticians assess estimation and dependence, and clinical owners define the decision needing

support. Keeping those responsibilities connected makes a material or calibration change actionable: the affected model can be revisited before an old interval is reused in a new state.

Two contributions, each counted once

The target is a **single future result** from the recorded plasma procedure. Its working budget includes $u_{\text{Rw}} = 0.1231601$ mg/L from the twenty-run spread and $u_{\text{cal}} = 0.080$ mg/L from the individual calibrator lot's assigned value. The reference contribution $u_{\text{ref}} = 0.050$ mg/L is already inside the latter. Lot-to-lot shifts are stipulated to belong in the former. This is why understanding the records came before choosing the formula.

The example now assumes approximate independence of those two retained contributions. Their **covariance**, which describes how they vary together, is treated as zero. It also assumes approximately Gaussian, or bell-shaped, behaviour for the later coverage convention. The table has not demonstrated either assumption. Contributions that move together need their dependence included; adding only their squares could misstate the width.

Under the stipulated independence, the variances add. Squaring each standard uncertainty puts both on the variance scale, in $(\text{mg/L})^2$; the square root returns the sum to mg/L. The result is the **combined standard uncertainty**, u_c , where the subscript means combined:

$$u_c = \sqrt{(u_{\text{Rw}}^2 + u_{\text{cal}}^2)}; = \sqrt{((0.0151684211 + 0.0064) (\text{mg/L})^2)}; = 0.1468619 \text{ mg/L}.$$

The routine term is the earlier variance, **0.0151684211**, represented by 0.1231601^2 ; the calibrator term is $0.080^2 = 0.0064$. Their sum is **0.0215684211** in squared mg/L before taking the root. Adding the two standard uncertainties directly would use a different calculation. Adding the reference's 0.050 again would duplicate a source already included.

For this example the **coverage factor** is $k = 2$. Multiplication by that factor produces **expanded measurement uncertainty**, U :

$$U = k u_c = 2(0.1468619) = 0.2937238 \text{ mg/L},$$

Under the stated Gaussian and estimation assumptions this is described as **approximately 95% coverage**. It is not an exact small-sample guarantee, assurance that every error lies inside the interval, or a standard-derived acceptance rule. The width is displayed as **0.29 mg/L**, rounding at the end.

For one in-scope result $y = 10.00$ mg/L, the source calculation retains $U = 0.2937238$ mg/L: $10.00 - 0.2937238 = 9.7062762$ and $10.00 + 0.2937238 = 10.2937238$ mg/L. The displayed interval is **9.71–10.29 mg/L**. It concerns values attributable to that specimen's defined concentration under the model. It is not a confidence interval for the twenty-run control mean.

That last distinction explains why u_{Rw} is **not divided by $\sqrt{(20)}$** . The twenty controls estimate the procedure's dispersion. The future patient result is one measurement, not an average of twenty measurements of that patient's specimen. A square-root-of-twenty reduction would address uncertainty of the control mean from repetition, while making the single-result interval misleadingly narrow.

Reviewer Lens. Correct arithmetic and a complete uncertainty model are separate judgments. Here the units, component boundaries, root and final rounding give a coherent conditional result. Real use would still need support for dependence, material behaviour, distribution and omitted sources. The review can accept the two-term arithmetic while naming those unresolved relationships precisely.

Consultant Lens. Each component needs provenance and an overlap check from the measurement owner; the statistician assesses dependence and distribution. The clinical owner identifies where the decision needs the interval. Evidence for omitted effects or a narrower scope can then be planned deliberately. If later lots show shifts outside the studied pattern, the routine-spread estimate and its combination need reassessment before the same width is reused.

What has the calculation earned?

For **mass concentration of Analyte Q in human plasma**, the recorded configuration, stipulated material behaviour, independent included components and approximate Gaussian convention yield $u_c = 0.1469$ mg/L and $U = 0.29$ mg/L with $k = 2$ near **10 mg/L**. This is a coherent, conditional working estimate for that concentration layer. Quantity, units and component boundaries align, so the model can be evaluated and used for the later fictional rule under its assumptions.

Its unresolved extensions remain specific: patient-sample bias, reference-material commutability, collection and separation coverage, wider-range behaviour and clinical-decision fitness. Those gaps do not erase the calculation. They delimit what it can support.

For PathSight, the same sequence of questions would begin with a defined colour quantity or scale, then image acquisition, reference assignment, repeat-image conditions and component boundaries. Its present average-colour comparison and unassessed weak-stain boundary supply no uncertainty width. A supported colour interval would still leave worklist timing and diagnosis separate. Imaging and claim owners would need to connect the submeasure to the workflow claim.

For HelixBridge's reported variant identity, the concentration model stops at the result-layer question. If a *separate count* were proposed inside its sequencing workflow, the assay owner would need the counted entity, unit, reference, specimen range, repeated-run conditions and decision link. That prospective assessment would remain distinct from identity performance and invalid-attempt monitoring. The unnamed

service's 2/40 calls for an attempt-to-report investigation with its denominator preserved; it supplies no count-uncertainty component or named-product history.

How far can the result travel?

A different control material

The frozen pool is replaced by a **freeze-dried pool**, record **Q-POOL-B**. Such *lyophilised* material must be reconstituted before examination. It too has an assigned value of **10.000 mg/L**, but no evidence is supplied about reconstitution, stability or whether its precision behaviour in the recorded configuration represents typical human plasma.

Decision

Does the quantitative model cease to fit this kind of result, or does a particular evidence relationship need repair? Which relationship, and can the earlier numerical estimate be reused?

What the material change reopens

The reported property remains mass concentration, so a quantitative measurement model still fits. What needs support is the new material's fitness for estimating the procedure's dispersion. Equal assigned concentrations do not establish equal behaviour after freeze-drying and reconstitution. The earlier pool retains its useful result, while transferring $u_{\text{Rw}} = 0.1231601 \text{ mg/L}$ or $U = 0.29 \text{ mg/L}$ requires evidence. Material and study owners can address reconstitution, stability and representative precision in a new applicability record. Copying the number would skip that relationship.

A different reported property

A separate generic case replaces concentration with a **nominal variant identity**, naming the sequence variant reported. No separate quantity or count is defined. Someone proposes “ $\pm 0.29 \text{ mg/L}$ ” as the identity's uncertainty.

Decision

What is wrong with that proposal, and which validation question remains even if the proposed interval does not fit?

What changes when the result is an identity

Mass concentration has a magnitude to which mg/L and its interval attach. A nucleotide identity has none. Its evidence still needs a fit reference assignment, relevant materials and variation, and categorical performance. A separately specified quantity or count inside the pipeline could receive its own measurement assessment, but none is given here. The reviewer withholds transfer of this model while preserving useful identity evidence; the consultant routes a later defined quantity separately. The material case concerned applicability within a quantitative model. This case concerns whether that kind of model fits the output at all.

Can the new result receive the label?

Return to the recorded plasma configuration and its conditional model. A fresh **single result is 10.15 mg/L**. The calculation uses $U = 0.2937238 \text{ mg/L}$, retaining that unrounded width until the endpoints are displayed to two decimal places.

For this fictional exercise the operational specification is **10.00 mg/L**. The local label **above specification under this rule** is permitted only if the **lower endpoint is strictly greater than 10.00 mg/L**. Otherwise the label is withheld and the case goes to the fictional protocol. Here y means the new concentration, $y \pm U$ its interval and $y - U$ its lower endpoint. The rule supplies no clinical action, retesting instruction or real laboratory policy.

Decision

1. What are the unrounded endpoints and their two-decimal display? 2. Does the lower endpoint satisfy the exact rule? 3. What does the ten-result reference-material check establish about patient-sample bias and commutability, and what remains open? 4. Does this interval cover uncertainty of a study estimate and every unresolved evidence question? 5. What additional basis would a real clinical or regulatory decision rule need?

How the rule treats this result, followed by a positive contrast

Follow the lower endpoint

Subtracting and adding the specified width gives:

$$10.15 \pm 0.2937238 = [9.8562762, 10.4437238] \text{ mg/L.}$$

The displayed interval is **9.86–10.44 mg/L**. The unrounded lower endpoint, **9.8562762 mg/L**, is below **10.00 mg/L**, so the rule **withholds the**

above-specification label. That decision follows the rule. It does not prove that the underlying concentration is below the specification.

The **+0.050 mg/L** difference belongs to the reference material. Patient-sample bias is not established, and commutability remains unresolved. The interval represents the two included measurement components under their assumptions. It is neither a confidence interval for all model parameters nor a numerical allowance for unstudied collection, material or clinical applicability.

A real rule would need a specification justified for its use, an applicable model and coverage assessment including relevant uncertainty sources, evidence for the patient-material and reference relationships, and explicit allocation of decision risk in the appropriate clinical or regulatory framework. The convenience of the fictional boundary supplies none of those judgments.

Only now consider a positive contrast: another fresh result is **10.50 mg/L**, exactly at the upper boundary of the stipulated *central-result* interval. The same width gives **10.2062762–10.7937238 mg/L**, displayed as **10.21–10.79 mg/L**. Its lower endpoint is greater than **10.00 mg/L**, so **the local label is supported under the fictional rule**. This positive outcome retains the model's assumptions and does not remove error risk, establish clinical adequacy or repair material gaps. Although its interval extends beyond **10.50 mg/L**, that endpoint supplies no validation of a wider measuring interval: the constant absolute width was assumed for the central result.

Figure 7.1: Conditional uncertainty at a decision rule shows the two post-commitment intervals against the fictional 10.00 mg/L rule. The visible relationship is between a lower endpoint and a stated rule, not between this exercise and a patient-care decision.

Book 1C – Chapter 7

Section 07 – Conditional uncertainty at a decision rule

QMP-A mass concentration, mg/L

Key takeaway

Same U , different lower endpoint, different rule result.

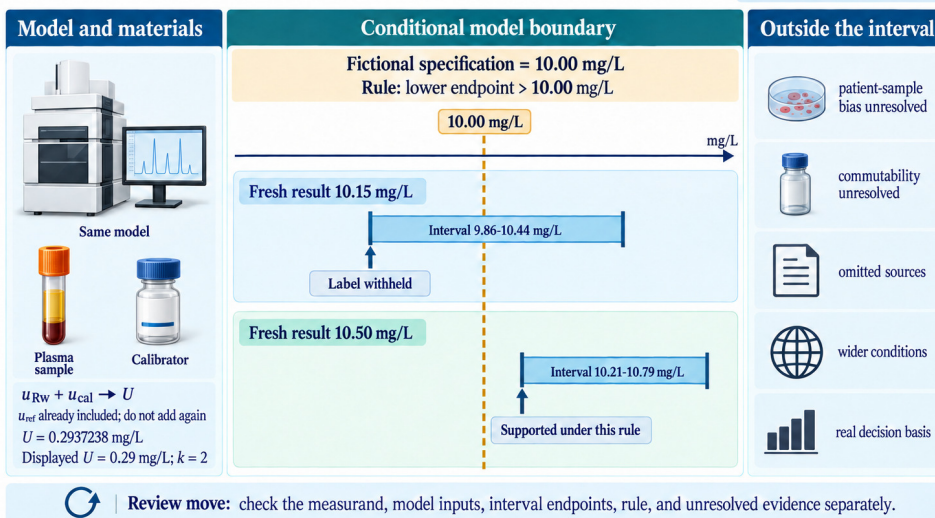


Figure 7.1. Conditional uncertainty at a decision rule – Technical review plate.

Reviewer Lens. The unrounded lower endpoint decides the result. A rounded display cannot replace it, and “greater than” cannot become “greater than or equal to.” The 10.50 result earns the local label within the stipulated model. The 10.15 result withholds that label without establishing the opposite condition. This is how a narrow negative and a narrow positive decision can both be sound.

Consultant Lens. The decision owner approves the actual use-specific specification and risk rule before results arrive; metrology and clinical owners establish model and material fitness. A defined route for results that do not qualify belongs with that prospective rule. The example provides teaching arithmetic, so it cannot become policy simply by copying the table into a procedure.

A record that preserves both the calculation and its limits

The two-page measurement record can now be read as a connected argument: first the measured object and evidence, then the decision and remaining limits. The completed table is a teaching record, not a real validation file.

Part of the record	What this example supports
Result layer	Mass concentration of Analyte Q in human plasma, mg/L
Procedure and configuration	QMP-A, SOP 1.2; QMP-A-CONFIG-A, AQ-017, software 1.2

Part of the record	What this example supports
Reference roles	Proposed reference procedure/material/end-user calibrator arrangement; no real qualification or SI claim
Material and entry	Prepared frozen plasma pool; collection and separation not evaluated here
Precision design	Twenty runs, two named operators and paired lot conditions; confounding and distribution assumptions retained
Comparison	+0.050 mg/L on the reference material; patient-sample bias not established
Uncertainty model	Distinct routine-spread and calibrator terms; zero covariance stipulated; reference contribution counted once
Coverage	$U = 0.2937238$ mg/L, displayed 0.29 mg/L; $k = 2$, approximate coverage
Decision	Fresh-result interval and exact fictional rule kept distinct
Limits and next evidence	Commutability, omitted sources, representativeness, wider conditions and use-specific adequacy

The comparison row is a useful test of the table. Its **+0.050 mg/L** belongs to ten reference-material measurements, while the precision row concerns twenty pool measurements. Their roles stay distinct when the uncertainty row combines the routine spread with the calibrator contribution. The result and configuration rows identify what the new **10.15 mg/L** means; the material and reference rows show where its support stops. Only then does the rule yield its decision. Filling the final row with explicit limits keeps that decision from quietly becoming a broader claim.

What is ready for a professional decision?

The positive result here is a reproducible, conditional calculation for a specified concentration layer. A reviewer can assess whether the claimed property matches that layer, the reference and specimen route serve the question, each contribution appears once, and the conclusion respects its scope. Real-product adequacy remains a different judgment.

The laboratory or metrology specialist supplies material entry, varied conditions, certificate inclusions and omitted effects. The manufacturer connects the measurement claim to its intended use, justified specification and controlled configuration. A consultant can then put the remaining work in a useful order: qualify material and

reference relationships, study relevant variation or bias, revise the model where necessary, and seek the decision owner's approval of bounded wording. Editing a formula cannot supply those facts.

European Union documentation has its own locations for the resulting evidence. The In Vitro Diagnostic Medical Devices Regulation places trueness/precision information in **Annex II §6.1.2.1** and calibrator/control value traceability in **§6.1.2.4**, with related **Annex I §§9.1(a), 9.3** provisions. These are evidence locations; they neither establish that the fictional exercise meets them nor provide an invented express measurement-uncertainty item. The model here comes from the metrology sources. Jurisdiction-specific conformity needs its own review. (Regulation (EU) 2017/746, dated consolidation.)

Does the same reasoning fit the tissue report?

HelixBridge Dx is the fictional tissue-oncology sequencing assay whose broad purpose concerns variants associated with eligibility for a named therapy. The therapy, exact population and variant classes remain unspecified. This occurrence concerns **only the reported variant identity**. No separately defined concentration, count or allele-fraction result is provided.

A proposed evidence row says: “Transfer QMP-A's calibration hierarchy and expanded uncertainty to support this output. Agreement with a comparison test will demonstrate trueness.”

Decision

What result property, reference question and kind of performance evidence belong in a replacement row? Which facts are still needed and which conclusion must be withheld? No concentration, threshold, count or clinical claim can be borrowed from the plasma example. If an underlying quantitative component were separately established later, would that change what the reported identity itself is?

How a reviewer replaces the proposed evidence row

The property is nominal variant identity. The plasma value-transfer chain and mg/L interval cannot attach to it. Agreement with a comparison test could still be useful if that test's role is justified; it cannot simply be renamed measurement trueness.

A useful replacement asks how identity is independently assigned for relevant specimen and variant conditions, which configured assay produces the report, and how incorrect, missed, indeterminate and invalid outputs are counted. The reference process, categories, study materials, configuration, observed results and criterion would permit

a reviewer to assess a bounded identity-performance conclusion. No numerical result or acceptance criterion for that comparison is supplied here. The consultant helps study and claim owners select a fit categorical comparison and the record that could close its question.

A separately defined count or quantity inside the pipeline could need a metrological assessment if proposed and supported. It would leave nucleotide identity as an identity and would not prove therapy eligibility. The useful judgment preserves three possible evidence jobs: the quantitative subproblem, reported identity and clinical meaning. “Insufficient” alone would obscure which one needs support.

Return to the opening precision and correlation summary

The two favourable phrases can remain when their designs and limitations are available. Precision requires stated repetition conditions. The comparison requires a justified reference role and paired differences near the intended decision. A measurement-uncertainty statement requires the quantity, model, inputs and coverage. The decision then requires its own justified rule and use context. Neither phrase supplies that complete chain by itself. The plasma calculation is useful because its positive result and conditional links can be inspected together.

Chapter 8 follows the result into detection and reporting boundaries. The exact output, reference role, specimen path and measurement limitations travel with it. Software's ability to compare numbers quickly does not establish its threshold or final reporting category.

Questions that test the reasoning

Decision

1. What turns “Analyte Q” into a measurand? 2. How can a traceable value still be unsuitable for its intended use? 3. Why does a single new result use the twenty-run standard deviation instead of the standard error of the control mean? 4. What would let the reference material's +0.050 mg/L difference inform patient-sample bias? 5. Which relationship needs reassessment if calibrator and operational uncertainties overlap? 6. Why do the freeze-dried material and nominal-identity cases need different responses?

The relationships a complete answer preserves

The quantity, plasma system and unit specify what is measured. Its reference relationship does not settle fitness for use. One specimen result is not an average of twenty measurements of that specimen, so the control-mean reduction answers another question. A fit patient-sample and material comparison is needed before the reference-material difference can transfer. Components require reassessment of overlap and dependence before combination. Finally, a changed control material creates an evidence-applicability question within a quantitative model, while nominal identity changes whether that model fits the output property.

Source trail and scope

The technical sources below support bounded definitions and distinctions. Every QMP-A name, specimen condition, certificate, measurement value and decision rule is original fiction. The twenty-row and ten-row data tables are not reproduced from a licensed standard, and none is a clinical protocol or product-acceptance criterion.

1. **Quantity results and traceability:** ISO 17511:2020, §1 and §§4.1–4.5.9. Its scope includes defined numerical and counting outputs and selected categories built from quantitative relationships; it does not make nominal identity a quantity. The fictional value-transfer route is not a conformity or real traceability claim.
2. **Metrology vocabulary:** Joint Committee for Guides in Metrology (JCGM) 200:2012, International Vocabulary of Metrology, especially terms 1.30; 2.1, 2.3, 2.9; 2.13–2.26; 2.39–2.42; and 5.15. Definitions are applied to the specified result layer, not every numeric-looking output.
3. **Distinct reference-system roles:** ISO 15193:2026 §1 and §4.1 addresses reference measurement procedures; ISO 15194:2026 §1 and §4.3 addresses higher-order certified reference materials; ISO 15195:2018 §1 addresses calibration-laboratory competence. The two 2026 editions replace their 2009 predecessors. None qualifies a fictional object.
4. **Precision and comparison:** CLSI EP05, fourth edition, §§1.1–1.1.2 and §1.3, supplies bounded quantitative precision scope; CLSI EP09c, third edition, §1.1 and §6.2.1, supplies patient-sample comparison scope and the correlation caution. QMP-A's twenty observations are not an EP05 protocol, and its ten material measurements are not an EP09 patient comparison. FDA's 2007 diagnostic-statistics guidance, §§3 and 6.1, distinguishes categorical agreement from target-condition correctness within its stated setting.
5. **Bounded uncertainty calculation:** ISO/TS 20914:2019, §§5.2–5.4, 6.2–6.6 and relevant Annex A clauses, informs the use of longer-term within-laboratory spread and calibrator-value uncertainty. The sample standard deviation, independence

assumption, root-sum-of-squares, $k = 2$ expression and all values are explicitly confined to the fictional model. No universally adequate width follows. The JCGM uncertainty vocabulary and JCGM 106:2012 support only the conceptual distinction and relevance of uncertainty to a stated scalar-limit rule, not a clinical cutoff.

6. **EU handoff:** Regulation (EU) 2017/746, dated consolidated presentation, Annex I §§9.1(a), 9.3 and Annex II §§6.1.2.1, 6.1.2.4. The consolidated text is a navigation aid; current amendments and product applicability require checking for real use. No conformity, harmonisation or dossier-sufficiency conclusion is made.

The local licensed ISO and CLSI editions were reviewed read-only at the Chapter 7 packet's bounded clauses; no licensed table or extensive passage is reproduced. HelixBridge, PathSight and the separate QMP-A arithmetic remain three distinct teaching records.

CHAPTER 8

Detection and Reporting Boundaries

A complete report can hide an incomplete route

Prepared tissue produces sequence data, and software finds a possible difference from a reference sequence. What would allow that finding to appear in a report? The procedure needs evidence that it can find this kind of difference in the relevant material and conditions. The report rules then decide whether the supported finding can be displayed, needs qualification or remains unresolved. A successful examination and permission to report a particular finding are separate steps.

That separation becomes visible when something remains unresolved. A failed run may leave no usable specimen result. A usable run may leave just one target without a call. A repeat may later recover that target. The final report can be useful while its history still matters.

Now consider an executive summary describing strong performance. Its table contains only final positive and negative reports. Elsewhere in the file are initially invalid examinations, targets without calls, borderline results and repeats. The arithmetic in the final table may be sound. Which question could it answer, and which question about the original specimens would still be open?

In digital pathology, the route ends differently: an image-analysis score helps determine display or priority, and a pathologist makes the diagnosis. The common question is what each step permits the next person to conclude. A threshold in both routes does not mean that both thresholds describe the same property.

What must be known before a boundary has meaning?

Chapter 7 separated a quantity from an identity and a numerical result from its interpretation. That distinction brings us back to the earlier claim, reference and material questions. A detection boundary needs something to detect, a procedure and conditions. A quantitative interval needs a quantity and unit. A report decision needs an output and intended action. None can be chosen reliably from a test name alone.

Decision

For the claim under consideration, what assertion reaches the user and what decision carries risk? Who or what assigns the comparison result, and with what limits? Where does the studied material enter the physical route? Is the output a quantity, identity, score or category, with which unit if any and which configuration? Which of those facts is still missing?

Figure 8.1: Result production and evidence accounting maps the two lanes you will connect. It is a general system map, not evidence for either named product or for an additional specimen branch.

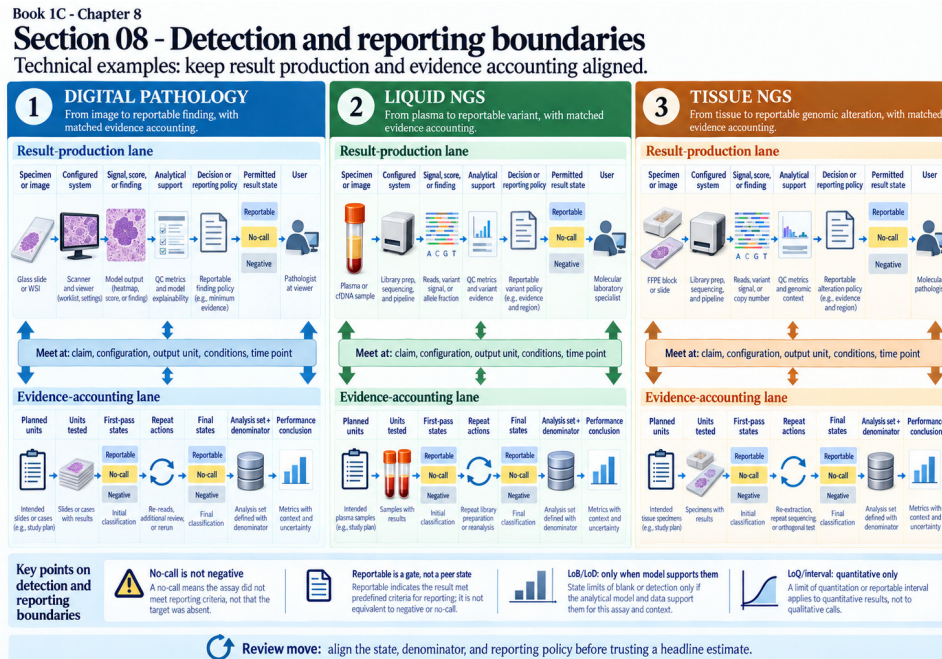


Figure 8.1. Reporting boundary as two linked lanes – Technical review plate.

What makes those earlier answers usable here

A useful starting point names the proposed user-facing assertion, the process assigning the comparison result and its limits, the physical material actually studied, and the result layer and configuration. “The test” identifies none of those relationships. When one is absent, the boundary remains a question for investigation. A familiar convention cannot supply the missing fact.

Follow both the result and the unit being counted

For one tissue specimen, the information route is **specimen** □ **controlled procedure and software** □ **signal or finding** □ **analytical support** □ **report decision** □ **state shown to the user**. Observing a sequence difference comes before deciding whether the relevant variant class is supported and whether the report rules permit its display. For an image, the same reasoning follows the score into the worklist and toward the pathologist, whose diagnosis remains a separate output.

To interpret a performance table, we also need the specimen's route through the study: **planned units** □ **actually tested units** □ **first-pass states** □ **repeat actions** □ **final states** □ **analysis set** □ **numerator and denominator**. A percentage of specimens with a first result, a percentage of targets with a call and a percentage of cases with a final report count different units. Their numbers become comparable only when the unit and stage are explicit.

A **boundary record** connects the two routes for each claimed output: specimen or image, configured system, underlying signal, analytical support, decision rule, permitted states, repeat path and units included in each estimate. This is the plain name we will use for the book's boundary and reportable-state evidence matrix. The routes must meet at the same claim, configuration, material conditions and time point. That is how another reviewer can reconstruct both what the system may say and which observations support saying it.

Reviewer Lens. Starting with one actual output makes a broad review manageable. The first trace reaches the displayed state; the second locates that specimen, target or case in the analysis. A correct final-result table deserves credit for its declared final subset. A first-pass claim needs the earlier states too. The focused question is which relationship is missing, because an undefined state or unmatched configuration is not itself an observed failure.

Consultant Lens. Before a study, the claim owner can name the assertion, assay or imaging owners can identify its signal, laboratory and software owners can define the state transitions, and the statistician can define the analysis units. Making their records agree then is easier than reconstructing excluded or repeated units afterward. A later change to a report rule or interface sends this comparison back to its owners even when the detector or model remains the same.

Is the low response enough to say anything?

A specimen with no target can still produce a background response. Material containing a small amount of target may produce a similar response. The problem is therefore more demanding than finding one low specimen that happens to look positive: the procedure must distinguish target from background under specified conditions.

A suitable **blank** contains no target, or a negligible residual amount for the particular low-end question. Calling that material present is **false presence**. Failing to call target present in the studied low-level material is **false absence**. A detection study needs justified control of both errors. It must identify the specimen matrix, target or variant class, input level, response or detected/not-detected model, instrument, reagents and software. The resulting boundary belongs to that procedure and those conditions. The detection-limit vocabulary in the International Vocabulary of Metrology ties the question to such a defined measurement context; it supplies no clinical decision rule.

This also clarifies two uses of “sensitivity.” A low-level detection probability concerns how often the specified procedure detects target-containing material at a stated level and under stated conditions. Clinical sensitivity concerns positive test results among people assigned the target condition by a fit clinical reference process. Repeated low-level material contains no substitute for that population, condition or clinical assignment.

Three questions at the lower end

Watching blanks establishes the background problem. Watching independently assigned low-target material then asks whether the procedure can distinguish presence reliably despite that background. If the procedure also reports a quantity, a further question follows: are the numerical estimates good enough for their stated use? These are the jobs of **limit of blank** (LoB), **limit of detection** (LoD) and **limit of quantitation** (LoQ).

The following table helps decide which job an offered low-end result can answer.

Boundary	Question the evidence must answer	What defines the answer
Limit of blank	How high can suitable blank responses reach under the chosen false-presence control?	Background behavior and specified error probability
Limit of detection	At what low target level can presence be distinguished while controlling false absence and respecting false-presence control?	The procedure, material and conditions
Limit of quantitation	How low can a quantity be reported with the required numerical performance?	The quantitative procedure and goals for bias, precision or total error

The detection row builds on the blank row: a low response means little until background behavior is understood. The quantitation row adds another demand, so a procedure may support detection before it supports a sufficiently good number. The largest response in a small blank set is not automatically its limit of blank, and one

positive low specimen is not a limit of detection. No row supplies a universal concentration, replicate count or pass rule.

The detection-capability guideline, CLSI EP17-A2, addresses clinical laboratory measurement procedures and distinguishes these jobs in its scope and bounded §§3.2, 4.3–4.4 and 5.5. Suitable qualitative final outputs can still have relevant underlying responses; molecular procedures may use detected/not-detected hit rates. Thus Chapter 7's nominal variant identity can have detection-capability evidence without becoming a mass concentration. Quantitation, however, needs a quantity and specified performance goals. The final label alone cannot choose the method.

The same material, two possible outputs

Here is a separate fictional procedure, unrelated to the named products. The table supplies every displayed response from three material groups. The two nonblank groups have separately assigned analyte levels; the instrument's **arbitrary response units are not concentration units**. Eight observations per group are teaching data, neither a detection-guideline protocol nor a recommended sample size.

Material	Eight responses, in arbitrary response units
Suitable blank	1, 0, 1, 2, 1, 2, 0, 1
Lower nonblank level A	1, 2, 2, 3, 1, 3, 2, 4
Higher nonblank level B	3, 4, 3, 5, 4, 3, 4, 5

Reading a row means following eight response observations for that one material group. It does not mean eight patients or an ascending concentration series within the row. For this demonstration alone, the detection rule is **response >2**: a response equal to 2 does not satisfy it.

For material at level B, a separate quantitative reporting mode produces **1.1, 1.5, 1.8, 2.0, 2.0, 2.3, 2.7 and 3.4 quantity units**. The assigned quantity is **2.0 units**. An explicitly fictional screen requires **every displayed estimate** to lie from **1.6 through 2.4 quantity units, inclusive**. This observed-record screen is not a complete limit-of-quantitation method or general acceptance rule.

Decision

How many responses in each material group satisfy the detection rule? How many B quantity estimates satisfy the separate numerical screen, and what does each fraction count? If every B response is detected, does B meet the all-estimates screen? Which formal limits, if any, have these small sets established?

Why the two sets of eight lead to different conclusions

The blank has 0/8 responses above 2. In the lower nonblank row, the two 3s and the 4 satisfy the rule, giving 3/8. The higher nonblank row gives 8/8. Each denominator is the eight displayed response observations in its own group. The lower nonblank values 1 and 2 overlap the blank's 0–2 range; a 2 is outside the strict detection condition even though it is on the boundary.

The quantitative mode asks a different question about level B. 1.8, 2.0, 2.0 and 2.3 are inside the inclusive 1.6–2.4 screen. That is 4/8 **quantitative estimates**; the other four are outside. The requirement was every estimate, so **B fails this fictional quantitative screen despite 8/8 observed detections**. Both fractions are correct because their numerators describe different outputs. Extra decimal places would not reduce the estimates' dispersion.

The eight detected responses are a promising observation for this tiny set. They do not establish a detection probability at B or a formal limit of detection under justified error control. The blank maximum of 2 does not establish a limit of blank, and one failed quantitative screen does not locate a limit of quantitation. None of the observations establishes clinical sensitivity. The useful conclusion preserves both the good observed detection count and the failed numerical screen.

For a real tissue-sequencing detection claim, the corresponding record would identify the target class, sequence region, preparation, independently assigned low-level material and valid and invalid attempts. Success for one class need not cover another. An image score controlling worklist display instead needs evidence for that score and workflow function; calling the boundary a detection limit would not establish an analyte question.

A separate FDA tissue tumor-profiling decision record illustrates why this specificity matters: its analytical-sensitivity evidence separates variant types and contexts, uses different signal axes and records invalid libraries. Those distinctions can guide a review question. That device's values, methods and status supply no HelixBridge result or universal rule.

Reviewer Lens. The useful first check is what the numerator counts and which observations the denominator retains. A well-designed detection study can support a specified low-level claim, even when it supplies neither quantitative adequacy nor patient-condition sensitivity. Blank and low-target materials still need to fit the proposed route and output class, and invalid attempts belong beside the hit-rate result. That preserves the credit due to successful detections while showing where the conclusion ends.

Consultant Lens. Assay owners identify classes and material levels; statistical owners select the error question and design; claim owners decide whether the output is a hit, a

quantity or a clinical category. Agreeing those choices and criteria prospectively prevents a favorable count from becoming an improvised claim. A new specimen route or variant class calls for an applicability decision before the old limit is reused.

What is the rule allowed to send to the user?

Chapter 7's fictional plasma concentration had a defined quantity, unit and conditions. For such a procedure, we can ask over which concentration values it produces measurements with stated performance. That is its **measuring interval**. The lower end need not be the level at which target can first be detected: a detectable response can remain too variable for quantitative use. Even a sufficiently good concentration value still needs a separate basis for its clinical interpretation and permitted reporting.

The interval question concerns quantity values. The metrology vocabulary and the in vitro diagnostic labeling standard ISO 18113-1:2022, §3.1.52, describe that quantitative meaning. The labeling standard also uses *reportable range* in a qualified quantitative sense. Those words require care when they reappear in sequencing.

For a tissue-sequencing identity, **genomic reportable range** concerns sequence regions examined **and reported at acceptable quality**, as used by the sequencing guideline CLSI MM09, third edition, §1.4.2. The axis is position in the genome. The region and quality conditions must be explicit, and targeting a region does not prove that every position yielded a valid call. A genomic region and a plasma concentration interval answer different questions even when both are called a range.

An image score offers a third possibility. It may vary continuously inside the software and then produce a priority category. That internal number does not automatically define a metrological quantity or a measuring interval for the final decision. The binary-output guideline CLSI EP12, third edition, §§1.1 and 2.1, distinguishes continuous internal responses, decision algorithms and visual reads within its two-output scope. Whether a separately defined numerical component needs measurement assessment remains its own question.

All three routes eventually reach a gate: what state may the system emit under these supported conditions? The book calls the map of inputs, conditions and permitted states a **claimed reporting boundary**. This is our reasoning framework, not a quoted regulatory or standards definition. A product-specific policy may permit positive, negative, equivocal, partial, invalid or no-call states. An interval or region list cannot express that whole policy.

The table locates the question when a file offers one kind of boundary as evidence for another.

Step toward the report	What must be established?	What remains a separate question?
Detection	Supported detection under named procedure and conditions	Quantitative accuracy, clinical actionability and report permission
Measurement, if applicable	Quantity values measured with stated performance	Reporting policy and clinical decision boundary
Cutoff or decision policy, if applicable	How a signal, score or finding becomes a category or region	Smallest detectable amount and complete clinical validity
Reporting boundary	Which supported states can reach the user, with which limits	Whether applicable studies support that policy in the claimed configuration

For the plasma procedure, the measurement row could be supported while the clinical decision remains open. For a sequence identity, the concentration question does not fit the output, while detection and report permission still matter. Each row is a question, not a mandatory separate study: one record may help answer more than one row if its reasoning supports each use.

The European Union's in vitro diagnostic regulation similarly locates measuring-range/detection-limit documentation separately from assay-cutoff documentation in Annex II §§6.1.2.5–6, with wider performance evidence in §6.2 and Article 56. These are evidence locations, not universal numerical limits or a conformity conclusion for either fictional product. One analytical result cannot settle the whole performance evaluation.

Reviewer Lens. A “range” becomes assessable when its axis is stated: concentration, genomic coordinates or permitted output states. A qualified genomic region deserves credit at its stated quality and variant-class boundary; unresolved positions still need a disposition. For an imaging score, the useful question is what the score changes for the user. This lets a supported component receive a positive finding while its further report permission is examined separately.

Consultant Lens. Assay or imaging owners identify the axis, laboratory owners define release permission, and claim owners connect the output to the intended decision. An analytically supported value may still be withheld under a governed policy. Making that policy explicit early prevents later disagreement about whether evidence or reporting rules need repair. Changes to a target region, cutoff or display policy require both relationships to be reconsidered.

What happens near the cutoff?

The demonstration's response **above 2** was a counting rule. A real **cutoff** assigns a defined signal, score or finding to a category or region for its specified purpose. Its placement may serve a different question from the smallest reliably detectable amount. A supported weak signal may be withheld from a positive report, and a concentration inside a measuring interval may still be clinically equivocal.

That possibility raises the next question: must a near-boundary observation be forced into one of two answers? A **gray zone** can reserve such observations for an equivocal state, review or another defined action. This can be a justified way to handle uncertainty. It needs a population, specimens, reference assignment, planned error tradeoff and operational response chosen for the intended question. A gray-zone observation is not automatically a failed examination.

A permitted third state must then survive into the evidence account. FDA's final, nonbinding diagnostic-statistics guidance, §§2, 5 and 6.2, explains why its binary measures do not directly apply to a nominally binary test whose instructions permit equivocal output. Simply removing those observations can bias interpretation. The guidance identifies the problem without prescribing one universal alternative analysis.

For an image, a hypothetical threshold might determine highlighting or priority; for tissue sequencing, a hypothetical rule might permit a supported identity to appear in the report. These examples supply no undisclosed PathSight or HelixBridge policy. Either real rule would need known inputs, outputs, configuration, fit reference process, planned criterion and intended user action. A later changed rule needs comparison with the studied one even if the assay or model name is unchanged.

Reviewer Lens. Reading the permitted states before the performance table reveals whether the table describes the actual product behavior. A justified gray zone can earn credit as an explicit policy. If its observations disappear from an analysis, the original counts and reference-result strata become the focused request. Those records make it possible to judge a secondary binary question without pretending the third state never occurred.

Consultant Lens. The claim owner defines the clinical or workflow purpose, reference and statistical owners define the study and tradeoff, and software or laboratory owners implement the corresponding state and user message. Closure requires their governed rule and an analysis that retains its states. Choosing a threshold after seeing which retained subset looks favorable would leave the claimed policy without the prospective basis it needed.

Follow the no-call through its repeat

The output is supported only if the conditions for using it were met. But what should a study count when those conditions fail, or when processing succeeds without a defensible call? A separate fictional **one-output procedure** makes the difference visible. It is unrelated to HelixBridge, PathSight and any standard-derived performance result.

There are **11 planned specimen IDs**. One specimen never arrives, so **10 are actually tested**. The first examination gives eight complete results. For another specimen, processing is usable but no supported call can be assigned to the requested output. The remaining specimen fails a required specimen-level control. These are three different first-pass states.

Here **valid** means that the stated conditions for using an output were met; it does not mean positive. The control failure makes the requested result **invalid**. Usable processing without a defensible category is a **no-call**, which cannot be substituted for a negative. These operational labels need product-specific definitions in a real record.

Now the no-call specimen is examined again using the **same prepared material**, and it yields a complete result. That **repeat** is an action from an earlier state: the record has both an initial no-call and a final recovery. The control-failed specimen is not repeated and remains unresolved. The added examination is another attempt on an existing specimen ID.

The first-pass numerator is **8 complete results**, and its denominator is **10 tested IDs**. Thus $8 \div 10 = 0.80 = 80\%$ **first-pass complete**. After recovery, $9 \div 10 = 0.90 = 90\%$ **final complete among tested IDs**. The final rate answers what the stated pathway eventually produced; the first rate answers what it produced initially. Neither erases the other, and neither is a universal acceptance threshold.

A third question starts before testing. Of the **11 planned IDs**, how many reached a complete final result? $9 \div 11 \approx 0.818$, or **about 81.8% planned-to-final complete**. The never-received specimen belongs in this account but was never a tested failure. Changing the denominator from 10 to 11 changes the question, not the observed final results.

A multi-output procedure adds another distinction. Some supported targets may be released while others are withheld, producing a **partial result** with the available and missing portions identified. It is neither automatically whole-invalid nor whole-complete. An **exclusion** instead concerns whether a unit enters a specified analysis, with a reason. It is not a new name for invalidity.

The stage table below is useful when a performance fraction cannot be reconciled with the original units. Its rows follow time; they are not buckets to add together.

Stage	What the record must retain
Planned	Analysis unit and all planned units
Not tested	Count and reason, separate from tested units
First pass	Each tested unit's complete, partial, invalid, no-call, equivocal or other permitted state; any overlaps
Repeat	Eligibility, action, material used and units not repeated
Final	Complete, partial, unresolved or another defined terminal state
Analysis	Inclusion/exclusion rules and reasons, numerator, denominator, configuration and first-pass or final basis

For the fictional ledger, the first two rows reconcile $11 - 1 = 10$ tested IDs. The first-pass row reconciles $8 + 1 + 1 = 10$: complete, no-call and invalid. The repeat row moves the one no-call to recovery without adding a specimen. The final row reconciles $9 + 1 = 10$: complete and unresolved. The last row can now identify precisely which of the three fractions is being reported. Excluding the never-received ID from an estimate among tested specimens leaves its place in the planned-to-tested account intact. Another claim may justify another declared analysis set.

The level of a problem also matters. A failed run can affect all its samples; a specimen failure may affect just one sample; an unresolved target can leave other outputs usable. The original FDA tumor-profiling decision record describes run-, sample- and analyte-level controls for one real device. Its particular state labels, thresholds and recovery paths are examples from that device, not HelixBridge instructions.

This same accounting preserves the two separate deployed observations from Chapter 1. The unnamed tissue service recorded **2 initial invalid statuses among 40 consecutive specimen attempts: $2/40 = 5\%$** for that interval. Final recovery and cause remain unknown. The unrelated pathology service recorded **1 late worklist arrival among 25 supported-image cases: $1/25 = 4\%$** , with pathologist review completed. Its other-stage counts and cause are not supplied. The first needs initial-to-final traces; the second needs arrival and review timing. These operational percentages establish neither clinical sensitivity nor detection probability, and they add no deployment event to HelixBridge or PathSight.

Reviewer Lens. The final 9/10 is a real recovery result under the fictional pathway and should remain visible beside the first-pass 8/10. The hard question in a multi-output file is often how far an unresolved state reaches: did one target disappear or the whole report stop? In pathology, a late arrival likewise needs its own interpretation before

anyone calls it a missed diagnosis or failed analysis. Numerator, actual divisor, unit, exclusions and stage make the judgment reproducible.

Consultant Lens. Laboratory and software owners can make state transitions and messages executable while the statistician defines the analysis sets and the claim owner chooses the reportability question. Their systems need to retain initial, repeat and terminal records, including the material used. A later cause investigation or narrower operational claim may close a concern. Moving the event out of the denominator cannot provide that evidence.

Which boundary can the HelixBridge file support?

HelixBridge Dx identifies tissue variants associated with eligibility for a named therapy. The selected Chapter 8 packet contains an initial package, an analytical-validation view and an unidentified commercial configuration. The table preserves those views of the same fictional tissue case. It introduces neither a new release chronology nor evidence for the separately proposed liquid branch.

The exact application and rule labels appear in the table so that the records can be compared. In the discussion they are the validation application, the validation report rules and the knowledgebase used in analytical validation.

Record	Supplied facts	Open fields
Initial package	Tissue-oncology sequencing; application 7.0; report rules R19; cloud knowledgebase described as “current”	Population, variant classes, therapy, treatment-facing role, confirmatory language, limitations, wet-lab identity, knowledgebase identity and commercial release identity
Analytical-validation view	Application 7.0; report rules R19; knowledgebase DB-18	Wet lab, instruments, reagents, other computational dependencies and match to commercial configuration
Analytical-validation study	A study with that job label is identified	Design, sample count, classes, acceptance criteria, failures, strata, detection boundary, output axis and result-state accounting
Commercial view	A commercial configuration is referenced	All exact component versions, release date, deployed sites and authorization state
Detection and reporting	No boundary or policy details supplied	Detection limit; cutoff; measuring or reportable interval; permitted states; invalid, no-call, partial and repeat policies; exclusions; denominators; criteria; defensible performance conclusion

The first two rows show how to read the views without merging them. The application and report-rule labels occur in both. The knowledgebase is identified in the analytical-validation view; “current” leaves the initial view's exact knowledgebase unidentified. The study row names a job without supplying its methods or results. The commercial row names a proposed destination for evidence without supplying its exact component state.

For each output or state, the boundary record asks eight connected questions: **what is claimed** □ **which construct applies** □ **which specimen and configuration** □ **what evidence is present** □ **what is absent or uncertain** □ **what are the first and final dispositions** □ **which numerator, denominator and exclusions** □ **what conclusion is permitted?** A construct genuinely inapplicable to the stated output needs a reason. A missing record needs to remain missing. Those are different entries.

Two shorter records beside it

The recovered no-call. A draft protocol defines an initial no-call and permits repeats. Its proposed analysis places recovered units only in a final-result table. Initial no-calls, repeat eligibility and material, units not repeated and unresolved final states are absent. The final table may be correct for its declared set. What additional account would be needed to support first-pass reportability or workflow burden?

The missing third state. A nominally binary test's instructions permit positive, negative and equivocal outputs. The draft analysis removes equivocal observations before calculating binary measures. It supplies neither their reference-result strata nor their operational disposition. Can that table stand alone, and what record would reveal whether the selection changes its interpretation?

Decision

For each plausible HelixBridge output or state, how do the eight connected fields resolve, including explicit unknowns? Does the packet support a detection boundary, quantitative interval, cutoff or equivocal policy, report-state policy, or first-pass/final performance conclusion for the unidentified commercial configuration? What can be credited? For the recovered-no-call record, what initial and final account is needed? For the third-state record, what changes when equivocal observations disappear? Which precise evidence request would resolve the boundary, configuration, state or denominator that most affects the judgment?

The initial judgment rests on the available packet. A missing fact, an inapplicable construct and an observed adverse result should be distinguishable in the answer; “more validation” leaves that distinction unresolved.

How the eight questions lead to a bounded conclusion**The result, its unresolved states and the evidence that could travel**

The eight entries form a decision sequence, not eight required studies. The first three concern the reported output and its rule. The next three concern what happens when the expected output is unavailable. The last two identify the observations and product state to which evidence can attach. Across all eight, the packet supplies no first-pass/final counts, exclusions, numerators, denominators, acceptance criterion or complete commercial configuration. That shared absence cannot become a pass or failure in any row.

1. What identity is the product claiming to report?

The broad tissue purpose concerns variants associated with therapy eligibility. The application, rules and validation knowledgebase give the question a partially identified product state. To select a detection construct, however, the reviewer needs the actual variant classes, signal and output axis. Population, therapy, treatment-facing role, specimen conditions, error conditions, wet-lab, instrument and reagent dependencies, and commercial state also remain unresolved. The stated purpose is creditable configuration-linked information; the packet has not established a detection boundary or supported commercial identity output.

2. Is there a separate quantity to assess?

Nothing supplied identifies a separately reported quantitative measurand, unit or interval. A concentration-like measuring interval is inapplicable to the nominal identity itself. If a separate quantity were later claimed, its material and configuration, units, model, interval, performance conditions, uncertainty, states and counts would need evidence. This leaves room for a genuine quantitative subproblem without inventing one. It also leaves qualitative molecular detection open to a suitable detection-limit study.

3. Which rule permits the finding to reach the report?

The validation report rules are identified, but their content is absent. Their label cannot disclose a cutoff, categorical rule or gray zone. Inputs, target-class differences, any cutoffs, permitted states, user actions and exact commercial rules state are unknown. The identifier helps reconstruct a configuration; it does not establish a reportability decision.

4. What becomes unavailable when a condition fails?

The packet supplies no HelixBridge definitions of invalid or no-call. A trigger at run, specimen or target level may affect different outputs and actions. The displayed message, next action, first and final states and their counts are all unstated. No invalid/no-call rate follows, and an uncalled target cannot be relabeled negative to make one.

5. Can a supported part still be released?

That depends on which result types the policy permits and which it withholds. No output-level permission, display, missing portion, specimen/configuration link, claim consequence or partial-state count is supplied. Thus an all-or-none policy and a partial-report performance claim are both unsupported. This is a different gap from merely knowing that a target was unresolved.

6. What route follows an unresolved first pass?

A repeat needs an eligibility rule, maximum attempts, material identity, timing and terminal states. It matters whether another attempt uses the same preparation, a new preparation or a new specimen, and what independence can be claimed for the observations. None of these details or recovery counts is supplied. With no first-pass or final denominator, the packet cannot establish recovery or workflow burden.

7. Which original units are behind a proposed rate?

The analytical-validation label supplies no prespecified counting plan. Planned and tested specimens, exclusions and reasons, invalid/no-call/equivocal/partial states, repeats, recovered and unresolved units, variant strata, numerical results and criteria are missing. No rate can be reconstructed from the job label. Even a favorable description would need numerator, denominator and stage before it could support a performance summary.

8. Which product state could inherit the conclusion?

The identified application, rules and knowledgebase specify part of the validation view. The wet lab, instruments, reagents and other dependencies remain open, as do every exact commercial component, release date, site and authorization state. First/final accounting is absent for both views. The validation-to-commercial comparison therefore cannot yet attach an established performance boundary to an identified commercial arrangement. That conclusion says nothing about whether the underlying study succeeded.

The positive conclusion is useful: tissue purpose and some controlled software and rule identities permit a precise request. They do not yet establish a limit, cutoff, state policy or numerical performance. A focused request could be:

Note

For each claimed output or variant class, identify the studied and proposed commercial configuration, specimen and signal axis; the applicable detection or measurement evidence; report-rule inputs and permitted states; first-pass, repeat and final dispositions; planned, tested, excluded and analyzed units with reasons; and criteria. Show how each supported conclusion applies to the proposed configuration. Distinguish justified nonapplicability from evidence that remains absent.

The assay, software, laboratory and statistical owners can assemble this relationship before deciding whether any further study is needed. Closure could support a narrower output and configuration. It need not be an all-or-nothing product verdict.

The recovered no-call keeps both moments

The final table may accurately describe outcomes after the repeat pathway for its declared set. A first-pass or burden claim also needs the unit and tested set, every applicable initial state, repeat eligibility, which units were and were not repeated, the material or preparation used, recovered and unresolved final states, exclusions and reasons, and separate first-pass and final denominators. Recovery is a positive later result. Retaining the earlier no-call explains what was required to reach it.

The equivocal observations keep their place

Removing the third state changes the analyzed population. It can bias interpretation, but the direction and size cannot be known without its counts and reference-result strata. The record needs the equivocal denominator, strata, operational action and any later disposition. A justified secondary binary re-expression may answer a specified question while preserving the original three-state observations. It cannot stand alone as the product's unqualified performance. The FDA diagnostic-statistics guidance, §6.2, identifies the concern without selecting one universal solution.

When the record becomes professional advice

Reviewer Lens. One output and one configuration keep the judgment bounded. An established detection condition, quantity interval, category policy or complete state account can receive an affirmative conclusion for its represented conditions and analysis set. Unshown states or another release remain open. Keeping the first-pass and final denominators beside the performance table makes that boundary visible; a remote footnote makes it easy to miss.

Laboratory view. The state account becomes useful when the record system retains which control or target failed, what the user saw, the material used for each repeat and when the case became terminal. Run failure, specimen failure, target no-call and partial

release may need different actions under the applicable manufacturer's instructions and governed local procedures. A low final no-result fraction can coexist with substantial repeats or late reports. This chapter supplies no universal repeat algorithm.

Consultant Lens. Product requirements can specify triggers, affected units, user-visible and machine-readable outputs, repeat eligibility, terminal states and claim effects before the successful-result table exists. Tests of those transitions need the exact software, references and interfaces; the analysis plan then represents the same states. A threshold, quality gate, report rule, repeat path or display can change the resulting states and denominators without changing core detector code or model weights. The closure record connects the changed requirement, risk, study, criterion and user-facing output so the manufacturer can make the corresponding decision.

A changed display between score and pathologist

PathSight Cloud prioritizes whole-slide images and highlights suspicious regions for pathologists, who make the diagnosis. This selected occurrence identifies the validation model as **M3.2** and a **predetermined change control plan** that includes threshold and scanner expansion. A reader-study summary describes improved average sensitivity without unacceptable loss of specificity. Reader population, case set, reference process, precise question estimated, denominators, uncertainty and complete workflow are not supplied.

After validation, the interface changes so that **low-confidence cases are collapsed by default**, while the identified model remains. The changed configuration describes its threshold as within the plan's performance range. A high-volume **simulation** records **two positive low-score slides exceeding expected turnaround**. Denominator, exact time, cause and clinical consequence are absent.

The record also leaves exact scanner, firmware, color, normalization, tiling, threshold, postprocessing, queue, interface, cloud and fallback states unresolved. Validation scanner counts, component versions, sites, case strata, commercial release and applicability of the reader study are not established. No numerical threshold, final diagnosis, authorization or preservation-of-performance result is supplied.

Decision

Following internal score/confidence □ threshold □ low-confidence display state □ queue effect □ pathologist-facing output, what does each stop require? Does any stop establish a limit of detection or a measuring interval? What support does the plan's range statement supply for the changed interface? Which evidence questions reopen? What exactly do the two delayed slides show, and which rate or consequence remains incalculable? The earlier unnamed service's 1/25 observation belongs to a different deployed case.

Where the display change reaches the evidence

From an internal score to a reviewed case

The developer needs to identify what the score and confidence mean and what the threshold does. The supplied route concerns a score or classification threshold, a confidence state, a display policy and a workflow disposition. It establishes neither the smallest detectable amount of an analyte nor a defined measured quantity with a metrological interval.

A separate FDA pathology-software decision summary, §§H–I, illustrates that category of question: the other product applies a predefined neural-output threshold to suspicious/not-suspicious classification, lists no measurand and retains the pathologist's diagnostic role. Its threshold, performance and authorization do not transfer to PathSight.

The plan's range phrase is a supplied case statement. It does not establish that the changed interface is authorized, covered or adequately validated. Collapsing a case can change what the pathologist notices and when it receives review, even with unchanged model weights. That reopens five questions: intended workflow and reliance, usability, delay-related risk, applicability of the reader study, and fallback behavior. The favorable reader-study summary remains useful for its incompletely described validation view; its applicability to the changed view still needs evidence.

The two positive low-score slides establish two **observed delayed cases in a simulation**. They establish no cause. Missing simulation denominator, exact elapsed times and workflow context prevent a delay rate, comparative-performance conclusion, harm finding or overall adequacy judgment. Substituting the separate deployed service's 25 cases would combine unrelated observations. The pathologist still owns diagnosis, while whether the changed interface changes reliance remains a question.

The table connects each part of the route to the evidence needed for this comparison.

Part of the route	What its record needs to preserve
Score or confidence	Its meaning and configuration; no assumed analyte measurement
Threshold	The rule and the function it serves
Visible state	Low-confidence visibility, collapse behavior, user message, queue effect and fallback
Counted cases	Every low-confidence and delayed case in the declared workflow analysis set
Earlier evidence	Comparison of validation and changed configurations against the five reopened questions
Diagnosis	The pathologist's role, separate from software classification

The visible-state row reaches both directions. Backward, it needs the rule that determined collapse. Forward, it needs the queue and display that the pathologist actually encounters. The counted-cases row would let timing be assessed without inventing a denominator. The earlier-evidence row then asks whether the studied reader configuration and high-volume workflow represent this changed route. The last row preserves the separate diagnostic actor throughout that comparison.

Reviewer Lens. The reader-study summary is a favorable statement tied to its validation view, and the simulation supplies actual delayed-case observations. Both deserve their bounded weight. The unresolved relationship is whether the earlier evidence represents the changed interface and use conditions. Neither the unchanged model name nor the stated plan range supplies that comparison.

Consultant Lens. Configuration owners can reconstruct the before/after route; pathology and human-use owners can define the reliance and usability questions; workflow and statistical owners can identify timing observations and their cases. Their results can support a bounded claim or identify work still needed. Naming an owner and a closure question makes this advice concrete while leaving authorization, release, cause and clinical consequence undecided.

Figure 8.2: PathSight low-confidence display change compares the selected before/after interface facts only after your transfer commitment. It does not supply a missing rate or a regulatory decision.

Book 1C - Chapter 8

After transfer commitment

PathSight Cloud transfer

Key takeaway

The transfer changes what is visible and how work is organized, while the pathologist retains diagnosis and key questions remain open.

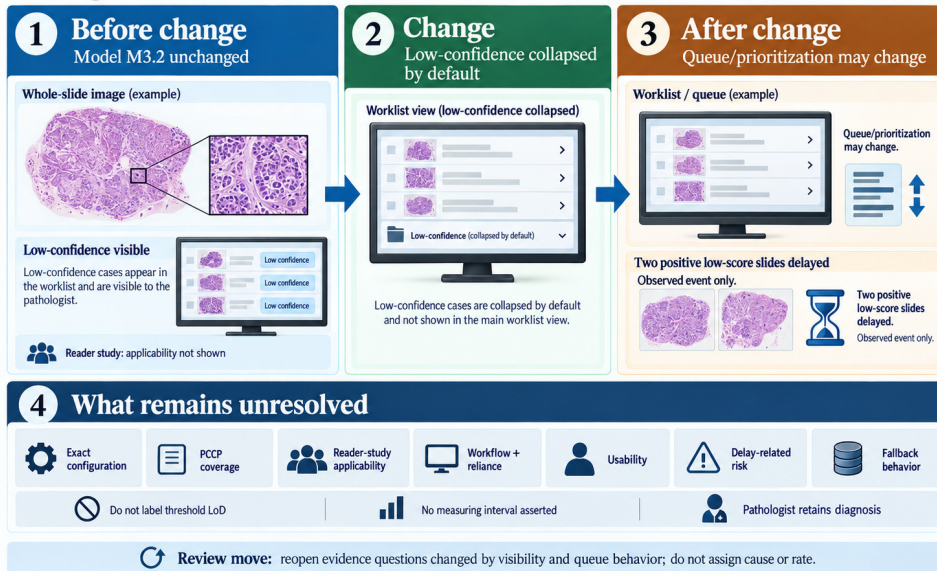


Figure 8.2. PathSight low-confidence visibility change – Technical review plate.

Return to the opening table

The executive summary's final-result table may have been correct for its declared subset. The unresolved issue was what happened to the original planned and tested units before they reached it. Initial states, repeats, final states and exclusions give that table a place in the whole account. A later recovery can be a good result and still have required an earlier unresolved attempt.

The result route now has equally distinct stops. Detection evidence supports a specified procedure and conditions. A measuring interval describes a defined quantity where applicable. A cutoff assigns a region or category. A reporting policy decides which supported states reach the user. Their connection lets the reviewer affirm what the evidence supports and identify the smallest missing relationship before a claim expands.

Chapter 9 asks whether the studied boundary survives consequential variation in days, operators, sites, lots, instruments and other conditions. A complete state account needs representative challenges, and a large study still needs defined states and denominators.

Questions that test the decision

For one claim and configuration, can the specimen or image be followed to the permitted report, and the planned units followed through first pass, repeat, final state and analysis?

1. Which output and underlying axis does each boundary describe?
2. Does the procedure support blank/detection-limit reasoning through an underlying response or hit-rate design? Is the quantitation or measuring-interval question actually about a quantity?
3. What justifies the cutoff or gray zone, and what action follows each state?
4. Did a failure occur at run, specimen, target or output level, and which other outputs remained usable?
5. What are each estimate's numerator, actual divisor, unit, exclusions, configuration and first-pass or final basis?
6. Which evidence earns a bounded affirmative conclusion, and what record would close the most consequential remaining gap?

The relationships a complete answer preserves

Recovery retains an initial unresolved event and a later final result. A score threshold is not automatically an analyte detection limit; a quantity interval and a genomic reportable region describe different axes. A missing field locates a question, not an observed failed study. The stronger answer can state both what is supported now and which specific evidence would change the next judgment.

Source trail and scope

The sources support the distinctions below, not the fictional teaching values or either named case's missing performance facts. No licensed table, protocol or numerical acceptance rule is reproduced.

1. **Detection and measurement vocabulary:** JCGM 200:2012, International Vocabulary of Metrology, terms 4.7 and 4.18, connects limits to procedures, quantities and conditions; it does not provide a device-study recipe or clinical rule.
2. **Detection capability:** CLSI EP17-A2, second edition, scope and bounded §§3.2, 4.3–4.4, 5.5, distinguishes blank, detection and quantitative capability and recognizes suitable qualitative-response or molecular hit-rate contexts. The eight-response example uses no EP17 formula, probability default, replicate prescription or criterion.

3. **Qualitative decisions and sequencing range:** CLSI EP12, third edition, §§1.1, 2.1, 3.1.1, has a two-output scope with defined internal-response and algorithm contexts; CLSI MM09, third edition, §1.4.2, uses a genomic reportable-range meaning. Neither supplies a universal PathSight threshold or HelixBridge state rule.
4. **Quantitative interval terminology:** ISO 18113-1:2022, §3.1.52, describes measuring interval and a qualified use of *reportable range* distinct from detection limit. The broader “claimed reporting boundary” here is explicitly this book's reasoning framework.
5. **Third states and analysis accounting:** FDA's final, nonbinding 2007 diagnostic-statistics guidance, §§2, 5, 6.2–6.3, addresses qualitative study reporting and equivocal-result bias without requiring one universal alternative analysis.
6. **EU evidence location:** Regulation (EU) 2017/746, Article 56 and Annex II §§6.1.2.5–6 and 6.2, distinguishes analytical range/detection, cutoff and wider performance evidence. The dated consolidation is a navigation aid; the base act and applicable amendments control. No case conformity or dossier-sufficiency conclusion follows here.
7. **Separate product-specific examples:** FDA's original P230011 summary, Test Output, Table 3 and analytical-sensitivity pages 62–66, illustrates nested NGS control levels and class/context-specific detection evidence; later supplements exist. FDA's DEN200080 summary, §§H–I, illustrates an image-classification threshold with the pathologist retaining diagnosis. Neither decision record transfers its values, labels, authorization or method to HelixBridge or PathSight.

CHAPTER 9

Variation, Challenge, Reagents, Instruments, and Time

What did another reading add?

A laboratory measures one prepared specimen again and again in one session. The same person uses the same instrument, reagent lot and software. A spreadsheet grows to one hundred rows. Its summary proposes a broad statement about dependable performance across routine laboratory conditions, but supplies no numerical results. Before opening a results table, what would we need to know about those rows?

Decision

Which part of the proposed statement could these repeated readings address? Which conditions never changed, and would more readings in this session address them? What bounded title would fit this study? If the broad claim remained, what first change to the design would be worth considering?

The repetitions may be valuable. Their value depends on what each new observation gives the study a chance to learn. That question will return after a planned comparison, a completed bench screen and two decisions about carrying evidence to different conditions.

Chapter 8 connected a supported signal to a permitted report and kept first attempts visible beside final results. Now the question is whether that boundary remains supported when the conditions producing it change. Instruments, people, specimens, reagents, scanners, software and elapsed time can affect the route even when its final output is a category.

A useful plan therefore connects the output and comparison to the conditions that change, those that stay still, the units newly sampled or assigned, and the rule that will govern the conclusion. It also follows plausible failure paths from exposure to output and control. This book calls that connected plan a **variation, challenge and sampling design**. The short canvas and challenge record developed below are teaching aids. They organize the reasoning; sample-size justification and statistical analysis still need their own basis, and no standard prescribes these particular forms.

Which earlier distinction matters here?

Decision

How does another measurement of one prepared specimen differ from a newly sampled patient specimen? How does uncertainty about one quantitative result differ from uncertainty about a performance estimate? How does finding a signal differ from producing a permitted report? And how does an unchanged software label differ from an unchanged report-producing configuration? Which reporting boundary from Chapter 8 could be sensitive to routine variation?

How the earlier distinctions shape a design

Another reading can improve knowledge of behavior under shared conditions; it does not sample a new patient or a new condition. Likewise, uncertainty about one measurement and uncertainty about a study's estimated performance concern different objects. The reporting distinction adds another step: detection can occur before a rule permits a report, and even a valid output may have a restricted category. Finally, the whole arrangement includes dependencies, settings and upstream inputs. A model name identifies only part of it. The relevant variation is whatever can reach the particular boundary being investigated.

Two machines, two people: what was compared?

Suppose a procedure runs on two identified instruments. A later difference in their results would be useful only if we knew what else differed between the measurements. Instrument is a condition that might affect the output—a **factor**—and the two particular machines are its **levels**. Those names identify the proposed comparison. They do not establish fair allocation, specimen counts or representation of every instrument of that kind. Two operators similarly represent two people, with no automatic inference about all future operators. [Sources 1–2]

What if the laboratory wants to understand the instrument difference without simultaneously changing its reagent? Keeping one identified lot can make that narrow comparison easier to interpret. It also leaves the behavior of other lots unexamined. This is a deliberate choice about scope. The following table helps expose such choices before the total number of observations hides them.

What happens to the condition?	Example	What must remain visible?
Deliberately varied	Both operators are assigned to both instruments	Why those levels matter and which comparison the assignment permits
Observed as it occurs	Specimen quality is recorded on receipt	Selection, imbalance and links with other conditions
Compared within a group	Aliquots from the same specimen are compared	Which shared difference the grouping controls
Held fixed	One identified reagent lot is used	Why it stays unchanged and which wider claim remains open

The one-lot row earns a positive interpretation: the lot restriction can help isolate the instrument comparison. It cannot establish between-lot reproducibility. The specimen-quality row makes a different choice. Merely recording quality does not distribute it evenly across machines; quality and instrument could still travel together. Matching portions of the same specimen across machines addresses a different source of ambiguity by giving the comparison shared starting material.

Here *held fixed* means practically unchanged during the study. A statistical model's *fixed effect* has a different job: it describes how observations are analyzed and can include a factor with several deliberately varied levels. The study's disposition and the model's terminology must be read separately.

Reviewer Lens. For tissue sequencing, the useful trace shows which patient specimens, preparations, lots, runs and report-producing versions varied, and what output or missing/invalid state each unit produced. For digital pathology, it shows whether scanner and tissue conditions were assigned together or only observed. A well-controlled one-lot comparison can be adequate for its instrument question. The narrow request is for the allocation supporting that question, followed by any evidence needed for the broader assertion.

Consultant Lens. Before collection, the assay or imaging owner can choose which condition matters, operations can check whether the assignment is feasible, and the statistician can identify the unit and comparison. A later failure pattern may reopen an unrepresented condition. It cannot retrospectively turn a fixed lot into studied lot variation. Closure requires either a feasible revised allocation or an explicitly narrower claim, with an owner and decision date.

Did each person use each machine?

If the first operator uses only the first instrument and the second uses only the second instrument, the two differences travel together. A difference in results cannot be assigned separately to person or machine. Those effects are **confounded** in this comparison. Adding either name to the study's factor list does not resolve that relationship.

If each person instead uses each machine, the comparison becomes possible within a person and within a machine. All four represented combinations are **crossed**. Some but not all combinations form a *partially crossed* allocation. These descriptions concern the arrangement of the represented conditions; they do not say how many observations exist or whether the design is sufficient. A focused comparison, a justified restriction or partial crossing may suit a bounded claim when complete crossing is unnecessary or infeasible. Any effects the chosen arrangement cannot separate still belong in its interpretation. [Sources 1–2]

Crossing also raises a new question. Is the second-minus-first instrument difference the same under each operator? If that difference depends on the operator, there is an **interaction**. Four boxes make it possible to ask the question; they do not supply a finding. Even with both people on both machines, using one instrument only in the morning and the other only in the afternoon could confound instrument with time. Allocation must therefore address order as well as names.

What is shared when an observation is repeated?

To make the within-person comparison useful, the laboratory could send matched portions of each specimen to both machines. Comparing inside a shared specimen grouping is **blocking**: the specimen's starting characteristics are shared across that comparison rather than allowed to masquerade as an instrument effect. The same tactic can use a day or another relevant group.

The observations also have relationships before the designer chooses a tactic. A particular run belongs to its day, and a scan belongs to its slide. Such observations are **nested**. Nesting describes where observations belong; blocking uses a grouping for allocation or analysis. One does not automatically imply that the other has been planned appropriately.

That distinction helps answer what a repeat adds. Rereading an existing image reveals behavior of that reread under its chosen settings; it adds no slide, case, scanner, patient or site. Rescanning the slide reaches more of the scan route while still sharing the same patient. Re-extracting a molecular specimen can introduce preparation variation while sharing the original material. Several variant findings from that specimen likewise do not become several independent patients.

The **independent unit** is the unit newly sampled or assigned for the effect being studied. Linked observations within it keep their shared history. Case □ slide □ scan identities make that history visible in pathology; patient/specimen □ extraction □ library □ finding identities do so in sequencing. Another reading of one aliquot can inform short-run scatter without informing patient-to-patient or lot-to-lot variation.

The separate deployed teaching notes from Chapter 1 illustrate why those identities remain useful after use: two initial invalids among forty tissue-sequencing attempts raise a possible material/run question; one late worklist arrival among twenty-five pathology cases raises a workflow question. Neither note identifies a cause, a variation estimate or a HelixBridge or PathSight event. They direct the next question to different opportunities for variation.

Where would each specimen go?

A separate fictional laboratory wants to compare two specified instruments operated by two specified people at one site. It wants the comparison both for routine concentrations and near a reporting boundary chosen in advance. These planned specimen groups are **strata**. A difference near the boundary could matter even when the average across all specimens looks reassuring.

The source identifies the machines as I-A and I-B and the people as O-A and O-B; below they are the first and second instruments and operators. This is a design sketch, with no supplied concentrations, clinical criterion, specimen count or statistical decision limits. It provisionally assumes a suitable reference assignment and enough stable material and volume for matched aliquots. Those assumptions still need establishment before the plan can become executable. It is neither a HelixBridge nor a PathSight study.

What answer is the plan trying to obtain?

The first page connects what could be calculated to the scope of the resulting answer. Its restrictions are as informative as its varied conditions.

Question	Entry in this plan
What is the output?	A quantitative result in the procedure's defined unit; any report-category effect considered separately
What difference is being studied?	Instrument and operator differences at one site, in the represented specimen strata
What is assigned to vary?	Two specified instruments and two specified operators
What is sampled and recorded?	Distinct patient specimens in the two strata, with relevant specimen characteristics

Question	Entry in this plan
How do specimens contribute?	Matched aliquots from each specimen across the represented combinations
What stays unchanged?	Site S-A, reagent lot L-A, software V-A and the specified preparation procedure
Why restrict those conditions?	To avoid adding a concurrent lot or version change to the selected comparison
Which breadth remains unexamined?	Other sites, lots, versions, concentrations and unrepresented specimen types
What still needs a decision?	Sample-size justification, exact levels, order, repetition, criterion, analysis and missing-observation handling

The output row determines the calculation: a difference in quantitative values. It does not create a report category without a separate mapping rule. The matched-aliquot row then explains how one specimen can contribute to several comparisons without becoming several patients. The fixed-state record identifies the one site, lot and software used; the conclusion cannot silently expand to other states because those restrictions made the immediate comparison cleaner.

The final row is consequently part of the scientific question. How many specimens would estimate the relevant difference precisely enough? What order would keep a material change from looking like a machine effect? What prospective rule would permit a favorable conclusion, and what would happen when a specimen yielded no usable result? Until those choices are made, the page is an intelligible plan rather than authority to begin or conclude a study.

Reviewer Lens. A clear canvas deserves credit for exposing a bounded comparison and its unfinished fields. A later performance report must preserve aliquot identity and report-rule linkage across the cells, as well as the scope exclusions. If that report claims reproducibility everywhere, the missing site, lot and version basis becomes a specific question. The canvas itself has not supplied a performance result.

Consultant Lens. The claim owner settles the instrument/operator question; laboratory and specimen owners establish aliquot fitness and feasible order; the statistical owner specifies count, model and decision limit before results exist. A later lot-related concern belongs to its own coverage decision. Relabeling the original plan would conceal the condition whose evidence is now needed.

How do the four cells make a comparison possible?

For each of the routine and near-boundary strata, every selected specimen is intended to be represented across this allocation.

Instrument	First operator	Second operator
First instrument	Selected specimens planned here	Selected specimens planned here
Second instrument	Selected specimens planned here	Selected specimens planned here

Reading across the first-instrument row compares operators on that machine. Reading down the first-operator column compares machines for that person. Once results exist, the second-minus-first instrument difference under one operator can be compared with the corresponding difference under the other. That difference between contrasts asks whether the instrument effect depends on operator. With only the first-person/first-machine and second-person/second-machine diagonal, person and machine would remain mixed.

Each cell shows **planned coverage**. It is not a measured result, a patient count or a promise of one observation per cell. A specimen identifier follows its matched aliquots across all four cells, because their measurements share a source. Repeated readings within one cell share additional conditions. Putting the near-boundary stratum in every cell also makes an instrument-by-stratum difference estimable in principle. Useful precision still depends on suitable specimen numbers, variability, allocation, missingness and analysis. Many readings on two machines do not turn those machines into a random sample of all instruments.

Order completes the allocation. If all first-instrument measurements occur early and all second-instrument measurements occur late, time or material change could produce the apparent difference. A feasible random or restricted order can reduce that ambiguity while respecting specimen stability. The protocol needs those order choices, intended comparisons and missing-result rules before outcomes arrive.

Figure 9.1: Structured variation and independent units compares the four-cell allocation with a diagonal-only allocation and traces the sampling hierarchy. The diagram is a planning aid: a filled box shows an intended comparison, not a completed result.

Chapter 09

Structured variation and independent units

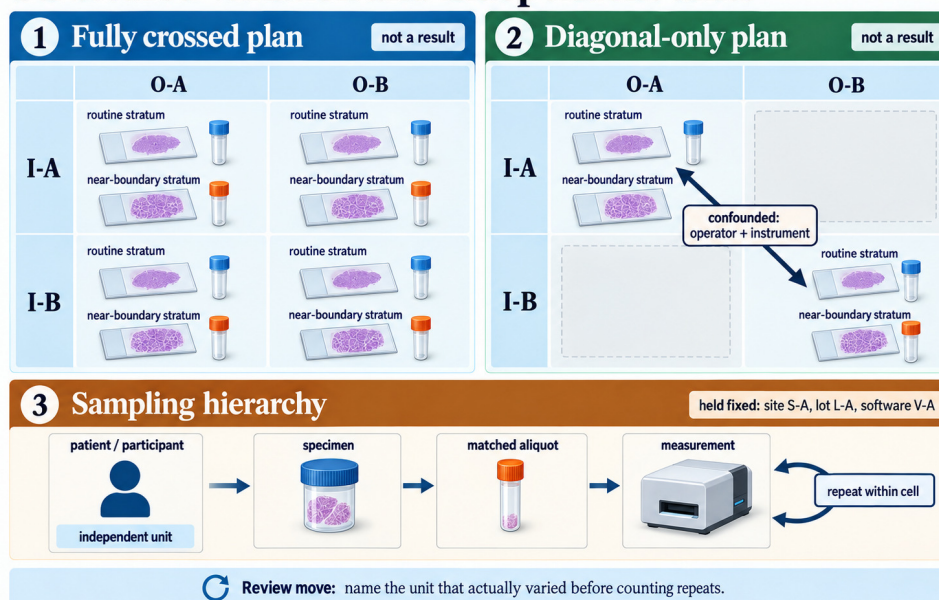


Figure 9.1. Structured variation and independent units – Technical review plate.

What difference would change the decision?

Suppose the concern is a shift that moves near-boundary specimens across the reporting rule. An average over routine and near-boundary concentrations could obscure precisely that shift. If the concern instead is scatter, the analysis needs the repeated readings within a run, between runs and under the represented instruments and operators. The output decides which analysis makes sense: agreement or error analysis may fit a category, while a quantitative coefficient of variation cannot simply be assigned to a nominal variant identity or pathology category. The Clinical and Laboratory Standards Institute's quantitative-precision standard, CLSI EP05, offers design options within that scope, without providing one universal schedule or threshold. [Source 1]

The decision requires both a consequential difference and a target for how precisely it should be estimated. An estimate is a value calculated from the study's observations. Uncertainty about it describes how well the study pins down the underlying effect; it differs from the fluctuation of a single measurement. Chapter 7's uncertainty components likewise remain attached to the conditions they represented.

If an analysis does not establish a difference under its chosen rule, the difference is nonsignificant in that analysis. That is not **equivalence**. Equivalence needs a justified margin and an analysis able to rule out differences beyond that margin. A neat

allocation or a nonsignificant result cannot supply the missing margin, precision or inference.

Did the intended comparisons survive collection?

The actual study may lose the specimens it most needed. If near-boundary specimens disproportionately fail, the surviving measurements no longer represent the planned stratum simply because exclusions were allowed. Every planned specimen and cell therefore needs a linked account of attempts, usable results, invalid states, exclusions, repeats and final outputs. A successful repeat can make a report possible while its first failure and delay remain real observations.

That account distinguishes two empty cells. One combination may never have been planned; another may have been attempted but unable to produce a result. The first records a coverage choice. The second can itself be performance evidence. A new scan still belongs to its slide and case, and a new sequencing library still belongs to its specimen. The unrelated two-in-forty invalid note calls for first-pass accounting, while the separate pathology delay concerns delivery and workflow timing rather than image color alone.

Reviewer Lens. The allocation and attempt ledger belong beside the result. They show which units and strata a favorable estimate can represent and whether apparently missing observations were failures. **Consultant Lens.** Laboratory, imaging, software and statistical owners can reconcile missing and repeat pathways against the planned analysis before closing the claim. Material fitness also constrains the design: an allocation that degrades its specimens defeats the comparison it was meant to clarify.

What does “stable wet lab” let us conclude?

The selected **HelixBridge Dx** record describes a tissue-oncology sequencing assay intended broadly to identify variants associated with eligibility for a named therapy. It describes the wet lab as “stable” and identifies an **analytical-validation** record. Neither description supplies the allocation, criteria, failures or results for a variation study. The first question is which material and report-producing state that record actually represents.

The following questions follow that route from entry to reporting, then ask what could change along it. Each row retains an unresolved relationship in the supplied record.

Question about the tissue route	Why the answer matters	What is not supplied
Which tissue specimens and preparations entered?	Incoming material can change what reaches the assay	Exact tissue route and represented preparation conditions

Question about the tissue route	Why the answer matters	What is not supplied
Which computation and reporting state produced the result?	A wet-lab description leaves the rest of the reporting route open	Complete tested and proposed-use identities
Which variant classes and sequence contexts were challenged?	One class or difficult context may behave differently from another	Classes, contexts, sampling and outcomes
Which abundance and reporting boundaries were represented?	An average detection result can obscure a consequential boundary	Applicable levels and boundary-specific results
Which lots, instruments, runs, people and sites varied?	Routine implementation can change the route	Allocation, independent units and included components

The first two rows set the ends of one evidence trace: what entered, and which **configured product** produced the report. Within that trace, the claim and plausible failure paths determine the relevant variant classes, contexts and abundance strata. Only then can an allocation show whether the routine conditions capable of changing those outputs were represented. The table makes relevance decisions explicit; it does not require a separate large study for every row. CLSI MM09 and the International Organization for Standardization's molecular-diagnostic standard ISO 21474-2:2022 provide bounded context for these questions. Neither supplies fictional results or one recipe applicable to every tissue, germline, singleplex and liquid method. [Source 3]

The proposed liquid branch has a separate route: whole blood □ separated plasma □ isolated cell-free DNA. These remain distinct materials and steps, rather than renamed levels of the tissue state. No liquid study is asserted here. The word “stable” also supplies no duration, tested condition or result establishing reagent shelf life, lot interchangeability or reproducibility.

Reviewer Lens. The specific request is for the actual validation allocation, tested state and first-pass failures. A documented tissue comparison can earn credit for the variant class, material, conditions and criterion it covers. The other rows remain open; a future release or an absent liquid result cannot fill them. The phrase in the file is a useful lead to evidence, with no supplied study pass or failure.

Consultant Lens. The assay owner identifies the consequential variant and specimen boundaries; laboratory and software owners establish feasible factors and controlled states; the statistical owner selects independent units and precision targets. These choices form a prospective plan. If a later invalid pattern arises, its material/run

question returns to the appropriate owner without a presumed cause. Closure attaches adequate additional evidence or a narrowed claim to an identified version.

Could a control miss the route that matters?

Suppose material from an earlier tube might reach a later preparation. A study labeled “robustness” leaves the essential questions unanswered: what moves, by which path, and what output could change? A **challenge** deliberately presents a plausible weakness under defined conditions and uses a comparison capable of revealing its effect. Its difficult specimens should reach clinically consequential boundaries. An arbitrary extreme outside claimed use may answer little about the actual decision, and a difficult sequence context is not automatically a difficult image or reagent condition.

Following the earlier-tube example earns the mechanism names. When unwanted material enters the later specimen or process, that is **contamination**. Asking where it came from may expose **carryover**: material from an earlier operation reaches a later one. The same event can therefore have both descriptions. One identifies unwanted presence; the other identifies the transfer path and order.

The next question is what the examination does with that material. A substance or property that changes its response to the intended material causes **interference**. A method that responds to something other than its intended target raises a **cross-reactivity** question. If transferred material distorts the response, interference may be the downstream effect; if it resembles the target, cross-reactivity may explain the response. These are possible relationships to investigate, with no claim that any mechanism occurred in a named case.

Where the change happens matters too. Changing the analyte before examination differs from distorting the analytical response to that analyte. The control must reach the relevant step in either route. CLSI EP07 distinguishes these pre-examination and analytical questions. The carryover standard CLSI EP47 has a narrower object: **reagent carryover** in its specified laboratory-method scope. It is not a procedure for sample carryover, all molecular cross-contamination, every multianalyte or algorithmic system, or every pathology image-chain effect. Its first-edition official record includes an April 2025 correction notice. Naming the standard cannot replace identifying the transfer path. [Source 4]

For tissue sequencing, source material entering a later preparation and a response to an off-target sequence call for distinct controls even if either could yield an erroneous finding. For pathology, stain and scanner-processing conditions can alter the digital input. These are routes for design, not observed interference or contamination results for HelixBridge or PathSight.

What would make this transfer test interpretable?

In the hypothetical earlier-source/later-recipient comparison, the recipient output needs to be near a boundary where a change could matter. A sequence without the proposed exposure gives the challenged sequence a comparison. The following record then connects that exposure to the control, sampling and decision rather than treating each as an independent checklist entry.

Question in the record	What the answer needs to identify
How could failure occur?	Source material reaching a later recipient through a specified instrument path
Which output could matter?	Recipient result or category near a consequential boundary
What exposure tests the route?	Justified source/recipient conditions and sequence, rather than an arbitrary high level
What comparison reveals the effect?	A condition that makes the proposed exposure interpretable
Where does the control travel?	Transfers, processing steps and outputs it actually experiences
What is newly sampled or repeated?	Independent materials, repeated sequences, order and sequence dependence
What would change the decision?	A prospective, use-specific criterion with uncertainty and failure accounting
What remains outside the test?	Other sources, paths, sample types or combined exposures

The control-path row can change the interpretation of the whole record. A control introduced after the suspected transfer may support the later analytical step when it passes. It cannot rule out transfer upstream of its entry point. Conversely, an exposure too mild to exercise the proposed path cannot close that path through a passing result. The record is a reasoned plan, not an executable EP47 procedure or an observation that the hypothetical transfer happened.

Which positive conclusion survives?

A synthetic input introduced after extraction gives this issue a concrete molecular form. If it behaves as planned, it supports the downstream computational or analytical segment under the tested conditions. It has never experienced extraction from patient tissue. A claim reaching that earlier route needs complementary controls or specimens that do experience it. A control can therefore succeed and leave a genuine evidence gap at the same time. [Source 3]

Reviewer Lens. Source, path, recipient, control entry and output form one trace. A favorable partial-path control deserves credit for the segment it traversed; the upstream question stays separate when the claim needs it. The unrelated invalid-attempt and late-worklist notes are clues for different investigations. They show neither reagent carryover nor a stain-related cause.

Consultant Lens. Risk and laboratory or imaging owners identify the plausible route; the study owner matches exposure and control paths; the statistical owner defines useful units and a prospective rule. After a passing challenge, closure names the covered segment and assigns the remaining segment to its responsible owner. Calling the control useless would discard support, while calling it end-to-end would conceal the next decision.

Figure 9.2: Mechanism-linked challenge traces source, transfer route, recipient, control entry and bypassed step. Use it to ask what a control really traverses; none of its illustrated paths is a result for either named product.

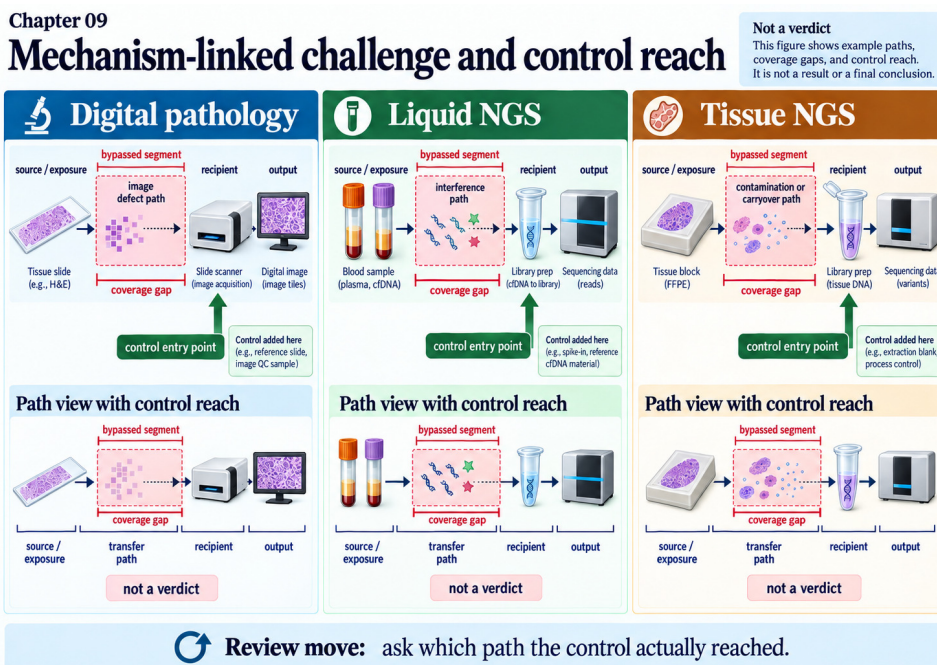


Figure 9.2. Mechanism-linked challenge and control reach – Technical review plate.

Which material has been aging?

A reagent bottle stored unopened for a year has experienced elapsed storage. It has not necessarily experienced repeated opening or the proposed temperature during use. The patient specimen examined with that reagent may have a different history again. To interpret time evidence, the first question is what material was in what state during the elapsed interval.

The table separates the clocks that can appear together in one laboratory record. It is useful when “tested over time” seems to answer more than one question.

Which clock?	What can the comparison address?
Short measurement sequence	Whether repeated readings or their order affect the output
Run or day	Variation in a specified within-laboratory result across runs or days
Elapsed unopened storage	Whether the reagent retains specified performance over its claimed shelf life and conditions
Time after opening or preparation	Whether that reagent retains performance throughout its stated in-use period
Transport exposure	Whether defined distribution conditions change the material before use
Configuration chronology	Which reagent lot and software/hardware state produced each record, and when

The unopened-storage row concerns the bottle's state throughout the duration. A record of several study days could instead use a fresh container each day and say nothing about an opened container aging between measurements. The first two rows concern repeated operation; the storage, in-use and transport rows need defined reagent states, durations, environments and performance attributes. The final row links the observation to the configuration that actually produced it, preventing an earlier result from quietly becoming evidence for a later state.

Once the clock is clear, a reagent-stability plan needs the product and lots, relevant attributes, observation schedule, comparison basis and acceptance rule before a trend is observed. A changing result could reflect the reagent, ordinary measurement scatter or changes in the test material used to evaluate the reagent. The design must distinguish those possibilities. Accelerated and real-time evidence have different roles: a convenient stress result alone does not assign an expiry date. ISO 23640:2011 requires real-time verification of an initial expiry claim based on accelerated or relevant prior evidence. [Source 5]

The scopes of CLSI EP25 and ISO 23640 concern in vitro diagnostic reagent stability. They do not establish patient-specimen or analyte stability, or validate scanner, instrument and software behavior. A collection device containing preservative may fall within reagent-stability scope, but evidence for the preservative does not automatically establish the fitness of every stored patient specimen. Chapter 6's material and handling route still needs its own support. In the dated source check on 24 September 2026, ISO listed the 2011 edition as current and a draft replacement under development; a draft was not a published replacement. [Source 5]

HelixBridge's "stable wet lab" supplies none of the unopened shelf-life duration, post-opening period, specimen-handling result or lot comparison needed to answer these questions. Along a pathology route, tissue storage, image production, scanner processing, software chronology and worklist timing likewise concern different objects. The unrelated deployed note's late worklist arrival with completed review is a timing observation, without a demonstrated cause or scanner result.

Reviewer Lens. A duration claim becomes assessable when material, state, conditions and measured output appear beside the elapsed time. **Consultant Lens.** Separate reagent, specimen and software/workflow owners can agree the needed time contrasts prospectively. A supported duration closes only the stated material and configuration question; extending another clock still needs a basis.

What can the completed observations earn?

The planned instrument/operator comparison has supplied no results. The next case supplies a completed, separate fictional bench screen. Its answer can therefore be affirmative in a way the plan's answer cannot. A second short case then returns to the distinction between study dates and material age. Neither is additional history for a named product.

The crossed bench screen

Three identified materials, M1–M3, are each read twice with reagent lots A and B on instruments I1 and I2 in one session. Software, operator, handling and reporting rule stay fixed. Order is counterbalanced; every planned reading is retained, with no replacement repeats. The materials are stipulated to be suitable for this narrow bench comparison. Their suitability for representing all patient specimens is not supplied.

The prospective decision concerns these observed combinations. For each material, three conditions must hold:

1. On each instrument, the absolute difference between lot B's and lot A's two-reading **cell means** is at most **0.70 units**.

2. The absolute difference between the lot contrast on I2 and the lot contrast on I1 is at most **0.25 units**.
3. Within every cell, the two raw readings differ by at most **0.40 units**.

“At most” includes equality. These are fictional engineering criteria, not clinical limits, population confidence bounds or a CLSI EP26 method. The first condition asks how much the lot changes the observed mean on each machine; the second asks how much that change differs between machines; the third asks how far apart the duplicate readings are.

The arithmetic follows those questions. Adding two readings and dividing by two gives a cell mean. Subtracting A's mean from B's mean on one instrument gives its lot contrast. Subtracting the I1 contrast from the I2 contrast compares those lot changes. The absolute value ignores direction for this screen. Duplicate gaps use the raw readings separately. Even a small contrast difference under this rule would not prove zero interaction in a broader population.

There are **three materials × two lots × two instruments × two readings = 24 readings**. This counts readings, with no claim of 24 independent patients, lots or sessions. Every number below uses the same fixed procedure's units; each cell contains a pair of readings.

Material	Lot A on I1	Lot B on I1	Lot A on I2	Lot B on I2
M1	4.8, 5.0	5.0, 5.2	4.9, 5.1	5.3, 5.5
M2	19.8, 20.2	20.0, 20.4	20.1, 20.3	20.3, 20.5
M3	79.8, 80.2	80.4, 80.6	80.0, 80.4	80.7, 80.9

For example, the first row's first pair belongs to material M1, lot A and instrument I1. Its next pair changes the lot while retaining M1 and I1. Moving to the third column changes the instrument while returning to lot A. Reading the table this way keeps each calculation attached to its material and conditions.

Decision

What are the cell means, the two lot contrasts for each material and their difference? Do all three inclusive criteria hold, including every duplicate gap? What one bounded conclusion is supported, and what two stronger conclusions would remain unsupported? If only lot A on I1 and lot B on I2 had been observed, what could no longer be separated?

How the readings support a finite pass

For M1 on I1, lot A's mean is $\$(4.8+5.0)/2=4.9\$$ and lot B's is $\$(5.0+5.2)/2=5.1\$$. The lot contrast is therefore $\$5.1-4.9=0.2\$$ unit. On I2, the respective means are $\$5.0\$$ and $\$5.4\$$, so the contrast is $\$0.4\$$ unit. Comparing the contrasts gives $\$0.4-0.2=0.2\$$ unit. The table applies the same sequence to all three materials.

Material	A / B means on I1	A / B means on I2	B – A on I1	B – A on I2	Contrast on I2 minus contrast on I1
M1	4.9 / 5.1	5.0 / 5.4	0.2	0.4	0.2
M2	20.0 / 20.2	20.2 / 20.4	0.2	0.2	0.0
M3	80.0 / 80.5	80.2 / 80.8	0.5	0.6	0.1

For M2, the two contrasts are both $\$0.2\$$, leaving $\$0.2-0.2=0.0\$$. For M3, lot B minus A is $\$80.5-80.0=0.5\$$ on I1 and $\$80.8-80.2=0.6\$$ on I2. Their difference is $\$0.6-0.5=0.1\$$. The greatest absolute lot contrast is thus **0.6**, below **0.70**, and the greatest absolute contrast difference is **0.2**, below **0.25**.

The duplicate condition is a separate check against the raw pairs. Its greatest gap is **0.4**, seen, for example, between M2's 19.8 and 20.2. That equals the **inclusive 0.40** limit and passes. All 24 planned readings are present. **The finite bench screen passes all three conditions for these three materials, two lots, two instruments and this session.** There is no reason to withhold that bounded positive result.

The result's limits follow from what was observed. M1's lot contrasts of 0.2 and 0.4 are visibly different; meeting the screen does not prove zero population interaction. Two readings in one session establish neither routine between-day precision, patient-material commutability, future-lot behavior, confidence bounds for all effects nor long-term stability. With only the diagonal A–I1 and B–I2, lot and instrument would change together, and the within-instrument lot contrasts displayed here would be unavailable. Crossing permits this specific attribution in the finite observed screen; it does not grant every further clinical inference.

CLSI EP26 addresses an end-user laboratory's assessment of an in-scope candidate reagent lot against its current lot for continuous-scale procedures. It has qualified application when a continuous response becomes a qualitative output; it does not cover purely qualitative or semiquantitative procedures. Its purpose is distinct from manufacturer validation. The fictional screen neither reproduces its method nor substitutes for a claim-matched lot evaluation. [Source 6]

Reviewer Lens. The retained readings, cell means and inclusive rules make the pass auditable. Their dependence and their material/session boundaries stay attached to the conclusion. Sequencing lot evidence might additionally need variant-class and

report-state coverage, while a pathology scanner comparison must reach relevant images and the assistive output. Neither can inherit this bench quantity's criteria.

Consultant Lens. Before a comparable study, laboratory and manufacturer owners need to agree whose decision is being made, which consequence the margin controls and which materials are suitable. After a favorable screen, the actual result remains useful. A wider precision, site or patient-representation claim goes to its owner with its own criterion, rather than being added to the local pass.

Return to the reagent's clock

A precision study collects results on several days with a **fresh reagent container each day**. It does not deliberately represent elapsed unopened storage or time since opening. The proposed conclusion is a longer period of acceptable use after opening.

Decision

What is missing from this calendar of observations, and what study question would address it while retaining any useful role for the precision results?

How a reviewer connects the duration to the container

The missing exposure is the stated time and conditions after opening. A useful next question identifies the container and reagent configuration, in-use temperature and handling, duration, performance attributes and comparison rule. The several-day precision observations can inform expected ordinary measurement variation. They do not supply aged in-use containers. Studying an unopened stored lot would answer another question—shelf life—without necessarily answering repeated-use stability.

Can more scans close the PathSight question?

PathSight Cloud has an assistive role: it prioritizes images and highlights suspicious regions for a pathologist, who diagnoses. The selected record identifies the validation model as M3.2 and claims support for the first and second scanner families, Scanner A and Scanner B. Validation images came mostly from the first family. In a postvalidation assessment, the second family applies proprietary **sharpening** before export. An average-color bench comparison does not reach weakly stained tissue near low-confidence boundaries.

Commercial **model weights** are described as unchanged. The complete end-to-end configuration and whether the changed state was released remain unresolved. These are supplied configuration and study-scope facts, with no observed patient outcome.

The team proposes closing scanner generalizability by repeatedly scanning the already available, strongly stained cases from the first scanner family.

Decision

What assistive output and decision does the claim reach? Which two or three factor relationships most need separation? How do case, slide and scan units depend on one another? Which conditions belong among those deliberately varied, observed, blocked or held fixed? What useful conclusion could the average-color comparison support, and where would it stop? What information would be needed to finalize a defensible design?

No second-scanner case count, interaction estimate, passing threshold, release event or regulatory outcome is supplied. The design judgment rests on that evidence state.

How a reviewer separates scanner breadth from repeat scanning

Follow the claimed output through the allocation

The proposed repeats could add useful scanning information under the first-family, strongly stained conditions. They add neither second-family coverage nor the weak-stain, low-confidence challenge. The claimed object is the assistive output under supported image-production and workflow conditions. Repeatable average color reaches only a submeasure of that route.

Scanner processing and tissue/stain conditions therefore need joint consideration. A scanner-associated difference could depend on tissue condition. That is an interaction question, not a demonstrated interaction or an effect estimable from this record. A possible design compares represented scanner/processing conditions across relevant case and stain strata, with feasible matching or blocking and recorded order and configuration. Clinical material constrains which arrangements can actually be used; no universal scanner count follows.

The units retain their hierarchy: case or patient $\square$ slide within case $\square$ scans of the slide. Additional scans do not become additional cases, and several slides from one case remain linked. The average-color comparison can retain support for its own quantity without establishing low-confidence assistive behavior or the clinical reader workflow. If the color quantity itself is used as evidence, Chapter 7's measurement questions apply to it.

To finalize the design, the owner still needs exact image-production and software states, the case spectrum, reference or annotation process, output and failure states, allocation, challenge strata, uncertainty, a prospective criterion and missing-data handling. The tested and proposed-use states must be compared before evidence can be

assigned to a release. Unchanged weights identify one component; they do not settle that relationship.

The U.S. Food and Drug Administration's FDA DEN200080 classification order illustrates a bounded source question. For its particular device type, technical/analytical provisions call for challenging cases and evaluation across relevant input-device, operator, scanner and site conditions. The order supplies neither PathSight data nor a universal digital-pathology recipe. [Source 7]

Reviewer Lens. The narrow repeat proposal need not be discarded. Its credit stops at the represented scanning conditions. A broader scanner claim needs the second-family image route and weak-stain allocation, linked through case/slide/scan identity, preprocessing, model/interface state, assistive output and reader action. Unknown release status remains unknown; this is not a supplied postmarket outcome.

Consultant Lens. The image-production owner identifies the sharpening state; clinical and pathology owners select case/stain strata and the reference process; the software owner controls the full configuration; the statistical owner plans the linked comparisons. A general claim closes only when these choices and criteria are set prospectively and the resulting evidence fits them. If only narrower support is feasible, that bounded conclusion and the owner of the remaining bridge stay explicit.

What would close the design question?

A reviewable allocation lets someone identify the effect it can separate, the independent unit and the reason for each fixed condition. Its attempt account shows whether the intended comparisons survived collection. A large total or a neat chart cannot substitute for those relationships. The consultant connects the next decision to the claim owner, the laboratory/imaging/developer owners of physical and computational states, the statistical owner of comparisons and precision, and the operations owner of specimen fitness.

That ownership also distinguishes a laboratory user's reagent-lot verification from a manufacturer's validation of product variation. Software, instrument, reagent and specimen identities remain linked to the final output in either decision. For applicable European Union in vitro diagnostic documentation, Regulation (EU) 2017/746 distinguishes analytical precision, interference/cross-reactions, stability and software configurations in the relevant Annex II provisions. It does not supply one all-purpose robustness study. This is a bounded documentation handoff; Book 2 treats the broader interpretation. [Source 8]

Two changes and one favorable comparison

The final case is separate anonymous fiction. An old controlled state pairs the old reagent lot with old instrument software; a proposed state pairs a new lot with new software. The source labels are L-old/V-old and L-new/V-new. Both components changed together, and only those two combined states were examined. The comparison meets its **prespecified aggregate endpoint for the tested combined state**. No separate lot effect, software effect or interaction assessment is supplied.

The team wants to call each component independently equivalent to its predecessor and exchange either without further assessment.

Decision

What useful positive conclusion does the supplied comparison support? What further basis would independent exchange need? Which next question follows from the affected mechanisms and proposed use, and how could it be addressed if an intermediate combination were unsafe or technically unavailable? Is a complete four-cell experiment necessarily the right requirement?

How a reviewer preserves the pass and limits the attribution

Which comparison has actually passed?

The aggregate result is favorable evidence for the two combined states, the endpoint and the conditions examined. It cannot assign a separate effect to lot or software. Opposing component effects, an interaction or a difference outside represented strata remain possible explanations compatible with the record, rather than findings. The table locates the unexamined relationships behind the stronger claim.

Reagent lot	Old software	New software
Old lot	Examined	Not examined
New lot	Not examined	Examined

The examined cells form the diagonal: the old pair and the new pair. More observations in those cells could refine their combined comparison, but would not create either intermediate comparison. The table exposes an attribution limit; it does not require an unsafe combination to be run.

The next question begins with the affected mechanisms, intended claim, prior evidence and proposed-use configuration. An intermediate contrast might separate effects where feasible and appropriate; other justified evidence could also support a bounded component conclusion. If an intermediate state cannot operate safely or technically,

that reason remains explicit while consequential challenges of the combined new state are examined. Independent equivalence and unrestricted interchangeability require an adequate separate basis.

Reviewer Lens. The aggregate endpoint keeps its declared pass. The request targets exactly what would support independent exchange, with unavailable comparisons stated as a limit. **Consultant Lens.** The risk owner takes the unresolved component question, the technical owner determines safe feasibility, and the claim owner chooses between focused additional evidence and a bounded combined-state claim. Closure preserves the useful result while keeping the stronger inference open.

Return to the hundred rows

The opening study's conditions have not changed while we have worked through the other comparisons. What title and conclusion now fit its repeated readings, and what design change would be needed for its broader proposed claim?

What the repeated-reading study can earn

A title such as “Short-run repeated-output behavior under one specified preparation and session” identifies the bounded question. A method appropriate to the output could characterize its repetition under that specimen, lot, instrument, operator, software state and session. No numerical result was supplied, so the record does not establish that the behavior passed a criterion.

Additional repetitions might sharpen that narrow estimate. They do not add another patient, lot, instrument, operator, site or use condition. A broader claim needs an allocation reaching its consequential variation. The first design change therefore depends on which missing condition the claim actually needs, followed by a choice of units and comparisons that can address it.

The useful chain now runs from claim and output through independent units, factor relationships and a plausible challenge to a prospective decision and bounded conclusion. Chapter 10 follows that reasoning through the tissue-sequencing route from specimen and wet lab to computation and reported finding. Its new questions leave this chapter's unsupplied facts unsupplied.

Questions that test the design reasoning

Can the following now be explained without relying on a study's title?

1. How can two operators and two instruments leave their effects inseparable?
2. Why can holding a reagent lot fixed be useful, and which inference does it withhold?

3. What different information comes from another scan, another slide and another patient case?
4. How can contamination and interference describe linked parts of one event without being synonyms?
5. Why do several study days not establish reagent shelf life or in-use stability?
6. What does a passing aggregate endpoint support after a combined lot/software change, and which further inference remains open?

How the six answers connect

The answers identify the actual allocation, unit, material and output. A diagonal can leave effects inseparable. A fixed lot can clarify an instrument comparison while withholding lot variation. The case/slide/scan hierarchy identifies what a repeated observation shares. Source $\square$ path $\square$ effect separates unwanted material from its consequence. The reagent's elapsed exposure distinguishes its age from study dates. And the combined-state pass retains its endpoint without assigning separate component effects. “More testing” becomes a design only when the proposed observations create the opportunity for variation that the decision needs.

Source trail

These sources support bounded concepts at their stated locators and scopes. The canvas, cases, allocations and numerical screen are original teaching synthesis, not prescribed forms. Licensed standards were consulted read-only in the source packet; no licensed method table or figure is reproduced.

1. **Quantitative precision design:** CLSI EP05, fourth edition, §§1.1, 1.3, 1.5.1–1.5.4.1 and Chapter 2. The chapter uses its design principles, not a copied protocol, universal count, variance-component formula or threshold. Official record.
2. **Design terminology:** National Institute of Standards and Technology (NIST)/SEMATECH e-Handbook, §5.7, entries on factors, levels, crossing, nesting, interaction, experimental units, replication, blocking and confounding. Official glossary.
3. **Molecular configuration and controls:** CLSI MM09, third edition, §§3.1–3.3.6 and §4.2.7; ISO 21474-2:2022, §§4.1, 4.3–4.5 and 5.1–5.2.5. Their scopes do not turn a germline, singleplex, tissue or liquid method into a universal NGS recipe. MM09 record; ISO record.
4. **Interference and reagent carryover:** CLSI EP07, third edition reaffirmed 2022, §§1.1 and 2.2.2–2.2.5; EP47, first edition, §§1.1, 1.6.1, 3.1 and opening of 3.2. The official EP47 record lists an April 2025 correction and excludes several

sample/multianalyte carryover questions from its detailed scope. EP07 record; EP47 record.

5. **IVD reagent stability:** CLSI EP25, second edition, §1.1 and §§3.1–3.2; ISO 23640:2011, §§1, 4.1 and 5.1. The latter's official ISO record, checked 24 September 2026, lists it as published/current with a draft replacement under development; publication-freeze status should be rechecked. EP25 record.
6. **Laboratory-user lot verification:** CLSI EP26, second edition, §§1.1–1.2. Its user/manufacture and output-scale limits matter; the fictional numerical exercise is not its method. Official record.
7. **Bounded U.S. DP example:** FDA DEN200080 classification order, pp. 4–5, technical/analytical validation provisions. It supplies a device-type-specific question, no PathSight result or authorization. Official order.
8. **Bounded EU documentation handoff:** Regulation (EU) 2017/746, consolidated text dated 10 January 2025, Annex II §§6.1.2.1(b), 6.1.2.3, 6.3 and 6.4. This is not a comprehensive current legal assessment.

No numerical performance result or release is asserted for either recurring fictional case. The separate deployed NGS and DP observations are teaching notes, not those cases' chronology.

CHAPTER 10

Oncology NGS as an End-to-End Examination System

Which product would this folder support?

A tissue examination ends with a finding in a report. Between the patient material and that finding, laboratory operations produce data, software identifies candidate differences, and rules determine what may be reported. Each step can have good evidence behind it. The further question is whether those records support the connected route that produced this output.

The fictional **HelixBridge Dx** package contains every major document. Its broad purpose is to identify tissue variants associated with eligibility for a named therapy. The laboratory work is described as “stable.” The record identifies application 7.0 and report rules R19, but calls the cloud **knowledgebase**—the reference information used by the analysis—“current.” At a release discussion, someone asks whether the analytical evidence supports the product.

Decision

Which product state would that question need to identify? Which identities are actually supplied, and which remain unknown? What can be said without inventing an extraction method, sequencer, therapy, reporting threshold or released state?

This is a return to a familiar case with a different task: joining its evidence. The initial judgment will return after the chapter's comparisons. The ordinary specimen route below is a generic teaching illustration; it supplies no missing implementation details for HelixBridge.

What would have to agree about one finding?

Suppose a study supports a reported sequence difference. Before applying that result, the reader needs to know what claim the finding serves and what assigned its expected status. A good comparison against the wrong expected finding would leave the claim unanswered. Even a fit comparison leaves another question: did the studied material pass through the same route as the material proposed for use?

That route brings the earlier chapters together. The following table is useful when a favorable result arrives without a clear account of what it supports. Each row asks about the *same prospective output*.

Earlier record	Question about this finding
Chapter 4: claim and study	Which exact claim and consequence does the result address?
Chapter 5: reference process	What assigned the expected finding, and where is that assignment limited?
Chapter 6: specimen route	Which collected and prepared material entered the examination?
Chapter 7: result type	Is the output a quantity, an identity or a category, and what evidence fits it?
Chapter 8: reporting boundary	What can be detected and reported, and what happens when a condition is unmet?
Chapter 9: variation	Which conditions, difficult groups, relationships and independent units were represented?

For the result-type row, a sequence identity does not become a measured quantity merely because the report contains numbers. That choice affects what the comparison can establish. The next row then asks whether the detected difference was permitted to appear in the report; the last asks which varied conditions the evidence actually reached. This linked view is a **coverage map**. It relates each record to the question and output it can support.

Decision

How could one entry from each earlier record refer to the same specimen, configuration and final output? Where would a missing link remain if the records described different material, variant classes or processing states?

What makes the records join

The claim, reference role, incoming material, laboratory and software states, output definition and analysis set must be compatible. A study of another variant class or an earlier processing state can remain useful for its own question. The shared assay title supplies no missing relationship to this output. The map can therefore show a supported component and an open connection at the same time.

A complete analytical argument also needs to identify which arrangement each record studied and what happened to planned outputs that never became complete reports. Those questions will give the coverage map its two companions. The resulting conclusion remains analytical and configuration-specific. Biological meaning, adequacy of a treatment-selection claim and regulatory outcome need further evidence.

How does tissue become a reportable finding?

The generic examination begins before software has anything to analyze. A laboratory accepts material suitable for its procedure and records specimens rejected before examination. For the accepted tissue, it needs DNA that can be examined. Releasing and purifying that DNA from the material is **extraction**.

The extracted DNA still has to reach the sequencing instrument in a usable form. It is prepared into pieces the instrument can read and link to their source; that prepared set is a **library**. The instrument then produces short observed strings of DNA letters with quality information. These **reads** are data about pieces of the material. They are not yet variants in a report. Different systems can use different materials, operations and names; this description remains generic.

How does a string acquire a location? The data and specimen identifiers pass to a controlled computing process that uses a specified **reference sequence** as its coordinate and comparison framework. Placing reads against it is **alignment**. Once placed, the readings can be examined for supported differences; identifying candidate differences is **variant calling**. Chapter 1's variant—a difference in DNA—now has a route from observed data to a proposed finding.

A candidate still has to survive the conditions for reporting. Quality checks and filters may reject an unreliable candidate or stop a specimen or run. The **report rules** select sufficiently supported findings within the permitted **reportable scope**. A region can have been targeted or examined while remaining outside that scope because of evidence limits, exclusions or rules. The knowledgebase can supply contextual annotations; report assembly communicates the selected finding and its limitations. The placement reference, contextual knowledgebase and selection rule perform different jobs, so their identities cannot substitute for one another.

Where should the evidence follow this route? The table locates the connections needed when a claim reaches from specimen to report. Its rows are questions for the evidence, not a requirement for seven separate studies.

Point on the route	What must the evidence connect?
Acceptance and preparation	Accepted material to studied conditions, with rejected specimens accounted for

Point on the route	What must the evidence connect?
Extraction	Claimed specimens, including difficult conditions, to suitable extracted material
Library and sequencing	Prepared material and intended targets to the instrument data
Data transfer and processing	Correct files and specimen identities to tested dependencies, settings and references
Quality and reporting decisions	Each rule to the result it permits, limits, stops or repeats
Annotation and report assembly	Identified sources and rules to the finding and limitations displayed
Released output	The actual report to the configuration and evidence asserted for it

Consider the data-transfer row. A downstream comparison is only attached to this specimen if the right files and identities reached the identified processing state. The next row can then send that specimen toward a valid, invalid, partial or repeated result. Those are different routes through the table, even if the final report eventually looks complete. One well-designed study can address several rows; a focused component study can reveal a mechanism that an average across the whole route conceals. Neither arrangement by itself establishes a treatment action.

Clinical and Laboratory Standards Institute (CLSI) MM09 treats relevant laboratory and computational operations as one test boundary, with quality checks before, during and after examination. It supports inspecting those connections rather than assuming two favorable component results join. [Source 1] A bounded legal example makes the same system span visible: for its defined United States tissue tumor-profiling device type, 21 CFR 866.6080 connects specimen requirements, components, stepwise quality criteria, information sources, analytical evidence and generated reports. Its device-specific scope supplies no general classification, method or authorization for HelixBridge or the proposed liquid branch. The source check on 24 September 2026 found the electronic Code of Federal Regulations showing Title 21 current through 22 September 2026. [Source 2]

Reviewer Lens. An actual reportable finding gives the review a starting point: back through its rule, filters, knowledgebase, reference, file identities and accepted specimen. The place where a study entered that route determines its direct reach. For digital pathology, the corresponding trace goes from case and slide through scanner and image preparation to assistance for a pathologist who diagnoses. The need for

linked evidence carries across; extraction and variant calling remain sequencing operations.

Consultant Lens. Before testing, laboratory, informatics, interpretation and report owners can agree what leaves each step and what the next accepts. That agreement makes a missing interface visible while it can still be studied deliberately. Later, an invalid sequencing attempt or late pathology worklist delivery reopens its particular material, processing or workflow relationship without establishing its cause. Closure needs an identified version and a supported connection, or a narrower conclusion and an owner for what remains open.

Where did the study enter?

The route also explains why generating a report at the end of a study does not identify where the study began. Stored files can enter directly into computation, bypassing collection, extraction, library preparation and a new sequencing run. A comparison can examine the processing those files encounter. To carry it back to patient material, a reviewer still needs to know whether the upstream route produces the input the computation expects, and whether the evidence represented that handoff. In the other direction, an extraction study leaves open what a later filter suppresses.

A control has the same entry-point problem. A synthetic input introduced after extraction can challenge the processing and reporting steps it traverses. Extraction from patient tissue lies before its entry point. Introducing a control earlier lets it encounter more of the route, but does not make it representative of every variant class or report state. Its entry, covered segment, observed output and bypassed segment show where complementary evidence belongs. A partial-path control retains its value.

Reviewer Lens. The useful comparison is between a study's actual entry and exit and the conclusion requested from it. A strong archived-file result may close a computational question even while the full specimen-to-report claim is open. Reprocessing a pathology file has a corresponding limit relative to preparing and scanning a slide; an observation about later worklist delivery owns another part of the route. "Pass" cannot replace those distinctions.

Consultant Lens. The component owner knows which inputs and interfaces were exercised and what stayed constant. The system owner knows whether those conditions join the proposed output. Their comparison should precede the choice of a focused integration study or further use of existing evidence. A narrow study does not automatically require a full laboratory rerun, and software producing a report does not enlarge the study's entry point.

Which findings does the evidence represent?

Even a joined route cannot test every possible finding one by one. That makes the choice of representative findings consequential. An easy-to-place difference may not challenge the same weakness as a difference in a difficult sequence context. Both may be called variants, while demanding different evidence.

Three questions therefore meet at the output. Can the method detect this kind of difference? Does the evidence reach the consequential genes, regions and difficult contexts in the claimed content? Can the claimed specimen material and handling support that result? MM09 distinguishes these complementary method, content and specimen questions. [Source 1] A defensible bridge to an untested condition needs both a clinical and a technical reason for the representation.

The count behind an overall result must follow the same distinction. Many correctly represented reference positions can coexist with few informative opportunities to detect a claimed variant class. Correct reference-base calls, expected variant calls, specimens with valid reports and patients represented are different observation units and denominators. The panel's nominal target list does not settle its reportable territory: classes, regions, contexts, specimen conditions, detection boundaries and exclusions still determine what it can say.

Reviewer Lens. The difficult question is often whether the strong result covers the output at the consequential boundary. In tissue sequencing, method, content and material need to reach that same finding. In pathology, a reassuring average over easy images leaves a question about scanner and stain conditions near a weak-stain boundary. The separate anonymous observations from earlier chapters—two initial invalids among forty sequencing attempts and one late arrival among twenty-five pathology cases—prompt accounting by actual unit and condition. They establish no rate or cause for HelixBridge or PathSight.

Consultant Lens. Assay and pathology owners identify the class or image condition that makes an error consequential; the reference owner explains how expected status is assigned; the statistician identifies independent units and denominators. This lets the team preserve adequate support for a represented condition and assign the missing condition to an owner. Calling a set “representative” without that reasoning leaves the bridge unexplained.

Figure 10.1: Route-guidance retrieval briefly recalls the distinction among incoming information, configured service state, reference snapshot, transformation, selection and communicated output. Figure 10.2: Configured NGS examination chain returns to the tissue examination just taught. The route analogy is a memory aid, not evidence that clinical computation or interpretation works like navigation.

Book 1C - Chapter 10

Retrieval cue - route guidance information flow

Key takeaway

A displayed instruction depends on upstream information state.

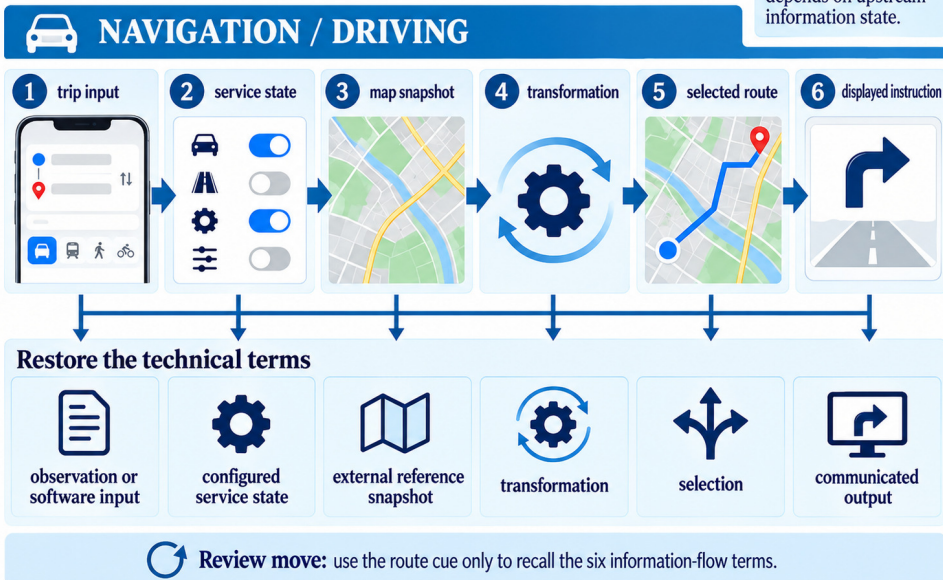


Figure 10.1. Configured NGS examination chain – Route-guidance transfer.

Book 1C - Chapter 10

Section 10 - Trace the NGS examination chain

A component study supports only the path it exercised.

Key takeaway

Do not join evidence across an untested interface.

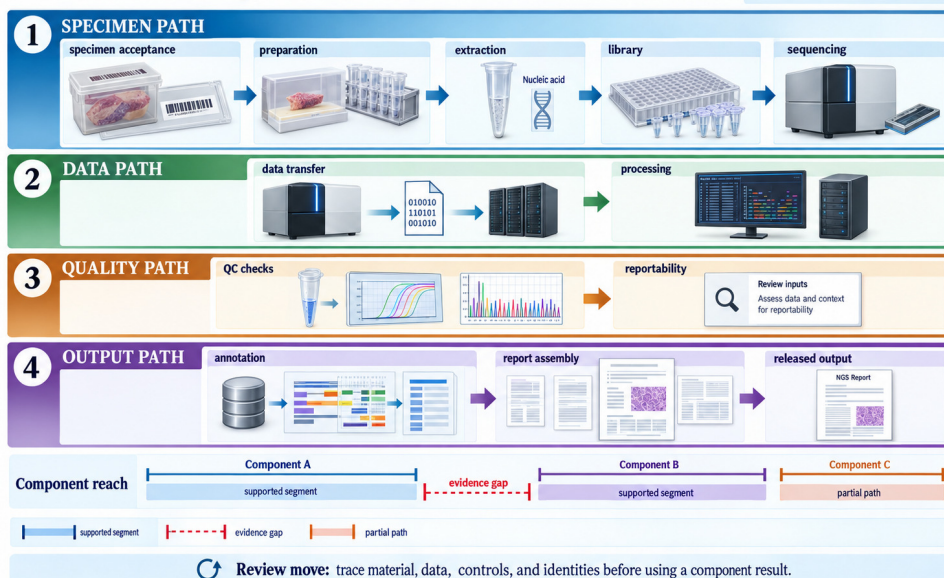


Figure 10.2. Configured NGS examination chain – Technical review plate.

Which state produced each record?

The Chapter 2 HelixBridge baseline now poses a precise integration question. Its analytical-validation record identifies **application 7.0, report rules R19 and knowledgebase DB-18**. A separate technical report identifies **DB-19** and describes strong overall agreement. The exact proposed commercial state remains unresolved. These are separate evidence and target states; the records establish no release sequence among them.

The plain names help keep the comparison readable: the **knowledgebase used in analytical validation** and the **knowledgebase in the separate technical report**. The latter report does not establish that every other component matched the earlier configuration or supply an affected-class denominator. The application identity therefore cannot stand in for a complete **configured product**, the working arrangement relevant to the comparison.

Attaching each record to its identified state creates a **configuration crosswalk**. A connection to the proposed use then asks whether material route, method, variant class, output and reference role also match. The supplier-managed publication job can publish a new knowledgebase without changing the application version. That capability establishes neither a particular publication nor a dated commercial release.

What would close the missing connection?

Suppose the crosswalk lacks the knowledgebase identity used by an old report. A new laboratory experiment cannot retrieve that old identity. The next useful action is to find its configuration record. If the state is identified but a summary hides class-specific observations, the needed action may instead be analysis of already collected data. Only a consequential question those records cannot answer necessarily calls for new evidence.

This gives the coverage map a practical order. The Chapter 4 claim leads to the Chapter 5 reference assignment: did it assign the finding this report makes? The Chapter 6 material route and Chapter 7 result type then identify what entered and what was produced. The Chapter 8 boundary and Chapter 9 variation ask whether the relevant class, rule and failed outputs were represented in that same state. A missing record calls for retrieval; an ambiguous version or material route calls for identity resolution; an uninterpreted subset calls for analysis; an unstudied claim-relevant boundary can call for prospective work. Record, software, laboratory, analysis and study owners have different parts of that task.

Figure 10.3: Evidence records attached to configuration states keeps DB-18, DB-19 and the unresolved commercial target apart. A drawn edge indicates a proposed evidence relationship to verify, not an automatic passing bridge.

Book 1C - Chapter 10

Section 10 – Attach records to configuration states

Evidence records do not join by product name.

Key takeaway

Useful records still need exact state links.

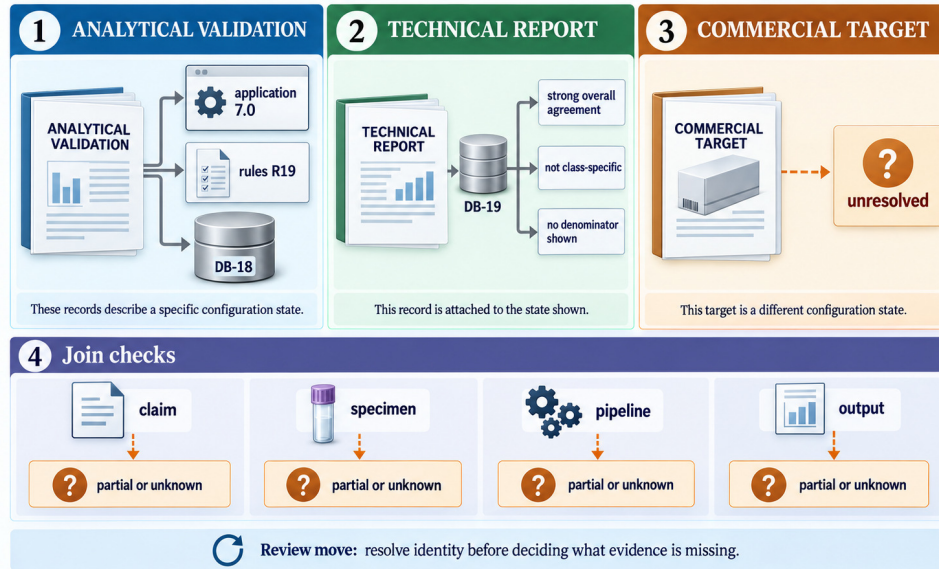


Figure 10.3. Evidence records attached to configuration states – Technical review plate.

What happened to the planned outputs?

Knowing the configuration still leaves a basic question: how much of the planned examination produced an answer? One specimen may be rejected before examination. A run failure can affect several specimens, while a specimen failure can leave others usable. Within one specimen, a region or output may remain unresolved while another can be reported. A later repeat may recover an answer without changing what happened initially.

Those paths need defined result states. A valid **negative** means that no reportable finding was detected under the defined procedure and conditions. Invalid, not assessed and nonreportable states do not make that assertion. Each product needs its own definitions, including when a partial report is permitted. A **disposition ledger** retains those states from planned output through first attempt, any repeat and final result.

MM09 supports checkpoints and exception or rerun handling through the sequencing route. The U.S. Food and Drug Administration (FDA) diagnostic-statistics guidance addresses transparent accounting within its two-outcome diagnostic-study scope. The following multilevel ledger is an original teaching example, not a form required by either source. [Sources 1, 3]

Eight patients and two planned classes

Eight anonymous patients enroll, with one specimen collected from each. The fictional protocol plans **two output classes, A and B, for every specimen**. The letters are placeholders, not named variant types. There are no supplied reference findings, clinical claims or acceptance threshold. The table answers what happened to each patient's specimen, from its initial state to any later result. A valid A or B could be positive or negative; the record does not specify which.

Patient's specimen	What happened first?	What happened afterward?
1	A and B valid	No repeat
2	A and B valid	No repeat
3	A and B valid	No repeat
4	A and B valid	No repeat
5	Examination attempted; neither class valid	Repeat produces valid A and B
6	A valid; B nonreportable under the fictional rule	No repeat; partial output retained
7	Examination attempted; neither class valid	No repeat; no final result
8	Rejected before examination	No examination or final result

Read the fifth row across: its first column describes the initial attempt, and its last column adds what happened on repeat. Both entries belong to the same patient's specimen. Reading only the last column would answer a different question from reading both.

Decision

How many specimens reached examination, and how many ended with complete, partial or no final output? How many class slots were planned, and how many were valid initially and finally? What does the repeat add? Which two denominators could describe complete-report availability, and why would neither establish correctness?

How the specimen and class counts reconcile

Reconcile the ledger

Specimens 1–7 reached examination: **seven examined**. Specimen 8 was rejected beforehand. At the final time point, specimens 1–5 have both planned classes valid, giving **five complete outputs**. Specimen 6 contributes **one partial output**. Specimens 7 and 8 contribute **two without final output**, although their routes differ: one was examined and the other was not. Five plus one plus two accounts for all eight patients' specimens.

The class count starts from the plan: **8 specimens × 2 classes = 16 planned slots**. At first, specimens 1–4 contribute **4 × 2 = 8 valid slots** and specimen 6 contributes one. The others contribute none, so **nine slots are initially valid**. Specimen 5's repeat fills its two existing slots, producing **eleven finally valid slots**. The five unfulfilled slots are B for specimen 6 and both A and B for specimens 7 and 8. Recovery is useful, but it creates neither a ninth patient nor new independently planned outputs. The first failure stays visible.

For complete-report availability, **5/8** asks how many of all enrolled specimens ultimately had both classes. **5/7** asks the conditional question among specimens that reached examination. Omitting the rejection changes the denominator and the question. Neither fraction is accuracy: the record supplies no reference process assigning which findings were truly present. “Valid” identifies a result state, not correctness. A first-pass availability question would also need its first-pass numerator; five final complete outputs cannot supply it.

Reviewer Lens. The recovered fifth specimen and the useful permitted partial result in the sixth deserve credit alongside the retained initial failure, attempted nonresult and pre-examination rejection. A sufficient ledger links patient, specimen, planned class slot, attempt, repeat, final state and analysis denominator. These fictional counts establish no HelixBridge result rate. The separate pathology note of one late case among twenty-five addresses delivery timing, not variant-slot accuracy.

Consultant Lens. Laboratory owners define states at run, specimen and target level; report-rule owners decide when partial output is permitted; analysts specify enrolled, examined, first-pass and final analysis sets before choosing a fraction. When operational failures return after use, that structure lets the team investigate the relevant denominator without erasing recovery or assuming a cause. Closure requires reconciled counts and an interpretation matched to the criterion.

How much can a favorable package support?

The ledger follows outputs through a study. Two independent anonymous cases now ask what conclusions the evidence itself earns. Neither supplies additional HelixBridge history.

The stored-file comparison

A team sends stored sequencing files through a candidate computational pipeline. File identities, the tested computational configuration and expected outputs are controlled. The comparison fits that processing question and meets its prespecified criteria. It evaluates no new collection, extraction, library or sequencing run.

Decision

What conclusion preserves this study's value? What relationship would be needed to use it within a wider specimen-to-report argument?

What the controlled file comparison earns

The bounded computational result is accepted: the candidate met its rule for the represented files and outputs from the controlled archived-file entry point. That is direct support for the processing question. The bypassed operations supply no new evidence here about what patient material would produce.

Its role in a wider change decision depends on a supported interface and still-applicable upstream evidence. If the consequential change is downstream, the first question is which output is affected and which earlier route still applies. An automatic laboratory rerun would skip that reasoning. Calling the file study a new whole-route validation would skip it too, even if its last step generated a report.

The matched analytical package

A second anonymous example stipulates a complete matched package: an explicit claim and final configuration; a fit reference process; intended specimens; relevant variant classes and difficult contexts; detection and reporting boundaries; and relevant variation. Justified criteria were set beforehand and met. Uncertainty is acceptable for the analytical claim, and all failures, repeats and limitations remain visible. The identified final configuration produces the evaluated output. No new benefit or treatment-selection conclusion is requested.

Decision

Can the specified analytical claim be accepted on these premises? Which boundary remains even if every premise is true?

When the analytical question can close

The bounded analytical conclusion can be accepted. This example supplies compatible links and adequate evidence for the specified configuration and claim. Other unanswered questions do not make that positive analytical conclusion provisional forever.

Its boundary remains the claim it supports. The package does not establish every finding's biological or clinical meaning, benefit in a new population, a therapy-selection claim or authorization. The two cases can therefore both be strong: the first supports its controlled file-processing question, while the second supports a joined analytical claim.

Reviewer Lens. Enough evidence means enough for the requested conclusion. The first study need not stand alone as whole-system validation to be useful; the second should close at its analytical boundary without borrowing that sufficiency for treatment selection. In pathology, processing archived images and establishing scanner-to-assistive-output performance under an intended workflow likewise require distinguishable evidence.

Consultant Lens. For the file study, software and system owners establish the input interface, affected output and continuing upstream applicability. For the matched package, the claim owner records the supported analytical wording and passes meaning and clinical-use questions to their owners. The practical difference is whether a connection still needs evidence or a supported question can be closed.

What does the added transcript change?

Return to the earlier HelixBridge crosswalk. The knowledgebase in the separate technical report added a transcript that makes a previously rare mapping anomaly execute more often. The earlier anomaly assessment concerned the knowledgebase used in analytical validation, where that transcript was absent. The validation application and report-rule identities are stated, but the source does not establish that every other component matched between the evidence states.

The separate technical report describes strong overall agreement. It supplies neither affected-class results nor a full analysis set or failure-and-repeat account. The proposed commercial state is still unresolved. Someone proposes closing analytical applicability using that aggregate statement together with the unchanged application label. This is

guided integration of a previously encountered case; the task is to make the relationships explicit.

Decision

What does the record establish, which earlier assumption changed, and which earlier or later evidence may remain useful? What smallest specific request could advance the conclusion? Which decision cannot yet be made without inventing a clinical error rate, a particular report-rule action, a released update or a regulatory outcome?

How the changed assumption sets the next request

Reasoning through the affected scope

The consequential difference is the new transcript, not the larger knowledgebase number. It changes a condition under which an existing anomaly executes. An assessment relying on rarity when that transcript was absent cannot simply be carried to the changed state. This establishes no affected patient result, clinical error frequency or failure of the assay's claim. Those would require observations the record has not supplied.

The first task is therefore to identify the compared arrangements: the analytical-validation state, the separate technical-report state and the proposed-use state. Version and dependency records belong to the configuration owner. Informatics can trace how the transcript reaches the anomaly and possible output classes; laboratory and interpretation owners can determine which specimen and finding contexts reach that behavior. Those connections identify the relevant part of the earlier anomaly assessment and the already collected later observations worth retrieving by class, context and report state.

Only then is the need for new evaluation clear. Earlier evidence may continue to support unaffected operations and outputs within its actual conditions. The separate technical report may contain useful observations too. Its favorable overall description, with the affected denominator unknown, does not establish performance in the reopened scope or identity of a commercial state.

A focused closure request therefore specifies the compared configurations, affected class and report pathway, fit reference assignment, prospective criteria, failures, repeats and uncertainty. If existing records cannot answer, the responsible owner can commission targeted new evidence. A knowledgebase update supplies no sample-size rule by itself.

The bounded conclusion is that **the change reopens the anomaly assessment and analytical-evidence applicability, while the full proposed-use configuration and**

adequacy in the affected scope remain unestablished. Both evidence records retain their supported roles. Overall agreement and one application label cannot close the broader claim.

Reviewer Lens. The review can preserve a supported unaffected segment while asking for the exact configuration and affected-output analysis, including its denominator and missing or invalid states. This gives the favorable records credit without assuming their summaries answer the reopened question. The supplier's publication capability remains a capability; it does not become a post-use release event.

Consultant Lens. Provenance, mechanism, tissue exposure, report consequence and analysis design have different owners. The consultant joins their answers at the same configured result, looking first for adequate existing evidence and then for the claim-relevant work still needed. Closure requires a documented proposed-use state and supported affected-scope decision. Agreement among teams on a favorable adjective cannot provide either.

What does the software answer leave for the laboratory?

The software specialist can return component and reference-data versions, affected modules, interfaces, regression-test inputs and the behavior actually exercised. The laboratory then has to connect relevant tissue inputs and analytical conditions to those behaviors. Interpretation specialists assess how a changed supported finding could reach or alter a report. Verifying code behavior does not establish clinical appropriateness; laboratory work does not identify the knowledgebase used by an old report.

MM09 ties change assessment to the nature and plausible performance effect of the change. That supports a scoped applicability judgment, with neither an automatic full rerun nor an automatic exemption. Book 1A develops the implementation questions further. [Source 1]

What could the tissue record contribute to a liquid branch?

HelixBridge Dx Liquid is a separate proposed branch related to the tissue program. Its broad purpose is still described as identifying variants associated with eligibility for a named therapy. Exact population, variant and treatment-claim elements remain unresolved. The tissue baseline retains its own identity.

The supplied liquid route begins with collected **whole blood**. Blood cells are separated from its liquid component, **plasma**. The analytical input is DNA isolated from that plasma without being contained in cells: **cell-free DNA (cfDNA)**. DNA originating from a tumor could contribute some of it. That possible tumor-derived fraction is **circulating**

tumor DNA (ctDNA); the case establishes neither its presence in a particular sample nor its amount.

The branch has a separate proposed configuration identity. Actual collection settings, processing, laboratory and computational components, knowledgebase, rules, release state and study results are not supplied. These facts describe a parallel design comparison, not a dated deployment.

Decision

Using that route and Chapter 6's material questions, what is changed, retained or reopened? How do collection and handling, derived sample and analyte, reference assignment, variant classes and analytical boundaries, first-pass and final states, and interpretation enter the comparison? For each possible reuse of tissue evidence, what relationship would need support? Does a continuing program name establish unchanged software, reference or clinical evidence?

Which relationships would justify reuse

Compare the two evidence maps

The established continuity is the program relationship and broad purpose. It supplies no equivalence of biological input, analytical behavior, report meaning or clinical performance. The table helps examine a possible reuse decision one relationship at a time; it does not order six independent validation studies.

Relationship	Tissue record	Proposed liquid record	Question for reuse
Collection and preparation	Tissue route incompletely specified	Whole blood to plasma	Which conditions are claimed and supported?
Input to examination	Exact tissue-derived input requirements unknown	Isolated cfDNA; ctDNA contribution unspecified	Which amounts, quality, backgrounds and input distributions matter?
Reference and study material	Exact methods and coverage unknown	No branch-specific package	Can reference, clinical and contrived materials support intended classes and conditions?
Detection and reporting	Actual class and context limits unsupplied	No branch-specific limits	What is detectable and reportable, with which conditions and failures?

Relationship	Tissue record	Proposed liquid record	Question for reuse
Examination configuration	Tissue states incompletely specified	Separate proposed identity; components unknown	Which operations and interfaces are shared, changed or unconfirmed?
Interpretation and claim	Broad therapy-associated purpose	Same broad purpose; exact clinical role unresolved	Does the meaning evidence address this material and use?

Take the first two rows together. A study starting with isolated cell-free DNA begins after collection and plasma separation. It can examine the downstream operation it enters but never observes those upstream steps. Tissue-input evidence cannot fill that gap. Nor does an excellent laboratory challenge material necessarily represent patient-input distributions or handling.

The input's biology also matters. Plasma cell-free DNA can come from more than one source, which affects how both a detected sequence and a negative result are interpreted for this specimen and use. A reference material can assign an analytical target without establishing how reliably that target occurs in plasma at the intended sampling time. A tissue detection limit or report rule cannot simply be renamed a plasma limit.

There is a possible non-tumor origin worth understanding. A lineage of blood-forming cells can acquire a genetic change and expand; this is *clonal hematopoiesis*. DNA from those cells can contribute to plasma cell-free DNA. Detecting a sequence there therefore does not automatically establish tumor origin. MM09 raises this background and reference question for liquid biopsy. It supplies no observed finding, frequency or mandated mitigation recipe for the fictional branch. [Source 1]

Even a genuinely unchanged downstream software operation needs a supported relationship to its new input. Reuse would require a matching configuration and evidence that the new inputs remain within the operation's supported range; neither is stipulated here. First-pass failures, repeats, partial results and final states need branch-specific definitions and accounting as well. Keeping a separate liquid map preserves these questions while allowing explicit links to tissue evidence where reuse is eventually justified. Planned liquid study identifiers remain plans, and the tissue summary retains its supported tissue scope.

Reviewer Lens. The collected, derived and isolated materials reveal where a study starts. A demonstrably shared downstream segment can receive credit when its input range and state match, while a study beginning at isolated DNA supplies no

whole-blood or plasma-preparation coverage. The record still establishes no released branch or tumor origin for a fictional result.

Consultant Lens. The specimen owner handles collection and plasma preparation; assay and reference owners define input, target and background assignments and reporting boundaries; informatics identifies shared operations. The claim owner decides what a liquid positive or negative is intended to mean. Separate maps let those owners close each reuse question with evidence or a narrower statement before advancing the liquid analytical claim.

What is the next answerable liquid question?

“Validate liquid biopsy” gives a study planner little to test. The collection question is whether the proposed whole-blood and plasma-preparation conditions preserve the properties needed at extraction in the intended sampling context. A reference question asks whether a reference or contrived material challenges a consequential low-input or background condition. A reporting question asks what a negative, invalid or nonreportable result would mean in the proposed use. Each needs an object, conditions, output and prospective decision. No universal input amount, detection limit, patient count or follow-up rule follows from the branch name.

The FDA's 2024 circulating-tumor-DNA guidance concerns biomarkers in early-stage solid-tumor drug development. It is not general device-validation authority for this branch. Germline sequencing guidance and another reviewed tissue product likewise cannot supply its missing assay and claim definition. [Source 4]

Which question should the team close next?

The three records now work together as an **end-to-end sequencing evidence and gap map**. The configuration crosswalk identifies the studied arrangement; the coverage map connects evidence questions to its claimed output; the disposition ledger accounts for the units that did and did not reach an answer. Their relationships let a reviewer be selective. A path ending at “strong agreement” needs the affected observations and denominator. An unidentified knowledgebase needs provenance. An unstudied material condition needs a claim-matched evidence plan.

A consultant can translate those gaps into retrieval, reconciliation, analysis or new evidence generation before asking the team to collect more specimens. Software specialists return versions, dependencies, supported behavior and exception states. Laboratory specialists identify material and quality checkpoints actually traversed. Interpretation specialists keep the analytical finding distinct from the clinical assertion. Pathology uses the same discipline for scanner, image, model, interface and reader states, with different scientific methods. Book 1A's implementation work and Book 1B's

security and integrity questions can concern the same route without replacing its analytical assessment.

Under the European Union's In Vitro Diagnostic Medical Devices Regulation, analytical performance joins **scientific validity** and **clinical performance** in an intended-purpose-specific performance evaluation. The analytical map alone supplies neither complete clinical evidence nor conformity. The detailed European framework belongs to Book 2. [Source 5]

Return to the release discussion

The initial judgment asked what product state the folder would need to identify. After the comparisons, which supplied identities can be kept, and which relationships still need evidence?

What the opening folder identifies

The opening names a tissue assay, the validation application and validation report rules. It does not identify a complete tested and proposed-use state. “Stable” supplies no laboratory configuration or proof of every analytical characteristic; “current” supplies no reproducible knowledgebase identity. The later records make the unresolved relationship concrete: the knowledgebase used in analytical validation, the knowledgebase in the separate technical report and the unidentified commercial target need distinct places in the comparison.

The folder can contain a strong component result for its entry point. A sufficiently matched package can support a coherent configuration-specific analytical claim. Asking either to support another state or claim requires a further relationship. The positive conclusions remain useful; no blanket verdict on either fictional branch follows.

Chapter 11 follows a finding across that analytical boundary into the question of meaning. Its identity travels with specimen and clinical context, source and configuration, reference limitations, report state and uncertainty. Support for a sequence difference alone has not established that it identifies a person likely to benefit from therapy.

Questions to carry to the next finding

- Why can two passing component studies leave a whole configured claim unsupported?
- Which record separates a missing result from a valid negative?
- Why does the fifth specimen's successful repeat add no patient and erase no initial failure?
- Why does the application version fail to identify the knowledgebase?

- What changed in the earlier anomaly argument when the later knowledgebase added the transcript?
- Which continuity is established for the proposed liquid branch, and which relationships still need evidence?
- What must accompany a finding into its interpretation assessment?

The relationships behind the seven questions

The component studies need matched states and interfaces. The ledger needs defined negative, invalid and partial outputs and both initial and final accounting. A repeat stays with its original patient and planned slots. The knowledgebase has an identity separate from the application, and the added transcript changes an exposure assumption without supplying an error rate. The liquid branch carries a program relationship and broad purpose, not tissue equivalence. The finding entering interpretation retains its context and evidence limits. Each answer identifies a specific relationship that “more validation” would leave unexplained.

Source trail

The sources support bounded tissue and proposed-liquid questions, not a transferable product recipe. Named-case facts come only from governed series records. The generic route, micro-cases and eight-specimen ledger are teaching constructions; licensed standards were consulted read-only and no licensed worksheet, figure, protocol or threshold is reproduced.

1. **Integrated NGS examination and analytical coverage:** CLSI MM09, third edition, §§3.3.1–3.3.4, 3.5.1–3.6.3 and 4.4.4. It supports integrated wet-bench/computational evidence, methods/content/specimen distinctions, distributed controls, exceptions and change-dependent assessment, with separate liquid-biopsy considerations. The official CLSI record still lists the third edition as of 24 September 2026; no MM09 numerical value or outcome is transferred.
2. **Bounded U.S. tissue tumor-profiling example:** 21 CFR 866.6080, (a), (b)(1)(ii)–(iv) and (b)(2). The eCFR displayed Title 21 current through 22 September 2026 when checked. This is device-type-specific; it neither classifies HelixBridge nor supplies a liquid framework.
3. **Transparent subject and repeat accounting:** FDA, *Statistical Guidance on Reporting Results from Studies Evaluating Diagnostic Tests* (2007), §§5 and 6.2–6.3. Its two-outcome diagnostic-study scope does not prescribe the chapter's multilevel NGS ledger.
4. **Liquid pre-examination boundaries:** CLSI MM13, second edition, and ISO 20186-3:2019, at the bounded locators in the Chapter 6 source packet. They

distinguish whole-blood collection, plasma preparation, cfDNA isolation, quality and handling questions, not fictional device settings. MM13 record; ISO record.

5. **Bounded EU handoff:** Regulation (EU) 2017/746, consolidated text dated 10 January 2025, Article 56 and Annex XIII Part A §§1.1–1.3. The dated consolidation is a documentation aid; this chapter is not a complete conformity assessment.

No numerical performance, release or authorization is asserted for either named fictional branch. The separate deployed NGS and DP notes remain anonymous post-use teaching observations, not HelixBridge or PathSight chronology.

CHAPTER 11

From Finding to Meaning

What should follow the detected finding?

A tissue test has detected a DNA (deoxyribonucleic acid) sequence difference. The report now needs a statement about what that difference means for the population being tested. One source says it is associated with Solid Tumor Type S. Another says the evidence is insufficient to support that association. The report build pauses while a clinical-evidence lead examines the records. Which statement could the product use?

The first step is to be sure both sources concern the same finding. This fictional finding is **GENE-Q c.742A>G (p.Lys248Glu)**, detected in resected Solid Tumor Type S tissue DNA from the product's intended Type S population. A **gene**, as Chapter 1 explained, contains information that can be copied into an RNA (ribonucleic acid) **transcript**; some transcripts supply instructions for making a protein. The finding's written name describes a sequence difference within that route.

The coding part, *c.742A>G*, says that an A has changed to a G at coding-sequence position 742 relative to the specified transcript. Here that transcript version is **TX-Q.2**. Without it, “position 742” would leave another reader unsure which coding coordinate to compare. The protein part, *p.Lys248Glu*, describes a lysine-to-glutamate difference at protein position 248 in this fictional naming system. That description does not yet tell us whether the protein works differently.

A further reference, **FA-2**, supplies the coordinate framework used to locate sequence findings: the **genome assembly**. Changing the assembly or transcript could change how an observation is written or matched to a source. Keeping both identities beside the finding makes the comparison reproducible. None of these coordinates establishes a Type S association, disease causation or treatment implication.

Decision

What three questions would help the clinical-evidence lead assess the two source statements before choosing report wording? What would those questions reveal that the source's name, date or confidence alone would not?

This is the **conflicting-source case (Case K)**, an independent fictional teaching case. Its gene, variant, transcript, assembly, tumor types, product and evidence records provide no real variant classification or treatment advice. Chapter 10 brought a

configured finding to the analytical boundary. The new question concerns the evidence connecting that finding to a meaning statement; Chapters 12 and 13 will examine product clinical-performance and treatment-related claims.

What reaches interpretation from the analytical route?

Chapter 10 ended with a configured finding and the limits of its analytical support. Before considering either source's opinion, what exactly has that argument handed over? The earlier reference process and specimen-to-report route give the new association question its starting point.

Decision

How did Chapter 5's reference process assign the comparison answer, and what did its scope, dependence between methods and unresolved disagreement mean for a correctness claim? Along Chapter 10's specimen-to-report route, where does external reference information enter, and where does an identified finding become an association question? What do the tissue and proposed liquid branches actually share, and what would be needed to transfer a meaning assertion?

Where the earlier evidence stops

A reference process supplies an answer through an identified role, assignment basis and scope. Dependence and discordance handling affect what a comparison can establish. An external database is a configured information input; it does not own the product's conclusion. The separate proposed HelixBridge liquid branch shares a broad purpose and program relationship with tissue, but Chapter 10 supplied no liquid finding, association assertion, interpretation package or reporting rule. Its whole blood □ plasma □ isolated cell-free DNA route therefore has no supplied tissue assertion that can simply be carried across.

What would the meaning statement add?

Suppose the configured tissue examination detects the sequence difference reproducibly. That is a valuable answer about detection. To add “associated with Type S in this population,” the report needs evidence about a relationship between the finding and the stated condition. Contrary observations may challenge that relationship; limited or missing evidence may leave it unresolved. Neither outcome changes what the detection record actually observed.

Even a supported association leaves a further question about how the protein behaves. A demonstrated protein effect would then leave another question: does that change contribute to cancer development, called **oncogenicity**, or cause disease in a defined context, called **pathogenicity**? A sequence description, protein effect and disease relationship are different propositions. Evidence for one cannot silently fill the next.

The same care is needed when the proposed meaning reaches a clinical decision. **Prognosis** concerns a patient's future course. **Actionability** concerns whether a particular clinical action is warranted. Therapy eligibility adds its own treatment-facing question, while **clinical utility** asks whether using the information improves decisions or outcomes. The Type S association alone settles none of them. These names identify boundaries the completed case record will need; they are not a classification or treatment method.

Applying evidence to a finding in a stated product context and reaching a bounded conclusion is **interpretation** as the word is used here. This teaching description is grounded in sources without claiming to quote a standard's formal definition. The solid-tumor guidance from the Clinical and Laboratory Standards Institute (CLSI), MM23 connects interpretation to the testing indication and addresses methods, limits and source references in reporting. It supplies a way to ask for support, not a conclusion for the fictional finding. [Source 1]

The distinction also reaches **digital pathology**. An image system may emit a feature flag and a pathologist may review it; the flag's existence does not establish a biological state or improved clinical decisions. The separate unnamed service in Chapter 1 recorded one late **worklist** arrival among twenty-five supported-image cases, with review completed. That observation concerns workflow timing. The other unnamed service recorded two initial invalid statuses among forty tissue-sequencing attempts, an observation about result availability. Neither supplies interpretation evidence for PathSight or HelixBridge.

Reviewer Lens. The useful first separation is between evidence identifying and supporting the detected output and evidence supporting its proposed meaning. A traceable sequence finding deserves credit at its analytical level. An image flag likewise needs its definition and execution records before its own biological association can be appraised. A reviewer who treats an invalid-result or late-arrival note as meaning evidence has crossed to a different question without a supporting relationship.

Consultant Lens. Before evidence collection, assay, informatics and clinical-evidence owners can agree on the finding and the exact association being proposed. A therapy or clinical-use question then has its own place in the plan. After use, an invalid sequencing result belongs with result-state investigation and a late image with workflow review. The meaning record changes when new information bears on its assertion; this keeps a narrow operational concern from needlessly reopening every conclusion.

What does “validity” refer to in this source?

A review note might say “clinical validity established” while leaving its object unclear. Does it concern a molecular difference's association with Type S, or how well a test method identifies or predicts a classification in the intended population? That uncertainty matters before any source terminology is borrowed.

In the European Union's In Vitro Diagnostic Medical Devices Regulation, **scientific validity** concerns the analyte–condition or physiological-state association. Analytical performance and **clinical performance** are separate, related evidence questions. A supported Type S association would therefore leave both the product's ability to detect the difference and its clinical performance to their own evidence. [Source 2]

CLSI EP49 also uses scientific validity for an analyte–condition relationship. It uses **clinical validity** for a *test method's* ability to identify or predict clinical classification in its intended-use population. The U.S. Food and Drug Administration (FDA) 2018 public genetic-variant database guidance uses that same phrase while discussing genotype–phenotype relationships supported by database assertions. The wording must travel with its framework and exact claim: a relationship, a source's assertion about it, or a test method's performance. [Sources 2–3]

This chapter keeps those questions distinguishable. Book 2 takes up the fuller European performance-evaluation appraisal; no European conformity decision follows from this discussion.

How did a study become the displayed statement?

Before accepting a database statement, imagine following its evidence backward. A repository receives a primary study record and retains where it came from, its population, specimen context and date. Collecting and maintaining such records is **aggregation** in the FDA public-database guidance's vocabulary. The stored record can now be found again, but storage has not endorsed its proposed relationship.

To make the record useful for the Type S question, someone must select it, check the finding and population labels, annotate the context and connect any index entry to the original study. That continuing work is **curation**. It helps the next reader recognize relevant evidence and avoid counting an indexed copy as another study. The remaining question is what the whole evidence set supports.

An evaluator answers that question by applying an identified method to the selected record and other relevant evidence, including contrary observations. This is **evaluation**. The resulting scoped statement is an **assertion**. Its confidence cannot come merely from appearing in a database field: a contrary study could limit it, or a necessary inaccessible supplement could leave it unresolved. These terms describe work in this source's vocabulary; they are not mandatory labels for every laboratory. [Source 3]

Now the product has a further decision. Does the assertion fit its finding, specimen, population and **report rules**—the controlled rules governing what may be communicated? Its interpretation may be narrower than the source statement. New observations, a revised evaluation method or a different population can reopen the link. Support, challenge, limited scope and missing access can lead in different directions; there is no automatic climb from detection to clinical meaning.

Figure 11.1: Route-guidance retrieval recalls only that external information has a version and is an input to a configured service. It cannot demonstrate source truth or a clinical association. Figure 11.2: Finding-to-assertion branches restores the actual evidence task: support, contrary evidence, inapplicability and missing access may lead to different bounded states before a separate product interpretation and report.

Book 1C - Chapter 11

Retrieval cue - external reference input

Key takeaway

A reference input does not own the conclusion.

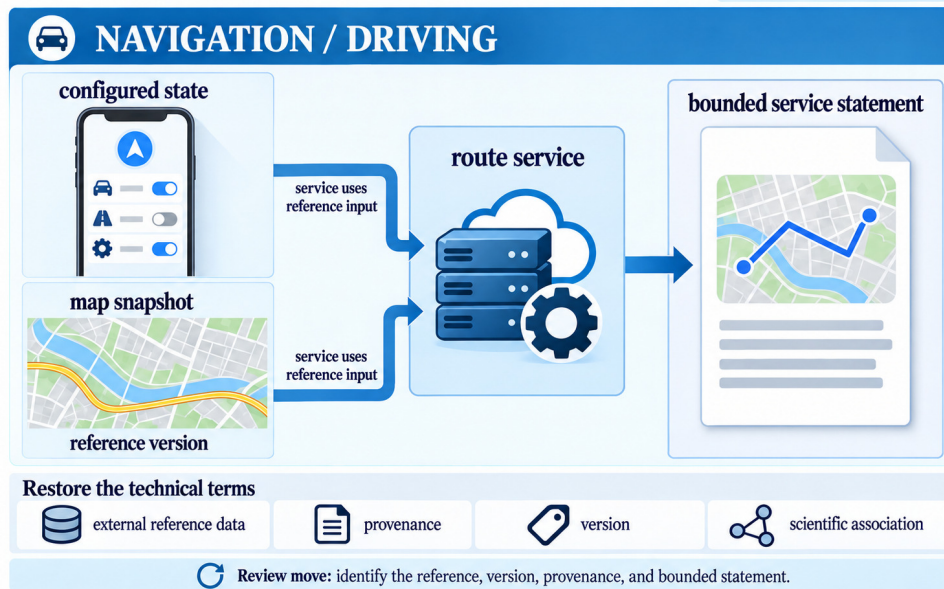


Figure 11.1. Evidence, evaluation, and bounded assertion – Route-guidance transfer.

Book 1C - Chapter 11

From finding to meaning

TECHNICAL MODEL

Key takeaway

Meaning is a bounded assertion,
not a property of detection.

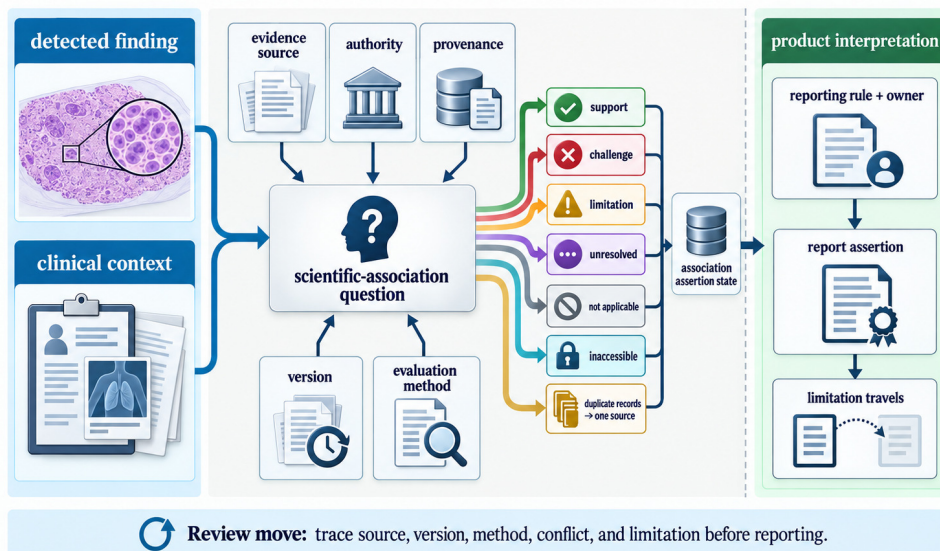


Figure 11.2. Evidence, evaluation, and bounded assertion – Technical review plate.

Reviewer Lens. A useful source review follows the primary observations through selection, duplication, evaluation and the final statement. An update notice establishes a changed source object, but the changed object still needs an appraisal. The same origin and method questions apply to an image-feature association using its own evidence. A sequencing source cannot supply that association simply because its provenance is excellent.

Consultant Lens. The source list, evaluator's method and decision, and product owner's report rule can be linked before release. That makes the next change manageable: a new study may affect the evidence, a revised method may affect the assertion, and a different product context may affect its application. Making duplicate and contrary records visible early allows a narrower report with an explicit limit to be considered as a legitimate endpoint.

Can an evidence level replace the method?

Suppose one source calls the association “high” and another assigns a numbered category to a therapeutic implication. Before comparing their strength, we need to know whether they classify the same proposition. The owner, admitted evidence types, handling of contrary evidence and method version all matter. A number used as a category name is not a portable quantity.

A reviewed crosswalk can connect specified categories within an exact scope; the later OncoKB example contains such a bounded FDA mapping. That does not make whole

hierarchies interchangeable or provide a method for the fictional Type S case. Here the useful stopping point is an identifiable assertion, evidence and method that a specialist can appraise for the next claim. A complete somatic classification or treatment-interpretation method lies beyond it.

Could another reviewer recover the same decision?

A regulatory decision may settle what action an agency took while having nothing to say about this Type S finding. Its **authority** concerns the role in which the source speaks. A primary study might address this finding and population closely yet have a limited design. That match is **applicability**. The source's standing and the proposition's fit answer different questions.

Once the question matches, the observations still need tracing. **Provenance** connects their origin to the displayed assertion. If an index republishes a study, it adds a way to find those observations. If two databases cite the same studies, their agreement does not create two independent observation sets. Even different author names cannot establish independence of the underlying patients, specimens or data.

Tracing the chain can fail at a very specific place. A landing page may open while the assertion text, evaluation method or necessary supplement remains unavailable. **Access** therefore belongs to an object and a decision date. Registered access may provide a complete preserved snapshot; a public page title may fail to reconstruct an earlier assertion. What matters is which object could be retrieved and which inference depends on the missing one.

Reviewer Lens. The issuer's professional or legal role is one field; the finding, tumor and population match are others. Counting studies comes only after their original observations have been traced. An image-system execution record can have excellent authority for its configuration and emitted output while saying nothing about biological meaning. A retrieval failure limits reconstruction; it does not establish that the inaccessible assertion was false.

Consultant Lens. The source reviewer can establish authority and scope, the curator can map originals and duplicates, and the records owner can preserve the decision-time assertion and method. A missing supplement then creates a precise retrieval request. Another broad search or specimen run would leave that same missing relationship untouched. Closure is possible when the required trace is present; otherwise the unresolved object and its product limit remain explicit.

Which change would alter the decision?

“Current database” cannot tell us what changed. A product update might alter which finding reaches interpretation, while a reference change might alter how the finding matches an external record. A source might add content without changing its method, revise one assertion, or change its evaluation rules. The product could separately change its communication rule. The following table locates the object whose version is needed when one of these questions arises.

Object with its own version	What that version identifies
Product configuration	The system producing and reporting the finding
Reference and naming state	The basis for identifying the finding consistently
Source content release	The collection available at that release
Individual assertion	The particular statement being appraised
Evaluation method	The rules used to develop or review the statement
Reporting rule	The product's controlled decision and communication logic

The middle three rows reveal why a single version number fails. The content release identifies a collection; the assertion version identifies a statement within it; the method version identifies how the statement was evaluated. A new collection cannot substitute for missing assertion text, and a later method listing cannot demonstrate that the old and new methods reach the same judgment.

Nor does a later version silently rewrite an earlier report. Reanalysis, reclassification and communication each require their own governed decision if an issued report is affected. The solid-tumor guidance CLSI MM23 supports tracing pipeline and source version or access date, limitations and report updates; it supplies no verdict about this particular product change. [Source 1]

What can a reconstructible assertion record conclude?

The evidence chain now gives a reason for an **interpretation provenance and support table**. One shared record identifies the finding; a separate row follows each distinct meaning assertion through its origin, intermediate indexes, evaluation method, versions and decision-time access to the product's communication decision. Keeping assertions separate prevents a source's reputation from replacing appraisal of its actual statement.

The four states below describe what such a row can establish. They are author-defined teaching states, not regulatory classifications or a universal variant tier.

State	What it says about this exact assertion
Supported with a boundary	The supplied evidence supports it within a stated limit
Unresolved	A material question or conflict prevents settlement
Unsupported	The supplied record lacks the support it needs
Not applicable	Its scope does not match the finding or context being evaluated

For instance, a proposed meaning with no applicable association evidence has missing support. An available assertion about another context has a scope mismatch. The former can be unsupported and the latter not applicable without either proving that a biological relationship is absent. The row's reason and limit do the explanatory work; the state label keeps that explanation visible.

What may the Type S report say?

The conflicting-source case now supplies its full packet. The finding was detected on **12 August 2026**. The identity table fixes what the two sources are being asked to address so that a specimen, coordinate or product change cannot slip into the comparison.

Field	Case K value
Finding	GENE-Q c.742A>G (p.Lys248Glu)
Transcript and assembly	TX-Q.2; FA-2
Specimen and context	Resected Solid Tumor Type S tissue DNA; stated Type S intended population
Product configuration	PX11-CFG-3.4
Analytical pipeline	PX11-AP-3.2
Reference-data version	PX11-REF-FA2.1
Nomenclature policy	PX11-NOM-1.0
Reporting rule	PX11-RR-2.2

For the transcript-and-assembly row, the question is whether a source addresses the finding written under those same references. The pipeline and reference-data rows locate the computational path; the naming policy identifies how the difference is written. The last row identifies the rule for the product's later decision. None of those

identities itself supports the Type S association. The question remains exactly that association in the stated population, without adding a biological effect, prognosis, therapy eligibility or match to any real database entry.

How many different observations are behind the sources?

The next table shows why counting citations would be misleading. Each record has a different relationship to the association question, and each access state is supplied for **15 August 2026**. These are fictional packet facts, not appraisals inferred from article titles.

Record and plain name	Origin and context	What it contributes	Access on 15 August 2026
E101 — direct study	Primary investigator record; Type S intended population	Supports the exact association	Complete archived record available
E102 — model-system study	Primary record outside the intended population; indexed by Source A	Indirect support with a context limit	Complete archived record available
E103 — contrary study	Independent primary Type S record; no shared authorship with E101 stated	Contrary to the exact association	Complete archived record available
E104 — duplicate index	Secondary index of E101; no new observations	Repeats E101	Index and original link available
E105 — inaccessible supplement	Citation states Type S; detailed population fields are in the supplement	Supporting chain cannot be reconstructed from supplied metadata	Restricted supplement inaccessible

The duplicate-index row is the simplest worked reading: its link is useful, but the observation count has not increased beyond the direct study. The contrary study is expressly supplied as independent. Its different authorship alone would not establish independence; that depends on the underlying people, specimens and data. Five record labels therefore do not mean five independent supportive studies. The inaccessible supplement leaves a reconstruction problem, not a negative result.

What does Source A actually assert?

Source A says **the exact GENE-Q finding is associated with Type S in the stated intended population**. It uses the direct study and the model-system study as support, identifies the contrary study, and lists the duplicate index. A has made the contrary record visible; whether its affirmative assertion is adequately supported remains the question to appraise.

The preserved source metadata makes the actual object available for inspection:

- **Operator and authority:** fictional Professional Consortium A; evidence committee; nonbinding, with no supplied legal or regulator authority.
- **Method and content:** A-METHOD-4.1; SA-2026.06 released 30 June 2026.
- **Assertion locator:** A-104.3 in preserved snapshot SA-2026.06-A104, record A-104, `association_statement` field.
- **Access:** public snapshot retrieved without login on 15 August 2026.

Those identities locate a statement about the specified finding and Type S context. They do not extend it to every possible use of the sequence label.

Is Source B making the opposite biological claim?

Source B says **the evidence is insufficient to support the exact GENE-Q finding–Type S association in the stated population**. It considers the direct and contrary studies already present in A's evidence; it includes the contrary record and identifies the inaccessible supplement. It does not count the direct study as establishing the requested association. A second database has appraised overlapping material, without replicating those observations.

Its wording matters: B asserts something about the *sufficiency of supplied support*, rather than demonstrated absence of the biological association. Its own decision-time object is identifiable:

- **Operator and authority:** fictional Laboratory Network B; review panel; nonbinding, with no supplied legal or regulator authority.
- **Method and content:** B-METHOD-2.7; SB-2026.07 released 18 July 2026.
- **Assertion locator:** B-77.1 in preserved snapshot SB-2026.07-B77, record B-77, `evidence_summary` field.
- **Access:** snapshot available through registered access on 15 August 2026; E105's required supplement inaccessible.

The association asserted by A and the insufficiency asserted by B both concern the same finding and Type S context. They remain different propositions when their states are assessed.

Which decision belongs to the product?

The clinical-evidence lead, identified in the packet as **PE-O1**, decides on **15 August 2026** under reporting rule **PX11-RR-2.2**. The rule permits reporting the detected finding but requires association meaning to be labeled **unresolved for the stated context**, with a supported limitation, when applicable sources make materially conflicting association assertions **or** a required evidence chain cannot be reconstructed. This is a supplied fictional rule, not a recommendation for a real patient's report.

Pause – make the source and product judgments

What state is justified for each source's *exact assertion*, and why? Can each be traced through its operator, object, locator and authority; original and intermediate evidence; content, assertion and method versions; release and access dates; supporting, contrary, duplicate and missing material; and applicability to the shared identity? What report statement does the product's rule permit, with which owner, date and limitation? Each sentence in that statement should be explainable through the particular evidence relationship that supports it and the information that could change it.

How the same finding supports different source judgments

The two source judgments and the report decision

Both completed records inherit the shared finding, specimen, transcript, assembly, pipeline and product configuration, including detection on 12 August. The clinical-evidence lead's decision is on 15 August. The tables keep this identity fixed while following different statements through the evidence to the same product reporting rule.

Source A: the affirmative association – record K-A

The first record asks whether A's affirmative association can be supported from the supplied packet. Its exact metadata remains consultable alongside its evidence relationships.

Field	Completed entry
Exact assertion and evidence job	Association of the exact GENE-Q finding, TX-Q.2/FA-2, with Solid Tumor Type S in the stated population; association support, not treatment selection
Source and locator	Professional Consortium A; SA-2026.06-A104; record A-104, association_statement
Authority and method owner	Fictional nonbinding consortium; evidence committee; no supplied legal or regulator authority

Field	Completed entry
Version and time	Content SA-2026.06; assertion A-104.3; method A-METHOD-4.1; released 30 June 2026; retrieved 15 August 2026
Access	Preserved public snapshot and underlying E101–E103 records available
Supporting provenance	E101 directly addresses the population; E102 is indirect model-system evidence
Contrary and duplicate evidence	E103 is contrary; E104 adds no observations beyond E101
Applicability	Exact finding and Type S context match; E102 retains its separate context limitation
Row state and basis	Unresolved: relevant support exists, but the supplied contrary evidence and disagreement with the applicable Source B assessment have not been resolved
Product decision	PE-O1, 15 August 2026, applies PX11-RR-2.2; no affirmative association statement is released from this row
Required limitation	Do not extend to biological effect, oncogenicity, pathogenicity, diagnosis, prognosis, therapy, actionability, utility, another context, or another version

The applicability row can be positive while the assertion state remains unresolved. A addresses the correct finding and Type S population. The direct study gives relevant support; the model-system study retains its separate context limit. But the duplicate index adds no study, and the contrary evidence plus B's competing applicable appraisal have not been resolved. Keeping A unresolved preserves its useful support without releasing an affirmative association from a record that cannot yet justify it.

Source B: insufficiency of support – record K-B

The second record asks a narrower question: is B's statement about the insufficiency of the supplied evidence itself supported? It needs a separate row because that proposition differs from biological absence.

Field	Completed entry
Exact assertion and evidence job	Insufficiency of the supplied evidence for the exact GENE-Q finding–Type S association; appraisal of association support, not proof of no association

Field	Completed entry
Source and locator	Laboratory Network B; SB-2026.07-B77; record B-77, evidence_summary
Authority and method owner	Fictional nonbinding network; review panel; no supplied legal or regulator authority
Version and time	Content SB-2026.07; assertion B-77.1; method B-METHOD-2.7; released 18 July 2026; retrieved 15 August 2026
Access	Registered preserved snapshot available; E105 supplement inaccessible
Evidence provenance	E101 and E103 overlap Source A's underlying records; E103 is contrary; E105 cannot be fully reconstructed
Missing and duplicate control	Missing population details and supporting chain for E105 stay visible; overlapping records are not new independent studies
Applicability	Exact finding and Type S context match
Row state and basis	Supported with a boundary: the stated insufficiency is supported by the supplied conflicting and incomplete record; it does not establish absence of the biological association
Product decision	PE-O1, 15 August 2026, applies PX11-RR-2.2; retain unresolved association meaning
Required limitation	No negative biological conclusion and no downstream clinical, therapy, utility, other-context, or other-version claim

The state row is a **positive bounded assessment**. Contrary evidence and the unreconstructible supplement substantiate B's insufficiency statement. Its newer date did not make it win, and missing population details did not prove a negative biological result. If the supplement became available, a reviewer would examine its actual population and support chain before reassessing the row. Changing “inaccessible” to “available” would reopen the appraisal, not settle the association by itself.

What the product can communicate

Both the materially conflicting applicable assessments and the incomplete required chain engage the supplied reporting rule. The lead can retain the detected finding at its analytical level while carrying the limit on its meaning:

Note

GENE-Q c.742A>G (p.Lys248Glu), TX-Q.2/FA-2, was detected under PX11-CFG-3.4. Its association meaning for the stated Type S context remains unresolved because applicable source assessments conflict and a required supporting chain is incomplete.

This is a fictional decision excerpt, not a complete clinical report or a real reporting recommendation. The owned rule and dated decision make a useful bounded conclusion possible while preserving A's supporting evidence and B's justified insufficiency. Neither biological absence nor therapy response follows.

Reviewer Lens. Reading the shared identity once allows attention to stay on what changes between source statements. The direct study and B's insufficiency judgment each deserve credit for their own contribution. Contrary, duplicate and inaccessible paths remain part of the product decision record. The same reasoning form can organize an image-feature question, but it cannot transfer this sequence evidence into one.

Consultant Lens. The curator can resolve the direct-study/index overlap and seek the missing supplement; the clinical-evidence owner can address the contrary study and the competing methods; the product owner can preserve the permitted disposition. The bounded decision can close now with a current access record, justified limited source conclusion and controlled report excerpt. An affirmative future association would require resolution of the open evidence relationship. If a later change affects meaning after use, the comparison must still name this finding, context and versioned reporting rule before communication is considered.

What does a recognition decision recognize?

A citation to a recognized database seems to promise a shortcut: the source has already been reviewed. The useful next question is which part of the source that decision covers. The FDA's database-recognition program can recognize a whole database or a subset. Its public list, checked on **24 September 2026**, includes **OncoKB**. The historical decision **Q191007**, dated **7 October 2021**, identifies the recognized subset as the **FDA-Recognized Content** tab: defined tumor-specific somatic alterations at FDA evidence Levels 2 and 3. Material elsewhere on the same site cannot acquire that tab's recognition by proximity. [Source 4]

The decision also identifies the method reviewed at that time: curation **standard operating procedure (SOP) v2.0, March 2021**. Within the recognized tab, it reviewed a mapping from qualifying specified OncoKB categories into FDA Levels 2 and 3. This gives the category numbers a bounded relationship. It does not recognize all biological, clinical or therapeutic material elsewhere in the database, or every therapeutic-option

procedure in the broader method. Nor does recognition of source evidence authorize a test that uses it. [Source 4]

What, then, does a later release tell us? On **24 September 2026**, the operator's public information endpoint identified data release **v7.5 dated 28 August 2026**. This identifies the collection reported as current at that check. To decide whether a particular statement fits a product, the reviewer would still need its actual text, provenance, evaluation and scope. [Source 5]

The operator's method archive supplies another object: an SOP listing of **v6.3 dated July 2026**. This chapter does not use that edition's technical content to decide a variant claim. Neither that listing nor the later data-release number establishes that FDA reviewed every later process or item. The difference also proves neither expansion nor withdrawal of recognition, nor unchanged recognition in all respects. A product decision requires the exact decision-time assertion and method to be reconciled with the recognized subset and scope. [Sources 4–5]

The real example supplies no hidden evidence for the conflicting-source case: its consortium, network and GENE-Q are fictional and unrelated to OncoKB. FDA's historical tumor-profiling fact sheet is likewise a dated policy illustration, not a universal oncology tier system. Its 2018 public genetic-variant database guidance has a defined scope; it supplies no general therapy-direction or classifier-software method. No real therapy or evidence level is assigned to a fictional product here. [Sources 3–4]

Reviewer Lens. A correct recognition citation can be useful. Its contribution becomes clear when the recognized subset, assertion, decision-time release and reviewed method scope are identified beside the product's finding and population. The difficult judgment concerns that match, which a badge cannot settle. An image-derived feature needs its own modality-appropriate association source and authority; this genetic-variant recognition program cannot fill that role.

Consultant Lens. Separate records for the regulator's decision, operator's content release, operator's method listing and individual assertion make the responsibilities visible. The database-governance owner can reconcile lineage and access, the clinical-evidence owner can appraise applicability, and the product owner can control the final statement under the identified configuration and report rule. A reconstructible, applicable assertion can then receive credit as a bounded input without making the citation carry an unreviewed claim.

What changed: the context or the available source?

The next two records retain much of the conflicting-source packet while changing a different dependency. Each poses a fresh decision before its feedback. The unchanged fields remain useful; the question is which conclusion still follows from them.

The context-change case – Case L

The same fictional sequence description, transcript **TX-Q.2** and assembly **FA-2** remain. The specimen is now resected **Solid Tumor Type T** tissue DNA from the Type T intended population. The product record identifies configuration **PX11-CFG-3.5**, analytical pipeline **PX11-AP-3.2**, reference data **PX11-REF-FA2.1** and reporting rule **PX11-RR-2.2**.

Both sources retain their assertions, methods, content versions, release dates and access states from the earlier packet. Their assertions remain scoped to **Type S only**. The direct, model-system, contrary, duplicate and supplement records—E101–E105—retain their supplied provenance and access, and no Type T association evidence is added. The clinical-evidence lead **PE-O1** decides on **16 August 2026**.

Decision

What state fits each source's assertion for the Type T question? What may the product conclude, and which evidence relationship would it need to go further?

How a known scope mismatch limits the conclusion

Both assertions are **not applicable to the Type T association question**. Keeping the sequence notation the same preserves the identity of the written difference; it does not turn the sources' Type S populations into Type T populations. A's Type S assertion has not thereby become false. B's Type S insufficiency statement supplies no negative biological result for Type T either.

The product cannot report the Type S association as Type T meaning from these records. The missing relationship is an applicable, traceable Type T evidence chain. The detected finding can remain only under the product's own analytical and reporting conditions; these sources add no Type T meaning. If a later Type T source becomes available, the clinical-evidence owner must appraise its actual population and evidence. Changing the tumor label in the old record would not provide that bridge.

The unavailable-current-source case – Case M

This separate decision returns to **Type S**. The product record now identifies configuration **PX11-CFG-3.6**, with analytical pipeline **PX11-AP-3.2**, reference data **PX11-REF-FA2.1** and reporting rule **PX11-RR-2.2**. The same clinical-evidence lead **PE-O1** decides on **20 August 2026**.

Source A's current metadata identifies content **SA-2026.08**, method **A-METHOD-4.2** and assertion **A-104.4**. Its current assertion snapshot and method cannot be retrieved

at decision time. No current assertion text is supplied. The preserved earlier assertion **A-104.3** and method **A-METHOD-4.1** are explicitly superseded.

Source B remains assertion **B-77.1** in content **SB-2026.07** under method **B-METHOD-2.7**. Its snapshot is available; the required supplement **E105** remains inaccessible.

Decision

What can A's old record establish? What belongs in the current-source access field, and what state fits the current assertion? What product decision follows without guessing the unavailable text?

What the preserved history can still support

The old A snapshot reconstructs a **historical assessment**. Its value remains, but it cannot supply the current assertion text or evaluation method. Current A support is **unresolved** because those necessary objects are unavailable. That is different from “no current support,” “the same support,” or a known change in direction or strength. Metadata identifies the missing objects; it does not disclose their contents.

B's **supported-with-boundary insufficiency conclusion** remains usable within its unchanged scope. The required current chain still cannot be reconstructed, so the product's reporting rule requires unresolved Type S association meaning. Preserved superseded history and current retrieval failure remain separate facts. The open question concerns version and access, whereas the previous case concerned a known population mismatch.

Reviewer Lens. In the context-change case, the finding notation stays constant while the intended tumor population changes. In the unavailable-source case, the history and B's bounded current assertion remain useful while current A text is unknown. A defensible judgment identifies both the changed field and the usable evidence. The same reasoning distinguishes an image-feature definition from the context in which its meaning is proposed, using the image system's own records.

Consultant Lens. Population changes call for prospective scope checks; source revisions call for preserved historical snapshots. Here the clinical-evidence owner needs an actual Type T association search and appraisal, while the source or records owner needs A's current assertion and method for the separate Type S decision. The product owner can keep the unresolved report rule in force meanwhile. A new wet-lab run alone repairs neither gap. Each existing bounded decision can close while its precisely identified further question stays open.

Who can resolve the missing relationship?

The unavailable-source case makes the value of separate responsibilities concrete. A molecular specialist can check the finding's representation and biological context. That does not recover a missing source snapshot. A data or software specialist can reconstruct the relevant snapshot and product configuration, while the clinical-evidence specialist examines its association evidence, evaluation method and limits. The report owner then determines how the supported conclusion and uncertainty travel under a controlled rule. One person may hold several roles, but the questions remain distinct: pipeline detection, database presence and committee approval each establish different things.

This separation also helps when something changes after use. If the finding and analytical configuration stay the same while an association assertion changes, the first question concerns the interpretive effect; a full wet-lab rerun is not automatic. If a reference or mapping operation instead changes which finding reaches interpretation, source-content review cannot answer that upstream analytical question. The decision-time report and its supporting versions must be compared before deciding on reanalysis, report amendment or communication. No universal duty or interval is supplied here.

Which object is missing for the familiar products?

The HelixBridge tissue baseline in Chapter 2 has a broad purpose: identifying variants associated with therapy eligibility. It identifies the validation application **7.0** and validation report rules **R19**; the initial package calls its knowledgebase “current.” A supplier-managed publication path describes a capability. Neither that wording nor those component labels supplies an actual finding, association assertion, evaluated source, interpretation result, decision-time rule behavior or commercial release.

The knowledgebase used in analytical validation **DB-18** and the knowledgebase in the separate technical report **DB-19** identify two evidence states. Their mapping-anomaly contrast does not establish a change in a finding's meaning. The separate proposed **HelixBridge Dx Liquid** branch has a different material route: whole blood → plasma → isolated cell-free DNA. Circulating tumor DNA could be a tumor-derived fraction of that input. No actual liquid finding or assertion is supplied, and tissue support cannot become liquid support merely by changing its name.

Decision

For each branch, what must be identified first, which decision-time assertion snapshot, method and access state would follow, and which conclusion would depend on them? What makes that request specific to the branch?

Why a generic database request would miss the first dependency

For tissue, the first object is an **actual finding with its specimen context and product state**. An exact decision-time assertion, method, underlying evidence, access record and product reporting rule could then be requested to support its proposed meaning. For liquid, the intended claim, specimen and analyte path, configuration and finding must first be defined. Only then can a liquid-applicable assertion and evaluation chain be appraised. A tissue assertion is not that chain.

These requests are useful even though neither named branch has an interpretation conclusion available now. The independent fictional cases and real OncoKB example cannot populate their missing entries; neither can an event from Book 1B.

Reviewer Lens. The tissue purpose and component labels deserve credit at their supplied level. “Current” leaves version identity open, and an available publication path leaves assertion content open. For the proposed liquid branch, the changed specimen and possible non-tumor sources of cell-free DNA raise their own meaning questions. **PathSight** retains its assistive role and the pathologist's diagnostic responsibility. Its support claim and the separate late-worklist note supply no feature's biological meaning.

Consultant Lens. Product and clinical-evidence owners can define the branch's finding and claim before commissioning a source search. The records owner can then seek decision-time snapshots and methods, with the product owner identifying the report rule to which that evidence would attach. After deployment of an actual configuration, a source update calls for comparing the changed assertion with affected report claims. It cannot be assumed to change therapy meaning. Existing component identification stays positive while an unprovided assertion stays open.

What does an emitted image label establish?

The **image-feature case (Case N)** is another anonymous fictional fixture, separate from PathSight. A configured image system emitted “**IF-17 present**” for one stained-tissue image in **specimen class Z** on **22 August 2026**. Acquisition made the digital image, preprocessing changed its input representation, and a feature model produced a flag. A definition record says what that output label names; a reporting rule governs communication.

The following record identifies those steps so that an interpretation can be attached to the output actually supplied.

Identity field	Supplied value
Product configuration	IMG-CFG-5.6

Identity field	Supplied value
Acquisition state	IMG-ACQ-2.1
Preprocessing	IMG-PRE-3.0
Feature model	IMG-MOD-4.2
Feature definition	IMG-FD-1.3
Reporting rule	IMG-RR-1.1

The feature-model and feature-definition rows answer different questions. One identifies what produced the flag; the other identifies what its label names. The acquisition and preprocessing identities locate the input route. These fields identify a system state without establishing a biological association.

On **22 August**, the internal definition record **IMG-FD-1.3**, owned by the fictional product-configuration owner, defines the label. The internal execution record **IMG-RUN-88**, owned by the operations owner, traces the configuration and emitted output. Neither record supplies feature-detection performance or biological meaning.

Someone proposes the report statement “**IF-17 present indicates Biological State B in specimen class Z.**” No association source, evaluation method, underlying study evidence, favorable or contrary result, or applicability assessment is supplied for it. Decision owner **IMG-O1** applies the fictional reporting rule **IMG-RR-1.1** on **22 August**. The rule forbids turning the flag into a biological or clinical statement without an applicable, traceable association package and a separate product interpretation.

Decision

What provenance exists for this assertion, what is missing, and what state fits its proposed meaning? What may the product retain under its rule?

How output provenance supports a bounded decision

The completed record follows the proposed biological statement from the supplied internal records to the owner's decision. It keeps an identifiable output visible while asking what supports the additional meaning.

Field	Completed entry
Exact meaning assertion	IF-17 present indicates Biological State B in specimen class Z
Available provenance	IMG-FD-1.3 defines the label; IMG-RUN-88 traces output and configuration; both internal records available on 22 August

Field	Completed entry
Authority of those records	Fictional configuration and operations records, not scientific or clinical association evidence
Missing support	Applicable association evidence, its evaluation method and source versions, underlying study provenance, and favorable/contrary evidence appraisal
Applicability	Cannot be assessed from a missing association package
State	Unsupported meaning: output definition and execution do not establish the proposed biological relationship
Product decision	IMG-O1, 22 August, under IMG-RR-1.1: retain the configured feature output as such; do not report Biological State B
Limitation	Neither successful execution, feature-detection performance, biological meaning, nor clinical use may be substituted for another evidence job

The available-provenance row preserves a useful positive result: the internal records trace **which label was emitted and by which configuration**. They do not establish how often that label is correctly detected, or which biological state it indicates. The proposed meaning is **unsupported from this packet**, without being disproved.

A future association package would need to match class Z, the defined feature output and the exact proposed statement, followed by interpretation under a controlled product rule. This generic case supplies no modality-specific digital-pathology classification method. Successful execution, detection performance, biological meaning and clinical use keep separate evidence requirements.

Reviewer Lens. The definition and execution records can be accepted as scoped output provenance. An additional image-feature meaning claim would require its own applicable association package, alongside a separate performance record. A post-use late arrival addresses timing and visibility, not biological truth; biological evidence would not establish timely arrival either.

Consultant Lens. The image-system owner can preserve the feature definition and configuration trace, the clinical-evidence owner can define the exact State B question and source requirements, and the decision owner can control report wording. The present decision can close with the configured flag retained and the biological statement withheld. A prospective study or source appraisal is needed only if that additional meaning remains an intended claim.

Which opening questions now matter?

The paused tissue report began with two source statements about one finding. Which three questions now seem most useful before selecting wording? Do they reveal what each source asserted, the relationships among its observations and its decision-time evidence, and the separate product decision?

How the initial questions become an assessable claim

Useful questions identify the exact assertion; whether original observations overlap and contrary material is addressed; and whether the source, method and required evidence were accessible at the decision date. The product's reporting rule remains a separate question for its owner. “Which database is better?” becomes assessable when narrowed to whether A's preserved affirmative assertion under its stated method supports this Type S relationship in light of the contrary study and B's insufficiency assertion.

A reproducible decision requires no guessing about finding, population, version or assertion. The shared identity supplies the first part. Each assertion record supplies its exact claim and **evidence job**; operator, object, authority and locator; original and intermediate evidence; distinct content, assertion and method versions; access at the decision date; supporting, contrary, duplicate and missing material; applicability and justified state. The product adds its own rule, owner, date and limitation.

A duplicate index counted as a new study breaks one relationship; hidden contrary evidence breaks another. Replacing a preserved historical statement with current content, assuming authority proves applicability, or copying a source directly into a report creates different defects. Each calls for reassessing that particular relationship. There is no percentage score. B's sound, bounded insufficiency appraisal remains positive even while an affirmative association is unresolved.

Questions that test the distinction

Detection, association appraisal, interpretation and reporting now have connected but separate jobs. Can the following questions be answered while keeping each positive result at its own scope?

1. How can the available evidence support an insufficiency statement without proving no biological association?
2. What distinguishes a Type T-inapplicable assertion from an inaccessible *current* Type S assertion?
3. Why are two databases citing the same primary records not two independent observation sets?
4. Which three versioned objects were deliberately separated in the OncoKB example?

5. What does an image-output trace establish, and what would biological interpretation additionally require?

What the five distinctions preserve

An insufficiency statement concerns the support supplied; it need not prove biological absence. In the Type T case, the scope mismatch is known. In the later Type S case, current A's assertion is unknown because its text and method cannot be retrieved. Independence belongs to the original observations, rather than the number of databases or the difference between author names.

The OncoKB example separates the operator's data release, its method listing and the FDA-reviewed method and scope. The image records trace an emitted label and configuration; detection performance and biological meaning remain additional questions. A biological interpretation would need an applicable association package and controlled product interpretation.

A larger dataset that fails to address the meaning assertion cannot repair its missing evidence relationship. Chapter 12 follows the finding into a product clinical-performance study designed around intended use; Chapter 13 examines treatment-related claims and bridging.

Source trail

1. **CLSI MM23, second edition (2025)**, §§7.1.5.4, 7.2.1 and 7.3.2.4–7.3.2.8, supports bounded solid-tumor interpretation/reporting, version and access traceability, limitations and report updates. **MM09, third edition (2023)**, §4.2.8, supplies molecular-system context; its substantially germline-oriented material is not a complete somatic method. Licensed copies were inspected read-only; no proprietary table, figure or report example is reproduced.
2. **CLSI EP49, first edition (2026)**, §§1.4–1.4.1, gives its own terminology. The IVDR base act, Article 2, Article 56 and Annex XIII, supplies the separate EU evidence terms; the dated consolidated display is a navigation aid, not itself the controlling legal text. This is not a full EU performance-evaluation appraisal.
3. **FDA (2018)**, *Use of Public Human Genetic Variant Databases to Support Clinical Validity for Genetic and Genomic-Based In Vitro Diagnostics*, §§III–V and VI.C, is nonbinding guidance for defined public genetic-variant database uses and genotype–phenotype assertions, not a universal therapy or classifier method.
4. The FDA recognition list, checked 24 September 2026, and OncoKB decision Q191007, 7 October 2021, §§1–3.3 and 4.1–4.3, establish the listed, partial FDA-tab scope, bounded category mapping and reviewed SOP version. They do not authorize any fictional product or every OncoKB assertion.

5. OncoKB's information endpoint and SOP archive, checked 24 September 2026, supply the operator's v7.5 data date and v6.3 SOP-listing metadata only. They are not an appraisal of a real variant or proof of later FDA review of technical changes.

Cases K, L, M and N, their values and row states are original fictional teaching material. The recurring tissue and proposed liquid discussion uses only their separately supplied case facts.

CHAPTER 12

Clinical-Performance Study Design

What happened before this table reached your desk?

A patient reaches a clinical decision point. A proposed test examines that person's specimen and returns a result that might help classify a condition. Before a study can tell us how well that works, people must enter it, specimens must arrive, the specified test must run, and a separate process must establish whether the condition is present. A person can stop at any of those steps. The final table shows only the comparisons that reached it.

A fictional report opens with a neat positive/negative table and concludes that the test's clinical performance is established for its intended population. A colleague asks you to sign. The test being evaluated is the **index procedure**; its table cannot yet tell you who entered, how condition status was assigned or who disappeared along the way.

Decision

Which three records would you inspect first, and what finding in each could change your decision? What relationship would each need to reconstruct beyond the broad labels “protocol,” “data” and “statistics”?

The anonymous packet later in this chapter lets you verify the arithmetic and see what the arithmetic leaves unanswered. It is not a HelixBridge or PathSight study.

Follow the person behind the percentage

The first missing relationship runs from the patient's decision to the question the study intended to answer. Which people and setting did it address? Which configured procedure, specimen or image route and separate condition-assignment process would make its outcome relevant? These belong together in the **study definition**. They identify what the study hopes to learn before a spreadsheet selects anyone.

That definition creates a second question: did each enrolled person reach the intended comparison? A specimen may be unavailable, an attempt may fail, a valid output may remain nonbinary, or the reference may remain unresolved. A **disposition ledger** follows people through those stages and preserves both first and final outcomes. It explains how the table acquired its denominator.

Once the denominator is visible, a third question becomes possible: what can this calculation establish for the claim? The **analysis and applicability record** connects the selected observations, the calculation and its uncertainty to the population and workflow in which the conclusion will be used. It can give a bounded estimate full credit while distinguishing a recoverable reporting gap from a population the study never represented.

Together, these three records form the chapter's clinical-study design and analysis critique. They are an author-designed learning aid, not a form prescribed by the International Organization for Standardization (ISO), the Clinical and Laboratory Standards Institute (CLSI), the U.S. Food and Drug Administration (FDA), or a diagnostic-study reporting checklist. A useful critique explains which conclusion the evidence supports and which relationship remains open. It leaves study authorization and specialist clinical and statistical judgments with their responsible owners.

Which earlier boundary matters here?

Decision

How do intended use, role, patient, setting, user and decision from Chapters 1, 3 and 4 define this study question? What makes Chapter 5's reference a way to assign the target condition, rather than simply another method? How do Chapter 6's tissue route and separate proposed whole blood □ plasma □ cell-free DNA route differ, with tumor-derived circulating DNA only a possible fraction of the latter? Where do Chapter 8's valid equivocal, invalid, no-result, repeat and unresolved-reference states belong? How do Chapter 9's population, specimen, site, user and configuration differences affect the comparison? Why does Chapter 11's support for a finding's association leave the product's clinical performance to be established? Which records would you join first?

How the earlier boundaries connect

Intended use supplies the clinical question, while the reference determines what a comparison can mean. The specimen or image route and configuration identify the actual test; the result-state account shows who can contribute to an estimate. Variation and provenance then limit where that estimate applies. The proposed liquid route makes the need for those links visible: recalling its material path supplies no liquid study and carries no tissue conclusion across by itself.

What would a final positive result tell us?

Suppose the claim concerns people tested when a clinician must classify a condition, but the study selects only specimens that are easy to classify. Every examination could run correctly. Would that establish the claimed result for all eligible patients? The difficulty begins before calculation: the study may have observed a narrower group than the clinical question concerns.

The planning question is therefore what must connect the proposed use to the observations. For clinical-performance studies of in vitro diagnostics using human specimens, the good-study-practice standard ISO 20916:2019 gives requirements for planning, conducting and reporting. It connects intended use, objectives, endpoints, comparator, specimens and statistical planning. The edition was confirmed in 2025. Its scope excludes analytical-performance studies, and it provides no product-specific pass rule. The nonbinding clinical-validity framework CLSI EP49 helps organize the complementary population and analysis questions. [Sources 1–2]

The question survives even when an observation is missing

For the binary example, a clinical question might be: among people who have the defined condition at this decision point, what fraction would receive a positive *final* result under the specified test-and-repeat strategy? That is a **target quantity**. Its words identify the people, the one-person unit, the configured procedure, the condition-assignment process, the time and the result policy. “Final” belongs in the question because allowing a repeat after failure differs from asking what happens on one attempt.

Now suppose an enrolled person supplies no specimen. There is no paired observation for that person. The original question still concerns the intended people, but the group available for this calculation has changed. That group is its **analysis set**. Choosing a rule for combining the available observations—the **estimator**—does not silently redefine the clinical target.

Before selecting that rule, the following questions make the target precise enough to examine. Their order follows the patient from proposed use to an observed outcome and then to its summary.

Question about the proposed use	What needs to be specified
For whom should the conclusion hold?	The target population
What can carry one outcome?	A participant, specimen, lesion, result, reader or other stated unit
What does the test strategy include?	Configuration, output categories and intended repeats

Question about the proposed use	What needs to be specified
How will the condition be established?	The target condition and the reference process that assigns it
What will be observed, and when?	The outcome at the clinically relevant time
How should those outcomes be summarized?	The quantity that answers the stated question

The test-strategy row shows why the table matters. A strategy that permits another attempt after an invalid result can give a person a different final outcome from a one-attempt strategy. Both can be legitimate study questions, but the target must say which is intended. In a pathology-assistance study, the same questions also need the reader's role and the relevant reader–case unit.

The linked **analysis specification** then states the actual sampling frame and analyzed people, the estimator, any weighting, missing and nonbinary-state treatment, dependence among observations and an uncertainty method. *Estimand* is a formal name sometimes used for a target quantity. The target-and-analysis description here is an author synthesis, not a quoted standard schema, and a study-specific plan may need more detail.

For the missing-specimen example, a complete-pair estimate could be precise for people whose specimens arrived while giving a distorted answer for all eligible patients. The reason specimens were unavailable now matters. Capturing that reason and planning how to examine it before outcomes are known gives the analysis a basis that redefining the population afterward would lose.

What is observed for one person?

The study must record an outcome before summarizing a group. A participant's final positive or negative index result, paired with separately assigned condition status at the specified time, is an example of an **endpoint**: the observed variable addressing the study question. Asking what fraction of the reference-positive participants have a positive result gives a summary, **sensitivity**. The endpoint and its summary do different work. A question about an analyte, a clinical condition or a reader's decision may require different reference and endpoint definitions even when each table uses plus and minus signs.

The first attempt remains part of the story even when the endpoint is final. If a prospectively defined rule permits a repeat after an invalid attempt, the final-result measure can address that strategy. Keeping the first failure visible also shows what the person and laboratory experienced on the way to the endpoint.

Figure 12.1: Population, study selection and analysis reinforces the separation: the intended patient group narrows through eligibility and acquisition into an analysis set, while the target quantity and analysis method remain distinct. Its digital-pathology, tissue and liquid lanes illustrate selection routes; they do not supply a new named-product study or transferable liquid evidence.

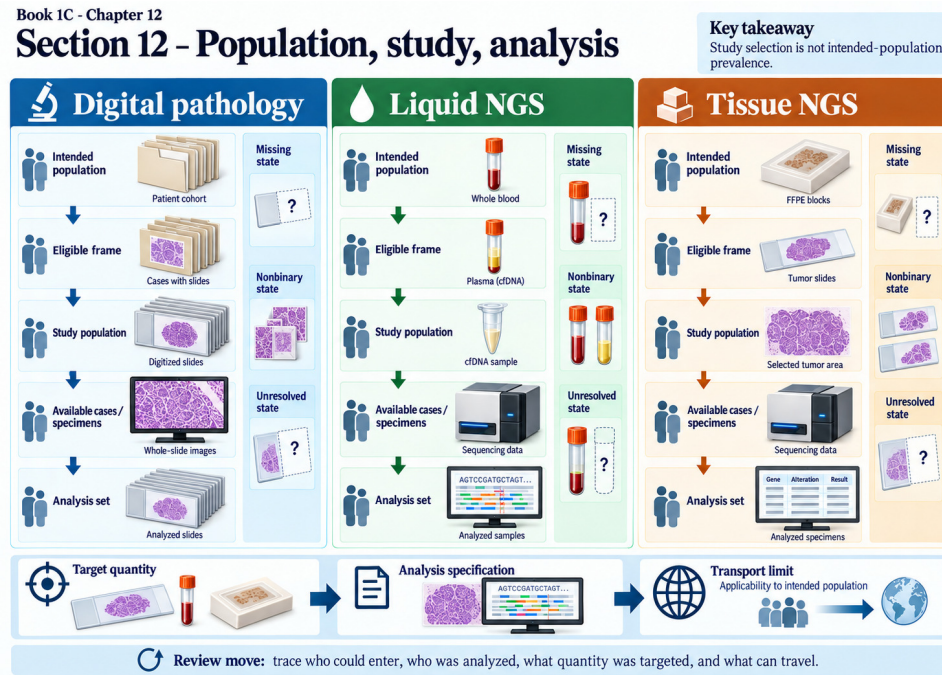


Figure 12.1. Population, study selection, and analysis set – Technical review plate.

Reviewer Lens. The first useful question behind a tissue-sequencing percentage is which clinical decision it addresses. A patient-level target, a specimen-level target and a finding-level target need different records, including a reference that establishes the corresponding condition. For assistive pathology, the pathologist's role and reader–case unit are equally consequential. A well-defined prospective study can be adequate for a limited question while broader population applicability remains open. Neither an analytical pass nor a supported association supplies the missing clinical-performance result.

Consultant Lens. Clinical, reference, configuration and statistical owners can settle the target and separate analysis specification while recruitment is still being planned. That is when first and final states, reference ownership and a supported bounded conclusion are cheapest to define. If specimens later fail to arrive or outputs fail, the owners can investigate what changed in the analysis set. Rewriting the target around the available file would conceal the very difference they need to understand.

Who could reach the calculation?

Imagine the proposed clinic test applies to all eligible adults at a stated decision point. Its claim concerns that **intended population**, with the relevant clinic type, disease stage and timing. The next question is whether the study can actually reach those people.

Recruitment may reach only adults attending particular hospitals who consent during a defined window. They form the **study population**. Even careful testing of every recruit leaves the gap between those hospitals and the intended care route to be explained.

A further narrowing can occur after enrollment. A complete-pair analysis may require an obtained specimen, a valid binary index result and a determinate reference. Its analysis set therefore contains only recruits meeting those conditions. One person can belong to the intended and study populations while contributing no pair. Each transition needs its own account of selection.

How were people invited in the first place? Inviting every eligible arrival during a defined clinic window is consecutive recruitment. An arrival log can then be compared with enrollment. That comparison still needs missed eligible arrivals, and the claim may include people who never reach that clinic.

Selecting people by a stated chance procedure from a roster makes recruitment random within that roster. It clarifies the selection mechanism, but a roster that omits relevant patients remains incomplete. Refusal after selection can change the resulting group too. The important improvement is knowing how selection worked, rather than relying on the label to establish representativeness.

Stored specimens introduce another entry route. Using whichever specimens are readily available is convenience sampling. It can answer a bounded question about that material, while storage and remaining-material requirements may have selected a different disease and handling range from routine care. To understand the result, the reviewer needs to know what those requirements kept out.

A study may instead deliberately add people with the condition or a rare subgroup to learn more about them. This **enrichment** can be useful when the represented range and analysis fit the question. It also changes the sample's positive fraction. That fraction cannot be treated as the ordinary population frequency merely because the added participants were tested well. Each recruitment route supplies something useful; none establishes fit to the intended population on its own.

Who is inside the positive and negative groups?

The fraction with the condition in a specified population and time is its **prevalence**. A selected sample's positive fraction describes its composition unless selection supports a population estimate. Even a matching fraction leaves a further question: which positive and negative patients were represented? That range is the **spectrum**. A sequencing study could include many positives while omitting difficult specimen or disease conditions. A pathology study could match the positive fraction yet underrepresent weak stains, unfamiliar scanners or users. These are possible design problems, not new facts about the named products.

This distinction will matter when interpreting predictive values. If only prevalence changes and sensitivity and specificity remain the same, the mixture of condition-positive and condition-negative people changes those predictive values by arithmetic. A change in disease stage, competing conditions, users, reference operation or workflow may change sensitivity and specificity themselves. Substituting a new prevalence cannot repair those differences. [Sources 2–3]

What did the archive keep out?

Suppose a tissue archive mostly retained specimens large enough for multiple tests. Its available material may yield clean results while the missing small specimens remain consequential to the intended use. That does not make archived material unusable. Collection, selection, handling, storage, remaining amount and linked clinical information determine whether it fits the new question. Prospectively gathered data can also miss an important group through site or eligibility practices. The decisive question is who and what could never enter, and why; the collection date alone cannot answer it.

Reviewer Lens. Following the intended patient journey into recruitment and then complete-pair selection reveals where the population changed. An archived tissue estimate deserves credit within a justified boundary when the unavailable-material pathway is understood. For pathology, readers, sites and images belong in that comparison alongside patient labels. The hard judgment is whether the missing part could change the claimed inference; sample size alone cannot resolve it.

Consultant Lens. The study lead can define eligibility and sampling before outcomes are reviewed; specimen or imaging owners can preserve reasons inputs were unavailable; the analyst can keep each analysis set identifiable. These decisions make later differences interpretable. If the sample misses a consequential positive or negative group, the available choices are a reasoned narrower claim, a justified argument for applying the evidence, or focused recruitment. Calling a large sample representative leaves that choice unresolved.

Could the comparison help manufacture its own agreement?

Suppose the index result decides who receives the examination that assigns condition status. The comparison has begun to select itself. This is why **bias** needs an operational explanation: what happened between selection and comparison that could distort the answer? The archive example supplied an earlier route. If readily processed specimens are overrepresented, difficult members of the intended population may be missing before testing begins. That selection or spectrum problem needs a description of the absent group and either a supported applicability argument or a narrower conclusion. Its label alone cannot establish the direction of error.

After recruitment, selection can happen again. If only index-positive patients receive the reference examination, many index-negative patients have no assigned condition status. The complete pairs now depend on the result under evaluation. This is **partial verification**. Reference ascertainment independent of the index result, or an explicit assessment of selective missingness and justified analysis options, addresses that mechanism.

What if everyone receives some comparison, but positives receive tissue assessment and negatives only clinical follow-up? The two routes can have different errors. This **differential verification** requires an explanation of assignment, comparability and consequences before their labels can be pooled as though they establish the same thing.

Even complete reference records leave another question: did the reference use the answer it was supposed to assess? A panel that sees and uses the investigational index result can make the index agree partly with itself. That is **incorporation**. A separately defined reference process, or an explicit account of the dependence, is needed at this step.

Information can cross the boundary in image review too. A pathologist who sees assisted output before an “unassisted” read may remember it or have extra reading time. The resulting judgment therefore depends on who saw what, and when. **Masking** controls access to information; justified sequence controls address order. The word “blinded” alone identifies neither arrangement.

There is also a boundary in time. Treatment between index and reference assessments may change the condition itself, so disagreement may reflect that change. The interval and intervening events need to fit the clinical question. Looking farther back, evaluation cases may already have shaped a model, cutoff or reporting rule during development. Appropriately separate evaluation data and disclosed reuse make that history assessable. More observations cannot by themselves remove either mechanism, and mechanisms can overlap. FDA's nonbinding 2013 pivotal-study guidance addresses these design and masking questions within its scope; it excludes studies establishing

companion-diagnostic clinical validity. Chapter 15 develops reader-study methods. [Source 4]

The table helps locate the operation that needs explanation when a comparison looks stronger than its design may support. It is a starting set of routes, not an exhaustive taxonomy or a finding that they occurred in the later anonymous packet.

What could change the answer?	What would address that route?
Easier patients or specimens enter preferentially	Justified sampling, a description of missing groups, and a narrower claim or supported applicability argument
Only selected people receive the reference	Planned reference ascertainment and assessment of selective missingness
Groups receive references with different errors	An explanation of assignment, comparability and consequences
The index result helps assign its own reference answer	A separate reference construct or explicit treatment of the circularity
A reader sees another result or gets different exposure time	A record of information and order, with justified masking and sequence controls
The condition changes between assessments	Justified timing and an account of intervening events
Evaluation cases shaped development	Appropriate separation, disclosed reuse and stated inferential limits

The third row is easy to miss: supplying a reference label for everyone does not establish that each label came through a comparable process. The tissue-assessment and follow-up routes need their own explanation. Identifying that route focuses the next request without deciding the size or direction of its effect.

Reviewer Lens. A useful bias question names what was selected or which information crossed a boundary, then asks which estimate could move and what constrains that inference. A sequencing reference that incorporates the index cannot support independent correctness in the usual way; a pathology judgment after aided viewing may carry that exposure into the next read. A described limitation can still leave a narrower conclusion supportable. The hard part is locating that boundary without guessing the direction of an unmeasured effect.

Consultant Lens. Before enrollment, the recruitment owner can preserve eligibility, the reference owner can define status assignment, the workflow owner can settle visibility and timing, and developers can separate evaluation from development. Once circular

assignment has happened, another fitted model cannot simply remove it. The analyst may assess missingness and prespecified sensitivity analyses, but closure still needs documented prevention, justified analysis or an explicit residual limitation for the actual mechanism.

What happened to the person without a binary pair?

A specimen's first examination produces no usable output under the stated rule. That attempt is **invalid or failed**. If the prespecified policy allows another examination of the same specimen, the **repeat** adds an attempt. It adds no patient. A later recovery therefore creates a final result while leaving the initial failure in that person's history.

Now consider why another specimen lacks a positive/negative result. Its examination may have run successfully and returned a category intentionally leaving the binary question open. This **valid equivocal** output records what the procedure reports in a difficult case. Calling it failed execution would hide the different decision facing its user.

The absence can occur earlier still. A person may supply no specimen, or an image may never reach the algorithm. There is **no result** at that acquisition stage. With no examination attempted, a negative or invalid test result would misdescribe the route. The gap concerns access or acquisition rather than analytical execution.

Finally, the index may produce a binary answer while the separate condition-assignment process remains unresolved. That **unresolved reference** prevents a complete condition pair without making the index a failure. Following the person through acquisition, examination and reference assignment shows which question each absence leaves open.

The protocol's prospective state definitions and user pathway make these histories interpretable. The analysis specification then determines who enters each measure. A final binary ratio can be useful for a defined final-result strategy, with a visible third category and first-attempt summary showing its operational experience. Alternative coding requires a scientific reason established independently of which treatment yields the best percentage. Even a discrepancy investigation that explains a conflicting pair must preserve the original index and reference observations. [Sources 2–3]

Does an alternative assumption change the conclusion?

The word *sensitivity* now has two distinct jobs. **Clinical sensitivity** asks what fraction of reference-positive people have a positive index result in a named binary analysis. A **sensitivity analysis** instead changes an uncertain assumption or permissible state treatment to see how the conclusion depends on it. That comparison can reveal fragility, but it cannot establish which assumption is true or recover an unobserved condition as fact.

Reviewer Lens. A single “validity rate” loses the history needed to interpret a sequencing study: first attempts, repeat eligibility, recovered outcomes, valid no-call or equivocal categories, missing specimens and reference failures. In assistive pathology, image rejection, a model flag, the pathologist's completed decision and worklist delay also occur at different stages. The separate unnamed 2/40 initial-invalid sequencing note and 1/25 late-arrival pathology note can prompt checks of their own routes. They estimate nothing for the named products or this chapter's anonymous study.

Consultant Lens. Laboratory and workflow owners define the states; the trial statistician decides prospectively how each enters a measure; the data owner preserves the links between attempts. That join becomes valuable when an invalid attempt recovers: the final outcome and its first failure can both be reported. A reconciled ledger and justified analysis policy can close the accounting question. Correct binary arithmetic alone leaves a report that hides valid equivocal outputs misleading.

Which people does each percentage describe?

A positive-result percentage could ask about people who have the condition, or about people whose index result is positive. Those are different groups. Before choosing between them, the next exercise fixes a narrow comparison: each participant supplies at most one specimen and one *final* paired index and reference result. The binary analysis is prespecified. An independent best available reference is stipulated adequate to assign the target condition. Counts are unweighted, with no repeated participants or clustering assumed. These are the exercise's conditions, not properties to assume for every real study.

The table first crosses what the index said with the condition assigned by that reference. A positive index and positive condition form a **true positive**; a positive index and negative condition form a **false positive**. The negative-result row gives the corresponding **false negative** and **true negative**. Their short labels—TP, FP, FN and TN—will keep the arithmetic readable. “True” depends on the stipulated reference role.

Index result	Target condition positive	Target condition negative
Positive	TP: true positive	FP: false positive
Negative	FN: false negative	TN: true negative

Read the positive index row: its left cell is a correctly positive result, while its right cell is positive despite a negative reference-assigned condition. Both cells contain *complete final pairs*. The sum of all four cells, $N = TP + FN + FP + TN$, therefore counts paired participants and may exclude some enrolled people.

Start with the condition, or start with the result?

To ask how often a person assigned the condition has a positive index result, the relevant group is the condition-positive column. Its total **TP + FN** divides the correctly positive count TP. That is sensitivity, abbreviated **Se**. Among people assigned no condition, the condition-negative column totals **FP + TN**; correctly negative TN divided by that group is **specificity**, or **Sp**. Each ratio is a proportion from zero to one. Multiplication by 100 expresses that same estimate as a percentage.

What if the question starts with the result just received? Among index-positive complete pairs, **TP + FP** is now the group and TP is the correctly positive count. Their ratio is the observed **positive predictive value**, or **PPV**. For an index-negative result, **FN + TN** is the group and TN the correctly negative count: the observed **negative predictive value**, or **NPV**.

One further ratio describes the analyzed sample itself: $(TP + FN)/N$ is its target-positive fraction. It does not automatically describe prevalence in intended care. Every divisor must be nonzero. The table is a locator for the question being asked before a ratio is chosen.

The question	Quantity and calculation	The people being counted
Of those with the condition, who tests positive?	Sensitivity: $TP / (TP + FN)$	Target-positive complete pairs
Of those without it, who tests negative?	Specificity: $TN / (FP + TN)$	Target-negative complete pairs
How many analyzed people have it?	Target-positive fraction: $(TP + FN) / N$	All complete binary pairs
Of positive results, who has the condition?	Observed positive predictive value: $TP / (TP + FP)$	Index-positive complete pairs
Of negative results, who lacks the condition?	Observed negative predictive value: $TN / (FN + TN)$	Index-negative complete pairs

The first and fourth rows share TP as numerator. Their divisors differ because one starts with the condition and the other starts with a test result. A change in divisor changes the question even when the same correctly positive people supply the numerator.

That reading also explains the reference boundary. If the other procedure is merely a non-reference comparator, its labels cannot assign who has the condition. Positive and negative agreement can still describe the comparison, but sensitivity, specificity and clinical predictive values require the condition-assignment role used above. [Sources 3, 5]

What if the population mixture changes?

Suppose, hypothetically, that a suitable study's Se and Sp apply in another population. Let p be the proportion with the condition; $1 - p$ is the proportion without it. At $p = 0.12$, those shares are 12% and 0.88, or 88%. We can now combine those population shares with the two conditional performance rates. The calculation starts with a hypothetical whole population, not a newly observed group of patients.

Within the condition-positive share, Se is the fraction the index calls positive. Thus $Se \times p$ is the modeled share of the *whole population* that is both condition-positive and index-positive. Within the condition-negative share, $1 - Sp$ is the fraction the index falsely calls positive. Thus $(1 - Sp) \times (1 - p)$ is the modeled false-positive share. Add those two shares: the sum is *all modeled index positives*, the denominator for predictive value among positive results. The true-positive share is its numerator.

Now perform the parallel operation for index-negative results. Among condition-positive people, $1 - Se$ is falsely negative, giving whole-population share $(1 - Se) \times p$. Among condition-negative people, Sp is correctly negative, giving share $Sp \times (1 - p)$. Their sum is *all modeled index negatives*; the correctly negative share is the numerator for modeled NPV. These conditioning routes yield:

$$PPV(p) = (Se \times p) / [(Se \times p) + ((1 - Sp) \times (1 - p))]$$

$$NPV(p) = (Sp \times (1 - p)) / [((1 - Se) \times p) + (Sp \times (1 - p))]$$

The expression uses **0.12** for 12%. Its products are proportions of a hypothetical whole population, and each modeled divisor must be positive. For those products to describe another actual population, the clinical context, units, configuration, reference basis and result policy must be compatible, and Se and Sp must transfer. Different disease stage or case spectrum may change those rates. The algebra changes the mixture; it cannot establish the required transfer.

An inferential report needs a justified uncertainty method and estimate as well. Several visits from a person, or crossed readings by readers and cases, cannot be treated as independent merely because results fit a two-by-two table. The packet below supplies **no interval method or estimate, acceptance threshold, or sample-size rule**. Its interpretation must keep those entries unknown. [Sources 2–5]

Reviewer Lens. The first check for a tissue-classification ratio is whether the reference assigns the stated condition and whether first and final no-calls remain visible outside its divisor. For assistive pathology, the first check may instead be what a pair counts: a patient, image, reader decision or repeated reader–case encounter. Correct arithmetic deserves credit within that contract. It cannot answer the dependence or broader claim question without the corresponding records.

Consultant Lens. Clinical and statistical owners can put the conditioning question beside a planned ratio while the protocol is still open. Linking person, attempt and

reference to an analysis unit makes the later choice of uncertainty method defensible. A projection creates another ownership question: the clinical owner must justify both population prevalence and transfer of component performance. Until then, the calculation remains a useful hypothetical with an unresolved applicability decision.

An anonymous study reaches the review desk

This is original fictional teaching data, not a HelixBridge or PathSight study, a real clinical investigation or an adapted standards example. The procedure has an unnamed binary classification role and no product-specific acceptance criterion. It does not establish a real clinical claim.

The *claimed* aim concerns binary clinical performance in a prespecified eligible population, with one prospectively collected specimen per participant. The *actual* sample combines consecutive enrollment with targeted enrichment, so its condition-positive fraction is not stipulated to be intended-population prevalence. For this exercise, an independent best available reference is stipulated adequate to assign the condition. The primary analysis is an unweighted complete-pair comparison of the prespecified **final** binary index result with determinate reference status in the sampled population. Whether that performance applies to the broader intended population remains another question.

First count people and specimens

The first record asks how people reached specimen acquisition. Its rows distinguish eligibility, enrollment and obtained material.

Record	Count
Potentially eligible participants	160
Not enrolled after prespecified eligibility review	8
Enrolled participants	152
Enrolled participants without an obtained specimen	1
Specimens obtained	151

The “without an obtained specimen” row belongs to enrolled people. The eight in the earlier eligibility row never entered that group. Keeping the two positions visible will matter when the final analysis is reconciled.

Then follow attempts into final states

The next record follows the obtained specimens through examination and then shows the separate reference outcomes. First attempt, repeat and final state occupy different positions in one specimen's history.

Stage	Recorded outcomes
First index attempt	143 binary; 5 valid equivocal; 3 invalid
Prespecified repeat rule	Only invalid first-pass outcomes are eligible
Repeat attempts	All 3 first-pass invalids repeated; 2 become binary; 1 remains invalid
Final index outcomes	145 binary; 5 valid equivocal; 1 invalid
Reference outcomes	150 determinate; 1 unresolved

The repeat-rule row applies only to an invalid first attempt. A valid equivocal result stays in its own category under this rule, and each attempt has one recorded outcome. The reference row belongs to condition assignment; it does not count another index attempt.

Join the two routes before selecting an analysis set

Which final index results can be joined to assigned condition status? This table crosses the two routes for each acquired specimen.

Final index state	Reference state	Participants
Binary	Determinate	144
Binary	Unresolved	1
Equivocal	Determinate	5
Invalid	Determinate	1

The binary/unresolved row contains a finished binary index result but no determinate reference. Reading the columns together prevents either route from standing in for the other. Likewise, a determinate reference cannot convert an equivocal or invalid index output into a binary one.

Of the two repeated invalids that recover a binary result, one has a determinate reference and enters the complete-pair table. The other has the unresolved reference and stays outside it. Their exact positive or negative cell allocations are **not supplied**.

The final paired results

The final record contains only participants with a binary final index result and a determinate reference. Rows identify the index result; columns identify reference-assigned condition status.

Final index result	Target positive	Target negative
Positive	54	8
Negative	6	76

The positive row contains 54 participants assigned the condition and 8 assigned no condition. It describes final paired outcomes. The preceding records explain how a participant could reach this table; none supplies a complete first-attempt cell allocation.

A separate algebra question uses hypothetical target prevalence $p = 0.12$. Transport of the packet's sensitivity and specificity to that population has **not** been established.

What judgment does this packet support?

The study definition and the analysis now need to be joined. The following questions concern the supplied record; an unknown first-attempt cell or interval remains unknown when considering them.

Decision

What claimed use and clinical role are supplied, and which product-specific details are missing? How does the intended population compare with the actual sampling frame? What are the index strategy, unit, endpoint and timing? What reference role and independence are stipulated, with which remaining limits? What exact target quantity follows from that definition?

Decision

Which estimator, weighting, unit and dependence assumptions apply? How do participants, specimens, attempts and result states reconcile? Who enters the primary analysis and what happens to each omitted state? What are the five observed ratios, their denominators and interpretations, and the two conditional projections at $p = 0.12$? Which uncertainty and acceptance information remains unsupplied? What conclusion is supported, what limits remain, and what repair would address them?

A complete judgment keeps the target separate from the estimator and follows screened, enrolled and obtained counts into first attempts, repeats, final index states, reference states and complete pairs. For each calculation, which unit, divisor and scale give it meaning? How far can its conclusion travel on the supplied evidence?

How a reviewer reconciles the packet and its calculations

Worked reasoning

Follow each unit to its next state

Of 160 potentially eligible people, eight were not enrolled: $160 - 8 = 152$ **enrolled**. One enrolled person supplied no specimen: $152 - 1 = 151$ **specimens**. The first attempts account for every obtained specimen: 143 **binary** + 5 **valid equivocal** + 3 **invalid** = 151 . Three additional attempts were performed *on the three initially invalid specimens*, so there were $151 + 3 = 154$ **attempts**, not 154 people or specimens.

Two invalid first attempts recovered to binary results, giving $143 + 2 = 145$ **final binary** results. The other first-pass invalid stayed invalid; five valid equivocal outputs stayed in their own category. Thus $145 + 5 + 1 = 151$ **final index states**. Separately, 150 **determinate** + 1 **unresolved** = 151 **reference states**. Their joint counts also cover all 151 specimens: 144 **binary/determinate** + 1 **binary/unresolved** + 5 **equivocal/determinate** + 1 **invalid/determinate** = 151 .

Only the 144 **binary/determinate** participants enter the final complete-pair table. Relative to the 152 enrolled, the eight outside that analysis are **one with no specimen**, **five with valid equivocal index results**, **one with a final invalid index result**, and **one with an unresolved reference**. These are *not* the earlier eight who were never enrolled. An analysis denominator counts what its calculation actually uses; it is not necessarily everyone eligible for the study or even everyone enrolled.

One recovered binary result enters the 144 pairs because its reference is determinate. The other recovery is outside because its reference is unresolved. Both remain recorded as initial invalids. Since the packet does not say which true/false cell holds the recovered complete pair, subtracting one from an assumed cell would fabricate a first-pass sensitivity or specificity. A first-attempt analysis would need the linked participant-level records that identify that cell.

Figure 12.2: Participant, specimen, attempt and paired-analysis flow shows the unit changes just reconciled. Its branches must not turn a valid equivocal result into test failure or a repeat into another participant.

Section 12 – Reconcile the flow

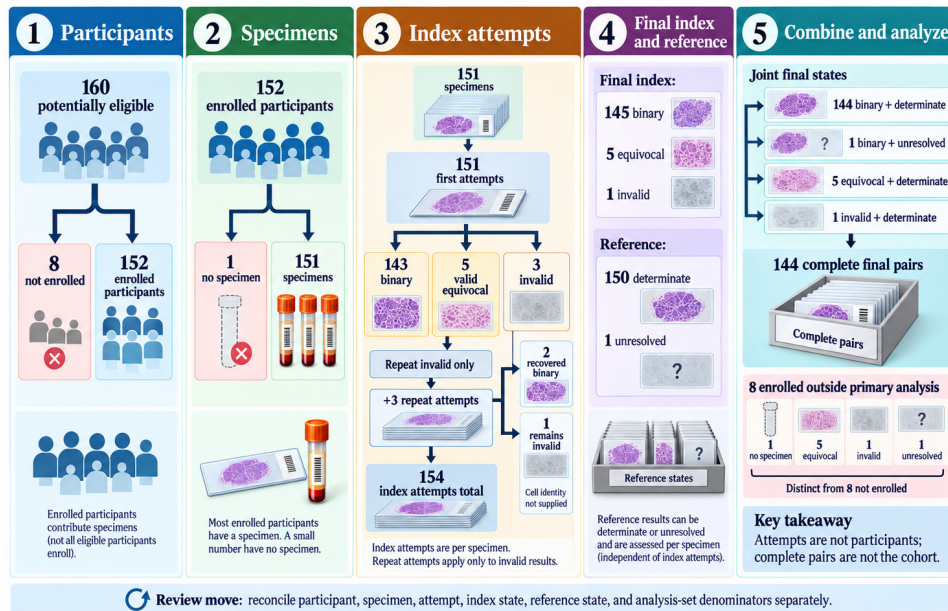


Figure 12.2. Participant, specimen, attempt, and complete-pair reconciliation – Technical review plate.

Calculate only what the paired table supplies

There are $54 + 6 = 60$ **target-positive complete pairs**; 54 of them have positive index results, so $Se = 54/60 = 0.900000 = 90.00\%$. There are $8 + 76 = 84$ **target-negative complete pairs**; 76 have negative index results, so $Sp = 76/84 \approx 0.904762 = 90.48\%$. Together the two condition groups make $60 + 84 = 144$ paired participants. The analyzed target-positive fraction is $60/144 \approx 0.416667 = 41.67\%$ —the enriched sample's composition, not established intended-population prevalence.

Condition instead on the index result. Of $54 + 8 = 62$ **index-positive complete pairs**, 54 are target-positive: observed $PPV = 54/62 \approx 0.870968 = 87.10\%$. Of $6 + 76 = 82$ **index-negative complete pairs**, 76 are target-negative: observed $NPV = 76/82 \approx 0.926829 = 92.68\%$. These are observed within the stipulated sample and final-result policy. The six decimal places show arithmetic, not six-decimal statistical certainty.

Quantity	Substitution	Result	Bounded interpretation
Sensitivity	$54 / (54 + 6)$	$0.900000 = 90.00\%$	Among 60 target-positive complete final pairs

Quantity	Substitution	Result	Bounded interpretation
Specificity	76 / (8 + 76)	$0.904762 \approx 90.48\%$	Among 84 target-negative complete final pairs
Analyzed target-positive fraction	60 / 144	$0.416667 \approx 41.67\%$	Composition of the enriched complete-pair sample
Observed PPV	54 / (54 + 8)	$0.870968 \approx 87.10\%$	Among 62 index-positive complete pairs in this sample
Observed NPV	76 / (6 + 76)	$0.926829 \approx 92.68\%$	Among 82 index-negative complete pairs in this sample

The observed positive-predictive-value row uses 62 positive index results as its group. The sensitivity row uses 60 reference-positive participants. Both are correct at their stated conditioning sets; neither is an all-enrolled result.

Now perform the *separate hypothetical projection* at $p = 0.12$, or 12% condition-positive, so $1 - p = 0.88$, or 88% condition-negative. Keep the exact table fractions $Se = 54/60 = 0.9$ and $Sp = 76/84 \approx 0.904762$ in the products rather than rounding percentages first. Among a hypothetical whole population, the modeled *true-positive share* is $0.9 \times 0.12 = 0.108$. The modeled *false-positive share* is $(1 - 76/84) \times 0.88 \approx 0.083810$. Their sum, ≈ 0.191810 , is the modeled index-positive share; it is the PPV divisor. Thus $PPV(0.12) = 0.108/0.191810 \approx 0.563059 = 56.31\%$.

For modeled negatives, the *false-negative share* is $(1 - 0.9) \times 0.12 = 0.012$. The *true-negative share* is $(76/84) \times 0.88 \approx 0.796190$. Their sum, ≈ 0.808190 , is the modeled index-negative share; it is the NPV divisor. Thus $NPV(0.12) \approx 0.796190/0.808190 \approx 0.985152 = 98.52\%$. The displayed decimal products are rounded; calculating with the unrounded component fractions yields the displayed percentages. The shares are proportions of a hypothetical population, not observations or a manufactured count of patients.

The observed PPV of 87.10% and conditional projection of 56.31% answer different questions. Under this algebra, Se and Sp were held fixed while the modeled condition mixture changed; it does **not** show a test deteriorating. Whether performance, reference operation, spectrum and result-state policy really remain comparable in another population is unsupplied. The projections therefore are not established predictive values for that population.

Put the arithmetic back into a study critique

The completed record connects each usable result to the boundary that remains. It helps distinguish a missing reporting link from missing evidence about the population.

Question for the record	What the packet supports
What claim and population are proposed?	Binary classification in a prespecified eligible population; actual sampling combines consecutive recruitment and targeted enrichment. Exact condition, product configuration and detailed eligibility are not an executable supplied protocol.
What is being estimated?	Participant-level sensitivity and specificity for the defined final binary strategy against determinate reference status in the sampled population; intended-population transfer remains open.
What assigns condition status?	An independent best available reference, stipulated adequate here; one reference remains unresolved.
How are observations combined?	Unweighted complete-pair ratios, one final paired result per participant and no clustering stipulated.
What happened outside the ratios?	Only initial invalids were repeat-eligible; first outcomes remain. Five valid equivocal, one final invalid, one unresolved-reference and one no-specimen participant remain visible.
Which numbers are usable?	Five observed ratios for the stipulated complete pairs; two conditional prevalence projections.
What inferential support is absent?	Interval method and estimate, justified acceptance criteria, a full sampling/applicability argument and claim-specific protocol.
Does this establish the broad population conclusion?	The table alone does not.
What would close the identified gaps?	Full flow and state history; study and analysis details; appropriate uncertainty; a prospective acceptance basis for a pass/fail claim; and a justified applicability assessment.

The “outside the ratios” row gives the positive calculations their proper boundary: the five final complete-pair estimates and two conditional projections are usable for the exercise questions they answer. They do not establish the report's broad intended-population conclusion. Recovering controlled records may close a reporting or analysis gap. Representing a population the study never reached may need a different evidence strategy. Identifying which gap exists comes before commissioning a whole new study.

Reviewer Lens. A correct final-result sensitivity can be credited as a complete-pair estimate while its tissue specimen flow, repeats, reference, uncertainty and spectrum remain under review for the claim. A pathology table similarly needs reader and case identities before its precision can be judged. Sampling uncertainty, selective entry, circular reference assignment and failure to apply elsewhere have different remedies. Treating them as one general doubt would make the request less useful.

Consultant Lens. The data owner can close the 152□151□154-attempt□144-pair account while retaining the recoveries. The statistician can verify the five ratios, label the projections conditional and specify uncertainty for the actual design. The clinical owner must decide which population and reference operation the claim needs. An existing prespecified criterion should be retrieved if available; setting it from these outcomes would answer a different question. The teaching critique can close its bounded numerical conclusion while broader clinical adequacy remains open.

How much information lies behind the estimate?

The arithmetic is now visible, but how much could the estimate vary under repeated sampling? An **interval method** and its resulting **interval estimate** address a specified kind of sampling uncertainty. The method must fit the independent units, weighting and dependence. This packet supplies neither method nor interval, and no acceptance threshold or sample-size rule. An attractive point estimate cannot fill those blanks or justify a pass based on a guessed interval.

That leaves a planning question: where will the necessary information come from? Both condition-positive and condition-negative groups matter, along with expected missing specimens, nonbinary outputs and the independent units actually available. A large total can contain few observations in a rare, consequential subgroup. A narrow interval can also describe the wrong patient spectrum precisely. Interval width therefore cannot repair biased selection, reference error or unsupported transfer.

Which subgroups deserve attention follows from the biology, specimen, user and setting questions that could change the claim. Testing every available column would obscure that purpose. A planned comparison can reveal a limitation; an exploratory observation can motivate further study. A nonsignificant comparison does not establish that groups are equivalent, particularly when information is sparse. The observations may still be compatible with consequential differences. Equivalence would require a justified margin for an acceptable difference and an analysis able to exclude differences beyond it. This packet supplies neither. [Sources 2–4]

Reviewer Lens. An uncommon sequencing target class needs its own evidence and missing-state route examined beside the overall total. For assistive pathology, reader–case dependence may determine whether the reported interval is meaningful at

all. Enough information for a narrow inference deserves recognition. Its precision and its applicability remain separate judgments even when the interval is narrow.

Consultant Lens. Before recruitment, the statistician and clinical owner can define the precision or hypothesis question, expected group frequencies, allowance for nonbinary states and unit of independence. After collection, a newly noticed subgroup pattern should remain exploratory while owners consider existing records, focused recruitment or a revised claim. Describing a failed significance test as equivalence would bypass that choice.

Two changes to the same numerical record

The next two records each change one part of the evidence. Both decisions come before their shared feedback. In each, which part of the study definition, disposition account or interpretation has changed?

A different comparison role

The comparison procedure in the main packet is now a named **non-reference comparator** that does **not** establish target-condition status. All counts, pairing, units, result states, sampling and analysis rules are unchanged.

Decision

Which metric names and conclusions does the same arithmetic now support?
What can no longer be claimed?

A different account of participant outcomes

Return to the original reference-based packet. The report replaces each repeated invalid with its final outcome, omits the five equivocal results from participant flow, and describes the binary table as **all valid study results**. All underlying observations and other design facts are unchanged.

Decision

What are the consequences of this account? What prospective primary and sensitivity presentation would fit the intended workflow and question, without selecting a way to handle result states because these observed outcomes make it favorable?

How the two changed records alter the conclusions

Compare your critiques

In the changed-comparison record, the cells can be called **a**, **b**, **c** and **d** in the same orientation, but the comparator does not establish whether a participant has the condition. Among $54 + 6 = 60$ **comparator-positive** complete pairs, 54 index results are positive: **positive percent agreement (PPA)** = $a/(a + c) = 54/60 = 90.00\%$. Among $8 + 76 = 84$ **comparator-negative** complete pairs, 76 index results are negative: **negative percent agreement (NPA)** = $d/(b + d) = 76/84 \approx 90.48\%$. These are conditional agreements. The unchanged numerators do not support sensitivity, specificity, clinical predictive values or an assignment of clinical truth from this comparison alone.

In the changed-reporting record, the four-cell arithmetic remains correct for its complete final binary pairs, but the report's account of the participants is wrong. The three initial invalids and three repeats vanished from the first-pass history, while five *valid* equivocal outputs disappeared from participant flow. A corrected account needs both original and final attempt records, every nonbinary and unpaired state, and the defined analysis set.

For a future protocol, a justified final-binary primary analysis can coexist with a visible third-category presentation and a separately defined first-attempt or other sensitivity analysis. The intended workflow and scientific question determine that choice before outcomes are known. For this completed packet, the actual prespecified rule remains the primary historical fact; further analysis must be labeled additional. A newly preferred rule cannot become prospective retroactively.

Turn a critique into a bounded decision

The reviewer can now give the table its due: it passes the arithmetic check for the restricted complete-pair question. The broader judgment still depends on the target, reconciled state ledger, reference role, dependence, uncertainty method and applicability to the population. These are distinct relationships to resolve, so the next action depends on which one is missing.

For the clinical or laboratory lead, the unresolved relationship may be what happens to someone who cannot supply a specimen, has a failed examination or receives a valid equivocal result. A conditional ratio cannot describe those experiences. Manufacturer owners can then address the protocol, capture of data, reference process, analysis or reporting. Controlled records may already contain an answer; a different gap may need a future design or focused evidence.

Scientific adequacy also leaves separate governance decisions. Ethics, consent, investigational-device obligations, authorization and clinical use require their own assessment for the actual jurisdiction and study; none is decided here. The **Standards for Reporting Diagnostic Accuracy Studies (STARD)** checklist helps make design, participant flow, results and limits transparent. Completion is useful reporting work, but cannot validate a biased or inapplicable study. [Source 6]

Reviewer Lens. For HelixBridge, “strong overall agreement” leaves the configured tissue system, reference, clinical decision, analyzed people and first/final outcomes behind a clinical claim to be identified. For PathSight, the corresponding inquiry also needs reader, case and workflow-unit dependence. Where the records warrant it, “useful bounded evidence, broader conclusion open” is a defensible disposition. It gives the usable result credit and identifies the exact extension still needing support.

Consultant Lens. The clinical owner controls the claim and patient pathway; operations owners trace specimen or image and reference routes; the data owner reconciles people and states; the statistician specifies estimation and uncertainty. This division allows a recovered record to repair reporting, a justified method to address analysis, or a claim limit or focused study to address representation. A completed checklist cannot choose among those remedies.

Return to the two named case records

HelixBridge's tissue-sequencing snapshot contains a separate technical report describing “strong overall agreement.” It supplies no clinical-performance study, target quantity, participant counts or sensitivity/specificity estimate here. The knowledgebase used in analytical validation and the knowledgebase in that separate report belong to their own identified records. No commercial release or liquid study follows from them.

PathSight retains its assistive pathology role and reader-study assertion, without reader–case counts, design or dependence supplied in this occurrence. Chapter 15 develops that study question. The unnamed post-use notes—2/40 initial invalid sequencing attempts and 1/25 late pathology worklist arrivals—belong to separate services. Neither adds an episode to the named products' chronology, and the anonymous numerical packet supplies no study data for them.

Decision

What precise target-quantity request fits each named case and the decision it would support? Which analysis details would be needed next, before a numerical conclusion?

The evidence requests behind the named claims

For tissue sequencing, the request begins with a decision- and population-specific clinical-performance target. The configured index, target-condition reference, first/final ledger, analysis set and uncertainty plan would make that target assessable. For assistive image review, it must additionally identify which pathologists, cases and workflow decisions are intended and how repeated or crossed readings will be analyzed. These requests identify evidence jobs without asserting that studies exist. A separate post-use note may prompt its own trace and denominator inquiry while leaving clinical sensitivity unanswered.

The same tissue configuration, a proposed population

This is another anonymous, nonrecurring example. Its specimen preparation is specified only for this exercise and must not be copied into HelixBridge's unspecified tissue preparation.

An unnamed tissue-based oncology sequencing configuration, reported output and classification role remain unchanged. Use is limited to **formalin-fixed, paraffin-embedded tumor tissue**: chemically preserved tissue supported in wax for sectioning. It is not a blood, plasma, cell-free DNA or circulating tumor DNA claim. The earlier study had an independent prespecified reference process and visible first-attempt and final-result accounting.

The earlier population was adults with previously diagnosed advanced solid tumors at tertiary oncology centers. Its analyzed sample represented the old eligibility criteria, disease stage, referral pathway, sites, users and available tissue path.

The proposed population is adults with earlier-stage solid tumors tested at regional oncology centers *before* the tertiary referral point. No patient-level values, subgroup estimates, new study, weighting scheme, precision result, bridge conclusion or acceptance decision is supplied.

Decision

What target quantity and analysis specification would describe the proposed population? What remains applicable, what requires a transport argument, and what is unknown or reopened? What bounded evidence recommendation follows?

How a reviewer assesses the proposed population

The unchanged configuration narrows the device-change question, while the new clinical decision point and patients define a fresh applicability question. A target for those patients needs a matching account of recruitment, reference, first/final policy, analysis set and uncertainty. The earlier result is still attached to its old patient and referral route.

The relevant comparison runs through old and proposed eligibility, disease stage and spectrum, condition frequency, sites, users, clinical timing and tissue availability. It also asks whether the reference assigns status comparably at the new point and whether missingness, equivocal and invalid outcomes, and the old estimator fit the new question. Existing subgroup records could help if their definitions and information cover that route. None are supplied, so neither a successful bridge nor its failure is established.

Applicable configured-system and earlier-study evidence remains useful. A documented transport assessment can identify which representation, reference and state questions controlled records answer and which need new evidence. That supports a selective next step without assuming either an automatic bridge or repetition of every study. Earlier chapters' separate proposed tissue-to-plasma liquid branch would reopen additional biological and handling questions; this transfer does **not** propose that branch.

Independent check: one person, three visits

A separate anonymous study seeks participant-level classification performance. It collects a specimen at each of **three visits per participant**. The report pools all visit-level binary pairs and uses an interval method that treats the pairs as independent observations. The target condition can change between visits. It does not say whether the clinical question concerns initial status, any positive visit or a trajectory. All original visit outcomes are retained.

Decision

Which target-quantity question must be settled before this estimator can be evaluated? What is the unit and dependence issue? What can the retained outcomes still support, and what bounded next action follows?

How the participant question governs the analysis

Initial status, any-visit positivity and change over time are different participant-level targets. Until the intended decision is named, the pooled estimator has no clear target. Three visits from one person may be correlated, and the condition may change; a method that counts them as three independent participants has not justified its uncertainty estimate.

Retained visit records remain useful descriptive evidence. Once the target and assumptions are specified, they could support an appropriate participant-level or longitudinal analysis. The bounded next step is to recover the intended clinical question and obtain a statistical plan compatible with that design. Keeping the data makes that work possible; it does not validate the pooled analysis.

Return to the report on your desk

The bounded disposition of the opening report

The report can now receive a bounded disposition. Its four-cell table supports the stipulated calculations, while the full intended-population clinical-performance claim remains open. The study definition identifies the patient decision and reference; the disposition ledger explains who became an analyzed pair; the analysis and applicability record distinguishes the arithmetic, its uncertainty and its use elsewhere. Those joins explain both why the table is useful and why the signature cannot stand for a broader conclusion.

Final reasoning check

The three records offer a final way to test the reasoning. Could the following questions be answered from their relationships?

1. How do the target quantity and estimator differ?
2. How can 152 enrolled participants, 151 specimens, 154 attempts and 144 complete pairs all be correct?
3. Why must a valid equivocal output remain distinct from an invalid attempt?
4. What changes when the comparison does not establish condition status?
5. What must be true before a prevalence projection describes another population?
6. What do a narrow interval and a nonsignificant subgroup comparison each leave unresolved about generalizability?

The relationships behind the six answers

The target identifies what to learn; an estimator describes how selected observations address it. Enrollment, specimens, attempts and complete pairs count different units or stages. Equivocal is a valid output category in this packet. A non-reference comparator supports agreement, not known target-condition accuracy. A prevalence formula assumes component performance transports. Intervals and subgroup tests answer specified inferential questions, not every bias or applicability question.

Chapter 13 addresses the more demanding claim relationship for oncology treatment-selection and bridging. Chapter 15 returns to reader-and-case structures that an elementary binary table cannot resolve.

Source trail

1. **ISO 20916:2019**, clauses 5.5.3.6–5.5.3.9 and 5.5.3.13, inspected in the licensed copy. The official catalogue, checked 24 September 2026, records confirmation in 2025. Its good-study-practice scope concerns IVD clinical-performance studies using human specimens, not analytical-performance validation or a universal protocol.
2. **CLSI EP49, first edition (2026)**, sections 3.3–3.3.2.4, 3.5.6 and 3.6: population, specimen, sampling, bias, comparison, accountability and analysis framework. The locally licensed pages were inspected; proprietary example data and tables are not reproduced.
3. **CLSI EP12, third edition (2023)**, sections 4.2.1–4.2.4.2: binary comparison terminology, performance versus agreement, uncertainty and equivocal-result considerations. It does not supply methods for every continuous, multilevel, clustered or reader-study design.
4. **FDA (2013)**, Design Considerations for Pivotal Clinical Investigations for Medical Devices, sections 2.2 and 8.6–8.8. Nonbinding, scope-limited guidance; its exclusions include studies establishing companion-diagnostic clinical validity. Chapter 13 uses dedicated companion-diagnostic sources.
5. **FDA (2007)**, Statistical Guidance on Reporting Results from Studies Evaluating Diagnostic Tests, sections 2 and 4–7. Binary reporting and reference/agreement boundaries; not a complete study-design or clustered-observation method.
6. **STARD 2015**, diagnostic-accuracy reporting checklist, items 3–27. A transparency aid, not a design-validity or regulatory-acceptance finding.

All numerical packets and the independent exercise are original fiction. The critique is an author teaching synthesis.

CHAPTER 13

Oncology Claims, Companion Diagnostics, and Bridging

What did the second assay actually establish?

A clinician is considering treatment. A test result may place the patient in a group for which a therapeutic relationship has been studied. The defining characteristic is a **biomarker**: something measurable that serves as a sign of a biological or clinical state. The assay's immediate contribution is a classification from the patient's specimen. What would let a second assay carry the first assay's clinical relationship to the people it identifies?

In the fictional comparison at the center of this chapter, a source assay already has a supported relationship to identifying such a group for a fictional therapy. A team has a paired classification report comparing a new assay with it. Its proposed conclusion says that the new assay identifies the **same patients** and **establishes the same expected treatment benefit for each identified patient**. Those added words reach beyond the classification in the report.

Decision

Which separate propositions does the proposed sentence contain, and what evidence would each need? What could an agreement comparison contribute to that judgment?

A useful judgment will preserve what the comparison establishes while locating the evidence needed for the rest of the sentence. The chapter concerns that relationship; it supplies no drug-development method, treatment recommendation or authorization decision.

What must arrive from the earlier evidence?

Decision

What question travels here from Chapter 10's configured examination and evidence map, Chapter 11's one-assertion provenance record, and Chapter 12's clinical-study definition, disposition and analysis records? How would those questions affect the proposed comparison?

What the three earlier records contribute

The examination map identifies the system and output actually compared. The provenance record asks whether the finding's meaning and clinical relationship are supported in that context. The study critique explains which people, result states and analysis underlie the comparison. A single agreement percentage cannot replace these relationships when the target claim depends on all three.

Which clinical question is changing?

The new assay's report begins with something concrete: a marker or alteration was found in a stated specimen under an identified **product configuration**. That detection or identification result can be well supported within its tested boundaries. To use it in care, the clinician needs to know which further question it answers.

If the question is whether the patient has a current condition, the proposed role is diagnosis or assistance with diagnosis. It needs defined patients and a separate way of assigning that condition. A marker–disease association may contribute, but a trustworthy diagnostic rule requires its own support.

The question changes when the clinician asks what course the disease is likely to take. **Prognosis** concerns that future course, with an outcome, time horizon and care context. If the question instead concerns a different response under a named treatment, **treatment-response prediction** needs evidence about response, benefit or harm in relation to treatment. A positive marker call leaves both relationships to be established.

Care can also involve repeating the test after treatment starts. A **monitoring** claim follows a defined state or quantity over specified times and identifies the decision a change might inform. One result and a rule for interpreting serial change ask different questions.

At the point where a result changes what the clinician does, the claim becomes **treatment selection**: a defined action for a positive, negative or other output in a

particular therapy and indication. The benefits, harms and alternatives of that action still matter, even when a predictive relationship has support.

Calling a device a **companion diagnostic** adds a further relationship: is its information essential to safe and effective use of a corresponding product under the named jurisdiction's conditions? That product-linked question is not answered by assigning one of the earlier clinical roles. The distinctions here help read claims; they are not a universal legal taxonomy.

The following table is useful when a report uses a broad phrase such as “clinical value.” It identifies the actual question that phrase would have to answer.

Role	Question the statement must answer
Detection	What marker or alteration does this configured examination detect, identify, or measure in this specimen?
Diagnosis or aid to diagnosis	How does the result support identifying or differentiating a current condition in this population and setting?
Prognosis	What future clinical course or outcome is associated with the result, over which time horizon and care context?
Treatment-response prediction	What difference in response, benefit, or harm is associated with the biomarker under a named treatment or treatment contrast?
Monitoring	What state or quantity is followed over time, and what decision is a change intended to inform?
Treatment selection	Which bounded result rule supports which inclusion, exclusion, or other treatment action?
Companion-diagnostic relationship	Does the device-to-product relationship satisfy the named jurisdiction's essentiality and use conditions?

The treatment-response row, for example, asks for a treatment-related difference in response, benefit or harm. It cannot be filled by a study that only reports a detected alteration. One product may have several roles, with evidence attached to each. A tissue sequencing assay may have strong analytical support while its therapy-selection role remains open. **PathSight** assists a pathologist by prioritizing or highlighting images; a highlight supplies neither an independent diagnosis nor a treatment directive. Neither product acquires a new claim here.

Which comparison distinguishes prognosis from treatment response?

Imagine marker-positive patients have worse outcomes than marker-negative patients under the care actually observed. That comparison may support prognosis. To ask whether one group benefits more from a named therapy, however, the evidence must relate the treatment contrast to marker status in a defined population and time horizon. Observing only people who received one treatment generally cannot separate a prognostic association from a differential treatment effect. A marker can have both relationships; the FDA–National Institutes of Health (NIH) biomarker resource explains why they need to be distinguished. [Source 1]

This does not require every marker group to receive every possible treatment. The exact claim, existing evidence, clinical development and ethics determine a justifiable strategy. What matters for the memo is which treatment contrast, outcome, population and uncertainty the evidence addresses. An absent contrast remains absent when the marker association is strong.

What action follows a result?

A rule that directs a clinician to include a patient in a therapeutic group needs to identify the product or bounded group, indication, eligible people and timing. It also needs a disposition for positive, negative, equivocal, invalid or partial outputs where applicable. Predictive evidence may help justify the action; benefit, harm, alternatives and clinical context determine whether that action follows. Chapter 16 examines this broader use decision.

Even then, a treatment relationship established for a group does not promise an identical outcome for every person. A test can identify the population to which supported evidence applies without measuring or guaranteeing each patient's benefit. The opening's added individual-benefit words therefore need a separate evidence question.

Reviewer Lens. The useful trace runs from the marker call through its interpretation to the exact patient-selection action. Each change of question needs a supporting relationship. In assistive pathology, the pathologist's role and clinical endpoint remain central when a claim expands beyond prioritization. A later invalid examination or late worklist arrival raises an operational question; neither creates a prognosis or treatment-effect estimate.

Consultant Lens. Before collecting evidence, clinical, assay and statistical owners can agree on three sentences: the observed output, the proposed relationship and the resulting action. A changed specimen, patient group or result-state meaning then has an identifiable owner. After use, a result or delivery failure may affect who reaches the intended decision. A bounded claim, targeted evidence or justified limitation can close that question without relabeling the original association.

Essential under which source?

Suppose the proposed claim adds “companion diagnostic.” The next question is which corresponding product requires this information and in what sense it is essential. The U.S. guidance and European Union regulation answer that question under different scopes.

The U.S. guidance: four possible roles

The U.S. Food and Drug Administration (FDA) 2014 nonbinding guidance describes an in vitro companion diagnostic device as supplying information essential to safe and effective use of a corresponding **therapeutic product**. Essential use concerns use required in the therapeutic labeling, beyond a test merely suggested as helpful. A result may identify patients most likely to benefit, or those at increased risk of a serious adverse reaction. These are two of the guidance's possible roles. [Source 2]

Its other two possible roles concern the treatment pathway itself. A test may monitor response so treatment can be adjusted for improved safety or effectiveness. It may also identify the population in which the therapeutic product was adequately studied and found safe and effective, when safety and effectiveness outside that population are insufficiently known. These four roles require the exact therapeutic-product and labeling relationship; they are not interchangeable proof methods. The guidance favors coordinated development and, in most circumstances, contemporaneous authorization while recognizing exceptions. It supplies neither a universal bridge recipe nor an outcome for the fictional assays. [Source 2]

The EU definition: two statutory purposes

Under the European Union's In Vitro Diagnostic Medical Devices Regulation (IVDR), Article 2(7), the corresponding product is a **medicinal product**. The device is essential to its safe and effective use for two stated purposes: identifying, before and/or during treatment, patients most likely to benefit, or patients likely to face increased risk of serious adverse reactions. Clinical usefulness alone does not establish this status. [Source 3]

The words “during treatment” matter when reading Recital 12. It excludes a particular use: monitoring treatment to keep concentrations of relevant substances in the body within the therapeutic window. It does not exclude every test performed during treatment. Nor can the U.S. response-monitoring and adequately-studied-population roles be added as two more EU statutory limbs. The U.S. guidance retains its four roles, and the EU definition retains its own purposes. Documentation, classification, consultation, conformity and change procedures remain jurisdiction-specific; Book 2 addresses the EU work. [Source 3]

What can a historical example contribute?

FDA's 2017 tumor-profiling sequencing fact sheet described three named authorization examples with different companion-diagnostic, clinical-significance and potential-clinical-significance evidence relationships. Its companion-diagnostic example involved analytical validity plus a clinical link or concordance route. This frozen example helps distinguish a marker call, a clinical assertion and a product-linked use. Its “levels” are not a current universal classification system, a somatic-tier method or evidence for HelixBridge. [Source 4]

Reviewer Lens. A companion-diagnostic proposal becomes assessable when it identifies the corresponding product, essential role, population, result rule and jurisdiction. An assistive image workflow may be clinically useful while lacking any supplied essential therapeutic relationship. A supported narrower role deserves credit, and a newly asserted product link needs its own evidence and authority.

Consultant Lens. Clinical and jurisdictional owners can compare the proposed diagnostic and therapeutic statements sentence by sentence. The U.S. record identifies which guidance role is claimed and how labeling requires use. The EU record uses the medicinal-product term and exact statutory purpose. The technical memo can close a supported relationship while the jurisdictional team separately decides the actual procedure.

What must remain true across the difference?

Return to the proposed second tissue assay. The first assay's supported result belongs to an exact assay, result rule, patient group, specimen and clinical relationship. That is the **source** of the proposed comparison. The **target** is the second assay and its exact claim. A **bridge** is the evidence argument that the supported source relationship remains applicable across their stated differences. A paired comparison can contribute to that argument; the claim determines what else it needs.

An agreement table leaves several possible explanations. Both methods may classify the relevant patients similarly, share a blind spot, or have been compared only on easy specimens. Selection, comparator fitness, boundary and nonbinary states, and uncertainty affect its meaning. The clinical relationship already supported at the source is the **clinical anchor**. A marketed comparator or an existing therapy supplies no such relationship by its existence alone.

Which object could change the classified patient group?

A new reagent, target-processing step, filter or positive/negative rule can change which result is reported. Those are assay differences, so their evidence needs to address the claim-relevant coverage and result states. If input also changes from tumor tissue to a blood-derived sample, the specimen path reopens collection, processing, analyte availability and negative-result meaning. A shared program name leaves those questions intact.

Now consider whom the result concerns. Earlier disease stage, another care setting or a different treatment line changes the population represented by the source. Another therapeutic product or a larger product group changes the therapeutic relationship even if the marker name stays fixed. In its narrow U.S. group-labeling scope, FDA's 2020 guidance considers indication, molecular alteration, same-specimen clinical evidence, mechanism and biomarker interaction, clinical experience, analytical coverage and clinical validity. Those considerations frame a scoped appraisal rather than an automatic bridge. [Source 5]

The apparatus and shared processing can change as well. A new instrument, signal-generation method, firmware or reusable base-calling software belongs primarily to the platform or technology comparison. A changed assay-specific filter during that move still needs its own assay row. The controlled change links the old and proposed states across affected rows; it does not absorb their questions. The table below helps allocate these differences. It is a reasoning map, not a regulatory taxonomy.

Object	Examples of fields to compare
Assay	Assay-specific chemistry, targets, extraction/library logic, filters, cutoff, scoring, classification, result states, reporting rule
Specimen	Collection, derived sample, analyte path, preparation, availability, handling, and biological context
Population	Disease, indication, stage, treatment line, setting, timing, spectrum, and eligibility
Therapeutic product or defined group	Product identity, indication, mechanism, biomarker relationship, and clinical evidence
Platform or technology	Instrument, signal generation, firmware, shared platform software, acquisition, and method family

The platform row illustrates the allocation: changed shared software belongs there, while a changed assay-specific filter belongs in the assay row. One paired study may inform both, but two citations do not create two independent studies. The comparison

directly observes who is classified alike or differently in its represented materials. It observes treatment outcomes only if the design includes them.

Within its specific-group scope, FDA's 2020 guidance identifies concordance with an approved companion diagnostic as one possible clinical-validity route, alongside prospectively defined retrospective clinical-sample analyses and outcome studies. These are possible routes, not a required sequence. Different technologies, cutoffs or filters can identify different people despite a shared biomarker; the exact claim and context determine the evidence needed. [Source 5]

What could an unchanged total conceal?

A marker signal near the reporting cutoff makes the consequence visible. A new filter might remove a call, taking a source-positive patient out of the target-positive group. A different patient might cross in the other direction and leave the total positive count unchanged. Paired identities reveal the exchange; a total does not. Boundary cases, alteration classes, nonresults and clinical strata belong beside those identities. A disagreement calls for investigation, while the locked observations remain unchanged.

Reviewer Lens. The claim-relevant crossings between patient groups matter more than an overall match. A tissue-to-liquid proposal reopens specimen and assay applicability even with the same reported marker. An image system's changed route to a pathologist needs its own image and workflow appraisal before a downstream patient claim is considered. Separate invalid-attempt or late-arrival observations can reopen their operational routes without becoming named-product treatment outcomes.

Consultant Lens. Assay, platform, specimen and clinical owners can each identify the fields they control, with one primary row per change and explicit dependencies between them. Before comparison, the plan can fix claim-relevant strata, original and final states, discordance handling and the source clinical relationship. This permits closure at the represented target while assigning an untested specimen or patient group to the owner who can address it.

That exchange matters because the people directed toward a therapy-defined group have changed. A high overall match can also conceal an affected alteration class or nonbinary state. The memo therefore preserves the original disagreements and clinically relevant strata rather than asking the aggregate to stand for them.

At the therapeutic interface, the remaining question is whether the target's exact alteration and result rule still identify the population covered by the clinical anchor. A broad gene name or shared mechanism cannot answer it. Reporting boundaries determine what appears, interpretation determines the attached assertion, and study design determines which people and outcomes the evidence represents. Investigation seeks a disagreement's cause and consequence; it does not make the newer or marketed method correct by default.

A record that keeps the two claims apart

A **claim boundary and bridging decision memo** makes the comparison reconstructible. It starts with the supported source clinical statement and configured examination, then identifies the exact target statement and proposed configuration. Each difference needs its relevant evidence. The resulting conclusion can preserve a supported relationship and identify an open part; if the target itself is unclear, the record preserves missing fields instead of inventing a precise remainder.

This is an author-designed reasoning tool. Its technical conclusions do not decide labeling, clearance, approval, conformity or consultation under U.S. or European Union law. Within that limit, a bridge can be fully supported for its bounded target. Unclaimed uses do not have to be invented before a useful positive conclusion can be reached.

Can another person reconstruct the same bridge?

The memo begins with whose evidence is being judged and under which authority frame. Its source entry records the exact claim and clinical anchor. The target entry records the proposed **claim**, configuration, therapy relationship and action. The following header keeps these identities and their limits available when individual evidence rows are read.

Header field	What the entry identifies
Identity and authority	Memo, evidence packet, jurisdiction, decision owner, and authority frame
Source	Exact source claim, configuration, clinical role, and supported anchor
Target	Exact target claim, configuration, clinical role, and proposed action
Therapeutic relationship	Product or defined group, indication, population, timing, and result rule
Boundaries	Supplied limitations and essential unknowns

The source row, for example, identifies more than an assay name: it states which clinical role and supported relationship will be carried forward. The target row then makes any added words visible. Boundaries preserve an essential unknown instead of treating the blank as a match.

Under that header, each affected object has a source-to-target row. A particular evidence design attaches to it through material, sampling, comparator fitness, locked pairing, analysis set, original and final result states, discordance, uncertainty and

relevant strata. The final sentence states the supported relationship in claim language, any residual or unresolved fields, and the owner of the next question. A complete form cannot itself supply those relationships; no universal margin or study design follows from the format.

Reviewer Lens. The target sentence is the route into the memo. Its changed objects, source anchor and paired evidence must address the patient group it names. An assistive pathology change instead starts with the image and pathologist workflow claim; a highlight or worklist metric cannot fill a therapeutic row. A supported technical row remains useful even when the broader target stays open.

Consultant Lens. Freezing source and target sentences before judging results prevents an attractive comparison from silently expanding the claim. Object owners can confirm changes and unknowns, while the analyst links each evidence item to its units and result states. The decision owner can then close a supported bounded relationship or assign the remainder to evidence or claim revision. Actual regulatory procedure stays with its jurisdictional owner.

What exactly remains unsupported?

A target can be judged only after its words and product state are clear. If the evidence addresses the whole bounded relationship actually proposed, the technical decision is **supported**. The context and limitations define that positive conclusion; a future unclaimed use does not make the present one incomplete.

If evidence reaches only specified parts or patient strata, the original target is **partially supported** and needs a narrower sentence. The supported part remains useful. Counting it twice cannot supply the unmatched part.

A clear target with a material required relationship contradicted or left without support is **not supported** by that packet. This judgment concerns the target's required relationship. It does not erase what a narrower comparison actually observed.

Sometimes the earlier problem is missing identity: the exact claim, configuration or decision basis cannot be reconstructed. That bridge is **not assessable** until the necessary fields are recovered. The table distinguishes these four author-defined teaching judgments; they are not FDA or IVDR dispositions.

Decision	Meaning
Supported	The supplied evidence addresses the fully stated bounded target relationship
Partially supported	Specified parts or strata are supported, but the target statement must be narrowed

Decision	Meaning
Not supported	Evidence conflicts with, or leaves unsupported, a material required relationship
Not assessable	Essential claim, configuration, design, evidence, or decision-basis fields are too incomplete to judge

The last row is different from the one above it. With a clear but unsupported target, the reviewer can identify the missing relationship. Without essential target fields, a precise remainder would be invented. The **residual unsupported claim** is the unmatched portion of a fully stated target.

It is not 100% minus agreement or a count of open rows. Even 99% source-to-target agreement may support a classification relationship while leaving a separate patient-outcome proposition unanswered. Conversely, a fully supported narrow target has no residual within that target. Those conclusions concern words and evidence relationships, not a completion percentage.

What do the fresh examinations show?

The opening comparison now supplies its full fictional packet. Disease, indication, variant, therapy, assays, platform, configurations, materials, adequacy premises and rule are invented for teaching. None is patient advice or product evidence.

The source clinical relationship is stipulated supported for this exercise. Its statement fixes the patients, material, result and therapy to which that support belongs:

Note

In adults with fictional Disease D1 under fictional Indication I1, Source Assay S on Platform P1 classifies Variant V1 in formalin-fixed, paraffin-embedded (FFPE) tumor tissue to identify the biomarker-defined population for fictional Therapeutic T1.

Formalin-fixed, paraffin-embedded tissue is chemically preserved tissue supported in wax for sectioning. The proposed target keeps that tissue context but adds an individual-benefit proposition:

Note

Target Assay T on unchanged Platform P1 identifies the same V1-defined population in the same FFPE tissue context and establishes the same expected treatment benefit from Therapeutic T1 for each identified patient.

The source configuration is practice-worked-source-cfg; the target is practice-worked-target-cfg. D1/I1 adults, FFPE tissue, V1, T1, clinical timing,

source clinical anchor, Platform P1 and the positive/negative classification language are unchanged. The *assay-specific* reagents, target processing, filters and classification implementation differ. This is not a platform transfer, and it is not HelixBridge.

Which results selected the patients, and which results are compared?

Older source-assay records select the patient panel: one specimen from each of **100 previously source-positive and 100 previously source-negative adults** in the stated disease and indication. These older labels fix two selection strata. They are not the fresh source results in the comparison table. The constructed 100/100 balance is not disease prevalence.

Allocated material from **every one of the 200 specimens** then undergoes a fresh source examination and a fresh target examination. Examination order is randomized; operators are masked to the old classification and to the other assay's result. Both fresh records are locked before comparison. This yields **400 fresh assay attempts**, but only **200 patients and 200 fresh pairs**. A nonbinary fresh result would stay in the account rather than being removed to preserve the original 100/100 layout.

In the observed panel, all 200 fresh source results happen to retain the older labels, and both assays produce binary fresh outputs for every patient. There are no fresh initial invalids, unresolved pairs, exclusions or repeats in this patient panel. An invalid fresh source result, had one occurred, would have stayed invalid rather than being replaced by its old selection label. The table therefore compares the locked fresh classifications, conditioning on Source Assay S within this selected panel.

Locked source classification	Target positive	Target negative	Total
Positive	98	2	100
Negative	1	99	100

The positive row contains 100 fresh source-positive pairs: 98 target results are positive and two are negative. The negative row contains 100 fresh source-negative pairs: one target result is positive and 99 are negative. The two positive-to-negative disagreements are identified as **D01 and D02**; the negative-to-positive disagreement is **D03**. Prespecified investigations confirm specimen association and locked configurations and locate observations near the classification boundary. The original disagreements remain in the table. Their investigation supplies neither replacement labels nor treatment outcomes, and the source comparison is not perfect clinical truth or an all-comer study.

The prospective rule assesses each row separately. **Positive percent agreement (PPA)** is the target-positive proportion among fresh source-positive pairs; **negative percent agreement (NPA)** is the target-negative proportion among fresh source-negative pairs.

Each receives a separate two-sided **95% Wilson confidence interval**, whose lower endpoint must be **greater than 0.90**. An overall match cannot replace either condition.

An interval describes sampling uncertainty for a proportion under a stated model. Across repeated comparable samples, its procedure has nominal long-run coverage. It is not a probability that this fixed patient group or an individual receives treatment benefit.

The arithmetic starts with one conditioning group: **x** agreements among **n** paired patients, giving observed proportion $q = x/n$. For a two-sided 95% interval, the normal-curve multiplier is $z \approx 1.96$. Squaring it means multiplying 1.96 by itself. The calculation below will be applied separately to each agreement row and to each assay's initial-invalid proportion.

The adjustment divisor is $d = 1 + z^2/n$. It gives the Wilson center $c = [q + z^2/(2n)]/d$. The added term and divisor pull an extreme observed fraction inward; they do not add patients or change the observed x/n .

The half-width is $h = z \times \sqrt{[q(1 - q)/n + z^2/(4n^2)]} / d$. Here n^2 is n multiplied by itself, and the square root returns a nonnegative size. The first term reflects variation tied to the observed proportion; the second keeps a finite-sample width even when q is zero or one. The endpoints are $c - h$ and $c + h$. Values such as 90% enter as proportions—**0.90**—throughout this calculation.

The National Institute of Standards and Technology (NIST) method supports this formula. It does not supply the fictional 90% or 2% rules. Nominal long-run coverage depends on the binomial-model assumptions; it does not establish an individual's benefit. [Source 6]

For this same-context teaching bridge, the source clinical anchor, sampling within each source stratum, independent-participant model, material fitness, boundary-spectrum coverage and consequence-based justification of the margins are **supplied as appraised prerequisites**. Their adequacy is not inferred from the high agreement percentages. These premises apply only to this bounded exercise, not as recommended real-world margins.

The patient comparison has a second evidence stream: a **separate 16-material challenge panel**. These prepared materials test specified assay behaviors; they add no patients. Both assays examine each material, with its expected initial state locked beforehand. That prevents equivocal, invalid, no-call or partial output from being recoded as a convenient negative. Only **Q13–Q16** have a permitted repeat from the retained preparation. The table identifies the state each group challenges.

Prepared materials (IDs)	Expected initial state	Observed source / target behavior
Q01–Q04	Two boundary-positive and two boundary-negative materials	Both assays return the specified state for each material
Q05–Q06	Equivocal	Both equivocal; no binary result substituted
Q07–Q08	Invalid, not eligible for repeat under this challenge rule	Both invalid and retained
Q09–Q10	No-call for V1	Both no-call; not counted negative
Q11–Q12	Partial report with V1 unresolved	Both partial with the unresolved target retained
Q13–Q16	Initial invalid, eligible for one repeat from the retained preparation	Both initially invalid; both recover the same two positive and two negative final states after one repeat each

The last row shows why initial and final states must both survive. Four materials are expected to begin invalid, then recover under an allowed repeat; a final-only record would lose half of that behavior. The earlier rows distinguish boundary-positive/negative, valid equivocal, invalid with no repeat, target-specific no-call and partial report with the target unresolved.

Sixteen materials on two assays give **32 first attempts**. One repeat on each assay for the final four materials adds **eight attempts**, for **40 challenge attempts**. These are separate from the patient panel's **400 fresh attempts**. The finite challenge rule asks whether every listed initial state and allowed repeat disposition matches its locked specification on both assays. It does not estimate the frequency of these states in routine patients.

A different question concerns a fresh initial examination's invalid probability in the **preclassified study population**. The unit is one selected patient per assay. Each assay has **zero initial invalids among its own 200 fresh examinations**, so its calculation is **0/200**. Pooling the source and target attempts into 400 would change the question. The challenge materials do not belong in this denominator either. Older selection labels did not guarantee fresh validity; all fresh states were retained.

Within each assay, the exercise permits pooling the two fixed older-selection strata by assuming independent participants and a **common initial-invalid probability for that assay across those strata**. Source and target are not assumed to share a probability. A

material probability difference between older strata would invalidate the within-assay pooling and require a suitable stratified analysis.

The prospective validity rule uses the other end of the interval: each assay's separate two-sided 95% Wilson **upper endpoint must be below 0.02**, or 2%. Both the source and target must meet that condition. Even zero observed invalids therefore needs an uncertainty calculation; zero observed is not a literal zero-risk finding.

This inference concerns selected, preclassified patients. It does not estimate the initial-invalid probability among all specimens screened before selection. The challenge panel's invalid and recovered states retain their value for the finite behavior question and stay outside the patient probability calculation.

The complete prospective decision is an **AND rule**. Both agreement lower bounds must pass separately; both patient-panel invalid upper bounds must pass separately; every challenge state and permitted repeat must match; and the supplied clinical and material prerequisites must hold. A high pooled match cannot compensate for a failed condition. The separate two-sided 95% intervals also do not provide simultaneous 95% coverage for the whole bridge or establish identical classification for every individual.

What evidence addresses the added benefit words?

For the target assay, no patient-benefit study, outcome, comparator, analysis set, effect-modifier assessment or uncertainty estimate is supplied. This is a separate evidence entry from paired classification, whose observations remain available for their own question.

Decision

What are PPA and NPA, and do both Wilson lower endpoints meet their strict rule? Do both assays' invalid upper bounds meet their separate rule, and does the challenge panel reconcile? Which object rows and supported relationship follow, and which target words remain open? Would the same AND rule accept 88 target-positive results among 100 source-positive patients if all 100 source-negative patients matched?

How the separate conditions support a bounded conclusion

What does the completed memo permit?

Which words belong to the source and to the target?

The completed header fixes the two propositions before their evidence is judged. The source identifies a population for the therapy in the supplied context. The target adds benefit for every identified person. Configuration and authority entries preserve this as a fictional U.S. teaching comparison, without giving either assay or therapy an actual authorization.

Field	Completed record
Source claim	S identifies the V1-defined population for T1 in the stated D1/I1 adult FFPE context
Target claim	T identifies the same population and establishes the same expected benefit for each identified patient
Source and target configuration	practice-worked-source-cfg □ practice-worked-target-cfg
Authority frame	U.S. teaching context; fictional stipulations; no actual authorization decision
Clinical relationship	Same stipulated source identification anchor; added individual-benefit proposition in the target
Memo responsibility	Exercise author/reviewer records the technical conclusion; actual diagnostic and therapeutic decision owners are not appointed by this exercise

Which changed object owns the evidence?

The clinical-relationship row of the header is the key reading: the stipulated identification anchor stays the same, while the individual-benefit proposition is added. That difference governs how much the assay comparison can support.

The physical and computational changes belong to the assay row: reagents, target processing, filters and classification implementation can move a boundary call or alter a nonbinary state. The platform, specimen and clinical context are explicitly unchanged. A platform-change row would invent a difference, and no separate lifecycle change event is supplied for a crosscutting change row.

The following assay record connects those differences to the fresh paired comparison, retained disagreements, separate state challenge, patient-panel validity bounds and same-context prerequisites. Its final entries distinguish the relationship supported by that package from the added proposition it never observes.

Field	Completed record
Invariants	D1/I1, adults, FFPE tissue, V1, P1, T1, timing, source anchor
Changes	Assay-specific reagents, target processing, filters, classification implementation
Plausible affected relationships	Detection/classification near a boundary; positive/negative assignment; invalid, no-call, partial, and repeat behavior
Evidence	98/100 positive and 99/100 negative agreement; retained three discordances; zero patient-panel invalids per assay; separate 16-material state challenge; specified Wilson rules and supplied claim-matched prerequisites
Supported relationship	T preserves the bounded V1 patient-classification relationship in the same stated context
Unsupported proposition	T establishes the same expected treatment benefit for each individual
Dependency	The clinical anchor remains limited to the source D1/I1, V1, FFPE, T1 relationship

How do the separate conditions decide the result?

The evidence row is read together with the dependency row: the comparison belongs to the stated source population, tissue, variant and therapy relationship. The supported/unsupported entries are conclusions of the following checks, not an arithmetic division of the claim.

For **PPA**, the conditioning set is **100 fresh source-positive pairs**. The target is positive in 98, so $q = 98/100 = 0.98$, or 98%. With $z = 1.96$ and $z^2 = 3.8416$, the divisor is $d = 1 + 3.8416/100 = 1.038416$.

Next, the adjusted center is $c = [0.98 + 3.8416/(2 \times 100)]/d \approx 0.962242$. For the half-width, put $0.98 \times 0.02/100$ and $3.8416/(4 \times 100^2)$ inside the square root; take the root, multiply by 1.96, and divide by d. That gives $h \approx 0.032256$.

Finally, $c - h \approx 0.929987$ and $c + h \approx 0.994498$, displayed as **92.999%–99.450%**. The **lower** endpoint, not just the observed 98%, exceeds the exercise's 90% condition.

For **NPA**, repeat the same sequence on the *other* conditioning group, the **100 fresh S-negative** pairs: $q = 99/100 = 0.99$. With n again 100, d stays 1.038416; $c \approx 0.971873$ and $h \approx 0.026360$. Their difference and sum give **94.551%–99.823%**. This lower endpoint separately exceeds 90%. These are agreements with fresh S in the selected context, not new clinical sensitivity or specificity against perfect truth.

For initial validity, use $x = 0$ **invalids** and $n = 200$ **fresh examinations** for S , then the same x and n separately for T . The observed q is zero, but $d = 1 + 3.8416/200 = 1.019208$. The Wilson center and half-width are each about **0.009423**, so the upper endpoint is **0.018846 $\approx$ 1.885%**. It is below the separate **2%** rule for S and for T under the exercise's independent-participant and within-assay common-probability assumption across older strata. The positive upper bound shows why zero observed is not proof of zero underlying probability. A material stratum difference would invalidate the pooled model, and this interval does not describe all-comer screening before preclassification.

The challenge check is not another 16-patient validity study. It retains **16 material pairs**, **32 first assay attempts** and **eight permitted repeat attempts** on Q13–Q16. Boundary-positive/negative, equivocal, no-repeat invalid, V1 no-call, partial V1-unresolved, and initially invalid then recovered behaviors all match their locked requirements on both assays. The challenge's observed invalids are neither pooled into the patient-panel 0/200 rates nor deleted. The three discordant patient pairs remain locked after investigation.

Together, the two separate agreement lower bounds, two separate validity upper bounds, challenge rule and supplied clinical/material prerequisites meet the fictional **AND** condition for the bounded *classification relationship*. A pooled **197/200** overall agreement would not replace those checks. These marginal two-sided 95% intervals do **not** amount to simultaneous 95% coverage for the entire multi-condition bridge, and the result does not prove S and T classify every individual identically.

In the proposed counterfactual, only **88 of 100 source-positive** patients are target-positive. PPA would be **88%**, already below the 90% lower-bound criterion. Perfect agreement among the source-negative patients would not rescue that separate failed condition. The planned rule stays fixed after the favorable row is seen.

The original target is **partially supported**. The bounded variant-defined classification relationship is supported in the stipulated adult disease/indication, preserved-tissue, platform and therapy context. The residual is the added statement that the target establishes the same expected treatment benefit for **each** identified patient. No target patient-benefit outcomes or analysis support that proposition here, and the source anchor cannot supply an individualized promise by itself. This is a limit on the actual words, not a universal requirement for a new outcome trial for every follow-on assay.

Note

Under the stipulated same-context evidence, Target Assay T preserves the source V1 classification relationship. The added individual-benefit proposition is not supported by the supplied evidence and is excluded from the technical conclusion.

Reviewer Lens. The separate lower bounds, state challenge, validity bounds and retained discordances support a positive classification conclusion within the stipulated population and result rule. The therapeutic-interface owner can either support the additional benefit proposition through an appropriate design or remove it. Neither the high pooled match nor the technical judgment grants companion-diagnostic authorization.

Consultant Lens. The assay owner preserves the locked pairs and investigates the three disagreements without relabeling them. The statistician preserves separate patient and material panels, the assumption behind each 0/200 calculation, and the absence of simultaneous whole-bridge coverage. The clinical owner can state the supported group-identification claim and make a distinct evidence decision about individual benefit. Closure takes the form of accurate claim wording or justified additional support.

Figure 13.1: Bridge only the relationship the evidence addresses is a post-commitment view of this memo: source clinical anchor □ fresh paired classification evidence □ bounded classification conclusion, with the added individual-benefit proposition visibly outside that conclusion. The table and explanation above, not the figure alone, carry the argument.

Chapter 13

Bridge only the relationship the evidence addresses

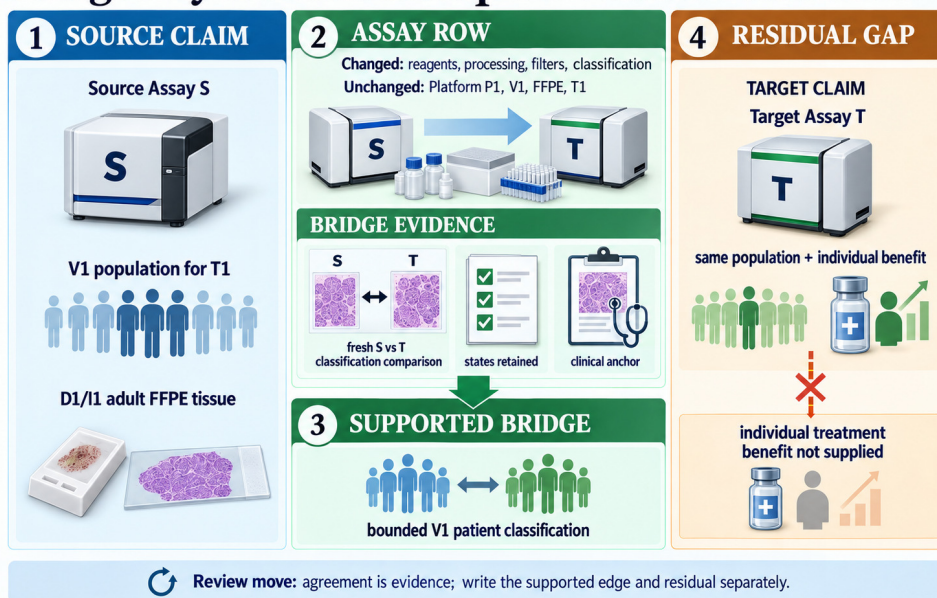


Figure 13.1. Bridge only the relationship the evidence addresses – Technical review plate.

What if only the target changes?

The next two packets change the target in different ways while retaining the earlier evidence. Both judgments come before their shared feedback; a familiar table does not make the new claim equivalent to the old one.

The narrower target – packet A

Replace **only** the worked packet's target statement. It now claims preservation of the V1-defined patient-classification relationship in the same population, indication, FFPE specimen, Platform P1 and Therapeutic T1 context. The individual-benefit proposition is removed. Every evidence premise is unchanged.

Decision

What source and target statements, object row, supported relationship and decision follow? Is there a residual within this stated target?

The liquid target – packet B

Return to the *original* worked target, including its individual-benefit proposition. Replace **only** the target specimen path with **venous whole blood** □ **plasma** □ **isolated cell-free DNA (cfDNA), with circulating tumor DNA (ctDNA) only when present**. The source remains the FFPE tissue assay.

No liquid preanalytical, assay-performance, reference, classification or clinical-anchor evidence is added. All other target terms and supplied evidence remain from the original packet.

Decision

Which object rows now matter, and what disposition follows for the full liquid target? Which relationship could the original tissue comparison directly support here?

Why the same evidence gives different target judgments

How the target wording changes the conclusion

A is supported within its exact boundary. The only change narrows the target to the patient-classification relationship already addressed by the adequate stipulated comparison and matching source clinical anchor. Its assay row closes that bounded target; there is **no residual inside that fully stated target**. It does not acquire individual benefit, a new population, another specimen or an authorization. Requiring every possible downstream claim before crediting this bounded result would misread the task.

B is not supported by this packet. The target is stated, but its new specimen route requires a specimen row *and* an assay-applicability row. Blood collection, plasma preparation, cfDNA input, possible ctDNA contribution, background, detection and failure routes can change which patients are called positive or negative. Retaining the assay and platform names does not supply data for those relationships. In particular, a negative plasma result cannot inherit a tissue-negative meaning.

No liquid analytical, patient-classification or treatment-selection relationship is directly supported by the FFPE comparison. The complete proposed liquid relationship, including its original added benefit proposition, remains unsupported. B differs from a case whose *target claim itself* is missing: here the words are clear, but the evidence does not reach them.

Reviewer Lens. Packet A permits a direct positive conclusion within its stated boundary. Packet B requires evidence about the changed material and assay routes before a clinical relationship can transfer; a shared biomarker cannot provide it. The

same question applies after a sequencing-input or image-workflow change: what new input and result meaning must the earlier evidence address?

Consultant Lens. The narrower packet is a bounded claim revision that preserves the existing patient and challenge evidence. The liquid packet assigns different work to collection, plasma-processing, cell-free-DNA assay and clinical-reference owners, with a separate decision about how the result would direct care. Closure needs actual claim-matched evidence or a narrower target. A tissue negative cannot be invented as a liquid control.

What do the familiar records actually permit?

HelixBridge Dx is the fictional tissue-oncology sequencing assay described as identifying variants associated with eligibility for a named therapy. The present record supplies neither the therapy identity, exact population, alteration set, clinical role nor complete claim wording. The worked fixture's disease, variant, therapy and study cannot fill those fields.

Its record identifies the validation application **7.0**, report rules **R19** and an initially described “current” knowledgebase. The knowledgebase used in analytical validation is **DB-18**; the separate final technical-report view concerns **DB-19** and describes “strong overall agreement.” Comparator, analysis set, strata, uncertainty and clinical-bridge basis behind that phrase remain unknown. The changed knowledgebase added a reference-transcript record that makes a previously rare mapping anomaly execute more often. No changed report classification or treatment outcome follows from that fact alone. Exact commercial configuration and release remain unresolved.

HelixBridge Dx Liquid remains a separate proposed branch. Its broad program purpose continues while its proposed route is whole blood □ plasma □ isolated cell-free DNA, with circulating tumor DNA a possible tumor-derived fraction. No tissue result rule, study, negative-result meaning, clinical anchor, release or regulatory outcome transfers automatically to it.

Decision

What can each memo header say from these facts? Which source and target fields remain missing, and can either branch yet have a precise residual unsupported claim?

Which missing fields prevent either bridge from being assessed

Both proposed clinical bridges are **not assessable** from the supplied snapshots. Neither has an exact target therapeutic relationship, so a precise residual would be invented. The tissue request therefore concerns therapy, indication, population and timing; configured examination, alteration and result rule; and comparator, analysis, uncertainty, clinical anchor and decision basis. A report described as “strong overall agreement” leaves these fields open.

The liquid branch additionally needs its own collection-to-cell-free-DNA assay path, patient context, every result-state and negative-result meaning, reference process and proposed reuse of tissue evidence. The program relationship locates the owners' questions without supplying a supported tissue-to-liquid relationship. Neither branch is established as a U.S.-authorized or EU-conforming companion diagnostic, and association with eligibility establishes neither essentiality nor benefit.

Reviewer Lens. The validation application/rules/knowledgebase tuple and separate technical-report state are useful provenance. They supply no source clinical anchor or commercial identity. The exact target statement comes before a numerical bridge request. In the liquid proposal, specimen and assay rows remain separate. A bounded not-assessable judgment identifies the actual obstacle when claim wording is missing.

Consultant Lens. The program owner can define tissue and liquid targets separately, with configuration and clinical owners supplying versions, population, therapy and result rules. The evidence owner can retrieve the actual comparison design and source relationship. The liquid evidence plan follows its own blood, plasma and cell-free-DNA route. Neither a precise residual nor a closure follows from the adjective “strong.”

Can one comparison cover a compound change?

This is a **different fictional exercise**, with no regulatory approval or connection to D1 or HelixBridge. In Disease D2, Assay A2 on Platform P2 classifies Alteration V2 from an unchanged tissue path for a bounded treatment-selection action involving fictional Therapeutic T2. The source claim and clinical anchor are stipulated supported.

The same clinical claim is proposed after moving A2 to Platform P3. The new platform changes signal generation, firmware and base-calling implementation, and the final classification also uses a changed assay-specific filter.

A paired P2/P3 comparison uses routine-input positive and negative materials for the **common V2 subtype**. The comparator is fit for those represented states, and their classifications satisfy a prespecified qualitative condition. But near-boundary inputs, another claim-relevant V2 subtype, invalids, no-calls, partial results and low-input specimens are outside the evaluable set. No claim-matched uncertainty assessment or

discordance investigation addresses those omitted relationships. The source anchor still matches D2, tissue, V2 and T2; no evidence says whether every claim-relevant classification survives the changed path.

Decision

What source and target claims, changed fields and evidence limits does this dossier establish? Which relationship is supported, what residual remains, which decision follows and who owns the next question?

What the represented comparison can carry across

How the two changes share one evidence item

The platform row compares the old and new signal generation, firmware and reusable base-calling route. The assay row compares the prior and changed assay-specific final-classification filter. A controlled-change row links their dependencies and reopened evidence. It cannot replace either object. The following table allocates the supplied comparison without counting it twice as independent evidence.

Row	Source target	Evidence conclusion
Platform/technology	P2 □ P3; signal generation, firmware, and platform base-calling change	Comparison addresses represented routine-input states only
Assay	A2's prior filter □ changed assay-specific classification filter	The same restricted comparison cannot establish preservation of omitted claim-relevant classifications
Controlled change	Prior configured state □ proposed P3/A2 state	Links the two affected rows, their dependencies, evidence limits, and responsible change decision

The platform row's positive result applies to the represented routine-input states; the assay row shows why that same comparison cannot establish omitted classifications. Its direct observation is that the studied new state meets the stipulated condition for **routine-input positive and negative classifications of the common variant subtype** with a fit comparator. It provides no observation of the other claim-relevant subtype, boundary or low-input conditions, or invalid, no-call and partial pathways. Those omissions leave support absent; they do not establish bad performance.

The decision is **partially supported**. The omitted subtype and input/state conditions remain unmatched target propositions, leaving the full clinical-claim transfer open. A claim-matched gap assessment needs suitable materials and reference/comparison

roles, uncertainty and state accounting, and evidence tied to the exact P3/A2 configuration. The useful routine-input comparison remains one bounded result; platform and assay rows cannot be added as scores. Chapter 17 addresses the wider lifecycle response, while the jurisdictional owner handles any actual regulatory route.

Reviewer Lens. The represented routine-input result deserves acceptance within its subtype and conditions. A focused request then concerns the omitted patient-classification routes. A pathology transfer would need its own changed scanner, model or display path and pathologist decision unit. The separate unnamed **2 initial invalids among 40 sequencing attempts** and **1 late worklist arrival among 25 pathology cases** are not observations for this fixture, HelixBridge or PathSight.

Consultant Lens. Platform, assay-filter, material/reference and clinical-claim owners have different decisions to make. The common comparison stays one evidence item linked to both object rows. A prospective plan can specify the omitted subtype, input and result-state coverage. Until the broader claim has support, a narrower represented relationship can be stated while an owner plans the missing work; passing routine inputs alone cannot close the compound change.

Who uses the bounded conclusion?

The reviewer needs a reconstructible source-to-target argument. In the main comparison, the supported classification and added benefit proposition separate. In the platform/filter transfer, the represented and omitted input states separate. A generic request for “more concordance” would lose the different evidence questions.

The diagnostic manufacturer assigns those questions to versions/report rules, signal/firmware, materials/reference processes, study strata and complete result-state capture. The clinical and therapeutic-interface team asks which patients the resulting rule identifies and whether the source relationship applies to them. Group identification remains distinct from an individual outcome promise.

The jurisdictional specialist receives the same bounded conclusion alongside actual labeling or intended purpose. A comparator's status and a signed technical memo do not authorize the target. Changing the audience cannot enlarge the evidence or resolve an unknown.

Return to the added words

How the opening conclusion separates

The opening sentence contained a possible patient-classification bridge and an added individual-benefit proposition. The worked evidence and stipulated anchor support the former within their exact boundaries; they do not support the latter. Packet A showed that a narrowed target can be fully supported. Packet B showed that the same tissue evidence does not travel automatically to plasma. The P2□P3 dossier exposed a compound platform-and-assay change, while the named tissue and liquid program lacks the precise claim fields needed even to write a residual.

Agreement is not weak by definition. Paired agreement, valid state accounting, uncertainty and a fit clinical anchor can be valuable bridge evidence when they address the exact target. The task is to state which relationship they address.

Questions that test the bridge

Could the memo now be reconstructed from its questions? The following distinctions test whether its conclusions remain attached to the right evidence.

1. what separates prognosis, treatment-response prediction, a selection action and a jurisdiction-qualified CDx relationship;
2. why a source clinical anchor is more than a marketed comparator;
3. when both assay and platform rows are needed;
4. why residual unsupported claim is not a percentage;
5. what change to the worked *target statement*, without adding evidence or an individual-benefit claim, produces a fully supported bounded decision;
6. why the sparse HelixBridge tissue and liquid branches cannot receive the worked packet's decision.

What each distinction protects

Those four roles ask about a disease course, treatment-related response, a clinical action and an essential device-to-product relationship under specified rules. A clinical anchor is a supported source relationship in exact patients, specimen and therapy context. A compound platform move with an assay-specific filter needs both object rows; change links them. Residuals are unmatched target propositions, not agreement complements. Removing the extra individual-benefit words yields Packet A's supported bounded classification claim. The named branches lack the precise target and clinical evidence premises that the fictional exercise supplied.

Chapters 14 and 15 turn to digital pathology: first the image-producing system, then the reader, algorithm and workflow relationship. Carry the same source–target and claim-boundary discipline, without assuming an attractive component comparison answers a different clinical question.

Source trail

1. **FDA–National Institutes of Health biomarker terminology resource, BEST**, Understanding Prognostic versus Predictive Biomarkers, published 22 December 2016. Used for the bounded prognostic/predictive distinction, not a real variant, therapy or universal trial design.
2. **FDA (2014)**, In Vitro Companion Diagnostic Devices, sections III–V; official final-guidance record, checked 25 September 2026. Nonbinding U.S. source for essentiality, four described roles and labeling context, not a universal bridge method.
3. **Regulation (EU) 2017/746 (IVDR)**, Article 2(7) in the 10 January 2025 consolidation, read with Recitals 10–12 in the Official Journal base act. Underlying EU law as amended is binding within scope; the consolidated display is a documentation aid without independent legal effect. Detailed procedure belongs to Book 2.
4. **FDA (2017)**, CDRH's Approach to Tumor Profiling Next Generation Sequencing Tests. Frozen historical illustration, not comprehensive current policy or transferable authorization.
5. **FDA (2020)**, Developing and Labeling In Vitro Companion Diagnostic Devices for a Specific Group of Oncology Therapeutic Products, sections II–IV, especially IV.1 and IV.4–IV.5; official final-guidance record, checked 25 September 2026. Narrow nonbinding U.S. group-labeling context with several possible evidence routes, not an automatic bridge rule.
6. **National Institute of Standards and Technology**, Wilson confidence-interval method, checked 25 September 2026. Supplies the binomial-proportion interval calculation, not the fictional sample, clinical prerequisites, margins or claim decision.

The numerical bridge exercise and all qualitative dossiers are original teaching fiction. Their sample counts, planned conditions, Wilson results and bounded dispositions are not prescribed thresholds, a clinical recommendation or real-product evidence. The memo and decision labels are author synthesis; current policy and actual jurisdictional procedure require separate assessment before real use.

CHAPTER 14

Digital Pathology: The Image-Production Chain

What image did the unchanged model receive?

A pathologist can look at a glass slide or at a digital image made from it. The digital route starts before the screen: patient tissue becomes a prepared slide, a scanner records it, software constructs and stores an image, and a viewer presents it. An assistive algorithm may receive a different version of that same slide's image. The important question is what each consumer actually receives.

PathSight Cloud gives that question a concrete setting. Its model is identified as M3.2, and its commercial **model weights**—the learned numerical settings—are stated to match those evaluated. The study images mostly came from Scanner A, the first scanner family. Scanner B, the second family, adds proprietary sharpening before export. Normalization was upgraded after validation, an average-color scanner comparison omitted weakly stained tissue near low-confidence boundaries, and the interface changed how low-confidence cases appear. These are supplied differences and limits, with their effects still to be assessed.

Decision

What route leads from tissue to the image seen by the pathologist and to the input seen by the model? Where might the earlier evidence need an applicability comparison? Which changes are known, and which effects remain questions?

This fictional assessment concerns changes after validation. It supplies no new study or verdict about actual release, diagnosis, harm or regulatory status. The later PathSight map returns to this same snapshot after the image-production steps have become clear.

Follow the object, then attach its evidence

Suppose the record shows that an image file arrived. What would connect that arrival to the claim that the correct case appeared to its intended user? First the route must be identifiable: what entered each step, which operation, version and settings acted on it, and what left. Those relationships make the **configuration record**.

The next question is what was tested on that route. An **evidence record** attaches a particular property to a reference, material and conditions, observations including failed states, a criterion if one was set, and a bounded conclusion. Together the two records form an image-chain map.

In the arrival example, successful delivery is useful evidence. Correct case association and the actual viewing path need their own observations before the claim can reach the user. Diagnostic performance asks a further question about the task the user performs. Following these joins keeps the valid transfer result intact while identifying what still needs support.

Which earlier boundaries belong on this route?

Decision

How does Chapter 2's component differ from the configured product? What makes Chapter 5's reference specific to a property? Where does Chapter 6's preparation enter? How do Chapter 8's first attempt, repeat and final state stay distinct? What could Chapter 9's missing combination of changed factors conceal? How does Chapter 12's clinical-study question differ from a technical comparison?

How the earlier boundaries join the image route

The scanner and model identify parts of the image-producing arrangement. Preparation determines what tissue reaches those parts, while a reference establishes only its specified comparison: a color target is not a reference diagnosis. A recovered rescan keeps the first failed scan visible and adds no patient. Similarly, a study dominated by the first scanner family can miss the second-family-by-weak-stain combination. These dependencies connect a technical comparison to its actual question. Chapter 15 will ask what the reader and algorithm did with the resulting image.

How does the tissue become an image?

A clinician sends patient tissue to a pathology laboratory so a pathologist can inspect its structures. Before scanning, the laboratory identifies that material and makes one or more glass slides. A small structure can appear in the image only if the physical preparation brought it onto the slide in a usable form. A sharper scan cannot recover a piece left out or reverse its preparation history.

Recording the prepared slide's optical information as digital data for later viewing is **whole slide imaging**. The U.S. Food and Drug Administration (FDA) 2016 technical

guidance treats acquisition and the workstation/display as configured parts of that system. It addresses surgical-pathology technical assessment, without supplying clinical submission data needed for clearance or approval or a method for every stain, fluorescence use or mobile viewer. [Source 1] The following route explains the work; it does not prescribe one preparation or processing sequence for all laboratories and scanners.

What made it onto the glass?

In a common fixed-tissue route, chemical preservation slows the changes that would otherwise continue in the collected tissue before examination. This fixation also affects how structures may appear. Knowing that it happened therefore leaves the particular preservation conditions to be identified.

The preserved and processed material needs support before it can be cut. Embedding gives it that support in a block; sectioning takes a thin piece from the block and places it on glass. What reaches the slide is the section actually cut and retained. A block can supply several slides, and a case can contain several blocks, so the slide is already a selected part of the material.

That section must then make the relevant structures visible. Staining gives selected structures contrast; mounting protects and presents the stained section for viewing. If preparation obscured the needed feature or the relevant area never reached the glass, a faithful scan will carry that limitation forward. These operations explain the route without defining a preparation recipe.

The record can now ask which tissue and clinical task the slide represents, how it was identified and prepared, which stain was used, and whether scanning accepted or rejected it. Fixation time, section thickness, stain, mount and rescan policy have no universal values here, and PathSight supplies none of those particulars. Each proposed application needs its own preparation and relevant tested range. Chapter 6's sequencing route asked the parallel question of which patient specimen and derived material entered extraction. The operations differ, but successful downstream computation cannot restore an unrepresented or mishandled input in either route.

What did the scanner sample?

With the slide ready, the scanner shines light through or onto it. Optics form an image of a small area on a sensor, which converts the light pattern into recorded values. Moving the slide or imaging field brings further areas into view. Software chooses the areas, detail and focal positions to sample. This is **acquisition**. To reconstruct it, a reviewer needs the scanner unit, acquisition software, scan mode, calibration and settings relevant to the claim; the scanner model name alone cannot identify that state. [Source 1]

The fields then need to cover the intended material and join into a larger image. The scanner may also save several resolution levels, often called a pyramid, so a viewer can open a low-detail overview before loading detailed regions. A successful overview shows that some data are available. It leaves open whether every relevant region was captured with adequate detail. Tissue detection determines what is included, focus determines which detail is recorded, and placement determines how neighboring fields meet.

Reviewer Lens. For a task involving small weakly stained regions, the decisive acquisition question is whether the studied slides actually contained those regions and whether they were assessed at the focus and resolution used. A late report by itself cannot locate an acquisition, delivery or viewing problem.

Consultant Lens. Slide identity, first scan, rescan, settings, image version and acceptance state can be linked while the study is planned. That trace makes a later unexpected image investigable and keeps another scan of the same slide from becoming another case. The same record lets owners distinguish an acquisition issue from what happened downstream.

Was the structure sharp, present and in the right place?

Suppose a small structure is hard to interpret in the digital image. If it was recorded but blurred, the question concerns **focus**. Even after its detail is resolved, the reviewer must ask whether all relevant slide material appears: **tissue coverage**. Once the material is present, neighboring acquired regions must also meet in their intended spatial relationship: **stitching**. The table helps locate which of these questions a reported “image quality” result actually answers.

Question	What the check needs to distinguish
Is the recorded structure adequately focused?	Included tissue whose needed detail is blurred
Is the relevant slide material present?	A region omitted from acquisition or delivered content
Are neighboring fields placed correctly?	A join that is misregistered, duplicated or discontinuous

The first row concerns an included field whose structure has merged edges because it was recorded at the wrong focal position. A check compares task-relevant fields at the recorded resolution with a defined detail reference or assessed slide feature. How those fields were selected matters: a sharp central field can support a central-field result while leaving other regions unexamined.

Passing that focus check still leaves the second row open. Tissue near the glass edge might never appear in the digital image. Coverage therefore compares relevant material expected on the physical slide with delivered content, including the region-selection rule. Sharpness elsewhere cannot establish that the missing area was included.

Even complete coverage leaves the third row. Two fields may contain the tissue but displace or duplicate a shared structure at their boundary. A stitching check examines that boundary against the intended spatial relationship. FDA's guidance treats these properties separately, alongside color, spatial resolution and timing. None becomes a universal diagnostic acceptance criterion. [Source 1]

What establishes the comparison answer?

A technical target with known feature positions can provide a reference for spatial placement or resolution under defined conditions. A defined color target can help assess color response. Neither tells the reviewer what tissue should appear from a particular glass slide; that coverage comparison needs the actual slide's relevant material identified. An expert region mark has an assignment process and uncertainty too. The reference must establish the property being tested before the conclusion can use it, and none of these objects automatically establishes the patient's diagnosis.

Reviewer Lens. The useful question is which object the reference established and whether the method tested the claimed property. A focus result can be adequate for selected central fields while weakly stained or edge regions remain outside its scope. Sequencing benchmarks have a comparable boundary: no call outside their qualified variant regions is not a demonstrated negative for every genomic position. The objects and methods remain different.

Consultant Lens. Preserving the field-selection rule, reference and failed fields before the summary makes that positive boundary defensible. It also identifies which additional regions or comparison could close a larger claim. A generic request for better image quality would hide the choice.

Figure 14.2 — Focus, coverage and stitching as separate technical questions keeps each property beside its own failure and reference question. It is a comparison aid, not a diagnosis or an acceptance criterion.

Chapter 14 - three image properties

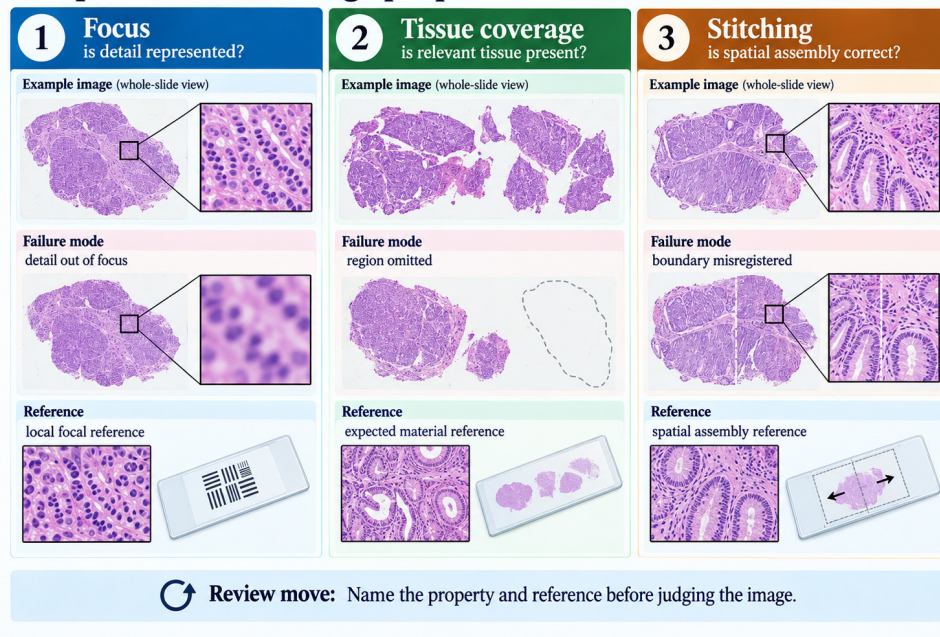


Figure 14.2. Separate focus, tissue coverage, and stitching – Technical review plate.

Could unchanged weights receive different pixels?

Suppose two otherwise comparable digital slides look different before they reach the model. The recorded fields have first been placed and joined into a slide representation, an operation often called composition. Further processing can change their values or arrangement. The product record needs the actual operations, order and settings before the difference can be explained. [Source 1]

If one image appears darker, a rule may remap its values toward a chosen color or brightness range. That is **normalization**, recalled from Chapter 1. The model then receives different values even with fixed weights. PathSight's upgrade identifies a change at this step; its actual rule, settings and effect remain unsupplied.

A different operation may make an edge more pronounced by emphasizing differences between neighboring pixels. This is **sharpening**. The second scanner family supplies it before export. Its presence identifies an operation to compare, while its strength and whether the relevant feature improves, degrades or stays unchanged remain open.

What if the image changes size or resolution? **Resampling** calculates values on a new pixel grid. A small feature can occupy fewer pixels or pixels with different values. Choosing a region before that operation need not give the model the same input as choosing it afterward. The tested sequence and resulting data therefore matter to the applicability of earlier input evidence.

Order also matters when the grid stays put. Consider one path that crops a region and then normalizes it, and another that normalizes the whole image before choosing the region. Their model inputs may differ despite identical weights. This illustrates a possible mechanism; it does not supply PathSight's actual order. The evidence record needs the tested sequence, version, parameters and resulting object to answer the input question.

Which branch reaches which consumer?

Before the stored image can be used, software converts its representation into usable image data. That is **decoding**. To put those data in front of a pathologist, a viewer selects a resolution, color/display handling and screen position. That further operation is **rendering**. Successful decoding establishes one step; the appearance of the rendered view belongs to the next.

The algorithm may take a different branch. It may receive small cropped regions, called **image patches**, at a selected resolution and perhaps after normalization. The pathologist may pan to a suspicious area while the algorithm's rule samples other regions. Sharing a slide therefore does not guarantee identical pixels or views. Any annotation also needs the right image version and coordinates. The crop/normalization-order comparison above shows why the branch must include its actual operations.

Reviewer Lens. Before accepting a reader-study conclusion, the review needs the image the person saw, the data the algorithm processed and evidence for the claimed combined workflow. If an upstream update precedes a changed report, this trace provides a testable input question without establishing model failure. In sequencing, unchanged calling or reporting rules can likewise encounter different readings, alignments or knowledgebase inputs. The analogy supplies no HelixBridge event and does not equate pixels with sequence readings.

Consultant Lens. A manifest can preserve both consumer branches and compare the evaluated input with a proposed input. Evidence remains useful for a bounded property when an appropriate comparison supports the match. Capturing the branch before a reader study makes a later change much easier to assess.

The file arrived. What has been preserved?

Large images may be compressed for storage and transfer. If a method preserves decoded pixel values exactly, it can be **lossless** for that object while using different compressed bytes from the original encoding. A byte-for-byte archive comparison instead tests the checked file bytes themselves. A **lossy** method discards some represented information. The object and property must accompany either label. Neither label alone establishes adequate diagnostic detail, orientation, metadata, case linkage or task validity.

The image can then be transferred, indexed, stored, retrieved and decoded. Suppose every checked retrieved file matches its source bytes. That supports preservation on the tested path. Whether it belongs to the correct accession, appears in the correct user's case list, decodes into the intended view and arrives in time requires further associations and observations. A rescan adds an image version of the same slide, so the record must also retain which version was reviewed and why the first was set aside.

Reviewer Lens. A good byte-preservation result deserves credit while accession–specimen–slide–scan–file associations and failed or retried deliveries are examined separately. A correctly transferred sequencing report has the same limit: delivery alone cannot establish the right specimen finding or intended recipient. Chapters 2 and 10 distinguish those system and result states.

Consultant Lens. The comparison object can be chosen to match each claim: source and retrieved files for integrity, linked identities for association, event timestamps for availability. Security implementation belongs to Book 1B. Here the evidence question is what reached the intended consumer, with its failed and recovered routes visible.

What did the pathologist actually see?

Even preserved and correctly associated data must pass through a viewing path. Review software decodes and renders them through computer graphics onto a display in a particular environment. Its settings, calibration and quality-control state can change what is presented. Two screens with the same size or nominal pixel count do not thereby establish equivalent views for the task. [Source 1]

For example, a viewing mode may enhance an edge while leaving the stored archive unchanged. The stored-byte test cannot answer what that enhancement does to a task-relevant feature. An algorithm receiving patches before display does not receive that display transformation as its immediate input, although the clinician's later response to the algorithm may still depend on the viewer.

Reviewer Lens. A “display passed” statement needs its output, reference, criterion and environment identified. A viewing concern arising in use calls for a configured reproduction before it is attributed to image generation.

Consultant Lens. Preserving the evaluated review software, graphics, display, settings and environment makes comparison with a proposed site or remote viewer possible. The decision is about whether those use conditions remain represented, not whether a screen specification still matches.

How far does a scanner result reach?

A color target may establish a scanner's response under specified conditions. That component result can locate a color mechanism. To know whether interacting parts deliver the required representation, the next question follows the assembled scanner-to-display system. To know what a pathologist does with it, a study needs the defined diagnostic or reader task. Each favorable result remains useful for its own evidence job, even when a later relationship stays open.

FDA's digital-pathology research program identifies the technical-to-pathologist, algorithm and clinical relationship as an ongoing methodological challenge. It supplies no validated universal conversion from a technical score to clinical performance. [Source 2] A usable “image quality” claim therefore names its property, reference, input, method, configuration and task.

Reviewer Lens. Component evidence can be adequate for its specified property without closing a whole-use claim. A qualified sequencing-read processing component has the same kind of boundary: it contributes to the evidence chain while leaving final interpretation and the clinical report to their own support.

Consultant Lens. The missing relationship determines whether an interaction assessment, reader comparison or applicability bridge would help. A later field observation reopens only the relationships its actual record plausibly touches. Treating every open link as a reason to repeat all testing would discard the distinction that made the original result useful.

Whose diagnostic use did the study represent?

Manufacturer characterization can connect scanner, file, display and assembled technical behavior for a proposed product. A pathology laboratory still needs to determine whether its own whole slide imaging system works for its intended diagnostic use, specimens and setting. These complementary questions have different owners.

The College of American Pathologists (CAP) guideline, developed with the American Society for Clinical Pathology and Association for Pathology Informatics, advises local intended-use validation resembling clinical practice, encompassing the system and addressing consequential changes. Its **three strong recommendations** differ from **nine consensus good-practice statements**. Those consensus statements are not

evidence-based recommendations, and they do not require a separate local clinical validation for each component. [Source 3]

Suppose the laboratory changes its viewing environment. Its earlier local result remains a record worth retaining, while its applicability to the changed use needs assessment. Manufacturer component evidence can help characterize the difference without standing in for every local condition.

Reviewer Lens. The owner and intended use attached to each result determine what can carry forward. Another laboratory's preparation or display cannot be silently entered as PathSight's.

Consultant Lens. Connecting bounded manufacturer characterization to the laboratory's changed-use assessment allows each retained relationship and each remaining check to be documented. The change need not erase the earlier local result.

What can a real product record or research tool show?

The frozen FDA decision summary **K232879** shows how specific an actual configuration can be. It describes one Roche Digital Pathology Dx arrangement: formalin-fixed tissue embedded in paraffin wax, slides stained with hematoxylin and eosin, a VENTANA DP 200 scanner, image management and compressed files in its named BIF format, and a specified display context. Its technical assessments remain distinct from the product-specific clinical comparison. [Source 4]

The useful lesson is the set of connected objects the record identifies. It is neither PathSight's arrangement nor a universal recipe or inventory of later Roche versions. Its results and thresholds stay with that real product record.

A software tool answers a narrower kind of question. **ValidPath** is an FDA regulatory-science tool that can generate image patches, map annotations and regions of interest, and assess binary model outputs. In the dated source record, its region-of-interest review output is compatible with Aperio ImageScope. The catalog distinguishes such research tools from recognized standards and qualified Medical Device Development Tools. [Source 5]

A metric it produces concerns the specified inputs and task. Running the tool does not establish tissue preparation, scanner, viewer, reader-workflow or whole-product validity. The Roche record is a frozen product decision; ValidPath is a scoped research tool. Each can help identify an evidence question without supplying a PathSight outcome.

Which handoff does the evidence belong to?

For any point on the route, the first relationship is **input** → **operation and configuration** → **output**. An evidence item can then attach its tested property to that identified handoff. The table helps locate a missing relationship; it does not require every system to contain every listed operation.

Stop on the route	What identifies the handoff?
Tissue and slide	Source, preparation, stain, mounting, quality state and identity
Acquisition	Scanner/unit, firmware, acquisition software, sampling, focus and coverage
Assembly and processing	Region placement, transformations, order, settings and output geometry
File and metadata	Encoding, compression, organization, associated identity and decoder
Delivery and retrieval	Transfer, storage, indexing, image version, availability and recovery
Viewing	Review software, workstation, graphics, display, settings and environment
Algorithm input, where applicable	Input version, patch or region route, preprocessing, model and result rule
Workflow interface	Case list, output visibility, timing, fallback and responsible user

The acquisition row identifies which scanning arrangement acted on a slide. Its unit and settings may be known without a performance result. The linked evidence entry needs the **property and claim, tested configuration and material, reference and method, analysis unit, failed or excluded states, result, criterion and uncertainty**. An unsupplied field remains unsupplied.

The first three rows follow material into an image; file and delivery identify the recoverable encoded version and its case association. The last three distinguish human, algorithm and operational consumers. A reviewer can follow a conclusion through that row and its downstream dependencies. A consultant can identify who owns a missing comparison and which result would close its claim. Filling the record with another product's values would obscure both decisions.

Return to PathSight with the image route in view

PathSight assists pathologists by prioritizing images and highlighting regions; the pathologist retains diagnosis. Chapter 2's baseline established that role before this detailed image route. The model identified in validation has unchanged commercial weights. Most validation images came from the first scanner family; the second adds proprietary sharpening before export. After validation, normalization was upgraded and low-confidence cases became collapsed by default. The exact evaluated and changed arrangements from beginning to end, and whether the changed arrangement was released, remain unstated.

The second scanner family's bench comparison examined **average color values** and omitted weakly stained tissue near low-confidence boundaries. Its numerical outcome and acceptance criterion are not supplied. Separately, a high-volume **simulation** recorded **two positive low-score slides exceeding expected turnaround**. No complete denominator, duration, cause, recovery, patient consequence or deployed-event conclusion is supplied. These are the observations available for the judgment.

What does the complete record now support?

Decision

Where does each supplied fact belong: input, operation or consumer? At each relevant stage, what identity is known and what performance is demonstrated? Which property and conditions did the evidence actually examine, which applicability question remains, and what record or comparison could answer it? Does the supplied evidence establish the full postvalidation image path and second-scanner-family claim? What narrow evidence remains useful, and which missing facts prevent a delay rate, average-color pass or image-quality/clinical-failure conclusion?

How a reviewer reads the PathSight configuration and evidence

Worked configuration and evidence map

The first table locates known and unknown configuration facts along the image path. The second attaches the supplied observations to their possible conclusions. Reading them together prevents an identity from turning into a measured property, or a measured property into evidence for the entire changed route.

Locate the configured object

Stop	What is known	What remains unresolved
Tissue and slide	Whole-slide images belong to the intended workflow	Preparation, stain, acquisition context, slide quality-control state and repeat/rescan rules
Scanner acquisition/export	Both families claimed; validation mostly first family; second adds sharpening	Exact identities, versions, settings, focus/coverage/stitching and represented strata
Processing	Normalization upgraded after validation	Old/new versions, parameters, operation order and effect on input
Algorithm input	Evaluated model weights retained	Actual decoded/normalized/tiling input, rule and full configuration equivalence
Storage and display	Relevant parts of the proposed evidence route	Exact file, metadata, retrieval, rendering, display and viewing conditions
Interface and timing	Low-confidence cases collapsed; two delayed positive low-score slides in simulation	Event linkage, denominator, timing, cause, recovery, reliance and actual release

The tissue row cannot identify an evaluated material range because PathSight supplies no fixation, embedding, section, stain or mounting detail. Moving to acquisition adds real facts—the two-family claim, mostly-first-family study images and second-family sharpening—while settings and difficult-material combinations stay unknown. The normalization upgrade identifies a change but lacks the two versioned input/output records needed to assess its effect. Retained weights establish the model's identity, leaving storage, display and interface conditions to be matched where the claim relies on their output. The final row locates the simulation at workflow presentation without assigning cause or a release relationship.

Match each observation to the conclusion it can bear

Supplied evidence	What it addresses	What it leaves open
Evaluated model package	A studied state with mostly first-family images	Identity of every changed input and workflow component
Second-family bench comparison	Average color was compared	Numerical result/criterion; weak-stain boundaries; focus, coverage, stitching, algorithm or clinical equivalence
High-volume simulation	Two stated timing exceedances occurred	Rate, mechanism, harm and full live-workflow conclusion
Weight identity statement	Evaluated model weights were retained	Preprocessing, input distribution, interface and whole-system applicability

The bench row supports a precise positive fact: an average-color comparison was performed. It contains no numerical result or criterion that would establish a pass, and its omitted weak-stain material remains outside the comparison. The evaluated package also stays useful for its represented state, while the mostly-first-family mix leaves the second-family boundary question open. The simulation establishes two late slides in an exercise and justifies tracing those events. Without its denominator and event path, it cannot establish a rate or cause. The weight statement preserves one component fact without identifying the images supplied to it.

The full postvalidation image-path and second-family claim is **not established by this record**. That leaves useful evaluated-state evidence intact.

Reviewer Lens. The next request needs the exact evaluated and proposed versions, the complete second-family comparison with material strata and reference/criterion, and the simulation's case-level timestamps and state account. A sufficient bridge would address changed inputs and relevant weak-stain boundaries for the proposed claim, with any remaining limit stated. Where it cannot, new or claim-matched assessment may be needed.

Consultant Lens. Scanner, processing-input and queue-interface owners can take those distinct comparisons. Each claim closes against the returned result; the case supplies no universal retest battery or conclusion about released use. This is how the earlier evidence can be retained without stretching it across an unanswered relationship.

Figure 14.1 — Trace the configured image from tissue to its two consumers follows the shared image lineage into the human-viewed and algorithm-input branches. Its

PathSight callouts are placed only here, after the map commitment and worked decision.

Book 1C • Chapter 14

Chapter 14 - configured image path

Trace one configured whole-slide image from tissue and slide preparation to two downstream consumers.

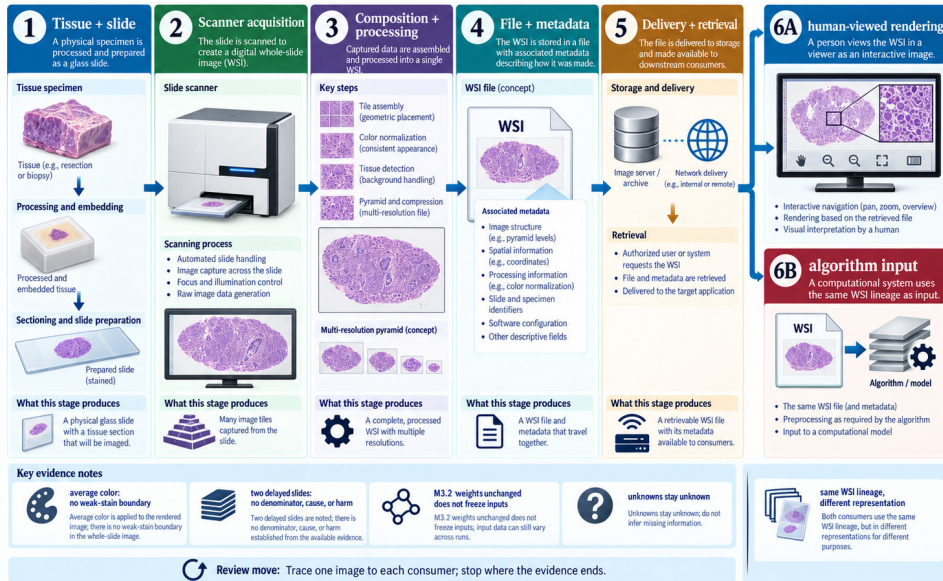


Figure 14.1. Trace the configured image from tissue to its two consumers – Technical review plate.

An archive comparison

This anonymous exercise is not PathSight. A prespecified set of image files travels through one recorded transfer, storage and retrieval configuration. Investigators compare each checked source file with its retrieved file for byte identity and verify the correct case/image-version association. The complete comparison and failure account meet the supplied criterion for **those two properties**. No scanner, decoder, rendered view, display, diagnosis or live queue is compared.

The proposed conclusion is: **the checked files and their case/image-version associations were preserved on this tested archive path.**

Decision

What are the input, operation, output, reference and disposition? Is this conclusion supported, and where does its support end?

The archive comparison at its stated boundary

The conclusion is **supported for the checked path and two properties** under the stipulated criterion. Source files and their recorded associations supply the reference; retrieved objects and associations supply the observations. The comparison therefore earns a positive preservation finding. It leaves decoding, rendered appearance, diagnosis and availability of another archive route under all loads untested.

Reviewer Lens. The precise preservation statement can be accepted. Adding clinical interpretation would ask this result to answer a task it never examined.

Consultant Lens. The archive-integrity item can close while viewing and live-service questions remain assigned to their own owners. A limited conclusion is useful when it accurately completes the question that was asked.

A proposed viewing mode

All facts from the archive comparison remain. A proposed viewing mode now enhances the image during display. Stored files and case associations remain unchanged, and no assessment of the new mode is supplied. A report proposes that the archive comparison also proves **equivalent viewed images for the diagnostic task**.

Decision

Which object changed? Which earlier conclusion survives, and what new relationship would have to be examined?

What survives the added viewing operation

The supported stored-file and association result survives. The viewed image passes through an additional transformation, and its task-relevant equivalence is **not established**. A prospective comparison would identify the enhancement version and settings, relevant input range and display path, then examine the effect with a property-appropriate technical reference and representative images. Whether further reader or clinical evidence is needed depends on the claim and findings. Nothing supplied demonstrates improved or harmed diagnosis.

Reviewer Lens. The archive statement can remain accepted while the expanded viewing statement is withheld. They concern different outputs.

Consultant Lens. A targeted viewing comparison can resolve the new relationship and inform the decision about further use-specific study. The existing preservation result need not be repeated merely because the viewing question is open.

How many observations came from one patient?

A second scan adds another image attempt while the patient stays the same. The full route explains why: one patient's case can include several specimens, blocks and slides; a slide can be rescanned; a scan can yield files at several versions or resolutions; a file can yield many regions and overlapping patches; and several readings can contribute to a report. A thousand patches from one image are not a thousand independent patient outcomes.

The question determines where to count. Tissue coverage concerns the material or regions expected under the defined comparison, so files that happen to open cannot supply its denominator. A focus conclusion needs the field-selection rule and how unacceptable regions affect the slide-level decision. An average over accepted regions can be useful while leaving the worst task-relevant region unresolved unless an aggregation rule justifies that extension.

A workflow claim starts earlier, with eligible arriving cases or slides. Its account follows attempted acquisition, failed scan, rescan, acceptance, processing, delivery, viewing and final review, preserving partial and unavailable states. An algorithm study that admitted only acceptable scans describes that selected input population. Its result cannot describe every incoming slide merely by being attached to the same model.

Reviewer Lens. A numerator and denominator must name their objects, including whether correlated scans or patches were treated as independent people. Sequencing has a parallel trap: extra readings and repeats add no patients, and a recovered report leaves the first invalid attempt in the record.

Consultant Lens. Linked identities and state transitions make a later delay locatable as an attempt, file, queue event or case observation. The modality-specific units differ, but the route to final disposition is what lets the evidence reach a defensible conclusion.

A proposed service built from an archive study

This dossier is separate from PathSight and the two archive/viewing exercises. An earlier study examined **accepted stored images** of fixed tissue stained with hematoxylin and eosin. The source scanner, recorded as **Q**, produced them under acquisition state **AQ1**. Images failing archive entry were excluded from the algorithm dataset, with their original dispositions retained separately. The evaluated population therefore consists of accepted images, rather than all arriving slides.

The recorded study chain identifies normalization **N1**, decoder **D1**, patching **T1**, model **M1** and result rule **R1**, with viewer **V1** in a documented workstation environment. In ordinary terms, these are the study normalization, decoder, patch selection, model, result rule and viewer. The study characterized its stated algorithm task on those

archived inputs. It observed no scanner downtime, incoming-case queues, retries, live delivery or reporting.

The proposed service retains the study normalization, patch selection, model and result rule. It adds newly scanned material from both the source scanner and an additional scanner, recorded as **R**, replaces the decoder with **D2**, and uses remote viewer **V2**. The additional scanner's acquisition/export state and remote viewing environment are unsupplied. Slides might require rescanning or reprocessing. The proposal requires every accepted case to reach its intended review destination under a defined workflow requirement, but supplies no live result or numerical timing target. Patient diagnosis, accuracy estimate, acceptance threshold and proof of equivalence are also absent.

What can travel from the study to the proposal?

Decision

How do the evaluated archive route and proposed live route compare? Which exact earlier observations remain useful, which changed operations need a property-specific bridge, and which live states need new observation? How would failed, retried, partial and unavailable cases remain visible? What bounded recommendation preserves the archive result without turning the unchanged model into whole-service equivalence?

How a reviewer separates retained evidence from the proposed service

Worked transfer record

The comparison below follows the accepted archive inputs to their recorded task, then asks what changes on the way to the proposed service. Each row identifies a relationship to assess; no row establishes a supplied failure or mandatory fixed design.

Relationship	What the study supplies	What the proposed service needs to establish
Physical and acquired inputs	Accepted source-scanner images under the recorded acquisition state	What incoming cases and the additional scanner's acquisition/export represent
Dataset entry	Archive acceptance rules and separate dispositions	Whether live entry matches, and what happens outside it
Decoding	The study decoder	Whether the proposed decoder preserves properties needed by patch selection, model, result rule and viewer

Relationship	What the study supplies	What the proposed service needs to establish
Normalization, patching, model and rule	The same component identities	Compatible actual inputs and order, including additional-scanner images
Viewing	The study viewer in its documented environment	The remote viewer's rendering, display, interaction and environment
Delivery and workflow	Static study with no live queue/recovery assessment	Arrival at the intended destination under represented load and failures
Accounting	Archive dispositions retained separately	Reconciled incoming cases, attempts, rescans, versions, partials, failed deliveries and final review

The decoding row makes the dependency concrete. Replacing the study decoder can alter the representation passed to retained components; their names cannot establish whether the necessary input properties survived. The preceding two rows ask whether the incoming population and additional scanner are represented at all. Downstream, the remote viewer changes the user's possible view, and live delivery adds time, load and recovery states absent from the archive study. The final account makes those states observable without erasing exclusions or treating retries as new patients.

The earlier algorithm result remains evidence for its defined archive-input task. Comparing the study and proposed decoders could support a particular decoded-property bridge. A scanner comparison could address the additional scanner and relevant tissue strata; a viewing assessment could address the remote viewer and environment; live observations could address arrivals, load, retries and destination. Sufficiency depends on the actual claim and returned results. A reader or clinical claim still needs the appropriate later evidence rather than an automatic conclusion from technical compatibility.

Reviewer Lens. The retained source-scanner archive result and the proposed service require distinct dispositions. Accepting the former gives credit to the supplied evidence while keeping the latter's new relationships visible.

Consultant Lens. Owners for the additional scanner/export, proposed decoder, remote viewer and live queue can define the property and workflow questions prospectively, then close the relationships answered by appropriate results. A qualified sequencing component can likewise stay useful when a specimen or data path changes, without carrying every earlier clinical or reporting conclusion across. This image exercise supplies no HelixBridge liquid result.

Define “in time” before timing it

One live-service question remains too vague to measure: what does “reach the review destination in time” mean? The **start event** could be slide availability, case acceptance or scan initiation. The **end event** could be file delivery, case-list visibility, completed review or final report. Each pair defines a different operational question, and this dossier supplies no measured interval or threshold.

A fast model can precede a late case-list appearance; an intact transfer can precede a delayed read. The timing definition therefore also needs its counted unit and whether rescans, failure, recovery and unavailable cases remain in the endpoint.

Reviewer Lens. A turnaround percentage stays with its event pair and included cases. Carrying it to a different start or endpoint changes the claim.

Consultant Lens. Specifying that pair and inclusion rule before measurement, then keeping timestamps for the stages, makes later delays investigable. Neither this dossier nor PathSight's two simulation slides identifies a cause simply because the service is remote or called cloud-based.

Who can close the remaining relationship?

The map now gives owners specific evidence questions. The pathology laboratory identifies its preparation, slide acceptance, scanner, rescan and review conditions. The developer or manufacturer identifies the image entering each component and connects evaluated versions and parameters with proposed ones. Informatics preserves case associations, image versions, queue events and recovery states so those owners can trace what reached each consumer. The version inventory identifies the objects to compare; performance still needs observations.

Reviewer Lens. A tested archive property or component result can be accepted where its criterion is met, while a changed acquisition, decoder, display or workflow remains under review for the target claim. Technical, algorithm, reader and clinical conclusions need their own boundaries.

Consultant Lens. Each open relationship becomes an owner, comparison object, prospective criterion where appropriate and decision record. If a later field report is surprising, those same identities help locate it before assigning a scanner, algorithm or user failure. This keeps one team's narrow success from becoming another team's unsupported whole-product statement.

The opening question after following the image

Return to PathSight's opening question

The unchanged model weights establish one component identity. They leave the input images, normalization, second scanner family, low-confidence display and live workflow to be compared. The average-color comparison remains a real, limited comparison, with its outcome and criterion unsupplied and weak-stain boundaries omitted. The two late positive low-score slides remain simulation observations. An event trace can investigate them; a rate, cause, harm or release conclusion would need more than these two observations.

The useful map preserves evaluated-state evidence and identifies the precise unresolved relationships before carrying a full postvalidation image-path or second-family conclusion forward. That leaves Chapter 15 a properly located question: what did the readers and assistive algorithm do with the image and workflow state actually studied? This chapter has not supplied a reader-performance decision.

Reasoning check

Decision

Why does the chain begin before scanning? How can focus, tissue coverage and stitching fail separately? What does byte-identical retrieval establish, and where does its evidence stop? How can fixed model weights receive changed inputs? Why can accepted archived images support a bounded task while leaving the live queue open? Why do rescan, patch and patient outcome require different counts?

The six relationships behind the map

Preparation determines which tissue becomes a slide. Focus concerns detail in included fields; coverage concerns whether expected tissue appears; stitching concerns the placement of joined fields. Byte identity preserves the checked bytes on the tested path, not their case interpretation, displayed appearance or arrival time. Processing before a fixed model can change its input. Archive selection and static evaluation leave live arrivals, retries, failures and user presentation unobserved. Scan attempts, patches and patients are linked but different units; none should silently stand in for another. A strong map lets another reader follow one important case from material source to final consumer and see precisely where each evidence item ends.

For detailed manufacturer work on the image/software boundary and integrated verification, see Book 1D, Chapters 3 and 11.

Source trail

1. **FDA (2016)**, Technical Performance Assessment of Digital Pathology Whole Slide Imaging Devices, §§I–III, IV.A.6–IV.A.11 and IV.B. Final, nonbinding, surgical-pathology WSI technical guidance; it does not supply clinical submission data, a universal acceptance value or PathSight evidence.
2. **FDA Office of Science and Engineering Laboratories**, Digital Pathology Program, research gaps and activities, checked 25 September 2026. Research context, not a validated technical-to-clinical conversion rule.
3. **CAP with ASCP and API**, active WSI guideline page and issuer teaching presentation, slides 16–18 (good-practice statements 1–5), checked 25 September 2026. The update appeared online in 2021 and in the journal in April 2022; its three strong recommendations and nine consensus good-practice statements have different evidentiary status. No numerical recommendation is converted into a Chapter 14 rule.
4. **FDA, K232879**, Roche Digital Pathology Dx decision summary, §§III–IV and VII. Frozen product-specific configuration and technical/clinical evidence illustration, not current entire-line labeling or transferable components and results.
5. **FDA, ValidPath**, regulatory-science tool record, function and limitation sections, checked 25 September 2026. Research-tool functions are not recognized-standard status, Medical Device Development Tool qualification or product validation.

The PathSight snapshot is controlled fiction. Micro-cases A/B and the Q/AQ1-to-live transfer are separate original instructional fiction; their stipulated results are not PathSight or real-product results. The image-chain map is an author-designed reasoning aid.

CHAPTER 15

Digital Pathology: Readers, Algorithms, and Workflow

Who performed better?

An unnamed team tests a fixed device that marks suspicious regions on eligible archived images made by whole-slide imaging. For this exercise, no pathologist reads the images in the test. The team compares the device's output with a reference process specified in advance for the study's endpoint and independent of the device output. The archive deliberately contains many uncommon positive and boundary presentations.

The summary calls the device-only result “favorable.” It supplies no endpoint, scoring rule, numerical estimate, uncertainty, denominator or account of images without an output. The proposed conclusion goes further: pathologists will perform better and the worklist will move faster. A device mark, a person's decision and a queue's movement are different events. What did this record actually observe?

What was observed, what remains unobserved, and how much of the proposed conclusion could this record support? This is a separate fictional exercise, not a PathSight Cloud study.

The distinction begins where Chapter 14 ended. That chapter followed tissue through slide, scanner, processing and display. Now someone may act on the image. A technically adequate file can reach the screen without telling us what the person did with it.

Two earlier questions help here. Why did Chapter 12 require a population and an explicit account of absent or unresolved results? And where, along Chapter 14's route from preparation through acquisition and processing, could the image shown to the pathologist differ from the data reaching the algorithm?

What the earlier chapters preserve

A percentage needs an outcome and a count of the observations to which its divisor refers. Agreement with a comparator is not automatically correctness. Missing images, invalid outputs and unresolved reference assignments need an explicit place in the study account. Conveniently retrieved images do not by themselves define the clinical population. The scanner, image-processing route, file, viewer and algorithm input can also affect what is being tested; naming the model alone does not freeze the configuration.

This chapter adds the user's information sequence. The same mark can appear before a case is selected, during interpretation or after an initial judgment. The product role changes even when the mark-generating code does not.

Follow the mark to the decision it changes

Suppose the device puts a mark beside a suspicious region. The mark is one output. A pathologist's interpretation after seeing it is another. If the service uses the score to move the case up a worklist, that movement is a third event. To interpret “performed better,” we first need to know which event was compared. The four question families below locate the actor; they are not stages that every product must pass.

When the study compares the fixed mark-generating device with a declared reference or comparator, it asks a **standalone-device** question. Its unit might be a region or a case. A sound result can support that output under the studied image conditions. There need not have been a reader in the study at all.

Once a person interprets the images, the observation changes owners. Readings without device output establish the **unassisted-reader** condition or a reader-alone result. That gives a possible control for the next question: what changes when the person receives the mark?

An **assisted-reader** comparison observes reader outcomes with and without the specified assistance in a design that permits the comparison. Even a correctly placed mark may tell a particular reader something already noticed; it may help, distract or go unseen. Its contribution lies in the observed reader difference.

If the score changes which case reaches the reader first, the question expands into the service. A **workflow** comparison follows cases and users through routing, review and completion against a specified operational alternative. One priority group can move faster while another waits longer. Diagnostic decisions and queue timestamps therefore need different endpoints.

The earlier name **estimand** makes this demand precise: what quantity is the study trying to learn, for which people or cases and under which conditions? “Helps

pathologists” leaves the quantity open. “Difference in final case-level diagnostic performance between assisted and unassisted readings for these readers and eligible cases” identifies a contrast, but still needs its endpoint and design. The table helps locate the appropriate question before interpreting a result; it is an author-created framework, not a regulatory classification.

Question family	Object and contrast	Boundary of a supported result
Standalone device	Fixed device output compared with the declared reference or comparator	Device-only performance for the studied input, configuration, population, and endpoint
Unassisted reader	Readers interpreting under a defined condition without device output	Reader-alone performance in that condition; not the effect of assistance
Assisted reader	Reader-plus-device outcomes compared with the prespecified reader control condition	Comparative assistance effect for the studied role, users, cases, configuration, and analysis
Workflow effect	Cases and users moving through a device-enabled process compared with a defined operational alternative	The routing, timing, completion, escalation, or other operational effect actually estimated

The assisted-reader row belongs to the reader-plus-device outcome. A standalone comparison in the row above cannot fill it, however useful that device result may be for its own question. The workflow row asks about routing or completion instead, so a diagnostic outcome alone cannot fill that row either.

The unit matters too. Several regions may come from one slide; several slides may belong to one case; several cases may come from one patient. Counting regions as independent patients would change the claim and make the numerical uncertainty look more informative than the sampling allows. Likewise, a worklist-level claim needs a queue or time unit, not just a case-level score.

Reviewer Lens. The first useful check is whether the conclusion has kept the study's actor, endpoint and unit. A precise device-only result may be enough for that question even while reader and queue claims remain open. **Consultant Lens.** Deciding the intended role early prevents a team from collecting one kind of result and discovering later that its claim requires another. Separate device, reader and operational comparisons are needed only where the intended use asks those questions. In sequencing, a variant caller's analytical agreement likewise leaves report interpretation and laboratory turnaround as different propositions.

Figure 15.1 — Put the result in the correct role is a text-linked reminder of these separate actors and outcomes. It supplies no result for the opening exercise.

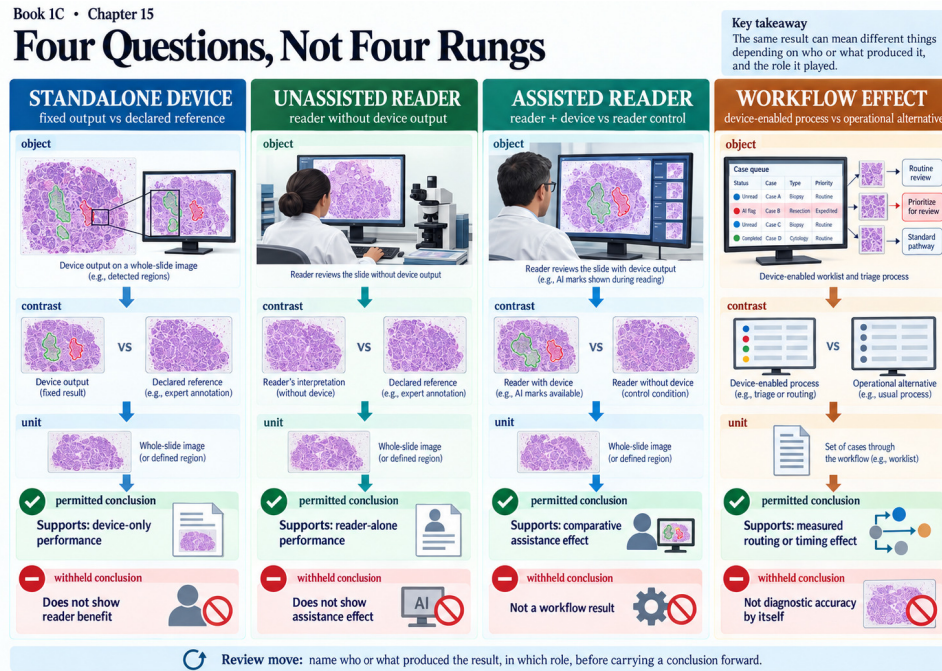


Figure 15.1. Put the result in the correct role – Technical review plate.

Put information in its intended order

The information sequence now becomes part of the intervention. For help after a preliminary interpretation, that interpretation must be recorded and locked before the mark appears. Showing the mark first would study concurrent assistance. Conversely, withholding a concurrent highlight until after the report would remove the experience the product intends to provide. The same mark-generating code can participate in different roles.

When a reader sees a case more than once, earlier exposure may be remembered. **Order** records which condition and case came first. Separate **sessions** and any **washout** try to reduce memory carryover; neither can prove memory is absent. **Training** and familiarization establish how readers learn to use the interface, but an unusually coached study might not represent intended use. These are choices with reasons, not a universal washout duration or mandatory reader count. The College of American Pathologists (CAP) guidance for whole-slide-imaging laboratory validation addresses trained pathologists, intended application, order and washout within its own scope; it is not a complete protocol for every artificial intelligence (AI) assistance claim. CAP issuer presentation, recommendations 1–3 and good-practice statements 2, 3, 6 and 9

Reviewer Lens. A reader difference is interpretable only alongside the recorded sequence: what had the reader already decided, and what information became available next? **Consultant Lens.** The inexpensive early choice is to specify that sequence and choose task-appropriate training and session controls, including what they cannot remove. If presentation or timing later changes, the same record shows where applicability of the earlier reader evidence needs examination.

Readers and cases both vary

Repeated viewing creates another dependency. When several pathologists read one difficult case, their responses share an image and reference assignment. When one pathologist reads many cases, those responses share experience and habits. When that person reads the same case with and without marks, both reader and case are shared. A pooled percentage can conceal these connections and overstate what its numerical uncertainty tells us.

Multireader multicase (MRMC) methods are designed to account for variation and correlation associated with readers, cases and study conditions. The label does not name one test suitable for every layout. A method has to match which observations were assigned and what conclusion is sought. Hillis and Scharztz, §§2.2, 2.4–2.6 and 5

First decide whom the conclusion covers. A **fixed-reader** analysis can make a useful, deliberately limited statement about the participating readers. A broader **random-reader** inference aims at a defined population of readers and accounts for reader variation under a model. The parallel case question asks whether the conclusion covers only the studied cases or a defined population from which they are meant to represent a sample. The statistical word “random” does not make a convenience group representative. Recruitment, experience, sites, task and case spectrum must justify the reach of the claim.

More reads from the same local group can improve knowledge about that group without adding the multisite diversity a wider claim needs. More easy cases can enlarge the dataset without representing difficult boundary presentations. **Reviewer Lens.** A well-supported fixed-reader result is worth retaining at its deliberate scope. **Consultant Lens.** Choosing the target readers and cases before recruitment keeps a later site or spectrum change visible; statistical terminology cannot repair a mismatch between the people studied and the people named in the claim.

Give every planned reading an address

Suppose a reader sees a case in an unassisted session and later sees that same case with marks. The two responses can be **paired** by reader and case; that pairing is not established merely by having equal counts in two columns. If every reader reads every case under each condition, the assignment is **fully crossed**. A planned **partially crossed** design instead gives different readers overlapping subsets of cases or conditions. It may reduce burden and remain analyzable, provided the method uses its actual assignments rather than pretending that all combinations occurred.

A response therefore needs an address: reader, case and condition, with session and sequence where relevant. That address also needs a state. A deliberately unassigned combination never called for a read; a scheduled read lost during an outage did. A completed response on a reference-unresolved case exists even when one endpoint cannot score it. The table is useful when the dataset shows blanks or excluded rows: it asks which history produced them.

Cell state	Meaning	What must remain traceable
Assigned and completed	A read was scheduled and a response was recorded	Reader, case, condition, session, response, and scoring eligibility
Planned unassigned	The design did not schedule this read	Assignment rule; this is not an observed failure
Assigned but absent	A scheduled read did not produce the required observation	Reason, timing, recovery or repeat, and analysis treatment
Response present, reference unresolved	A read exists but the comparison classification is unavailable	Both records; endpoint-specific scoring and denominator consequence
Response present, not evaluable under the rule	The response cannot enter a specified analysis as an ordinary scored result	Original response, reason, rule, and any separate analysis

Consider the outage row. Its scheduled reader, case, condition and attempted session should still be recoverable, along with the reason if known and any recovery. The planned-unassigned row instead points back to the allocation rule. For a reference-unresolved case, the image and reader response survive while the unavailable comparison classification limits scoring. Other nonevaluable responses retain their actual contents and the rule governing their endpoint treatment. The completed row then traces an eligible response into the analysis. A shared blank symbol would erase all these distinctions.

Reviewer Lens. The revealing document is the route from scheduled to observed to scored, not a pooled total alone. **Consultant Lens.** Distinct state codes chosen before enrollment make an outage, a planned split and an unresolved reference recoverable later. The sequencing parallel is an initial invalid attempt, an analyte-level no-call and an unattempted specimen: each has a different counting basis. Here a late or absent mark likewise needs an arrival and reading history; the state alone proves no error in a named product.

Choose the result before counting readers and cases

Before choosing how many readings to collect, decide what each reading will contribute. The **endpoint** is what is scored. For a final positive/negative decision, Chapter 12's sensitivity is the fraction of reference-positive cases with a positive reader decision; specificity is the fraction of reference-negative cases with a negative reader decision. Assistance asks how those outcomes compare with the prespecified unassisted condition. A localization score, report agreement or time to first review would require its own unit and rule. Agreement with a report still depends on the report's qualifications as a reference.

Readers might instead give graded suspicion scores. Moving the score cutoff changes which cases count as positive: a lower cutoff generally catches more reference-positive cases but may also call more reference-negative cases positive. A **receiver operating characteristic (ROC) curve** traces those true-positive and false-positive rates as the cutoff moves. Its **area under the curve (AUC)** summarizes discrimination across those cutoffs, commonly read as how often a reference-positive case ranks above a reference-negative one. A comparison of those areas asks a different question from sensitivity and specificity at the cutoff the service will actually use; it cannot silently substitute for that operating decision.

Next choose what difference would count as success. A **superiority** question asks whether assisted reading improves the specified endpoint over control. A **noninferiority** question instead asks whether any loss stays within a justified limit fixed before looking at the results. The margin is that prespecified limit, not the largest observed loss that happens to look acceptable. Changing from superiority to noninferiority changes the proposed conclusion and the calculation needed to support it, even if the same reader responses are collected.

The **estimator** is the stated calculation that turns those scored observations into the chosen estimate, such as an assisted-minus-unassisted difference in a defined reader outcome. **Power** is the chance that a specified design will meet its statistical comparison rule if the assumed effect and variation are true. **Sizing** works in the other direction: it chooses reader and case counts for a target power under those assumptions. Both need the endpoint, effect or margin, reader/case variation, dependence, crossing, anticipated missingness and any multiple comparisons. More

rows do not automatically mean more independent readers or cases; no universal count follows from the phrase “reader study.”

Hillis and Schartz distinguish the inference and design choices that feed such calculations. The U.S. Food and Drug Administration (FDA) iMRMC research-tool catalog describes software for analysis and sizing of multireader multicase studies, including non-fully-crossed designs and functions beyond its core area-under-the-curve comparison. Its availability does not select a suitable endpoint or establish that a particular study's assumptions are met. The catalog tool is neither a recognized standard nor a qualified Medical Device Development Tool (MDDT), and its documented limitations matter. Hillis and Schartz, §§2.4–2.6 and 5; FDA iMRMC catalog

Reviewer Lens. A software printout is useful when its estimate and uncertainty can be traced back to the declared endpoint, actual design and inference population.

Consultant Lens. Those choices belong in the protocol before the data arrive: target, contrast, unit, allowed missing states and defensible design inputs. A digital-pathology assistance claim needs that connection before a reader-benefit conclusion; a HelixBridge tissue variant score similarly needs its analyte or report endpoint before anyone proposes a clinical-utility interpretation. These parallels supply no product study result.

Figure 15.2 — Preserve the reader–case–condition observation address links assignment, observed state and analysis. Its record shape is useful even when a legitimate partial design leaves cells intentionally unassigned.

Preserve the Observation Address

Key takeaway

Every observation has an address: reader, case, and condition. Keep that address through the entire study.

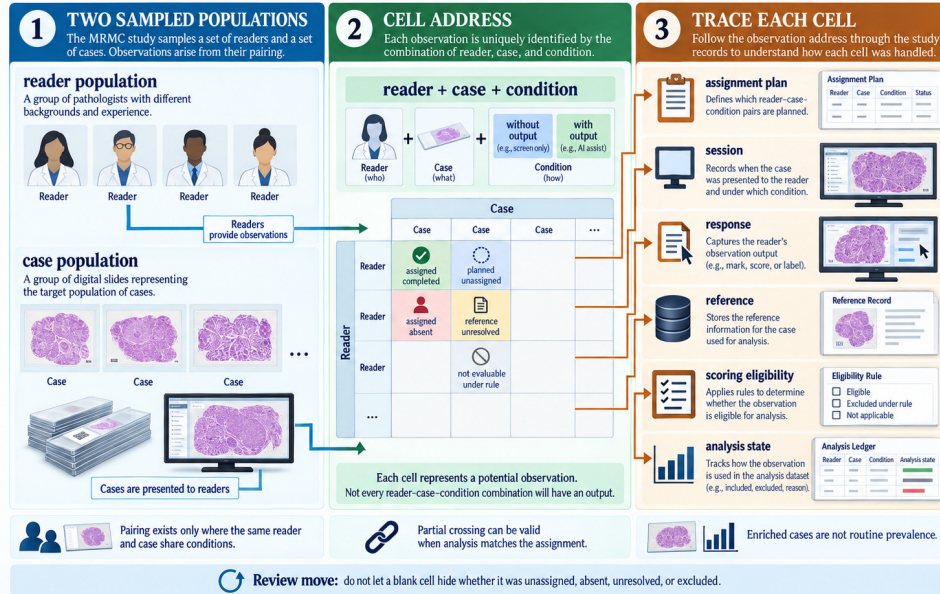


Figure 15.2. Preserve the reader-case-condition observation address – Technical review plate.

Five operations between a candidate image and a scored comparison

Now follow a candidate image into the study. Before a reader can contribute a scored response, someone must decide whether the image can enter and which comparison classification applies. After the reading, someone must connect the response to that classification. Calling all of this “ground truth” hides who knew what and when. The order matters most when a later panel disagrees with the original label.

At entry, **eligibility and quality screening** applies the prespecified tissue, image and technical requirements. The screener acts before outcome information could encourage removal of difficult cases. Exclusions and uncertain eligibility retain their counts and reasons. Rejecting an image for quality says something about eligibility; it has not diagnosed the patient.

For an eligible case, **reference construction** supplies the endpoint-specific comparison classification. A declared pathology process might examine defined material and permitted clinical information to assign a target category. Its record identifies contributors, their information, disagreements and unavailable labels, including whether device outputs or study reads can influence assignment. Independence and prospective planning strengthen the reference, but do not make it unquestionable truth.

The pathologist then provides the **reader observation** under an assigned condition. Marks, a prior report or an earlier read visible at that moment all belong to the observation record. The response remains a separate fact from the reference label. If a usable reference is later unavailable, the completed reading still happened.

Only then does **comparison and scoring** connect the relevant response and reference under the locked endpoint rule. A binary match may enter an accuracy table; an ambiguous mapping may require a prespecified state. Manual scorers' access to identities and earlier results matters because it can affect choices after outcomes are known. The record must let a reviewer follow every candidate through screening, reference assignment and observation into the scored set or its retained alternative state.

Suppose only disagreements now reach a panel that can see the device and reader results. This **discordance review** can uncover a valuable study problem, but its revised labels come from a selected, informed later investigation. Substituting them silently for the original independent reference would not turn a recalculation into an unbiased original estimate. A panel's prospective contribution for all relevant cases would instead belong to the original reference method. The first table makes that timing and information distinction inspectable. FDA diagnostic-statistics guidance, §6

Operation	Purpose and timing	Information and masking to specify
Eligibility and quality screening	Apply prespecified inclusion and image-quality rules before outcome-driven removal	Who screens; whether device, reference, and study-reader results are unavailable
Endpoint-specific reference construction	Establish the comparison classification under a declared process; lock it or define any prospective adjudication contribution	Permitted diagnoses, clinical data, methods, panel inputs, and independence from the candidate device and study reads
Study-reader observation	Capture each assigned reader's response in the specified session and condition	Device output and other information available at that point; access to prior reads or reports
Comparison and scoring	Apply the locked endpoint and scoring specification to the relevant records	Records available to manual scorers or software, identity masking, and handling of ambiguous mappings

Operation	Purpose and timing	Information and masking to specify
Discordance review or adjudication	Either contribute prospectively to the reference or investigate disagreement after reads; name which	Original labels, outputs, reads, ancillary data, and whether disagreement triggered access

The last row requires a decisive distinction: was the panel planned to contribute to reference assignment, or did disagreement trigger later access? The same specialists could participate, yet the evidentiary role would differ. The second table follows the resulting counts. It is useful when an apparently clean analysis omits candidates, unavailable labels or readings that the first table says should exist.

Operation	Coverage and retained uncertainty	Label and denominator consequence
Screening	Account for every candidate case, including unclear eligibility and quality failures	No silent removal; screening is not target-condition assignment unless a separate prespecified role says so
Reference construction	Identify all cases receiving the process; retain unavailable labels, disagreement, and limitations	Creates endpoint-specific comparison labels, not unquestionable truth
Reader observation	Separate assigned, unassigned, completed, incomplete, and missing cells	Reader responses remain observations; they do not overwrite reference labels
Scoring	Apply the declared rule consistently and retain uncertain mappings	Trace each cell into scored, excluded, unresolved, or other planned analysis states
Discordance work	Preserve original and reviewed sets, including cases not reviewed	Report discrepancy-selected revisions separately; do not silently substitute them for an unbiased original estimate

Take a case with a reader response but no final reference classification. Its reference row retains the missing label; its observation row retains the response; its scoring row explains the unavailable comparison and analysis treatment. **Reviewer Lens.** That trace can support a bounded comparison where the reference is complete and qualified, without pretending uncertainty elsewhere has disappeared. **Consultant Lens.** Setting records and masking before reads begin preserves both the original study and the separate value of a later investigation. In sequencing, a comparator's qualified region

similarly supports a local analytical comparison while uncovered regions remain visible; a pathology reader response has its own actor and claim.

What happens when the aid is late, absent or misunderstood?

A reader's final answer does not reveal how the reader used the device. They may have noticed a mark, ignored it, misunderstood it or changed their judgment because of it. A study of final diagnostic outcomes and a study of interaction or reliance therefore answer complementary questions. A successful nominal reading session also does not show what users do when an output is delayed or unavailable.

Consider a proposed priority worklist without a timely score for a case. The prospective plan must say whether that case returns to ordinary order and how staff would know. The chapter calls such a specified departure in output, timing, availability, presentation or recovery a **degraded state**, which may invoke fallback. This is an author-defined operational label, not a diagnostic method or standards term, and the example reports no PathSight event.

Use-error terminology needs its own evidence. FDA's August 2026 human-factors guidance describes a user action or omission different from the manufacturer's expectation. It must cause a result unexpected by the user, not be due solely to device failure, and have caused or been capable of causing harm. All of those conditions matter. FDA guidance, §3, printed p. 4

International Electrotechnical Commission (IEC) 62366-1:2015+AMD1:2020, clause 3.21, uses separate wording: a user action or omission produces a result different from manufacturer intent or user expectation. Its notes address inability to complete a task and distinguish an unexpected result caused by malfunction. This definition should not be silently merged with FDA's.

A deliberate override is not automatically a use error. Neither are a wrong diagnosis, a late report or a device malfunction without analysis of task, interface, action, system behavior and consequence. The Reviewer can request the role-specific observation and risk record while withholding a use-safety claim not studied here. The Consultant assigns human-factors and risk specialists to investigate reliance and fallback, and operations staff to capture actual arrival and completion timestamps if a workflow claim is proposed. In a sequencing service, a late variant report similarly calls for a specimen-to-report timing record, not a diagnosis of user error from lateness alone. Book 1A treats the deeper usability methods; this chapter keeps their results attached to the question they tested.

Connect the question to the observed record

The question, planned comparison, actual observations and permissible conclusion belong together in a **reader-study design and interpretation memo**. The memo makes a statistical plan and a human-factors protocol answer the same product question: who or what produced the outcome, in what role, for which people and cases? It does not replace either specialist plan.

Two records that must agree

The memo's first record connects the actor and arrival of information to the claim, population, configuration, comparison, endpoint and unit. Its reader and case frames set the possible reach of a result. A fixed-reader conclusion on eligible archived images can be useful while leaving wider-reader and live-queue questions open. The table shows those dependencies as inspectable fields; its unsupplied entries remain genuine gaps, not invitations to borrow values from another study.

Question record	Supplied information	Basis and remaining uncertainty
Role, workflow position, and target quantity	Not supplied	Not supplied
Claim, population, setting, comparator, and frozen image/device configuration	Not supplied	Not supplied
Endpoint, unit, scoring, hypothesis, estimator, and compatible analysis	Not supplied	Not supplied
Reader frame, experience, sites, and fixed/random inference target	Not supplied	Not supplied
Case frame, spectrum, enrichment, and target generalization	Not supplied	Not supplied

For example, the reader-frame row determines whom a result could describe; a local group cannot become a multisite population through an empty table cell. The second record then checks whether the plan actually produced the data its analysis needs. A scheduled read, missing response and unresolved comparison must each connect to the plan before the conclusion can be interpreted.

Design and accountability record	Supplied information	Basis and remaining uncertainty
Assignment, pairing, crossing, order, sessions, training, washout, and learning	Not supplied	Not supplied

Design and accountability record	Supplied information	Basis and remaining uncertainty
Five-operation reference and observation record	Not supplied	Not supplied
Missing reads, invalid outputs, deviations, exclusions, and denominators	Not supplied	Not supplied
Dependence, variance inputs, multiplicity, and sensitivity analyses	Not supplied	Not supplied
Workflow fidelity, reliance, degraded conditions, fallback, and specialist handoff	Not supplied	Not supplied
Supported conclusion, withheld conclusions, limitations, and next owner	Not supplied	Not supplied

The assignment row and denominator row must reconcile: an outage loss cannot vanish simply because only completed readings were scored. The last row then separates retained evidence from unmeasured claims and assigns the next question. If post-read marks become pre-read prioritization, both records can change—the quantity, units, sequence, missing states and specialist method. Chapter 14's technically sound image path remains relevant, but cannot supply these new observations.

Return to the unnamed archive

How do the opening archive facts populate these two records? The independent reference is a useful design feature. What can be concluded while readers, a numerical estimate, a pass criterion and routine prevalence remain unsupplied?

How the archive result reads

The actor in the opening record is the device, not a pathologist. The image set is enriched, not a measured routine queue. The planned reference is independent of device output, but its actual coverage and the comparison method still need to be examined. That positive design feature does not supply a missing endpoint or result. Read each worked row as a chain from supplied fact to permissible inference.

Memo component	Worked entry
Role and target	Standalone-device evaluation of a fixed configuration on eligible archived images
Readers and cases	No reader population was observed; the deliberately enriched archive does not establish routine prevalence or queue behavior

Memo component	Worked entry
Endpoint and analysis	Endpoint, scoring, estimate, uncertainty, missing-output account, and denominator are absent; “favorable” cannot be interpreted as a demonstrated pass
Reference process	Prospective and independent of device output as stipulated; actual construction, coverage, uncertainty, and scoring relationship still need review
Narrow conclusion	The record describes a device-only evaluation. An interpretable device-only result would require the missing design and result fields
Withheld conclusions	Reader benefit, operational improvement, reliance, use safety, clinical utility, patient outcome, and authorization
Next owners	Pathology for the task, spectrum, and reference; statistics for endpoint and analysis; human factors and risk for use questions; regulatory specialists for jurisdictional sufficiency

Reviewer Lens. The independent-reference, fixed-device archive is a useful starting record. The missing endpoint, scored counts, output-loss account, uncertainty and configuration report still prevent an interpretable device-only judgment. **Consultant Lens.** Obtaining that underlying report is the first narrow closure task. Separate assistance or operational studies follow only if the intended claim requires them. There is no reason here to discard the archive or demand every possible study.

PathSight: a reported conclusion and its missing study basis

PathSight Cloud remains an assistive digital-pathology product: it prioritizes whole-slide images and highlights suspicious regions, while a pathologist makes the diagnosis. Its controlled Chapter 15 occurrence reports the validation-linked assertion “**improved average sensitivity without unacceptable loss of specificity.**” Treat that as an assertion in the file, not as a reader study reconstructed from missing data. This occurrence supplies no reader frame, case frame, design, endpoint definition, numerical estimate, margin or analysis record from which to verify it.

“Average” needs a defined set of readers and cases and a method of combining results. “Sensitivity” needs reference-positive cases and a defined positive reader outcome. “Unacceptable loss” needs a prespecified specificity comparison and margin, with uncertainty. **Reviewer Lens.** The protocol and configured report must connect those words to the assistive role. **Consultant Lens.** The studied image/interface configuration, assignments and result-state ledger are the route to any supportable

narrower claim. No second-reader history, reliance or worklist benefit can be inferred. The following unnamed exercises remain independent records.

Three short decisions

Three separate records test whether the memo can keep the role, observed result and conclusion aligned. What judgment follows in each? Their shared interpretation follows all three questions.

A mark after the preliminary read • Record A

A pathologist records and locks a preliminary interpretation before device marks appear. After seeing the marks, the same pathologist may revise the final report. The device does not select cases, reorder the worklist or remove cases from pathologist review. No performance or use-related result is supplied.

Where does this assistance enter the workflow? What reader comparison would estimate its effect, and which conclusions remain unavailable?

A later panel changes labels • Record B

Eligibility and image-quality screening use prespecified criteria before study reads. An independent endpoint-specific reference is constructed before those reads. Readers provide locked observations, which are scored against that reference.

Only disagreements are later sent to a panel. The panel sees the original reference, reader response, device output and disagreement status. The team proposes replacing the original labels with panel labels and calling the recalculated table an unbiased corrected estimate.

Where do the five operations separate? Which original and revised records must remain visible, and what could the revised table legitimately describe?

An adequate bounded result • Record C

For this author-created exercise, stipulate that the task, configuration, reference, complete observation account, endpoint-compatible analysis and prespecified comparison have been independently judged adequate, and the comparison criterion is met. Its inferential target is deliberately fixed to the participating readers and a justified sampled case population. The proposed conclusion concerns only their diagnostic performance under the studied reading conditions. It makes no wider-reader, timing, safety or utility claim.

Should the memo retain the proposed conclusion? Does a fixed-reader target alone invalidate it? What would change if the proposed conclusion expanded to all intended pathologists?

How the three judgments differ

Compare your decisions

Record A is post-preliminary-read assistance. A comparison of final reader outcomes under specified assisted and control conditions could estimate an assistance effect if its design handles the sequence and repeated reader/case observations. Locking an initial response makes the information order visible; the opportunity to revise does not demonstrate benefit. Since the device does not route cases, no queue effect is observed.

Record B has five real operations: early screening, prospective independent reference assignment, locked reader observation, scoring and later selected discrepancy review. The last is not the original reference process. Preserve the original classifications, reads, table and denominator; preserve the panel's selected set and revised findings separately. The revised table can describe that investigation, but substituting it for the original labels does not yield an unbiased corrected estimate. A prospectively designed panel contribution would have been a different study method.

Record C **does retain** a positive, bounded fixed-reader conclusion under the exercise's explicit adequacy and criterion-met stipulations. The restriction is deliberate, not a flaw to cure by deleting a useful result. A claim about all intended pathologists would require a reader-population target and supporting recruitment and inference. This exercise stipulates adequacy; it supplies no actual product result or reusable numerical acceptance rule.

Independent decision: concurrent highlighting

In a fresh unnamed study, a fixed highlighting configuration is visible while pathologists interpret each assigned whole-slide image. The claim is improved assisted-versus-unassisted diagnostic performance for a defined multisite reader and case population. Timing is outside the proposed claim.

The recruited readers are a local convenience group that does not represent the stated multisite target. The plan does not resolve whether inference is restricted to those readers or reaches a wider population. The archive is retrospectively enriched for positive and boundary cases; routine prevalence and case mix in the intended population are not supplied.

Readers receive overlapping but nonidentical case subsets in both conditions, a planned partially crossed allocation. A system outage causes additional absent observations. The data file uses the same blank value for a deliberately unassigned cell and a scheduled read missing after the outage.

The reference process is documented, complete before study reads and masked to device and study-reader output. Some eligible cases nevertheless have no final

reference classification. The proposed analysis pools observed cells into condition-level percentages without accounting for reader and case variation, shared observations or the partial crossing. Unassigned cells, outage-missing reads and reference-unresolved cases disappear from the reported denominators without separate accounting. No timing, reliance, fallback, use-error, utility or patient-outcome observation is collected.

Does the proposed analysis support its claimed populations? What useful parts remain, what must be repaired before inference, and which effects remain unmeasured? The record supplies no basis for a new sample size or numerical effect; the useful decision is which evidence and owners could close the gaps.

What the study can retain and repair

Review the independent memo

This is an assisted-reader question. The current pooled percentages do not support the proposed multisite reader conclusion. Pooling treats dependent responses as if each were a fresh reader and case, while the blank code hides two different origins. Deleting reference-unresolved cases without an account changes the analyzed set as well. The record does not permit a corrected assistance effect, confidence interval or replacement sample size to be calculated.

There are useful parts to retain. Planned partial crossing is a legitimate design choice, not automatically a defect. The independent, masked reference specified before reads is a strength, although some cases still lack a usable label. The problem is the unclosed path from target and assignment to scored denominator and compatible analysis. Follow the rows as a sequence of decisions, not as seven reasons to abandon the study.

Memo issue	Necessary repair or boundary
Reader target	Resolve fixed-reader versus broader inference; justify recruitment against the target or restrict the conclusion
Case target	Describe the archive, enrichment, spectrum, and intended generalization; do not manufacture routine prevalence
Assignment and missingness	Recover the assignment plan and distinguish planned unassigned cells from outage-related absences
Reference-unavailable cases	Retain their eligibility and reader observations; specify endpoint-specific analysis and sensitivity treatment
Analysis	Select an endpoint- and design-compatible method that respects reader/case variation, shared observations, and actual crossing

Memo issue	Necessary repair or boundary
Denominators	Reconstruct the full scheduled-to-observed-to-scored account; justify every analysis set
Other effects	Withhold workflow, reliance, fallback, use-safety, utility, patient-outcome, and regulatory conclusions

The first two rows set the reach of inference: either justify a broader frame or limit the result to the studied people and cases. The next two reconstruct what was intended and what actually happened; they preserve outage-missing cells and reader responses on reference-unresolved cases. The analysis and denominator rows then ask the statistician to use that recovered structure, rather than fit a method to an already pooled percentage. The last row prevents any diagnostic comparison, however well repaired, from turning into a worklist or safety result.

The Reviewer can request the assignment plan, full cell ledger and a design-compatible analysis with an explicit reader/case target. If those records support a restricted fixed-reader statement, retain it rather than claiming that only broad generalization has value. The Consultant assigns the data owner to recover allocation and outage history, the pathologist to assess task and reference coverage, and the statistician experienced in multireader multicase studies to choose and justify the analysis. Those are prospective closure tasks tied to missing evidence, not an invented assertion that the fixed highlighting device performed badly.

Check the memo's reasoning

The following questions help diagnose a memo's reasoning. Each warning identifies an inference that would need repair; these are learning criteria, not study acceptance criteria.

Criterion	What the reasoning must preserve	Warning sign
Scientific question	Names the object, role, contrast, endpoint, unit, and population	“AI performance” substitutes for the question
Inferential structure	Retains reader/case targets and actual assignment	Pooled cells are treated as independent or random-effects terminology substitutes for sampling
Reference and accountability	Separates the five operations and all observation states	Relabeling or deletion silently improves the reported result

Criterion	What the reasoning must preserve	Warning sign
Conclusion	Matches the supplied evidence and its uncertainty	Device, reader, workflow, safety, and utility results are exchanged
Action	Assigns a specific unresolved question to a competent owner	“More validation” replaces an evidence request

The third row is particularly practical: a revised label can be valuable as an investigation, yet the memo should still show the original and revised records. The fourth row permits a positive, limited result like Record C without inflating it into a different outcome. The final row names who can close each question.

Put the same decision into professional language

Reviewer Lens. The estimand, assignments, actual states, scored denominators and analysis must form one route to the claimed reader and case populations. A disclosed descriptive summary may survive at a narrower scope. Later changes in interface or information timing create an applicability question; they do not automatically erase all earlier reader evidence.

Consultant Lens. Recovery begins with the data team's assignment and outage history, then pathology's judgment about archive and reference coverage, and statistics' method for the actual partial crossing. These results determine whether the multisite target can be justified or the claim should narrow. Guessed missing responses or a favorable method chosen after seeing results would conceal the original problem. A sequencing program needs the same ownership discipline between analytical variant detection, report interpretation and laboratory turnaround, even when all share a specimen path.

For the **pathology and human-factors specialists**, the questions are what the reader saw at each decision and whether that situation resembles the intended use. The study does not observe reliance or fallback. That absence is not evidence of either safe or unsafe use; it identifies the separate method needed if the claim depends on those conditions.

Changed-case transfer: an earlier queue decision

A different unnamed device is used after a preliminary read. Each pathologist records an initial interpretation before seeing device output and may revise the final report afterward. The baseline study question concerns the effect of that assistance on final diagnostic performance.

The proposed target role is earlier: the device processes cases before primary reading and changes worklist order. Every case still goes to a pathologist. For this exercise only, the image-producing and device configuration remain unchanged. The team proposes reusing the old reader-performance plan and conclusion without a separate queue or fallback evaluation.

No role-specific result, queue result, reliance result, degraded-state result, use-error result or clinical consequence is supplied. Queue rules, thresholds, arrival patterns, load, staffing, escalation, timing units, absent-output handling and recovery remain unspecified. Which operational comparison and units would the target role need? Which parts of the original evidence remain useful, and which specialist questions remain unanswered?

What the earlier queue role changes

Review the revised memo

The unchanged image and device configuration preserves technical information worth reviewing. It does not preserve the *role*: a post-preliminary mark informs a person's final judgment, whereas a pre-read score changes which case reaches that person next. The original reader-performance question therefore does not estimate a queue effect. Nor is all prior evidence automatically worthless; it may remain relevant to unchanged image and output properties and, if appropriate, a carefully bounded reader question.

First specify the operational alternative: the proposed device-enabled ordering compared with a defined baseline ordering under comparable service conditions. Then specify the intended outcome before measuring it. Is it time from arrival to first review, time to completed report, priority-group completion, or another quantity? The record has not chosen. A case, reader, queue, shift, site and time window can yield different denominators and different conclusions.

Next follow a case event by event: arrival, assignment, device processing, output availability, priority action, first review, completion, escalation and any fallback or recovery. Preserve cases still waiting, delayed or absent scores, and work shifted to other cases. Because every case remains assigned for pathologist review, the proposed change is ordering, not autonomous diagnosis or removal from review. A nominal reader result cannot stand in for queue behavior at peak load or during missing output.

The Reviewer can retain the earlier technical and reader evidence at its actual scope while asking for a new role-specific operational claim, event account and comparison. The Consultant asks operations and pathology to define arrival and staffing conditions, statistics to design analysis at the chosen units, and human-factors and risk specialists to assess presentation, reliance and fallback. If the team also claims that reordering improves diagnostic performance, that becomes a separate reader-performance

question; neither automatic transport nor automatic invalidation is justified by the facts supplied.

Let the result keep its owner

The opening team may eventually have a useful device result, an assistance effect and an operational benefit. The disciplined memo leaves those possibilities open while refusing to call one observation all three. Ask first, “Who or what produced this result, in which role?” Then ask which configuration, reader and case populations, reference, observation states and analysis made that statement supportable. Chapter 16 will connect supported performance to the further question of clinical use; it cannot make an unmeasured role appear retrospectively.

Closing retrieval

Why does each pair require a separate judgment: a fixed-reader result and a wider reader-population result; planned unassignment and an assigned missing read; prospective reference contribution and selected later discrepancy review; a nominal correct response and behavior with late or absent output; diagnostic assistance and worklist ordering? What would the opening of a memo say if it retained a useful bounded result while withholding an unmeasured broader effect?

The distinctions that carry forward

The pairs differ in population, assignment, independence, operational condition and target quantity. A sound memo states the studied result and its supporting record first, then identifies the separate question left open. Record C demonstrates that a positive, limited conclusion can be justified; the opening archive demonstrates why an unsupported adjective cannot substitute for its estimate and denominator.

Source trail

The four-question map, linked memo, five-operation teaching sequence, operational label “degraded state” and unnamed exercises are original teaching synthesis. They are not a protocol, numerical sample-size rule, clinical recommendation or additional PathSight result.

- **Hillis and Schartz (2018), *Multireader sample size program for diagnostic studies: demonstration and methodology*.** The primary methods article distinguishes reader/case inference targets and connects sizing to endpoint, hypothesis, design and assumptions in §§2.2, 2.4–2.6, 5.1 and 8. Its examples are not a pathology-wide reader or case count. The prior source packet records full-text

section review through Europe PMC XML; the paper's indexed primary text and publisher metadata were checked again here, while live PMC delivery returned a challenge page. Primary article

- **FDA iMRMC research-tool catalog**, checked 25 September 2026: core AUC comparison, other endpoint functions, non-fully-crossed and missing-data support, and stated limitations. The catalog does not replace endpoint-specific method selection or turn a tool into a recognized standard or qualified MDDT. Official tool record
- **CAP whole-slide-imaging laboratory-validation guidance**. The issuer's teaching presentation supports the chapter's limited discussion of application, trained readers, order and washout, especially recommendations 1–3 and good-practice statements 2, 3, 6 and 9. It is not recast as an AI-reader-study protocol; no full-text claim about the separately published article is made. Official teaching presentation
- **FDA diagnostic-statistics guidance (2007)**, §§3–4 and 6; **Clinical and Laboratory Standards Institute (CLSI) EP49 (2026)**, §§3.3.1 and 3.6; **International Organization for Standardization (ISO) 20916:2019**, clauses 5.5.3.8–5.5.3.13 and 8.2. These are separate, scoped supports for reference/comparator and study-accountability reasoning. Licensed-standard clauses were reviewed in the read-only source packet; no protected table or protocol is reproduced. FDA guidance
- **FDA human-factors guidance (3 August 2026)**, §3; **IEC 62366-1:2015+AMD1:2020**, clause 3.21. Their use-error definitions are attributed separately. The FDA's definition and issue date were checked against its current final document; IEC scope is from the licensed-source packet. Neither defines this chapter's operational label “degraded state.” FDA guidance

An unavailable source is not treated as inspected, and a reporting checklist is not a study method. Regulatory sufficiency depends on the applicable jurisdiction and complete product record; this chapter's conclusion is about what the supplied reader evidence can support.

CHAPTER 16

From Performance to Clinical Use

Evidence at a decision boundary

A diagnostic program accepts the fixed input in this exercise and produces a two-category review flag. For this one record, the flag matches an independent reference assignment made before the output and without access to it. The unnamed, locked configuration is recorded as **Delta**; here we will follow its **release record**.

A specialty clinician must decide whether to request confirmatory investigation during a scheduled multidisciplinary session. The stated requirement is for the released flag to be in a controlled result queue *before the session begins*. The audit record says, in order:

1. The input is accepted.
2. The flag is generated and its stipulated reference match is recorded.
3. The session begins and ends without a queue-release record.
4. Release is recorded after the session ends.

The record stops there. What appeared after release, whether the clinician saw or understood it, what decision was made and what happened to a patient are all unsupplied.

Decision

What exact proposition does each item support? Which downstream proposition remains unestablished? What can be said without adding a cause, consequence or rate?

This release record is source-neutral fiction, separate from both recurring products. One occurrence can expose an evidence boundary while leaving product performance unestimated. The later state comparison returns to this initial judgment.

What remains to be shown?

A report can establish what a program produced while leaving the clinician's use of it unknown. An **evidence-layer and residual-claim matrix** connects those two statements: for each item it identifies the product and context, the exact proposition tested, its comparison and observations, then the precise proposition left open. That last proposition is the **residual claim**. It names an unanswered question, rather than a percentage of validation remaining or a product-failure verdict.

A useful conclusion therefore has two linked parts: “This evidence supports [bounded proposition] for [configuration and context]. It does not establish [different proposition] because [specific observation, comparison or context link] is missing.” The second sentence identifies a decision someone can take. “Clinical evidence incomplete” leaves both the missing relationship and its owner unclear.

Which earlier evidence travels with this question?

Decision

What would the intended claim and configured product from Chapters 1–2 contribute? How do Chapter 11's detected finding and justified meaning, Chapter 12's clinical classification and study population, Chapter 13's treatment-facing role, and Chapters 14–15's image path and reader or queue role differ? Which already observes what a clinician did with the information?

What the earlier evidence can carry forward

Each earlier conclusion belongs to an exact claim, configuration, population and comparison. Detecting a tissue variant does not establish its clinical meaning. Classifying a condition does not show how a result was used. A treatment-eligibility purpose does not identify a therapy benefit for an individual. A reader study does not measure queue behavior unless designed to do so; a queue observation does not measure patient outcome. Keep valid earlier evidence at its own scope instead of asking this chapter to replace it with one new universal study.

Follow information into use, one action at a time

Follow an ordinary report from its source. A specimen or image enters a configured product; the product generates a result, a system makes it available and an interface shows it. A person may then understand and weigh the information, choose management and experience patient benefits, harms or burdens. What would show that each event occurred?

The comparative question reaches across that route: does a specified test-informed clinical strategy improve patient-relevant outcomes relative to the relevant alternative? This is **clinical utility**. It is not an automatic eighth event. The route is an author-defined teaching model; neither its order nor a separate experiment at every point is a regulatory requirement.

Start with the generated result. Its **technical correctness** depends on the exact system, specified input, claimed reliability and qualified reference. A tissue sequencing report may correctly represent a tested variant call while the variant's clinical meaning and treatment relevance remain separate questions. Likewise, a pathology mark can match a defined reference before any reader encounters it.

Reviewer Lens. This analytical or technical result deserves credit for its actual output question. The configuration, input, reference, failures and denominator identify that boundary.

Consultant Lens. Recording those fields prospectively makes the result usable when a later component changes. A complete record identifies what to compare; a broad assurance about “the test” leaves that decision uncertain.

A correctly produced result still has to reach the place where the decision needs it. A send record may precede arrival in a clinician's queue. **Availability** concerns the required destination and decision window. A generic sequencing report arriving after a treatment-planning meeting raises a different timing question from a pathology queue measured from arrival to first review. Neither example reports an event for HelixBridge or PathSight.

The requirement makes the timestamps interpretable. A reviewer compares the observed arrival with that specified window. A consultant can preserve release, access and fallback events from the start, so a later service delay can be located along its actual route.

Even arrival in the database leaves the next event open: what appeared on the intended screen? A report category, context or limitation may be surfaced, collapsed or absent. **Presentation** concerns that actual view. In a tissue report the relevant object may be the finding and its limits; in assistive pathology it may be the information visible when the slide opens. The released interface and user-task evidence let a reviewer examine that view. Recording settings and the task path lets a consultant locate which claims an interface change could affect.

Once the information is visible, what did the person take it to mean? A representative user can be observed and asked neutral questions about its meaning, uncertainty and next step under representative conditions. This examines **comprehension**. A pathologist might explain what a highlight does not mean; an oncology clinician might distinguish an eligibility-associated finding from a guaranteed treatment response. The users, task, actual interface and elicited understanding bound the observation.

Human-factors specialists can investigate difficulty in that task without inventing a universal pass rate.

Correct understanding still leaves a choice. A clinician may check the suggestion against other evidence, ignore it, overweight it or appropriately decline to act. **Reliance** describes how the information enters that decision. A pathologist's final responsibility does not reveal how a mark was used; a report's availability does not reveal how the oncology team weighed it.

Reviewer Lens. Usability and a correct explanation can be useful observations while role-appropriate reliance evidence is still absent. The claimed decision needs its actual information history.

Consultant Lens. Specifying what users may check, override or do when output is unavailable makes the human-factors question concrete before a study starts. A final decision alone can conceal those different routes.

The resulting action may be confirmation, treatment change, monitoring, referral or deliberate continuation of the existing plan. More action is not necessarily better care. A recorded before/after plan change shows a change; other new information may also have influenced it. To attribute a treatment choice to a sequencing report, or earlier review to a pathology alert, the comparison must address what would have happened under the alternative information or workflow. The reviewer separates observation, attribution and appropriateness. The consultant can specify that clinical alternative and action trace before an order or acknowledgement is mistaken for an outcome.

What happened to the patient remains a further question. A benefit, harm or burden needs a patient-relevant definition, follow-up, analysis population and strategy comparison. Time saved may matter, and a therapy change may help, harm or make no measured difference. Their consequences have to be observed and attributed at the scope claimed. Clinical and statistical owners can define that outcome before looking for a favorable number. A later service signal may reopen an existing claim for review while still supplying no patient-outcome result.

The table is a compact locator for these operations. Read each row as “what was actually observed?” followed by “what does that still not answer?” Its last row is the spanning comparative question, not an eighth inherited measurement or eight mandatory experiments.

Question	Evidence it asks for	A conclusion it does not automatically supply
Technical correctness	Whether the exact configured product generates the specified output correctly and reliably from the specified input	That the output is available or clinically useful

Question	Evidence it asks for	A conclusion it does not automatically supply
Availability	Whether output reaches the required destination within the claim-relevant decision window	That the intended user sees or understands it
Presentation	Whether the released interface surfaces the result, context, units, warnings, and limitations as intended	That the user interprets them correctly
Comprehension	What intended users understand about the result, uncertainty, limitation, and next step	How they actually rely on it
Reliance	How users check, weight, ignore, override, or defer to the information	That their resulting action is appropriate
Action	What management changes because of the incremental information relative to the specified alternative	That the change benefits patients
Outcome	Patient-relevant benefits, harms, and burdens over the declared follow-up	That the test-informed strategy caused an observed difference
Clinical utility	Whether using the specified test-informed strategy meaningfully improves patient outcomes relative to the relevant alternative	Transport to another population, pathway, setting, configuration, or management option

The availability row illustrates why the rows cannot be exchanged: reaching the required queue in time answers a release question, while the next row asks what appeared to the person. The final row instead compares outcomes under care strategies. No identical user action is required in every patient pathway.

A further distinction sits before that final row. Technical correctness concerns production of the output; **clinical validity** concerns its relationship to the clinical state of interest in the relevant framework. Chapter 11's meaning appraisal and Chapter 12's classification study remain separate inputs. Utility asks what using the incremental information adds to outcomes against the available care alternative.

A direct total effect need not reveal every mechanism

Suppose eligible people are allocated to two defined strategies: a test-informed management rule and the existing rule. A sound outcome comparison may estimate the **total effect** of assignment to those strategies. It can answer that question while presentation, comprehension and reliance remain separately unmeasured. The absence of those measurements identifies unknown mechanisms; it supplies no finding that the mechanisms succeeded or failed.

Now consider a different record: one study supports classification, another records plan changes and a third examines a management option. Its conclusion depends on links among populations, decisions and outcomes. A continuous arrow on a diagram cannot supply a missing or mismatched connection.

Reviewer Lens. The actual strategy design and its remaining validity limits determine whether a direct total-effect result is credible. Requiring a separate comprehension survey merely because the route contains comprehension would change the question being judged.

Consultant Lens. A linked argument needs the particular evidence that connects its parts. Mechanism studies are useful when the claim, risk or proposed transfer depends on them; they are not eight automatic additions to every outcome study.

Keep each source within its own question

The Clinical and Laboratory Standards Institute (CLSI) EP49 framework concerns evidence of **clinical validity** for medical-laboratory test methods. Its noninterventional clinical-performance studies do not use the candidate result to manage participants, so they do not estimate outcomes of test-guided management. EP49 distinguishes analytical validation, clinical validity and utility, but expressly does not provide a method for developing clinical-utility evidence. A classification study can be valuable without being an outcome-strategy study. CLSI EP49, §§1.1 and 2.1–2.3

The International Medical Device Regulators Forum (IMDRF) N41 uses its own vocabulary for **Software as a Medical Device**: valid clinical association, analytical or technical validation, and clinical validation. This helps ask whether software output is generated as intended and whether it is clinically meaningful for the intended use. It is nonbinding guidance scoped to that software category, not a universal replacement for laboratory or jurisdictional terms. IMDRF N41, overview and §4

In European Union (EU) law, the In Vitro Diagnostic Medical Devices Regulation (IVDR), Regulation (EU) 2017/746, describes scientific validity, analytical performance and clinical performance as elements of performance evaluation. This chapter's clinical-utility question is an instructional downstream lens, **not** a fourth IVDR pillar or an IVDR conformity decision. Its detailed legal application belongs to Book 2.

Human-factors and reliance methods belong to Book 1A. IVDR, Article 56 and Annex XIII

Figure 16.1 — Locate observations; keep utility as a spanning comparison uses generic records only. The text and table govern the reasoning; the figure does not assign a state to Delta or any product.

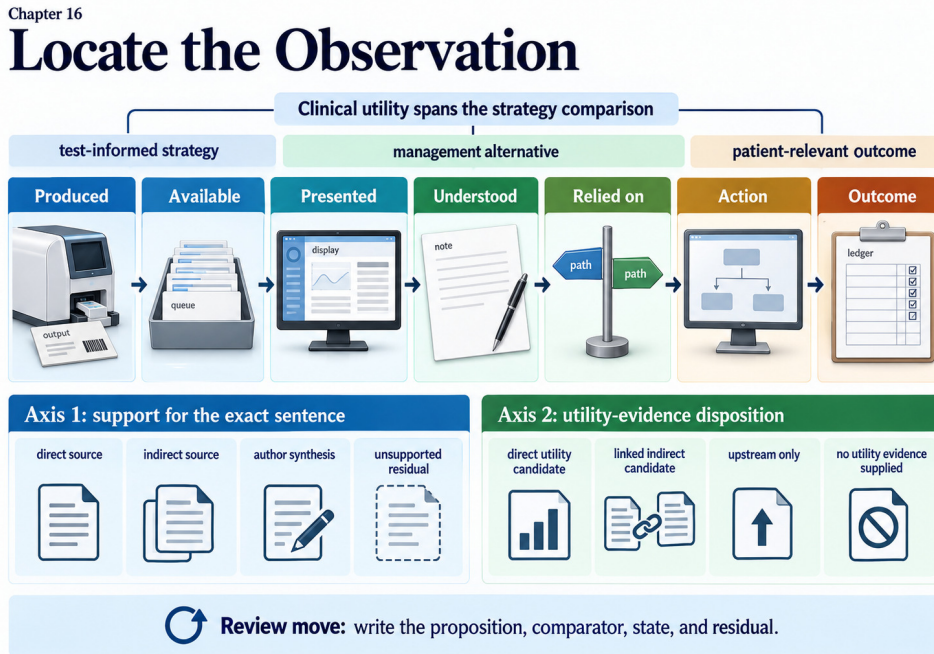


Figure 16.1. Locate observations and keep utility as a spanning comparison – Technical review plate.

Choose the alternative that answers the question

Hold one clinical decision in mind. The laboratory may have assigned a variant correctly, the test may classify a clinical condition, and a clinician may have acted on a report. Each statement compares different objects. Choosing the comparison starts with which of those events the proposed claim concerns.

For **output correctness**, the comparison needs the reference or specification appropriate to the output itself. A generic tissue sequencing caller's variant call, for example, can be compared with a qualified sequence reference in its covered region. That comparison concerns the produced call; it does not assign a patient's clinical state.

For **clinical classification**, the comparison asks whether the test separates patients by a defined target condition. An independently assigned clinical reference may require different material and judgment from the sequence reference. Its coverage and possible error limit the classification conclusion even when the variant call was produced correctly.

For **reader assistance**, the comparison concerns a pathologist's outcome with the aid against a prespecified reading condition without it, for the same task and an analyzable reader/case design. Agreement between a digital-pathology mark and a reference region does not tell us whether showing that mark changed a reader decision.

For **availability**, the comparison concerns observed release or access with the stated decision-relative requirement or operational alternative. A generic priority worklist might need a score before primary reading; a score arriving later could still exist technically but not meet that particular timing question. The requirement must be defined before interpreting timestamps.

For **utility**, the comparison concerns the test-informed *care strategy* with the relevant strategy without its incremental information. A tissue-sequencing-informed decision might be compared with existing care that already uses other tests, imaging and follow-up. The alternative is not automatically “no care,” and neither an analytical reference nor a reader control can stand in for its patient-outcome comparison.

Proposed question	Comparator role to name
Was this output generated as specified?	Reference assignment or specification appropriate to that output
Does the test classify the relevant clinical state?	Endpoint-specific clinical reference or explicitly limited comparator
Does assistance improve reader performance?	Prespecified reader control condition
Does the output arrive in time?	The defined decision-relative availability requirement or operational alternative
Does using this strategy benefit patients?	The relevant management alternative without the incremental information, including its existing tests and care

Read down the table while holding one intended clinical decision in mind. The first comparison can tell us whether a category was produced as specified; the second whether that category relates to the target state; the third whether showing it changes a reader outcome; the fourth whether it arrives for the decision; and only the last asks about patient outcomes under different care strategies. No earlier comparator can silently answer the last row.

The Medical Device Innovation Consortium (MDIC) 2019 framework uses a test-to-decision-to-outcome question and a clinical-management alternative. It is a nonprescriptive United States planning document, not payer policy or a guarantee of regulatory or coverage success. A replacement test's lower burden or a faster test's

earlier result may contribute to a useful strategy, but neither quality proves patient benefit by itself. The Reviewer asks whether the comparator matches the proposed claim; the Consultant defines the available alternative and the clinically relevant timing before selecting evidence. MDIC framework, §4.2.2.1

“Direct” can mean two different things

Imagine an exercise record stating that representative users correctly explained a report limitation. The record directly supports the narrow sentence “these users explained that limitation under these conditions.” It does not directly test whether a test-informed clinical strategy improves patient outcomes. One sentence can therefore have **direct source support** and still be **upstream evidence with no utility inference**.

The first question is about attribution: does the cited locator actually support this exact sentence? **Direct source support** means it states the bounded proposition. **Indirect source support** means it supports an adjacent component. **Author synthesis** joins supported parts transparently. An **unsupported residual** is the target proposition the supplied record has not established. These descriptions concern the relationship between source and sentence; they do not grade study quality.

Source-support value	Meaning
Direct source support	The exact bounded proposition is expressly supported at the cited locator
Indirect source support	The source supports a related component or relationship, not the full proposition
Author synthesis	The explanation combines bounded components into a stated interpretation
Unsupported residual	The target proposition is not established by the supplied material

For the comprehension example, the direct-support row covers the sentence about what those users explained. Replacing that sentence with a patient-benefit claim would change the proposition, and the same locator would no longer directly support it.

How does the item contribute to the utility argument?

The second question is about the utility argument. A **direct utility candidate** tests the specified test-dependent strategy against its clinical alternative through patient-relevant outcomes. A **linked indirect utility candidate** uses a nonisolating outcome observation or connects separate performance, management and outcome components. **Upstream evidence** answers an earlier question without itself observing or credibly connecting the whole comparative outcome proposition. **No utility-relevant evidence supplied** describes a defined set with no responsive item, leaving benefit unknown. Each judgment must name the item or set it classifies.

Utility-evidence disposition	Meaning in this book
Direct utility candidate	A design directly tests and isolates the specified test-dependent strategy contrast through patient-relevant outcomes
Linked indirect utility candidate	Nonisolating outcome evidence or a transparent assembly of separate links addresses the utility proposition, with assumptions and mismatches exposed
Upstream evidence; no utility inference	The item supports an earlier proposition but does not itself observe or credibly connect the full comparative outcome proposition
No utility-relevant evidence supplied	The defined set contains no responsive item; this is an absence of supplied evidence, not evidence of no benefit

The upstream row fits the isolated comprehension observation. A clinical-classification study can occupy that row alone and contribute to a larger linked argument when management and outcome evidence are also supplied. Those are judgments about different named sets, so they need not share one category.

Both classifications are author-defined, informed by MDIC's bounded direct/indirect discussion. Neither decides bias, precision, applicability, clinical importance, benefit, sufficiency or authorization. A reviewer needs both axis names to know which judgment is being made. A consultant can use the utility category to locate the missing link or direct comparison needed by the intended claim.

Replace an arrow with a sentence

A drawn arrow from result to action leaves its claim unstated. The full sentence might be: "The incremental result caused this management difference in this population and setting." A plan change after the report is observable, while causation may remain uncertain because other information arrived or clinician selection changed. The sentence identifies which relationship the evidence would have to answer.

A relevant recorded event is **directly observed**. A transparent connection between separate observations is **indirectly linked**. An unobserved connection taken as a premise is **assumed**. These ordinary descriptions say what carries the claim.

Three other states distinguish lack of information from conflicting information. **Absent** means the needed item or link is not supplied; **not assessed** means the record did not evaluate the question. **Contradicted** requires an actual observation that conflicts with the exact proposition. A timing contradiction cannot also contradict an unassessed patient-benefit claim. A reviewer preserves that distinction; a consultant can then address the actual absent or assumed link without relabeling every empty field as a failure.

From an evidence item to a defensible sentence

Before deciding what an item proves, its record must say what was examined and what happened. The questions below form two linked parts of the matrix, an author-designed decision aid. An attractive classification cannot repair an unreported outcome in the underlying evidence record.

What is the evidence item?

- Which item, source, date, authority and exact locator supply it?
- Which product and output, configuration/version and lifecycle state does it concern?
- What claim, population, input, user, setting and decision does it represent?
- What design, sampling, denominator, failed states, missingness and timing produced its observations?
- What endpoint, comparator, result, uncertainty, harms and applicability are supplied?

An unsupplied field remains **not supplied**. The design question, for example, needs the assigned or sampled population and its failed and missing states before a final outcome percentage can be interpreted.

What sentence can that item support?

- What exact proposition at a layer or transition is attached to this item?
- What is its source-support judgment, and why?
- What is its utility-evidence judgment, for which named item or set?
- What is the proposition's state, context match and assumed transport?
- What do quality, sufficiency, uncertainty and competing explanations permit?
- What residual claim, missing link, explicit unknown and prohibited inference remain?

Again, an answer missing from the supplied record stays **not supplied**. The item/set question prevents a single classification result from inheriting the outcome-study role of a larger evidence set.

For a worked exercise, the stipulated fact sheet can directly support a sentence *about that exercise*. It is not external clinical evidence. One item may warrant several rows only if each row names a different exact proposition; one item does not change its utility category merely because the writer changes adjectives. A Reviewer should be able to move from a sentence back to its locator and forward to the residual. A Consultant can then choose among obtaining a result already measured but not reported, repairing an evidence link, designing a targeted comparison or narrowing the claim. An outcome trial is not automatic for every residual.

A strategy design with an incomplete result record

The first fictional strategy record identifies a locked assay and report as **Alpha**, with its specimen, population, setting and management option carrying the same local label. Qualified clinicians use the two-category report under a prespecified rule to choose that management option or existing management. This **strategy design record** is an exercise, not an empirical study.

Eligible participants are randomly assigned with concealed allocation either to the report-dependent management rule or to the existing strategy without that report in the same care setting. The investigators follow a prespecified patient-relevant outcome, with burdens and harms recorded separately. The **clinical-strategy comparator** is the existing management pathway—not the laboratory reference used to assess the assay's output.

All assigned participants belong to the assigned-strategy analysis account. If the report is invalid or unavailable, that is a consequence of being assigned to the test-dependent strategy, not a reason to pretend the person was never assigned. Some protocol deviations and missing outcomes remain in the account, but their extent, causes and balance are unsupplied. The record says outcomes were analyzed; it gives no direction of difference, effect estimate, interval, harm result or benefit conclusion. Presentation, comprehension, reliance and each person's action were not separately observed.

Decision

Which utility-evidence category describes this design? What prevents a benefit decision from the supplied record? Does a missing comprehension measure change the design's category?

What the strategy design establishes

The assignment contrast directly tests the test-informed strategy against its clinical alternative through follow-up. That makes the strategy design a **direct utility candidate**, not an established benefit. Randomization helps isolate the strategy contrast, but missing outcome and deviation information can still affect validity; an unreported comparative result and uncertainty cannot be appraised. The two linked records show where the design's strength stops.

Record	What the design record establishes
Evidence item	The stipulated strategy-study record; teaching authority only
Configuration and context	Locked assay/report and its specified specimen, population, qualified clinicians, setting and two management rules
Design and denominator	Concealed random assignment; all assigned participants retained in the strategy account, including invalid or unavailable reports
Endpoint and timing	Prespecified patient-relevant outcome over declared follow-up; duration not supplied; burdens and harms recorded separately
Comparator	Existing management without the incremental report, not an analytical reference
Result and uncertainty	Analysis stated, but comparative effect, interval and harm results absent; deviations and missing outcomes unresolved

The first table credits a suitable comparison and denominator while refusing to invent an effect. In the classification table, “direct source support” applies only to the fact sheet's explicit design sentence. “Direct utility candidate” describes that design's path to a strategy outcome. Neither adjective says the missing results were favorable.

Classification	What can be concluded
Exact proposition	The design directly tests the specified strategy contrast through outcomes
Source support	Direct support from the stipulated design field for that exercise sentence
Utility evidence	Direct utility candidate
Unmeasured mechanisms	Presentation, comprehension, reliance, and individual action remain unmeasured

Classification	What can be concluded
Quality and sufficiency	Not established merely by randomization; missingness, deviations, harms, precision and applicability need review
Residual claim	Whether the test-informed strategy meaningfully improves outcomes remains unresolved because the comparative result and uncertainty are absent and the impact of missingness and deviations is unknown

The Reviewer can retain the direct design classification and ask for the actual assigned-strategy outcome and harm analyses with missingness and deviation assessment. The Consultant can retrieve and appraise those analyses before commissioning a different study; a comprehension survey would not substitute for the missing comparative result. Unmeasured intermediate mechanisms remain unmeasured, not failed.

A separate study with complete outcomes

A separate fictional study is recorded as **Alpha-P**. It asks the same bounded strategy question, but its observations belong to this **complete outcome study** and cannot fill the first record's missing fields. Eight hundred eligible people are individually randomized with concealed allocation: 400 to the test-informed strategy and 400 to the existing strategy. The locked assay/report, population, setting and management rules match the proposed comparison. Outcome assessors are masked to assignment; clinicians cannot be masked to receipt of a report. The prespecified analysis concerns assignment to the strategy, rather than perfect adherence or the effect of one intermediate action.

Both groups undergo the same specimen procedure and scheduled encounters. All 800 have the specified 30-day outcome records. In the test-informed arm, 12 of the assigned 400 have an invalid or unavailable report, and six *other* participants do not receive protocol-directed management. All 18 remain in that arm's 400 and their outcomes count. Eight deviations in the existing-strategy arm also remain among its 400. None of these people is moved to the other arm or deleted after the fact.

The predefined **benefit endpoint** requires a person to be alive, back to their recorded pre-episode daily-activity level and without unplanned urgent care by day 30. The separate **harm endpoint** is at least one event from a locked list of serious adverse events by day 30. The same person may meet the benefit definition and have a listed harm, so the rows cannot be added to make mutually exclusive patient groups. These are fictional endpoints, not a recommendation for any disease.

Complete assigned-strategy account	Test-informed strategy	Existing strategy
Participants in each outcome denominator	400	400
Benefit endpoint met / not met	300 / 100	240 / 160
Harm endpoint present / absent	40 / 360	36 / 364
Missing outcome	0	0

Read the table by arm. The 300 and 100 benefit states add to the test-informed 400; the 240 and 160 add to the existing-strategy 400. The harm present/absent pairs separately add to those *same* arm denominators. The 12 report problems and six other deviations are included in the test-informed outcome cells, but no cross-tabulation tells us which benefit or harm cell each occupies. Their allocation to those cells remains unknown.

Before calculating, the source exercise stipulates a prospective clinical and statistical judgment: these two endpoints, their decision limits and assessment of other material burdens address this deliberately narrow strategy claim; no extra visits or procedures occur in either arm; material bias, model assumptions and applicability concerns have been reviewed for the represented context. Arithmetic cannot establish those clinical premises, and the limits are not universal benefit–risk rules. The joint rule requires **both** a benefit-difference lower bound *above* 0.05 and a harm-difference upper bound *below* 0.06. Every difference is test-informed minus existing strategy; equality does not pass either strict inequality.

First see what the observed fractions mean. For benefit, the first arm's proportion is 300 divided by its 400 assigned people; the second arm's is 240 divided by its own 400. For harm, use 40/400 and 36/400. The difference between two proportions is measured in **percentage points**, not a percentage of the other arm's rate. An interval adds uncertainty around that comparison under a statistical model. Its lower endpoint is used for the benefit rule; its upper endpoint is used for the harm rule because an increase in harm is the concern.

Here is the specified **Agresti–Caffo** interval recipe for two independent groups, as documented by the National Institute of Standards and Technology (NIST). Let x_1 and x_2 be the number meeting the endpoint in the test-informed and existing arms, and n_1 and n_2 their assigned-arm denominators. The observed fractions remain x_1/n_1 and x_2/n_2 . For the interval calculation only, adjust them to $a = (x_1 + 1)/(n_1 + 2)$ and $b = (x_2 + 1)/(n_2 + 2)$. The added ones and twos stabilize the calculation; they are **not** added real participants or observed events.

The interval's **center** is $a-b$. Its **standard error (SE)**—the model's estimate of how much this difference could vary across comparable samples—is $SE = \sqrt{a(1-a)/(n_1 + 2) + b(1-b)/(n_2 + 2)}$. Multiplication by $z = 1.959964$, the supplied normal multiplier for a two-sided nominal 95% interval, gives the **half-width**. Subtract that half-width from the center for the lower endpoint and add it for the upper endpoint. The square root converts the summed variance terms back to the same proportion-difference scale. This approximate method assumes independent binomial groups; it is not a paired or clustered analysis and does not promise exact coverage. NIST method record

The exercise report supplies a benefit interval of **0.085285 to 0.213223** and a harm interval of **−0.030962 to 0.050862**.

Decision

What are the observed differences, and how does the interval recipe relate to them? Do both strict bounds pass? What bounded conclusion and remaining uncertainty follow? How would the decision change if the harm upper bound were 0.08?

The two interval calculations and their joint decision

Start with the observed benefit proportions. Test-informed benefit is $300/400 = 0.75$; existing-strategy benefit is $240/400 = 0.60$. Their observed difference is 0.15, or **15 percentage points**. The interval calculation uses $a = 301/402 \approx 0.748756$ and $b = 241/402 \approx 0.599502$. Its center is about 0.149254, and its standard error about 0.032638. Multiplying by 1.959964 gives a half-width about 0.063969, so the endpoints are **0.085285** and **0.213223** after rounding. The lower endpoint exceeds 0.05.

For harm, the observed fractions are $40/400 = 0.10$ and $36/400 = 0.09$, an observed difference of 0.01, or **one percentage point**. The interval uses $a = 41/402 \approx 0.101990$ and $b = 37/402 \approx 0.092040$. Its center is about 0.009950, standard error about 0.020874, and half-width about 0.040912. The rounded endpoints are **−0.030962** and **0.050862**. The upper endpoint is below 0.06. The adjusted centers need not equal the raw observed differences; both still use the real 400-per-arm denominators for the actual outcome account.

Both prespecified conditions pass under the exercise's supplied clinical, model and context adequacy premises. Retain a bounded positive conclusion: assignment to this test-informed strategy improves the stated 30-day benefit endpoint while meeting the defined comparative harm criterion **in the represented context**. This is affirmative clinical-utility evidence for that narrow strategy claim, not a universal test benefit or a real clinical result. It is not “zero harm”: the respective harm counts are 40 and 36, and

the interval remains compatible with an increase. Two separate nominal 95% intervals do not give simultaneous 95% assurance for the entire strategy or product.

If the harm upper bound were **0.08**, it would fail the second condition even while the benefit condition passed; the joint rule would fail. The 12 report problems and six other test-arm deviations, and the eight comparator deviations, cannot be deleted to manufacture a more favorable analysis. Presentation, comprehension, reliance and individual actions were not separately measured, but that absence does not undo a credible total-effect result or make those mechanisms observed. Longer-term effects, rare harms and transport to a different population or care system remain outside the bounded conclusion. The Reviewer can accept the narrow outcome finding and request separate evidence only for a broader claim; the Consultant can retain the complete assignment account and plan a claim-specific applicability or post-use check. The first and complete strategy studies are both direct *candidates* by design; the complete study differs because its results and stipulated adequacy basis can be inspected.

Two independent evidence records

Two more independent dossiers concern different unnamed systems. Their individual items and defined sets need separate judgments. They add no facts to either strategy study, the opening release record, HelixBridge or PathSight.

A dossier containing three study records

This **three-record dossier** identifies a locked assay/report configuration as **Beta**, with the population, setting and management option carrying that local label. Clinicians choose that management or the existing alternative. The utility question compares that decision with versus without the report's incremental information. The dossier has three items: **A1, the classification record; A2, the plan-change record; and A3, the management-study record**. Their identifiers do not make them steps in a proved causal chain.

Item	Supplied evidence
Classification record	A prespecified evaluation of all eligible records for the stated specimen against an independent target-condition reference. Invalid, equivocal and failed outputs remain in the account; their distribution is not supplied. The report states that the intended clinical-classification proposition is supported, but gives no criterion, estimate or precision. Reference error and performance outside the stated population/input remain unresolved. Analytical validation is not separately supplied

Item	Supplied evidence
Plan-change record	Prospective observation of every returned assay report and the clinician's documented pre-result and post-result management plan. Cases without a returned report remain visible. Plans sometimes change after review within the decision period. Attribution, appropriateness and patient outcomes are not established; missing plan records, other influences, clinician selection and reliance remain unresolved
Management-study record	A randomized study of the specified management option versus the existing management alternative in a different population, recorded as Gamma , using all assigned participants. Eligibility, baseline care and follow-up differ from the dossier population. Deviations and missing follow-up remain visible but their extent is not supplied. The outcome was analyzed; no comparative direction, effect estimate, interval or benefit conclusion is given

Each row compares different records. The classification study uses a target-condition reference and has no numerical basis for independently appraising its “supported” summary. The plan study compares before/after management plans, without concurrently assigned clinical strategies. The management study examines options in the different population; it includes neither this dossier's test-selection pathway nor a supplied outcome result. No study count or effect permits a calculation.

An observed interface task

The **interface record** identifies a final or production-equivalent interface as **Kappa**, with its representative setting carrying the same local label. Representative intended clinicians review fixed result records at the claim-relevant workflow point. Each report contains a two-category result, a limitation and a follow-up instruction.

Every enrolled user completes every specified scenario. Difficulties, incomplete tasks and knowledge-task responses stay in the account. The comparisons are the interface specification and the intended interpretation set in advance for each scenario. Applicable information is observed on screen, and the stipulated interpretation is observed when understanding is elicited neutrally. One type of navigation difficulty occurs; its cause is unresolved and no scenario or response is removed. No actual clinical reliance, management action, patient outcome, use outside these users and conditions, duration criterion or numerical result is supplied.

What does each item—and each complete dossier—support?

Decision

For the classification, plan-change and management-study items, what exact proposition, comparator and two-axis judgment applies to each? What applies to their defined three-item set? For the interface record, what presentation and comprehension were observed, and what residual remains? How do quality and applicability differ from directness in both complete judgments?

How the two dossiers reach different boundaries

Review the two classifications

The three-record dossier forms a **linked indirect utility candidate**, not a demonstrated utility result. The classification record supports only the reported clinical-classification proposition for its specified population and input; no supplied number or criterion permits independent sufficiency appraisal. It must not be renamed analytical validation or proof that a particular output was technically correct. The plan-change record observes some after-report plan changes, but does not isolate the report as their cause or establish appropriateness. The management-study record studies management outcomes in a different population and eligibility path; even an interpretable result there would need a justified bridge to the dossier's test-selected patients, and here the comparative result itself is missing.

The interface record reaches a different boundary. Its final or equivalent interface, representative users and conditions, complete scenario account, observed surfacing and neutrally elicited interpretation support **bounded presentation and comprehension**. The navigation difficulty remains an observed limitation with unknown cause. Nothing in the record observes actual reliance, an action or a patient outcome. That is an upstream positive result, not evidence that the product has no utility.

Item or set	Source-support judgment	Utility-evidence judgment
Classification record	Direct support for the stipulated exercise description; not independent proof of its unstated quantitative basis	Upstream evidence by itself
The plan-change record's observed plan changes	Direct support for the described observation, not for causal attribution	Upstream evidence by itself

Item or set	Source-support judgment	Utility-evidence judgment
Management-study description	Direct support for the stated design and absence of a supplied comparative result	Linked indirect utility candidate for the dossier test-strategy proposition; it needs the test-selection and context links
The three-item set	Joining the records is author synthesis with explicit unresolved links	Linked indirect utility candidate
The interface record's presentation and comprehension observations	Direct support for those bounded exercise propositions	Upstream evidence; no utility inference

Why is the management-study record a linked candidate **even alone** when the evaluated proposition is the dossier test strategy? It supplies a management-to-outcome study component; connecting it to selection by the dossier test information and transporting it from the different management-study population would require additional evidence. The classification and plan-change records are upstream when judged individually. The defined set can be a linked candidate because it contains proposed components of a test-to-decision-to-outcome argument, but the components are not a completed bridge. These are not contradictory labels: each row names a different item or set and its target proposition.

The residual for the three-record dossier is specific: the set does not establish that the dossier test information causes appropriate management and patient benefit in the dossier population. The plan-change record is nonisolating; the management-study record has no supplied comparative result or demonstrated transport; and the classification record's reference uncertainty, failures and unstated estimate still limit appraisal. The residual for the interface record is different: the observed interface presentation and understanding do not establish reliance, attributable or appropriate management, outcome, harm or utility in actual use. The Reviewer retains the classification record's bounded report statement and the interface record's observed user result while withholding the separate strategy claims. The Consultant assigns the dossier's missing outcome report and transport argument to clinical/statistical owners, and the interface navigation cause and any role-dependent reliance question to the human-factors owner, rather than ordering an undifferentiated new outcome trial.

Check the reasoning, not the number of filled cells

The reasoning is sound when the clinical reference, before-result plan and management strategy keep their different comparator roles; when exact source support differs from utility directness; and when the item or set, failed and missing states, and exact unobserved link remain visible. Calling action an outcome, source reputation exact support, randomization automatic benefit or an empty field proof of no benefit would break those connections. These are qualitative learning criteria, not numerical study thresholds.

Return to the recurring products

For **HelixBridge Dx** tissue, the broad purpose remains identification of treatment-eligibility-associated variants. The validation report rules, the knowledgebase used in analytical validation and the knowledgebase in the separate final technical report—recorded as R19, DB-18 and DB-19—identify bounded evidence records. They do not show a patient report reaching a clinician, changing management or improving an outcome. Neither record establishes deployment or the commercial configuration, and the exact claim elements remain unresolved.

A reviewer can preserve the analytical evidence while determining which specific clinical-use claim is proposed. A consultant can map release, interpretation and any strategy comparison once those elements are fixed. This fictional file supplies no actual used result.

PathSight Cloud remains assistive, with diagnosis by a pathologist. Its Chapter 15 reported reader-study assertion is not an observation of later reliance or patient outcome. A separate high-volume **simulation** observed two positive low-score slides beyond expected turnaround; no denominator, exact delay, cause, output correctness, recovery, live release or clinical consequence is supplied. A reviewer can identify a timing question without labelling those slides technically wrong or harmful. A consultant can trace the configured image, queue and user conditions for an applicability assessment while leaving the simulated observation separate from the opening record's one reference-matched flag. Neither recurring product inherits the separate complete strategy study's outcomes.

Translate a bounded conclusion for three colleagues

The **clinician** asks what decision the new information changes relative to the available alternative for the intended patient. The matrix can make the proposed action and outcome discussable without giving individual medical advice or assuming that taking more action is better.

The **Reviewer** asks which exact source or observation supports each sentence and where the current claim stops. An adequate total-effect result like the complete outcome study can be accepted at its bounded scope without requiring every mediator to have a separate study; a classification statement like the first dossier item cannot be promoted to patient benefit simply because its source is reputable. Scientific adequacy and jurisdictional sufficiency remain separate decisions.

The **Consultant** asks which of four choices is justified next: retrieve an existing result, build and test a missing link, narrow the claim, or design a new comparison. Human-factors specialists own representative understanding and role-dependent reliance methods; clinical and statistical specialists own strategy/outcome and causal questions. A product change or new post-use signal can reopen a specific inference, not automatically erase every earlier result.

The same flag under two release histories

Return to the unnamed opening release record. This is a paired teaching comparison, not a dated deployment. The **before-session record (State A)** has a single two-category flag that matches the independent reference assignment and is released into the required queue before the decision session begins.

In the **after-session record (State B)**, the input, independent reference assignment, locked configuration, flag, intended user, decision, destination and required window are unchanged. The flag still matches its reference. Release is recorded only after the session ends.

Both records require release **before the session begins**. Neither supplies a clock time, delay duration, later visibility, notification, comprehension, reliance, management, recovery, patient outcome or harm.

Decision

What does each record support against its reference assignment and release requirement? What clinical-strategy comparison remains absent? Which one proposition changes evidentiary status? What residual follows without completing the clinical story?

Which proposition changes with the release history

Review the state comparison

Both states have the same **single-record reference match**. The before-session record's release meets the stated decision-relative requirement; the after-session record's does not. The result-level reference assignment and the release requirement are different comparators. Neither state supplies the clinical-management alternative or patient-outcome comparison needed for a utility conclusion.

Proposition	Before-session record	After-session record
This stipulated flag matches its independent reference assignment	Directly observed for the single exercise record	Directly observed for the same bounded comparison
Queue release met the specified decision-relative boundary	Directly observed	Contradicted by the ordered release record
The clinician saw and understood the output	Not assessed	Not assessed
The output influenced an appropriate management action	Not assessed	Not assessed
The test-informed strategy improved a patient-relevant outcome relative to its alternative	Not assessed; no such comparison supplied	Not assessed; no such comparison supplied

The exact exercise facts directly support the corresponding bounded sentences on the **source-support axis**. On the separate **utility-evidence axis**, each state is upstream evidence with no utility inference. For the after-session record, an accurate residual sentence is: “This single stipulated occurrence has a reference-matched flag and release after the required decision window. It does not show what was presented or understood, how the clinician relied on the flag, what management occurred, or patient benefit, harm or utility.”

Only now is “correct but delayed” a permissible shorthand—and even “correct” means only the stipulated one-record reference match, not product-level analytical or clinical validity. The after-session record **contradicts** the on-time-release proposition; it does not prove harm. The before-session record **meets** that release proposition; it does not prove benefit. This paired exercise supplies no rate, delay duration, cause or clinical consequence.

Figure 16.2 — Compare the reference match with the release boundary is linked only after both Delta commitments. It depicts the same bounded comparison as the table, not an additional observation.

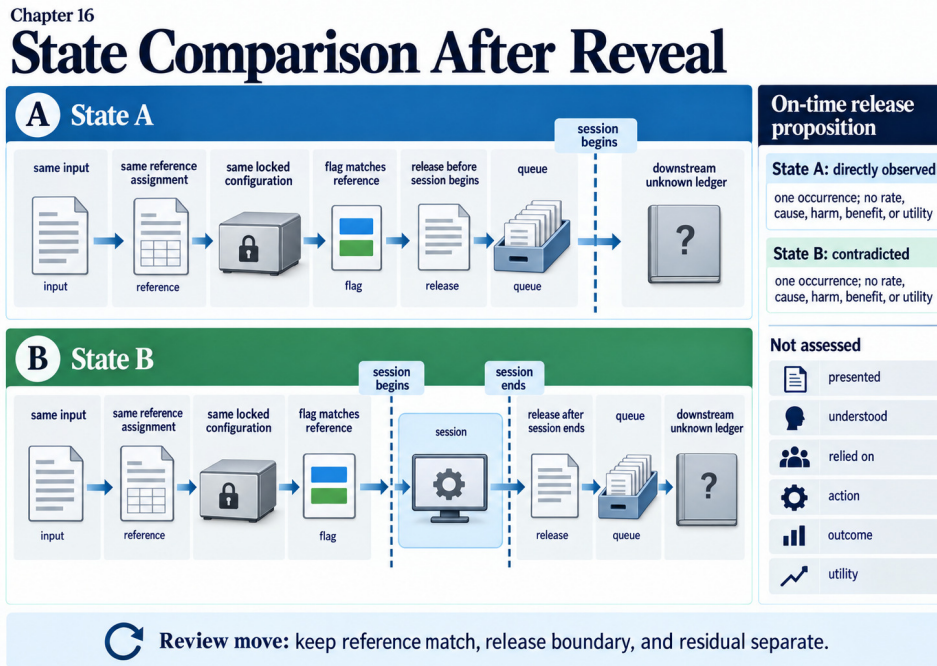


Figure 16.2. Compare result match with the decision-relative release boundary – Technical review plate.

Stop each claim where its evidence stops

The matrix gives credit without inheritance. It can preserve a well-supported classification, an observed comprehension result or a credible comparative strategy outcome while preventing any one from answering a different question by implication. A residual claim should point to an exact unsupported proposition and an owner or choice, not a vague fraction of work left. Chapter 17 will ask whether these bounded results still apply after a configuration, specimen, population or workflow change; that future question does not make them worthless today.

Closing retrieval

Decision

How can a source directly support a sentence while the evidence remains indirect or upstream for utility? Why can a strategy comparison remain direct without a comprehension measure? What does a before/after management-plan change fail to isolate? Why does “not supplied” differ from “no benefit”? What makes a residual claim actionable?

Five distinctions that keep the claim bounded

Source support concerns the exact sentence and its locator; utility directness concerns the study's path to a comparative clinical outcome. A credible total-effect strategy result can answer that contrast while leaving mechanisms unmeasured. A plan change alone may not isolate the report's contribution from other influences. Missing evidence is not a negative outcome. An actionable residual names the exact claim and context, the evidence already credited and the missing observation, comparison or transport link.

Source trail

The seven local observation questions, spanning utility proposition, two classification axes, matrix and fictional exercises are original teaching synthesis. Alpha, Alpha-P, Beta, Gamma, Kappa and Delta are not empirical products or clinical studies.

- **CLSI EP49, first edition (2026), §§1.1 and 2.1–2.3.** Its medical-laboratory clinical-validity framework distinguishes noninterventional clinical-performance studies from test-guided utility questions and explicitly excludes guidance on developing utility evidence. The licensed clauses were reviewed in the read-only packet; no protected procedure, table or figure is reproduced. Official CLSI record
- **MDIC, *Developing Clinical Evidence for Regulatory and Coverage Assessments in In Vitro Diagnostics* (2019), §§4.2.2.1 and 4.2.3.** The publicly available framework supports the clinical-management comparator, test-to-decision-to-outcome question and bounded direct/indirect discussion. Its United States planning context is nonprescriptive and is neither law nor payer policy. Official framework record
- **IMDRF N41 (2017), overview and §4.** Its final, nonbinding SaMD clinical-evaluation distinctions remain within SaMD scope; they do not prescribe this chapter's full use-to-utility model. Official document
- **NIST Dataplot, difference-of-proportions confidence interval.** The Agresti–Caffo adjusted-Wald formula supplies the mathematical method for Alpha-P's invented

counts only. NIST does not supply the endpoints, sample size, margins, adequacy or utility conclusion. Official method record

- **IVDR Article 56 and Annex XIII; U.S. Food and Drug Administration (FDA) 2026 human-factors guidance; International Electrotechnical Commission (IEC) 62366-1:2015+AMD1:2020.** These support narrow jurisdictional terminology and human-interface handoffs, not product-specific clinical utility. The licensed IEC material is read-only and safety-related in scope; detailed methods and jurisdictional sufficiency are outside this chapter. EU legal text, FDA guidance

An evidence class in this chapter is not a regulatory approval, a coverage decision, or a real-patient recommendation.

CHAPTER 17

When Evidence Must Travel

Which result still applies?

A laboratory has evidence that a particular version of a tissue test produces its intended report. Then a reference file changes. The earlier study still describes what it tested. To know whether its result applies to the new combination, we need the dependency between that file and the report: which output can the replacement affect, and which earlier observations actually tested that path?

HelixBridge Dx's analytical-validation record identifies knowledgebase DB-18. A later **final technical report** identifies DB-19—a different evidence object and knowledgebase state. The controlled record says the newer knowledgebase added a **reference-transcript record**, a selected reference description of a ribonucleic acid (RNA) sequence used to interpret a deoxyribonucleic acid (DNA) finding. It makes a previously rare mapping anomaly execute more often; frequency and clinical consequence are not supplied. Application 7.0 and report rules R19 are stated across this comparison, but complete identity of the remaining configuration is unestablished. The newer report describes strong overall agreement without an affected-class design, denominator, acceptance rule or result.

Does the identified evidence still support the **affected transcript-selection part** of the broad therapy-eligibility variant-identification purpose? Which two configurations and evidence records should be compared first, and what would test that narrow question? The exact therapy and claim details remain unsupplied, as do commercial release, quality action and regulatory outcome.

Chapter 16 left useful results attached to their exact propositions. A change now asks whether one of those propositions still holds for a new input, process or use condition. The answer may differ across a product's claims: a changed dependency can reopen one conclusion while leaving another supported.

What earlier evidence would matter to a tissue-report change, and what would matter to an assistive-image change? Which claim, configuration, input, reference and use condition must remain attached to it? Where would Chapter 16's residual claim stop a technical finding from becoming patient benefit?

What the earlier evidence must retain

The configured object and evidence plan come from Chapters 2 and 4. The reference, material, detection and end-to-end constraints from Chapters 5–10 tell us what the observation covered. Chapter 13 supplies the bounded bridge question; Chapters 14–15 preserve the image and reader conditions; Chapter 16 preserves the unproved clinical-use proposition. These records have different jobs, but each keeps a result attached to its tested object. Unchanged weights leave changed images and reader behavior possible, just as an unchanged application label leaves database-dependent output unsettled. Neither technical finding demonstrates clinical benefit.

Freeze the comparison before classifying the change

Suppose a tissue sequencing service replaces its reference file while retaining its wet-lab procedure. The first comparison follows the studied specimen-to-report path and the proposed path beside it: input material and population, instrument and processing, reference and report rules, user and report destination. That gives the replacement an address in the evidence. For an image service, the route instead begins at slide and scanner and continues through processing, model input, display and the pathologist's task. A file or weight identifier describes only one part of that route.

Each asserted state also needs its source and time. A package called “current” at one date may not identify what a site actually installed. Missing deployment lineage leaves the target unresolved. The comparison header is useful before judging whether evidence transfers, because it distinguishes the exact decision from the product name and the documented change from an imagined release.

Comparison field	Entry to establish
Decision purpose and claim component	The exact proposition and decision being assessed
Prior configured baseline	Product, components, input path, population, workflow, effective state and source
Proposed or observed current state	The same fields, including explicit unknowns
Change or signal	What is documented, when, and from which record
Prior evidence	Study ID, exact tested configuration, design, reference, endpoint, criterion, results and limitations

Comparison field	Entry to establish
Responsible owners	Who can decide evidence applicability and each downstream action

A source-file replacement fills the change row. Without installation history, the target-state row remains unknown; it does not inherit the replacement by assumption. **Reviewer Lens.** Missing lineage is the narrow request that precedes a deployed-product applicability judgment. **Consultant Lens.** Capturing versioned inputs, effective dates and evidence identities when a change occurs makes later comparisons reproducible. A precise record can still contain explicit unknowns.

Classify facts, then ask about combined effects

The application labels in the opening comparison match at 7.0. That supports sameness at **label granularity**, while leaving behavior and unrecorded components open. The knowledgebase content changed between the two identified records; even a later successful test could not undo that historical fact. Where no installed knowledgebase is identified, deployment is uncertain. The table distinguishes these fact statements before any performance judgment is made.

Base state	Meaning	Common misuse
Unchanged	Traceable information supports sameness at the defined level	Treating an omitted field or unchanged label as proof of full behavioral identity
Changed	A documented difference exists; its known attributes are recorded	Calling it unchanged because it later passes an acceptance rule
Uncertain	The record cannot establish the relevant state or attributes	Quietly treating uncertainty as no change

The changed row would remain changed after an acceptable result; the uncertain row cannot become unchanged through silence. Now ask a second question: could one component's effect depend on another component or operating condition? An unchanged application can execute a new database path; an unchanged assay can receive a different specimen background; fixed weights can receive differently normalized images. This **interaction question** identifies a mechanism worth evaluating. It does not announce that an interaction or cause has been demonstrated.

The combined question needs the affected facts, plausible dependency, evidence, evaluation state and disposition. “Plausible and open” means something different from “evaluated within this scope”; a conclusion that no plausible dependency remains also needs its reason. **Reviewer Lens.** The relevant object is the combined path as used.

Consultant Lens. A targeted combination challenge earns its criterion before the result appears, and its qualified owner decides what passed. A later anomaly can reopen that conclusion without retroactively proving the suspected interaction.

A format changes; what must survive?

A separate fictional export service makes the distinction tangible. Its older format, F1, becomes F2; the assay inputs and clinical claim are traceably held constant for this exercise. The narrow promise concerns numeric value, unit, result state, identity and order. The fictional unit U names no clinical quantity. This is an export comparison, with no result assigned to HelixBridge, PathSight or a patient decision.

Before execution, the bridge-plan owner defines exactly eight records, one fixed decimal-serialization conversion and the listed fields as the finite domain. The plan's identifier is BP1. A redundant trailing zero may disappear: 10.000 and 10 have the same decimal value here. Rounding, truncation and unit conversion are forbidden. UNAVAILABLE and INVALID must remain distinct and valueless; identity and order must persist. The resulting report, BR1, contains paired decoded observations inspected under the identified schema. The table therefore permits field comparisons that a favorable export log would not.

Record, in order	F1 value / unit / state	Decoded F2 value / unit / state
E01	0.000 / U / VALID	0 / U / VALID
E02	0.005 / U / VALID	0.005 / U / VALID
E03	9.995 / U / VALID	9.995 / U / VALID
E04	10.000 / U / VALID	10 / U / VALID
E05	99.995 / U / VALID	99.995 / U / VALID
E06	100.000 / U / VALID	100 / U / VALID
E07	no value / U / UNAVAILABLE	no value / U / UNAVAILABLE
E08	no value / U / INVALID	no value / U / INVALID

The third row tests more than visual similarity: its 9.995 must retain that exact value. The second and fifth rows likewise challenge digits a rounded conversion might lose. The first and fourth permit only redundant trailing-zero removal; the sixth retains its value and unit. The last two require different no-value states. All eight planned records appear once in order, without retry or exclusion. Their independent comparison uses

exact decimals, not rounded screen values or a floating-point tolerance. Before deciding what this result closes, it helps to distinguish the possible roles of earlier evidence.

Decide what an earlier observation can answer now

An evidence disposition belongs to one observation and one part of a claim—a **claim node**. A prior tissue study in a qualified reference region may directly address the same tested input and output question. For a new transcript-dependent interpretation path, it may instead be a **bridge candidate**: useful prior support that needs a defined connection to the changed question. For liquid preanalytics, that tissue study answers a different material question. The export comparison shows why an actual accepted connection deserves its own disposition.

The table is useful when assigning a study to a current claim component. One study can receive different judgments against different components. These are the book's decision aids, not permanent study grades or mandated regulatory categories.

Disposition	What it means for the current node
Directly applicable	The evidence addresses the current claim, configuration, population, input, workflow and decision without a material unresolved mismatch
Bridge candidate	A bounded argument and, where needed, targeted data could connect prior evidence to the new question
Applicable through an accepted bridge	An identified bridge was performed against an owned prospective rule and accepted for the stated scope
Repeat or new evidence needed	The relevant node is not represented, or the uncertainty cannot be credibly resolved through the proposed bridge
Nonresponsive	The item may be sound, but it answers a different question
Unresolved	Baseline, lineage, data quality or change characterization is insufficient for another conclusion

The accepted-bridge row requires a performed comparison and an owned prospective rule; merely naming a possible bridge leaves it in the candidate row. A separate question remains even for relevant evidence: is it **sufficient** for the decision, with adequate design, coverage, precision and limitations? Direct applicability alone cannot answer that. A sound scanner-color comparison may be nonresponsive to weak-stain reader performance. **Reviewer Lens.** Credit the answered claim and locate the missing support for the next. **Consultant Lens.** Additional work is defensible when it has a gap, owner and prospective decision rule, not simply because a whole dossier can be rerun.

A proposed bridge is not an accepted result

A bridge has to move from possibility to evidence. Once the claim component and two configurations are identified, its method and materials must challenge their difference. An accountable owner sets the rule before inspecting results. Execution then supplies observations, failures and uncertainty for a qualified review, which accepts or refuses an exact scope. A candidate with no plan, a plan not run, a scoped pass, a limited pass, a failure, an inconclusive result and an unresolved record are different states. In the export example, the finite rule and paired decoded observations let the reviewer inspect that sequence.

Even a completed passing activity may leave the claim open. A technical comparison can preserve exported values while leaving clinical interpretation unexamined. The claim is **open** without an accepted adequate basis, **limited** when a stated portion is supported with its remainder explicit, and **closed** only when reviewed applicable and adequate evidence answers the named decision. Closure records a reviewer, date, supported boundary and condition for reopening. Planned/completed/pass/fail/inconclusive describes the activity; none of those badges authorizes release.

For example, a file-decoding bridge in assistive pathology could close that property while leaving scanner-domain and pathologist-workflow questions open. The reviewer therefore needs both what was challenged and what was accepted. Separate fields for activity result, evidence applicability, claim closure and release let the consultant preserve a real success without allowing one “passed” badge to settle another decision.

With those distinctions in view, should the export format be called unchanged? Does the supplied comparison support closure of the export-preservation claim for the plan's finite scope?

What the finite comparison closes

The format remains **changed**. The bridge report preserves value and unit for all six valid records, the two distinct no-value states, and all eight identities in order. The stipulated authorized reviewer accepts this finite domain, so the export-preservation node closes **through an accepted bridge**. That positive conclusion rests on comparison across a difference. It supplies no field error rate, result for every possible export, or validation of other encodings, consumers, interfaces or the assay.

Two independent variations show where the acceptance ends. If the third record instead became 10 rather than 9.995, it would violate the no-rounding rule. If the eighth instead became valid zero, it would violate preservation of INVALID with no value. Neither is an actual observation in the bridge report, and the other favorable rows could rescue neither. **Reviewer Lens**. Retain the actual finite acceptance while rejecting either failing variation. **Consultant Lens**. A new schema or consumer creates

a new applicability question for an owner; it does not automatically require repeating the whole assay, nor close an untested dependency.

Keep follow-on decisions with their owners

After an evidence question closes, a different owner may still have work. A changed reporting rule can need verification of its implementation and validation for intended use. A changed manufacturing or service process instead asks whether that process still produces its specified output under its own criteria. Product performance and process capability are related through the route, but a result about one cannot settle the other.

Risk management asks whether new information changes a hazard, estimate, control or residual-risk judgment. Quality action asks a different question. A **correction** may fix a detected instance, while **corrective action** addresses the cause of an existing nonconformity and **preventive action** addresses the cause of a potential one under the applicable process. An enhancement or evidence gap is not automatically a nonconformity. Some organizations group the latter two under “corrective and preventive action” (CAPA), but that local label does not make their triggers identical.

Finally, the authorized release owner decides whether to release, hold or limit use; a regulatory specialist separately assesses each applicable jurisdiction. Neither decision follows mechanically from a bridge result. The table records these outputs in distinct streams so an evidence reviewer can route a question without impersonating an owner.

Decision stream	Question and owner boundary
Product verification or validation	What analytical, clinical, software, usability or workflow evidence answers the reopened node?
Process revalidation	Has a production or service process changed or lost demonstrated capability under its applicable revalidation criteria?
Risk management	Does new information affect hazards, risk estimates, controls, residual risk or prior risk-management work?
Corrective action	Is there an existing nonconformity whose cause requires corrective action under the applicable process?
Preventive action	Is there a potential nonconformity whose cause requires preventive action under the applicable process?
Release or use limitation	Which release decision, hold, scope limit, labeling constraint or fallback is warranted and authorized?

Decision stream	Question and owner boundary
Regulatory assessment	What submission, notification, report or other action is required under each applicable current jurisdictional framework?

The product and process rows illustrate the separation: an unaffected specimen/report result can remain useful while a changed interpretation path needs product evidence and a changed service process raises its own capability question. **Reviewer Lens.** Route those questions, risk implications and unknown release configuration to the relevant owners. In pathology, sound first-scanner evidence may survive while changed processing and presentation leave second-scanner and reader claims open. **Consultant Lens.** Coordinating the prospective comparisons and owner decisions is more useful than one all-purpose revalidation signature. A post-use signal can begin the same routing without proving its cause.

International Organization for Standardization (ISO) 13485:2016 distinguishes design changes, production/service process validation and corrective/preventive action; ISO 14971:2019 addresses review of production and post-production information for risk-management relevance. Their clauses inform different questions, not this book's table as a mandated form. In the United States, the Food and Drug Administration (FDA) Quality Management System Regulation (QMSR) became effective February 2, 2026 and incorporates ISO 13485:2016 by reference; the applicable current regulation controls. The FDA's 2017 general device-change guidance concerns its stated premarket notification (510(k)) scope and unintended-consequence assessment, not a global bridge rule, a software-change methodology or a verdict for either fictional product. FDA QMSR implementation; FDA 2017 change guidance, §§III–IV.

Let monitoring reopen the right question

Suppose a service revises how it captures invalid-result notices and then records more notices. The count is a real observation. Whether the invalid-result rate rose needs the relevant examination denominator and a comparable capture definition. The notices also need event identities: are several notices about one attempt, and which configurations and periods do they cover? What would the older capture rule have recorded? These questions preserve the signal without assigning a device fault, cause or patient consequence.

A useful signal record starts with provenance and scope: what source produced the observation, and which specimen, image, configuration, site, population and period does it concern? Then it names the indicator and its owned rule, the denominator and missingness, the quality of capture, alternative explanations and an investigation

owner. This is a route to a decision, not a reason to wait for complete causal certainty when an applicable safety or reporting process calls for action.

Signal field	Required question
Provenance	What source and collection process produced the observation?
Scope	Which claim, configuration, specimen, site, population and time window are affected?
Indicator and rule	What was measured, against which owned baseline or decision rule?
Accountability	What are the denominator, missingness, ascertainment and data-quality limitations?
Interpretation	Is the observation credible, and which alternative explanations remain?
Response	Who investigates, by when, with what escalation and documentation path?
Decision links	Which evidence, risk, complaint, quality, release and regulatory owners need the information?

For the notice example, the accountability row cannot be read until source definitions and event identities are recovered. Scope then identifies the service and window; the indicator row asks whether an owned rule existed; the denominator and capture quality determine what comparison is interpretable. **Reviewer Lens.** The increased count warrants investigation while rate and cause remain open. **Consultant Lens.** Prospectively defined exposure units and capture rules, including unfavorable and missing records, make the concern usable by risk, quality and release owners. It might reopen invalid-state/availability evidence in sequencing or a worklist question in pathology; neither generic parallel adds a named-product event.

What is the first investigation question for the notice example? Is a rise in the invalid-result rate established, and which observations must survive the investigation?

What the increased notices mean

The count changed; the rate and cause are unknown. Recover the number of relevant examinations, configuration and time-window identities, old and new notice definitions, capture completeness and unique event records. Preserve the notices and investigate possible device and process contributions. Do not discard the signal for lack of a denominator, and do not manufacture one from unrelated convenience records.

Real-world data (RWD) are data routinely collected about health or health-care delivery; **real-world evidence (RWE)** is clinical evidence generated by analyzing such data for a particular question. The FDA's December 2025 guidance asks whether the data and analysis are relevant and reliable for that use. Its stated scope does not address adverse-event reports or nonclinical data; it does not prohibit those sources under every other process. A complaint count or drift dashboard is not automatically clinical real-world evidence because it was observed after deployment. Its first job here is to initiate a sound investigation; a later clinical evidence question needs a fit-for-purpose design. FDA RWE guidance, §§I and III.

Connect the change to a decision

These relationships form a **change-impact, bridging and revalidation plan**. It begins with two identified states and one decision, then follows earlier evidence to the exact claim component it addressed. A transcript-selection question, specimen reporting path, scanner domain or decision-relative availability requirement may each need a different evidence judgment. The plan connects their facts, comparisons and owners without turning each claim node into a separate product approval.

Three linked records

The comparison header gives the records a common object. The first record identifies facts and possible combined effects. The second connects an earlier observation to a claim component and the actual state of its bridge. The third states what that component supports and who still decides downstream action. Comparison and node identities link the records so their conclusions cannot drift apart.

Fact and interaction record	Supplied information
Fact or dependency; prior/current values; source and date	Not supplied
One base state and its granularity	Not supplied
Interaction state, basis, affected facts and aggregate question	Not supplied
Interaction evidence, evaluated scope and disposition	Not supplied

The granularity row would permit sameness of an application label without asserting sameness of every component. The interaction rows then keep a plausible dependency separate from an observed effect and identify any evaluated scope. This prevents a tidy fact inventory from masquerading as a whole-system test.

Evidence and bridge record	Supplied information
Evidence ID, tested configuration/population and applicability question	Not supplied
Evidence disposition and exact reason	Not supplied
Bridge status, method and evidence IDs	Not supplied
Prospective rule, owner, result and reviewer decision	Not supplied
Residual scope, limitations and next evidence question	Not supplied

For the export, the prospective plan and paired report would occupy the method/evidence rows, followed by the exact eight-record rule, observed pass and finite reviewer acceptance. HelixBridge has no analogous supplied bridge result. Those fields must stay unsupplied in its record, even though the same form could hold a successful result.

Closure and owner record	Supplied information
Node state: open, limited or closed	Not supplied
Closure basis, authorized reviewer/date and reopen condition	Not supplied
Verification/validation decision	Not decided
Process-revalidation decision	Not decided
Risk-management decision	Not decided
Corrective-action decision	Not decided
Preventive-action decision	Not decided
Release/use-limitation decision	Not decided
Jurisdiction-specific regulatory decision	Not decided

The export's finite claim can be closed while its release or regulatory owner rows remain undecided. “Not decided” preserves the absence of authority and evidence in the exercise. **Reviewer Lens.** The useful conclusion names the supported component and residual. **Consultant Lens.** Each remaining row needs the appropriate owner and a specified missing result or decision, not a copied acceptance badge.

Return to the two HelixBridge evidence objects

The analytical-validation study used the earlier knowledgebase; the later technical assessment used the changed one. Strong overall agreement in the latter report is still distinct from a result for the affected class. The added transcript and more frequent execution of a rare mapping anomaly are documented, while harm, an error rate and deployment remain unestablished. Application and report-rule labels match only at that level. The exact therapy, population, variant/transcript class and claim elements remain unspecified within the broad therapy-eligibility purpose. The exhibits keep those evidence identities consultable for the decision.

Exhibit	Supplied record
EX-17-01: Configuration	Analytical validation identifies DB-18; the later technical report identifies DB-19. The added transcript makes the previously rare mapping anomaly execute more often. Application 7.0 and R19 labels remain stated across the comparison; the full component match is unknown
EX-17-02: Evidence	The analytical-validation study belongs to the DB-18 state. The DB-19 final technical report describes strong overall agreement without the affected-class design, denominator, criterion or result
EX-17-03: Decision	The affected node is transcript-selection evidence within the broad therapy-eligibility variant-identification purpose. Exact therapy and claim elements, complete commercial configuration, affected-class scope, bridge result and downstream owner decisions are not supplied

The evidence exhibit assigns the analytical study to its older configuration, while the decision exhibit leaves the commercial target unknown. Is either item directly applicable and adequate now for the changed knowledgebase's affected transcript-selection behavior? What fact states, interaction question, item/claim dispositions and bridge status follow? What remains supported, what stays open, and which next evidence question belongs to which owner?

How the HelixBridge comparison reads

Review the HelixBridge plan

The knowledgebase content changed. The application and report-rule labels are unchanged **only at label granularity**; complete application, service, supplier, inputs and reporting behavior have not been established as identical. The commercial target and other configuration remain uncertain. A new transcript could direct the same labeled application through another mapping path, but a mechanism of error has not been demonstrated. The fact table separates that possible combined effect from each known state.

Row	Base fact state	Separate interaction question
Database content and affected execution context	Changed	Does DB-19 content interacting with application 7.0 and R19 alter the affected output or interpretation? Plausible and open; no result supplied
Application and report-rule labels	Unchanged at label granularity only	Does the combined configuration behave equivalently for the affected class? Complete identity and aggregate behavior remain unresolved
Other components and commercial target	Uncertain	Missing lineage can conceal relevant dependencies; no effect or release state is established

The first row combines a known content change with an unanswered performance question. Label continuity in the next row cannot answer it. The final row leaves deployment unresolved: neither knowledgebase may be assigned to commercial use from these exhibits. The evidence table then asks how each item bears on those exact unresolved claims.

Evidence-item/node pair	Disposition	Bridge and node state
DB-18 analytical study for affected-class behavior under DB-19	Bridge candidate, subject to recovering the study and defining a credible bridge question	Candidate-unplanned; no method, rule, assigned owner, run result or acceptance; node open

Evidence-item/node pair	Disposition	Bridge and node state
Unchanged labels as support for complete configured behavior	Unresolved	No bridge selected; no acceptance basis; complete-configuration node open
DB-19 overall-agreement statement for the affected-class claim	Unresolved; full study and stratified basis not supplied	No accepted bridge or adequate targeted result shown; affected-class node open

The older study remains useful analytical evidence for its identified state; its contribution to the changed-class question is only a candidate. The missing comparison needs both complete configurations, the affected transcript/variant domain, output and reference, and the earlier study design. A qualified owner could then set a targeted method and acceptance rule before seeing results. If the question cannot be represented by that bridge, new targeted evidence may be needed. The later report separately needs its denominator, failures, affected strata, criteria, uncertainty and full results. An aggregate favorable sentence cannot supply them.

The supplied records do not establish affected transcript-selection performance with the changed knowledgebase or its relationship to an unresolved commercial state. They also leave valid earlier analytical evidence available for other demonstrated questions. **Reviewer Lens.** These two sentences prevent an open affected claim from becoming wholesale rejection. **Consultant Lens.** A minimal affected-class comparison needs specified configurations, a responsible owner and a prospective criterion. Product/service owners establish lineage; validation, risk, quality, release and regulatory owners decide their own outputs. The three exhibits supply none of those decisions and establish no corrective-action trigger.

What would close this node?

Closure needs an applicable and adequate reviewed basis for the exact affected node: recoverable prior evidence, an accepted bounded bridge, targeted new evidence, or another justified basis. Record reviewer and date, supported scope, residual limits and a reopen condition. A later relevant database change, newly affected class, configuration mismatch or credible signal could reopen the conclusion; these are proposed conditions, not new HelixBridge events.

Figure 17.1 — Reopen only the evidence node the change affects is a text-linked map of this already reviewed comparison. It marks the changed database, label-level continuity, uncertain configuration and open affected node; it adds no test result.

Book 1C - Chapter 17

Section 17 - Reopen only the affected node

Technical example: a DB-19 change can keep some labels while reopening one evidence question.

Key takeaway

A changed database stays changed; evidence closes only through an accepted basis for the named node.

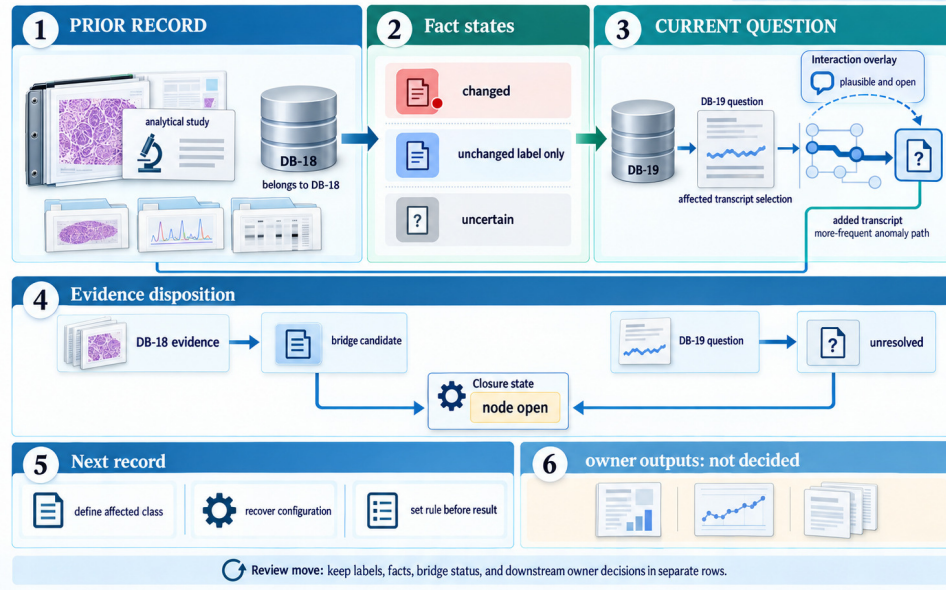


Figure 17.1. Reopen only the evidence node the change affects – Technical review plate.

Two short comparisons

The next records begin at different uncertainties. One lacks a deployment event; the other proposes a different material path. What first question would each require? Neither adds a release or outcome.

A supplier can publish • Record A

A supplier-managed job *can* publish a new HelixBridge knowledgebase while the validation application label remains 7.0. The controlled record describes a capability, with no dated publication. The commercial record supplies no deployed knowledgebase, release date, sites or complete dependency set.

What is known about deployment, and what combined-effect question is only conditional? What must be established before prior evidence can be judged against an installed configuration?

Why capability leaves deployment unresolved

Whether publication occurred and altered a deployed knowledgebase is **uncertain**. Capability is not an event. An unchanged application label could interact with a changed knowledgebase *if* such a deployment occurred, but that is a dependency question, not a fact about this timeline. First retrieve the publication and deployment history, actual installed configuration and decision owner. The Reviewer withholds a deployment-specific applicability judgment; the Consultant arranges lineage capture and a claim-specific comparison if a real change is established. No bridge, release, corrective action or regulatory outcome follows from the capability alone. Unlike the DB-18/DB-19 evidence comparison, even the deployed comparison object is missing here.

A proposed liquid path • Record B

HelixBridge's tissue baseline remains present. A separate *proposed* liquid branch retains the sparse broad program purpose but starts with venous whole blood. Processing derives plasma, followed by isolation of cell-free DNA, fragments in that plasma. Circulating tumor DNA may be a tumor-derived fraction within the cell-free DNA; no tumor origin is assumed for all isolated material.

The liquid configuration is separate. A tissue-evidence-applicability activity is required in scope, but has neither design nor result. Product-specific collection devices, handling, stability, background, input, references, reporting, population, claim details and studies remain unsupplied. Which facts differ, which interactions remain open, and can a tissue study directly answer a liquid claim from these exhibits? What evidence question comes first?

What the tissue result can retain

The broad purpose is unchanged only at its sparse stated level. The specimen-to-sample path, analyte context, configuration and workflow branch differ; their product-specific conditions and effects remain uncertain. Material and handling may interact with input background, detection and result states, but no result of that interaction is supplied. The tissue study remains useful for its tissue question, not directly sufficient for a new liquid claim. The Reviewer preserves that positive tissue scope and records the unaddressed liquid node. The Consultant obtains the liquid claim, material path, assay configuration, reference process and result-state accounting, then defines a proportionate applicability study or new evidence plan with its owner. The planned activity is not a completed bridge.

Whole blood, derived plasma and isolated cell-free DNA are different material states. A challenge material resembling the final isolated DNA may leave collection and handling unrepresented. A possible tumor-derived fraction adds no separate specimen stage and

no basis to rename tissue validation as liquid validation. Once the new claim and path are known, the question is which evidence can be retained or bridged; universal repetition of tissue studies does not follow.

Reconstruct an assistive image change

The next package uses neutral aliases for its configuration, evidence and decision records. Recognition of a familiar pattern cannot supply missing results. How far does the evidence reach when both image preparation and presentation differ? The configuration record supplies the route before the evidence items are judged.

The two image states • EX-17-T-01

Assistive Image System Q prioritizes whole-slide images and highlights suspicious regions; the pathologist, not the system, makes the diagnosis. Its package names support for Scanners A and B.

The evaluated state, Configuration P, used model M3.2 and a validation set mostly from Scanner A. Exact allocation and the complete technical path are unknown. The changed state, Configuration Q, retains those evaluated weights, uses upgraded normalization, collapses low-confidence cases by default and includes Scanner B's proprietary sharpening before export. Normalization changes the appearance supplied to fixed weights; sharpening can change the image before it reaches them. Default collapse can change what a reader sees even if score computation were unchanged. Exact component versions, release state, fallback and site conditions remain unknown. These are plausible dependencies, not observed effects.

Four evidence items • EX-17-T-02

The four records concern an overall endpoint, reader-related performance, average color and simulated timing. Their table identities let each remain attached to its own question; none supplies missing observations from another row. In particular, the color comparison omits weakly stained tissue near the low-confidence boundary and supplies no numerical outcome.

Item	Supplied statement
E1	A prespecified overall endpoint was met; definition, scanner strata, failures, denominator and precision are not supplied
E2	Improved average sensitivity without unacceptable loss of specificity is reported; reader, case, reference, estimand, denominator and workflow details are not supplied

Item	Supplied statement
E3	A bench comparison examines average color values; weakly stained tissue near low-confidence boundaries is not assessed
E4	Two positive low-score slides exceed expected turnaround in a high-volume simulation; denominator, exact time and clinical consequence are not supplied

The decision now proposed • EX-17-T-03

The endpoint row reports success without its underlying study account; the reader row lacks the design needed for transfer; the color row omits the difficult domain; the last row records two late slides without a rate, mechanism or patient consequence. The proposed decision concerns scanner support and the assistive workflow in the changed state relative to the evaluated state. No scanner equivalence, accepted interaction evaluation, reader-workflow transfer, release, quality or regulatory outcome is supplied. What base states and combined questions follow? How should each item be judged against the claim it is meant to support, and what is the smallest defensible next evidence question?

How the image evidence applies

Review the reconstruction

The aliases describe the controlled PathSight Cloud record, with no new study or chronology. The evaluated model weights remain unchanged at the stated level. Normalization and default low-confidence presentation changed. The second scanner's sharpening path differs from the mostly-first-scanner validation representation; its earlier processing details and exact allocation remain unknown. Full configured identity and the relevant study basis are still uncertain.

Follow what reaches the next operation. The second scanner sharpens an image; normalization adjusts appearance; fixed weights calculate a score; default presentation collapses a low-confidence case; a pathologist works through the queue. Weak stain, processing, presentation and workload therefore form a plausible combined question. That route helps choose an evaluation but does not show that an interaction occurred or caused either simulated delay. The table judges each supplied evidence item against the proposed current claim.

Item	Disposition against the proposed current claim
E1	Unresolved for Q and the specified scanner/tissue nodes; the aggregate assertion lacks the needed configuration and stratified basis

Item	Disposition against the proposed current claim
E2	Unresolved for the changed reader workflow; the role, interface, design, cases, readers and analysis needed for transport are missing
E3	Nonresponsive to closure of the weak-stain/low-confidence node. It documents the scope of an average-color comparison; no favorable numerical result or clinical equivalence is supplied
E4	Responsive as a bounded simulated timing observation, but unresolved as support for current workflow adequacy; it establishes no rate, cause, recovery, clinical consequence or live-use performance

The overall endpoint's “met” can remain a genuine positive result for its original question while the difficult second-scanner domain stays open. The favorable reader assertion similarly remains in the record without establishing an effect under changed presentation. The average-color comparison gives useful scope information, but no outcome and no near-boundary test. The simulated timing observation is responsive: two positive low-score slides were late in that exercise. Missing denominator, timing and recovery context prevent a rate or live-workflow conclusion. No bridge is shown as planned, passed or accepted.

Complete evaluated and changed configurations, together with the underlying endpoint and reader-study records, come first. Qualified owners can then target the second-scanner/weak-stain domain, changed interface, reader-workflow applicability and timing/fallback behavior. For example, whether changed presentation alters the intended reader task is the question; its design, endpoint, reference, readers, cases, analysis and rule must be specified before interpreting new results. Related checks may share a justified design. The table does not require a separate experiment for every row.

Reviewer Lens. Preserve the bounded favorable statements and simulated timing observation while leaving unsupported scanner and workflow claims open. **Consultant Lens.** Pathology, product and validation owners need a defined image/reader comparison and configuration lineage. Risk and quality receive the signal and unresolved scope; release and regulatory owners decide their own actions. Neither role can assign the delays to normalization, infer harm or award a release from this record.

What a defensible reconstruction preserves

A strong response holds one base state per fact, records interaction as a separate question, and assigns each evidence item against a named node. It does not transform unchanged weights into unchanged product, missing fields into “no change,” or an overall favorable result into an unrepresented scanner-stratum result. A signal prompts investigation without becoming a causal verdict. If a later, genuinely accepted bridge closes one node, keep the documented component difference classified as **changed** and retain other open nodes.

Hand each owner a decision-ready question

The validation lead needs the reopened claim and smallest adequate comparison, not a demand to rerun a historical checklist. The laboratory or pathology specialist needs the exact specimen, image, reference and workflow mismatch. The statistician needs the target quantity, design and accounting for invalid, partial or missing results. Product and service owners need a reproducible change and deployment history. Risk and quality need the new information without a premature cause label; release and regulatory owners need the accepted scope and remaining dependencies.

Book 1A develops software, artificial intelligence (AI) and human-factors change methods; Book 1B addresses cybersecurity and resilience reassessment. Book 2 owns detailed European Union in vitro diagnostic lifecycle and performance-evaluation procedures. These are specialist handoffs, not substitutes for the local evidence judgment.

Evidence travels with conditions

Two shortcuts are tempting: preserve every old conclusion because the product name is familiar, or repeat every study because one component changed. The defensible route is a specific applicability judgment tied to claim, input, configuration, conditions and evidence quality. “Smallest defensible” means the smallest set of evidence and dependency closures that *actually supports the decision*; it does not mean least cost, fewest specimens or thinnest documentation. It can contain retained evidence, an accepted bridge, targeted new work, a justified limit and continuing monitoring, each with its own basis.

The next chapter uses a different, protected evidence package. Carry the method, not a guessed answer.

Closing retrieval

How do these five pairs differ: component identity and configured behavior; documented difference and performance consequence; bridge candidate and accepted bridge; evidence closure and release; monitoring signal and causal conclusion? What exact future change or observation would reopen a legitimately closed claim?

What makes reopening specific

A useful reopen condition names the dependency and routes it to an owner. “Review periodically” alone does not say what would change the conclusion; “evidence is valid forever” ignores its tested scope. A retrospective applicability review can use existing evidence when selection, timing, bias and limitations are explicit. It cannot turn that evidence into a prospective study or rescue a result by choosing criteria afterward. Keep initially unanswered questions separate from evidence reopened by a documented change. For manufacturer monitoring and change assessment in a different local case, see Book 1D, Chapter 15.

Source trail

The fact-state taxonomy, interaction overlay, evidence dispositions, bridge and closure labels, linked plan, finite export exercise and monitoring exercise are author synthesis, not regulator-prescribed forms or universal thresholds. Fictional case facts are not external authority.

- **ISO 13485:2016**, clauses 7.3.9, 7.5.6, 8.4, 8.5.2 and 8.5.3: licensed local PDF pp. 24, 27, 33–34 inspected for distinct change, process, data-review and corrective/preventive questions. No protected procedure or checklist is reproduced. Official ISO record.
- **ISO 14971:2019**, clauses 9 and 10: licensed local PDF pp. 23–24 inspected for production/post-production information review and risk-management follow-through, not a mandated evidence-transfer method or acceptable risk level. Official ISO record.
- **FDA QMSR implementation page**, effective-date and incorporation context; the current regulation governs application. Official page.
- **FDA general device-change guidance (2017)**, §§III–IV: its stated 510(k) scope and risk-based, unintended-consequence principles, not a case outcome, software-change method or current Part 820 text. Official PDF.
- **FDA RWE guidance (December 2025)**, §§I and III: RWD/RWE, question-specific relevance and reliability, and its stated exclusions. It supplies no automatic clinical-evidence status for a dashboard. Official PDF.

- **Clinical and Laboratory Standards Institute (CLSI) MM09 and MM13, and ISO 20186-3:2019** retain the bounded molecular and cfDNA-path foundations developed in Chapters 6 and 10. They supply no HelixBridge operating value or tissue-to-liquid equivalence.

The source packet's formalin-fixed paraffin-embedded tissue and ctDNA drug-development references are not used to define either recurring product's tissue preparation or clinical claim.

CHAPTER 18

The Integrated Validation Review

One judgment you can defend

An unfamiliar evidence package now puts the book's reasoning to work. What is the product being proposed, what exactly is claimed, and what do the supplied observations support? The review calls for one bounded judgment and the smallest additional work needed to defend it. Neither document count nor the number of defects found answers that question.

The sequence has three decisions: an initial judgment, a revised judgment after an additional exhibit, and a final judgment before the worked review. Each version remains intact. Later evidence should change only the parts of the reasoning it actually answers.

Each decision has two pages: a system-and-evidence map and a decision sheet. Together they hold one judgment, at most three decision-critical gaps, the dependencies among them, retained evidence and limitations, and the smallest defensible closure plan. The task is scientific review; it grants no release or regulatory authorization and supplies neither a consulting proposal nor clinical instructions.

Retrieve without a case

Decision

What identifies an exact claim and configured object? What makes an evidence item relevant? When is an uncertainty critical? How can useful evidence survive while a broader conclusion remains unestablished?

These preliminary questions sit outside the two-page response. They concern the method, without choosing a familiar product, diagram, source, study design or expected answer. Their usefulness can be judged after the final commitment.

Three complete versions of one judgment

The three complete responses are identified as **initial (V1)**, **revised (V2)** and **final (V3)**. They belong to one attempt, with its identifier, date and time, available exhibits and prior version recorded for each. Every field is reconsidered at each stage; a deliberate carry-forward is marked “reviewed and unchanged,” rather than left blank. Each version remains separately preserved when the next begins. That history is what makes a change in reasoning visible.

The additional exhibit follows the complete initial judgment. The worked review, rubric, figure and source trail follow the complete final judgment.

Sealed packet

EX-01 through EX-04 form the initial packet; their order does not rank their importance. Their product, configurations, observations, values and criteria are fictional teaching facts. A stipulated appraisal or acceptance belongs to this exercise record and is not an actual approval of the manuscript.

EX-01

An outpatient clinic needs an additional diagnostic result for adults with signs and symptoms of **acute pharyngitis**—an inflamed throat, with symptoms such as sore throat. The device in the packet, recorded as **N**, tests for a bacterial target called an **antigen**. A reader measures light from a fluorescent signal as part of that examination. The proposed claim is detection of **group A streptococcal antigen** in throat swabs as an aid to diagnosis for those adults.

The intended operators are clinic health-care staff without laboratory training, using the instruction package recorded as **I2** and its defined orientation. They are not lay self-testers. A clinician interprets the result with other information and arranges appropriate follow-up. A negative or invalid output cannot alone exclude infection or determine treatment. Contagiousness, treatment monitoring, a standalone treatment decision and improved patient outcomes are outside the claim.

Staff collect the throat swab identified as **S1**. The extraction procedure, **P1**, prepares its material; an **aliquot**, a measured portion of that preparation, goes into a cartridge. The **R1** reader with **F3.1** firmware controls development time and measures the fluorescent signal. The **T1** decision rules lead to a positive, negative or invalid report shown by the instruction/display package. Invalid is a distinct output state, not negative. The timing claim covers automatic timing only.

The **target cartridge (C2)** has a different raw-material supplier for its extraction buffer from the **study cartridge (C1)** used earlier. The declared formulation specification stays the same. Across this comparison the records identify the swab and extraction

procedure, reader and firmware, decision rules, instruction package, intended population and automatic timing as unchanged: S1/P1, R1/F3.1, T1 and I2 remain the recorded identities.

The review question is:

Note

Does the supplied scientific evidence support the intended-user diagnostic-classification claim for the target configuration under the stated conditions?

No regulatory jurisdiction, authorization, waiver, conformity, release or commercial-use decision is supplied.

EX-02

The **analytical dossier (A0)** evaluated the study cartridge with the recorded swab/extraction, reader/firmware, decision rules and instruction package. It examined whether that configuration performed as specified across its stated material and operating conditions. It did not examine a clinician's later use of the report.

For this exercise, the dossier's methods, materials, references, acceptance criteria, complete results and uncertainty have been independently appraised and accepted as adequate for its represented study-cartridge analytical scope. It covers specimen preparation and handling; storage and reagent stability; detection and behavior near the classification signal boundary; precision; relevant cross-reactivity and interference; lot and instrument variation; declared environmental conditions; controls; invalid states; and automatic timing. Experienced laboratory personnel performed this work.

That bounded adequacy is a supplied fact. The summary supplies no numerical analytical limits. What does its stated scope contribute to the review? A demand to repeat the work needs a reason beyond the fact that the dossier is summarized.

EX-03

The **clinical study (C0)** enrolled **200 distinct eligible adults** from the represented outpatient population. Candidate and reference swabs were collected from each person at one paired visit, in randomized collection order. The reference process cultured the material and then identified the species. It was specified before candidate testing, masked to candidate output and applied to every participant. It assigns this study's comparison classification with its stated limitations; it is not asserted to be perfect clinical truth.

The study used the study cartridge. Experienced laboratory personnel performed the test, with study-team support, on the same clinic side-room counters and within the proposed operating envelope. Candidate results did not guide participant management.

The excerpt reports **199 paired final binary results**. Binary means positive or negative rather than invalid. In this table, a row identifies the reference classification and a column identifies the device output among the included results. Each person occupies one cell.

Reference classification	Device positive	Device negative
Positive	48	2
Negative	3	146

The positive-reference row contains 48 device-positive and two device-negative results. The excerpt ends, “The study’s criteria were met.” That sentence and this table are the clinical-study records available at this stage.

EX-04

The **user record (U0)** includes **12 clinic health-care staff without laboratory training** from the intended operator group. After the defined instruction-package orientation, they examine fixed positive, negative and invalid report screens. All correctly explain the displayed meanings and follow-up instructions, and every response is retained.

Each session is a desk interview with those screens on a monitor. The log contains the screen, participant explanation and interviewer response. These are the complete activities and observations in the user record.

The two-page decision record

The two pages organize material relationships rather than an exhaustive inventory. A clear written sequence can show them; an illustration is optional. Before the final commitment, only the neutral exhibit identifiers support the gap judgments.

Page 1

The first page asks:

- What is the clinical need and exact claim?
- What product and configuration does that claim concern?
- What is the specimen-to-decision chain?
- What job does each evidence item do, where does it apply, and which exhibit supplies it?
- Which uncertainties are material?

These fields begin unanswered. The evidence question connects an item to its proposed use in the judgment; it does not assign an exhibit a role in advance.

Page 2

The second page asks:

- What one supported-claim judgment follows?
- Which, at most, three gaps could change that decision?
- How do those gaps depend on one another?
- Which evidence is worth retaining, and what limitations remain?
- What is the smallest defensible closure plan, and what acceptance logic would judge its result?

These fields also begin unanswered. The plan must follow the selected questions and their dependencies, rather than precede the judgment.

A **critical gap** is an unanswered fact or evidence question that could change the judgment, a necessary limitation or the validity of a proposed closure. For each selected gap, the record connects its exact question, impact, exhibit, dependency, smallest closure and limitation while open.

The three-gap maximum limits communication; it does not permit an independent blocker to disappear. If the limit cannot hold the required reasoning, the judgment must be narrowed or identified as unclosed. No cost, schedule, numerical protocol or regulatory requirement is supplied for invention here.

Initial decision

Decision

Using EX-01 through EX-04, what complete two-page judgment is defensible? How does the record distinguish documented differences from consequences, and reported conclusions from the observations available for appraisal? What supports the claim judgment beyond document presence?

The initial version has every field complete and cites exhibit identifiers only. It remains intact before the additional exhibit becomes available.

Additional exhibit – after the initial judgment

Additional exhibit

This stage follows the preserved initial judgment. It discloses **existing contemporaneous records**: no new participant, revised reference label, criterion chosen after seeing results or erased earlier observation is introduced.

EX-05

EX-05 contains a **bridge report (BR1)** and the detailed account from the clinical study.

The bridge report compares the study and target cartridges for the extraction-buffer supplier change, using the same exact swab/extraction, reader/firmware, decision-rule and instruction-package path. Its method and acceptance rules were owned before execution. For this exercise, the method adequately covers the bounded change through paired representative patient-matrix materials, negative, near-cutoff and positive states, challenged conditions, justified material suitability, handling and stability, relevant interacting factors, and complete accounting for failures, missingness and uncertainty. *Patient-matrix* means material representing the relevant patient-sample surroundings, beyond a purified target alone.

Prespecified criteria were met, and an authorized reviewer accepted the bridge for analytical- and classification-evidence applicability within the represented laboratory-user context. Experienced laboratory personnel performed the bridge. The study cartridge remains the source configuration and the target cartridge the target.

The clinical account separates each initial result from its final state after the permitted repeat rule. Rows identify reference-classified participant groups; device-state columns count people at that stage. A further attempt does not add another person to a row.

Reference classification	Initial positive	Initial negative	Initial invalid	Participants
Positive	47	2	1	50
Negative	3	144	3	150
Total	50	146	4	200

In the positive-reference row, 47 positive, two negative and one invalid result account for its 50 participants. The locked protocol permits **one repeat**, using the original retained extract, after an initial invalid. All four initial invalid results were repeated; no other result was. The reference-positive participant's repeat was positive. Of the three reference-negative participants, two repeats were negative and one remained invalid. The final states are:

Reference classification	Final positive	Final negative	Final invalid	Participants
Positive	48	2	0	50
Negative	3	146	1	150
Total	51	148	1	200

The negative-reference row now contains three positive, 146 negative and one invalid result among its 150 participants. No other participant was excluded, missing or left with an unresolved reference classification. No reference label was revised after comparison with the candidate result.

The prespecified analysis asks three different questions. **Final-evaluable sensitivity** is the fraction of reference-positive people with a positive final device result among those with a final positive or negative device result. **Final-evaluable specificity** is the fraction of reference-negative people with a negative final device result among those with a final binary device result. The **final invalid proportion** instead uses *all enrolled participants*. The initial-attempt account is reported separately. These quantities do not share one denominator merely because they came from the same study.

The supplied two-sided 95% Wilson score intervals express model-based uncertainty around each fraction. For this exercise only, independence and sufficient homogeneity for the binomial calculation, the clinical design, represented patient spectrum, reference process and its limits, analysis and endpoint-specific criteria have been independently appraised as adequate **for the clinical study's represented laboratory-user context**. This stipulation supplies no universal method or population-wide conclusion. The original fictional criteria below govern the later revised and final judgments.

Quantity	Result and two-sided 95% interval	Prespecified exercise criterion
Final-evaluable sensitivity	48/50 = 96.0%; interval 86.54%–98.90%	Lower bound greater than 85%
Final-evaluable specificity	146/149 = 97.99%; interval 94.25%–99.31%	Lower bound greater than 93%
Final invalid proportion	1/200 = 0.50%; interval 0.088%–2.777%	Upper bound less than 3.5%

In the sensitivity row, the lower interval endpoint is compared with 85%; the specificity row compares its lower endpoint with 93%. The invalid row instead compares its upper endpoint with 3.5%. “Greater than” and “less than” are strict: equality would not satisfy

the corresponding rule. These are original exercise cutoffs, not standards, clinical recommendations or regulatory thresholds.

Revised decision

Decision

With EX-05 available, what changes in the complete two-page record? For each initial gap, is it resolved for a defined scope, still open or no longer material to the bounded decision—and which exhibit and reason support that state? What changes in the map, applicability judgment, dependencies, limitations and closure plan? Which fields remain reviewed and unchanged?

The revised version starts from both initial pages, with the initial version preserved. Every field is reconsidered. An earlier gap stays only if still material; a favorable result closes only the question it answers. The complete revised record remains intact before the final judgment.

Final commitment – after the revised judgment

Final commitment

Decision

What complete two-page decision answers EX-01's exact question after the revision? Does the three-gap limit leave an independent blocker hidden? Does the conclusion claim any unsupplied clinical or jurisdictional outcome?

The final version includes any correction to the revised reasoning and identifies both changes and reviewed-unchanged fields. It is complete and preserved before the worked review; subsequent feedback does not alter that recorded judgment.

Worked review, rubric and sources – after the final judgment

Worked review

The additional records change two parts of the evidence account while leaving a different condition unobserved. The analytical dossier and clinical study deserve credit within their stated study-cartridge scopes. EX-05 supplies an accepted bridge that makes their relevant evidence applicable to the target cartridge in the **represented laboratory-user context**. The proposed claim names clinic staff without laboratory training, and complete execution under that operator condition has not been demonstrated.

The user record also earns a positive conclusion: all **12 intended-group staff** explained the fixed screens and follow-up instructions correctly. What they did was a desk interview. The record observes no collection, extraction, loading, controls, actual invalid run, repeat or complete result production by those staff. It therefore leaves operator-dependent diagnostic performance and use-related assessment unanswered. It demonstrates no failure by the staff.

Note

The analytical and clinical evidence supports its represented scope and applies to the target cartridge through the accepted bridge. The proposed intended-clinic-user claim remains unestablished because complete execution and performance under that user condition have not been demonstrated.

That is a positive limited conclusion and a precise residual, not an assertion that clinic staff will fail or a release decision.

Page 1: one defensible reconstruction

The first page starts with the clinical need: an additional result for symptomatic adults. That role leaves diagnosis and treatment with the stated clinical pathway. Following one swab through preparation, testing, possible repeat and report then identifies where each evidence item attaches. The table makes those relationships consultable after they have been reconstructed.

Field	Example final entry
Clinical need and claim	Additional diagnostic information for adults with acute-pharyngitis signs/symptoms; group A streptococcal antigen detection as an aid, with the stated follow-up limits and proposed clinic staff

Field	Example final entry
Bounded product	Swab and extraction, target cartridge, reader and firmware, decision rules, automatic timing, result presentation, controls and invalid-repeat path
Specimen-to-decision chain	Collection □ extraction □ cartridge loading □ controlled development/reading □ result/state assignment □ prescribed repeat where applicable □ clinician review with other information and follow-up
Evidence jobs	Analytical dossier: study-cartridge analytical scope. Clinical study: reference-based classification and result availability on that cartridge with laboratory personnel. Bridge: accepted applicability to target cartridge. User record: fixed-screen comprehension by intended clinic staff
Material uncertainty	Complete intended-user execution/performance with the target cartridge and instruction package under representative clinic conditions; no independent utility, use-risk adequacy or regulatory outcome supplied

The invalid branch belongs in the sequence. A simple swab-to-positive/negative arrow would hide a report state and its repeat route. Automatic reader timing does not automate swab collection, extraction, loading or controls. The reference process assigns the study's comparison status through culture and species identification; it has its own limits and does not establish that detected antigen caused all symptoms, that the person is contagious or that a particular treatment is right.

Where the evidence attaches

The **analytical dossier** has a stipulated independent appraisal and adequate represented scope. That is more than document presence: it supplies an accepted basis for the specified analytical question on the study cartridge. A reviewer can credit that result, and a consultant can preserve its configuration, material and conditions as the source side of a later comparison. A summary alone is no reason to redo work expressly stipulated adequate.

The **clinical study** adds reference-based classification in adults, performed by laboratory personnel with support. The same side-room counter and operating envelope preserve useful environmental context while leaving the operators unchanged. Its observations still belong to the study cartridge.

The **bridge report** addresses the cartridge supplier difference with a prospective owned rule and accepted result. It carries relevant analytical and classification evidence to the

target cartridge for the laboratory-user scope. This changes applicability without rewriting either original study as direct evidence collected on the target cartridge.

The **user record** instead observes intended staff explaining fixed screens. The task it recorded determines its scope, even though its participants have the intended job description.

Reviewer Lens. The screen explanations can be accepted while complete collection-to-report execution remains unobserved. Neither the right staff title nor the same room substitutes for the missing task observations.

Consultant Lens. The next question concerns that task under its actual conditions. Another fixed-screen interview would add observations of the same narrow activity while leaving the full operation unanswered.

The result account

EX-05 reveals the existing clinical-study protocol and records. Begin with **people**: 50 reference-positive plus 150 reference-negative participants equals 200 distinct enrolled adults. Each has an initial candidate attempt. Four initial results are invalid, so $200 - 4 = 196$ **initial binary results**. The protocol repeats those four once from retained extracts, adding four *attempts*, not four people. Three acquire a binary result and one remains invalid: **199 final binary participants**, one final-invalid participant, and **204 total examination attempts**.

Population or event count	Reconstructed account
Enrolled participants	200
Initial valid binary results	196
Initial invalid results	4
Repeat examinations	4
Participants recovering a binary result after repeat	3
Final valid binary results	199
Participants still invalid at the final endpoint	1
Total examination attempts	204

Now follow the **reference-positive** row. Initially 47 device positives, two device negatives and one invalid account for 50 people. The one invalid becomes positive after its permitted repeat, yielding 48 positive, two negative and no invalid final results. Among **reference-negative** people, initial 3 positive + 144 negative + 3 invalid = 150.

Two of those invalids become negative; one remains invalid, yielding 3 positive + 146 negative + 1 invalid. No reference classification was changed to improve agreement, and no missing participant was dropped.

Final-evaluable sensitivity therefore uses the **50 reference-positive people with final binary results**: $48/50 = 96.0\%$. Final-evaluable specificity uses the **149 reference-negative people with final binary results**: $146/149 \approx 97.99\%$. The remaining reference-negative invalid is not counted as a correct or incorrect *binary* classification. It remains in the all-enrolled final-invalid quantity, $1/200 = 0.50\%$. Initial binary availability is $196/200$; final binary availability is $199/200$. These are availability fractions, not sensitivity or specificity, and they do not test the proposed clinic-staff workflow. For contrast, first-pass evaluable groups would be $47/49$ and $144/147$; they answer a different, pre-repeat conditional question.

Reproduce the reported intervals

The interval method is supplied in this exercise. For each of the three fractions separately, let **x** be its numerator, **n** its own denominator, and **p** = **x/n** the observed proportion. Let **z** = **1.959963984540054**, the normal multiplier for a two-sided nominal 95% interval. The Wilson calculation uses an adjusted divisor, center and half-width:

$$d = 1 + (z^2) / (n); c = (p + z^2/(2n)) / (d); h = (z * \sqrt{(p(1-p)/n + z^2/(4n^2))}) / (d).$$

Here the square z^2 adjusts for finite counts, the square root combines the two terms inside it, and division by **d** adjusts both center and width. The lower and upper endpoints are **c-h** and **c + h**. This is a calculation rule, not proof that the selected population, reference or user condition is appropriate; those premises are separately stipulated for the clinical study's represented context.

For sensitivity, substitute **x** = **48**, **n** = **50**, **p** = **0.96**. Then $d \approx 1.076829$, $c \approx 0.927180$ and $h \approx 0.061781$, giving **0.865399–0.988961**, or about **86.54%–98.90%**. Its lower endpoint exceeds the exercise's strict 85% rule.

For specificity, use **x** = **146**, **n** = **149**, **p** $\approx$ **0.979866**. Then $d \approx 1.025782$, $c \approx 0.967805$ and $h \approx 0.025324$, giving **0.942481–0.993129**, or about **94.25%–99.31%**. Its lower endpoint exceeds the strict 93% rule.

For the all-enrolled final-invalid proportion, use **x** = **1**, **n** = **200**, **p** = **0.005**. Then $d \approx 1.019207$, $c \approx 0.014328$ and $h \approx 0.013445$, giving **0.000883–0.027774**, or about **0.088%–2.777%**. Its upper endpoint is below the strict 3.5% rule. The three endpoint-specific criteria therefore pass **under the exercise's stipulated design, model and context**. These are three separate nominal 95% intervals, not simultaneous 95% assurance for an entire product or evidence bridge; an invalid result is not reclassified as a binary error to make the calculation work.

What the additional record does not do

EX-05 discloses records that already existed: protocol, bridge and study results. It adds neither participants nor a new study after the initial judgment. This distinction determines the next action. A missing appendix may be recoverable; a task never performed needs actual observations before it can enter the evidence account. Here no criterion was chosen after favorable values appeared, and no initial attempt or reference label was erased.

The bridge's comparison remains the extraction-buffer supplier change in the represented laboratory-user context. Arriving alongside the clinical account cannot extend it to full clinic-staff execution. A reviewer preserves its accepted scope; a consultant keeps the configuration comparison separate from any proposed user comparison. Changing a report title would change neither the observations nor their applicability.

Page 2: judgment and closure

Three questions could legitimately control the initial decision. The **cartridge-applicability question (G1)** concerns carrying earlier evidence across the buffer-supplier change. The **clinical-account question (G2)** concerns what the 199-result table and “criteria met” sentence actually represent. The **operator-task question (G3)** concerns whether the clinical study and user record represent the proposed staff condition. These labels identify the worked review's questions; other wording can express the same dependencies.

Initial question	Why it could change the decision	Dependency
Does study-cartridge evidence apply after the supplier change?	Retaining the formulation specification alone does not establish relevant performance compatibility	Establish the configured basis used in later conclusions and studies
What do the final-only clinical counts represent?	State, repeat, uncertainty and criterion details determine how the summary can be interpreted	Understand clinical evidence before relying on it or its bridge
Does the intended clinic-staff workflow support the claim?	The user record observes an interview; the clinical study has different operators	Use the resolved configuration and retained scope to define the remaining task question

The first row changes because the later packet supplies a prospectively planned, accepted comparison of the supplier difference. The second changes because the complete clinical account, stipulated appraisal and three satisfied criteria supply the

missing basis. EX-05 therefore closes the cartridge-applicability and clinical-account questions **within the represented laboratory-user scope**. The two cartridge configurations remain different.

Those closures supply the basis for the third row; they do not observe its task. The final version needs no invented extra gaps to fill its three available slots. Its remaining question can now be stated against the resolved configuration and retained evidence.

Final decision-sheet field	Example final entry
Judgment	Retain analytical/clinical evidence and the accepted target-cartridge bridge for the represented laboratory-user scope; full performance with proposed clinic staff remains unestablished
Critical gap	Performance and use when intended clinic staff execute the complete instruction-package task with the exact target configuration under representative conditions
Evidence basis	EX-01 supplies users/task; EX-03 uses laboratory personnel with support; EX-04 observes fixed-screen comprehension; EX-05 adds no complete intended-user execution
Dependency	Accepted cartridge applicability and clinical accounting supply a bounded basis for the remaining task question
Limitation while open	No intended-user equivalence, adequate full-use performance, standalone exclusion, clinical utility, waiver, authorization or release claim
Smallest closure	Claim-matched intended-user evidence across collection/extraction, reporting, controls and invalid-repeat handling on the target cartridge; prospectively owned methods and decision logic; separate performance and use-related conclusions

The critical-gap row contains related performance and human-use questions. A clinical-classification endpoint examines the output against its reference; task observations examine how the operator performs the route; use-related risk concerns difficulties and possible consequences. Their shared task does not make them one interchangeable endpoint. A coordinated evaluation may address them efficiently when justified, while retaining separate observations and analyses.

Reviewer Lens. The existing screen-comprehension result remains credible for its task. The missing full execution cannot be inferred from it or from one favorable accuracy measure.

Consultant Lens. Laboratory, clinical/statistical, human-factors and risk owners can define the next questions together against the accepted bridge and clinical account. A shared plan is useful only if each conclusion still has the observation needed to support it.

Figure 18.1 — Bounded evidence closure after the final commitment is linked only here, after V3 and the worked decision. It depicts the same bounded closure and remaining question; the prose and records above govern the answer.

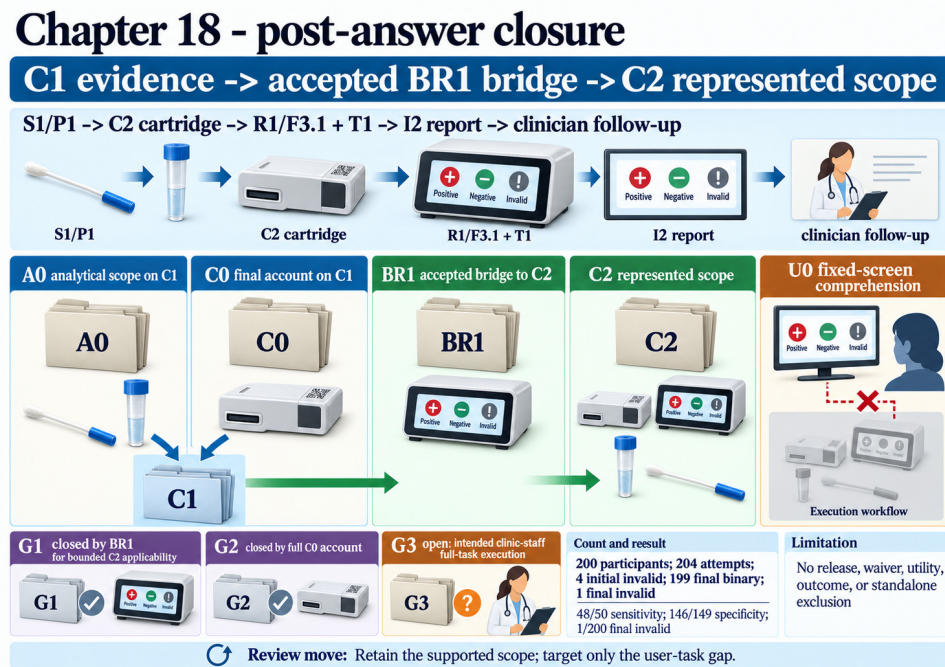


Figure 18.1. Close G1 and G2 while keeping the intended-user task open – Technical review plate.

Design the next question before a study

The next comparison follows from the remaining difference: **who performs the complete test, and under what conditions.** It concerns clinic staff using the target cartridge and instruction package to collect, extract, load, operate controls, handle invalid states, repeat as instructed and report under representative conditions. The planned work must identify which performance and use propositions each observation will test.

Laboratory and clinical specialists choose materials and conditions representing the claim, including challenging signal and result states. Statistical specialists specify the target quantity, reference comparison, sampling, result-state account, uncertainty and prospectively owned criteria. Human-factors and risk specialists evaluate actual task execution, difficulties, interface and instruction effects, controls and possible

consequences through methods fit to their own question. These responsibilities may be coordinated, but none is replaced by an assumed numerical sample size, detection limit, training duration, clinical margin or compulsory regulatory study invented here.

Note

Planned evidence should address the exact operator-dependent claim and target configuration under representative conditions, with prespecified endpoints and criteria, complete result-state accounting and relevant uncertainty. Material use-related dependencies should remain visible alongside the accuracy estimate.

This is prospective acceptance logic, not a detailed protocol or a promised result. Adequate favorable evidence could support its specified scope. Inconclusive or unfavorable evidence would leave a choice among a limitation, revised claim, further targeted work or inability to close. Effort alone changes none of those branches. The narrower statement supported now by the analytical dossier, clinical study and bridge is legitimate; it grants no permission to relabel, launch or release the device.

Compare the initial, revised and final judgments

The preserved versions now show how the additional exhibit changed the reasoning. Did it alter cartridge applicability and clinical accounting while leaving the operator-task question separate? Was a missing fact added only when its contemporaneous record appeared? Did any conclusion exceed the bridge's accepted context?

- **Unchanged refusal:** all analytical and clinical work is demanded again despite accepted bounded evidence.
- **Unbounded acceptance:** the complete proposed claim is closed because three laboratory-context criteria passed.
- **Selective revision:** only the questions answered by EX-05 close; study-cartridge provenance and initial invalid states remain, and work targets the remaining condition of use.
- **Retrospective rewriting:** an earlier version is changed after the additional exhibit or answer, hiding the reasoning transition.

The selective response demonstrates the intended evidence skill. An excellent final sentence cannot repair an overwritten initial response.

Rubric: what an excellent review shows

The rubric judges evidence relationships, rather than matching words or favoring a positive product conclusion. Each row pairs an operative requirement with the change that would make the reasoning inadequate.

Dimension	Strong response	Material revision needed
Clinical need and claim	Keeps the adult population, aid role, intended users and follow-up limits	Invents a standalone exclusion, treatment or utility claim
Product reconstruction	Includes preparation, cartridge, reading/rules, report, controls and repeat path	Reduces the product to the reader or assay
Evidence mapping	Gives the analytical, clinical, user and bridge records their different jobs	Treats document presence or common product identity as applicability
Configuration continuity	Preserves study-cartridge provenance and the accepted bridge to the target	Relabels the clinical study as direct target-cartridge data or leaves all target applicability unresolved after accepted bridging
Clinical accounting	Reconciles 200 people, 204 attempts, initial and final states and analysis populations	Deletes invalids, treats repeats as independent participants or calls 199 all enrolled
Uncertainty and criteria	Uses the supplied intervals/criteria only within the stipulated context	Converts a numerical pass into whole-product sufficiency or a universal threshold
User evidence	Credits fixed-screen comprehension but distinguishes full execution	Assumes automation or the right participant job title establishes the whole task
Gap selection	Selects genuinely decision-critical questions and changes their states when evidence changes	Chooses by document count, hides independent blockers or fills three slots unnecessarily
Closure plan	Uses the accepted evidence and targets the remaining question with appropriate owners	Repeats everything, promises success or invents detailed study requirements

Dimension	Strong response	Material revision needed
Judgment and limits	Gives a bounded scientific conclusion and explicit unsupported outcomes	Grants release, authorization, waiver, conformity or a clinical outcome
Revision integrity	Preserves complete initial, revised and final versions with the available exhibits recorded	Backfills an earlier response after later disclosure

The clinical-accounting row turns on the unit being counted: a repeat adds an attempt, while the person remains in the enrolled account. The user-evidence row turns on the activity observed: explaining a screen preserves a positive finding without observing the full test. Missing either that operator condition or the invalid states requires revision, however fluent the prose. Different wording can be excellent when it preserves those same evidence relationships, dependencies and limits.

What this capstone adds

This review brought separate tools to one unfamiliar claim. The additional record required a revised decision because it answered existing uncertainties at a defined scope. Recognition of a disease modality supplied no missing evidence. Configuration change, clinical accounting, accepted bridging and user-task representation remained different questions throughout.

A dossier may contain strong evidence and still miss a consequential condition of use. Conversely, a later record can close a genuine uncertainty without making all earlier work worthless or closing the entire claim. A supported limited conclusion is often more useful than blanket reassurance or blanket rejection.

Carry the method into practice

In a professional review, the decision and configured product give each study its question. Actual observations, failures, uncertainties and alternative explanations then determine what can be supported. A changed dependency calls for an applicability judgment. The useful final record joins supported scope, unresolved proposition, consequence for the decision and smallest defensible next question so the next owner can act on the same facts.

Software, artificial intelligence (AI) and human-factors depth belongs to Book 1A; cybersecurity and resilience to Book 1B; computational product lifecycle to Book 1D; and detailed European Union (EU) In Vitro Diagnostic Medical Devices Regulation (IVDR) practice to Book 2. A specialist handoff is useful only when the receiving owner

sees the same claim and configuration. The book ends with a habit, not a certificate: **keep evidence attached to the exact claim, product and conditions it can defend.**

Post-commitment source trail

The case, numerical data, criteria, acceptance stipulations and worked answer are original teaching material. No real product's instructions, results, authorization or clinical outcomes were transferred to Device N. These sources appear only after the final commitment and support limited method or modality points, not the fictional verdict.

- The Food and Drug Administration's (FDA's) historical K171976 record, indications and summary §§5.8–5.9, shows only that reader-based fluorescent group A streptococcal antigen testing from throat swabs is a plausible modality. It describes a different product; its operating details, labeling and regulatory decision are not assigned to Device N.
- The National Institute of Standards and Technology (NIST)/SEMATECH e-Handbook, §7.2.4.1 supports the Wilson method used to recompute these original counts with $z = 1.959963984540054$. It supplies no diagnostic criterion, clinical-adequacy judgment or transport to new operators.
- International Organization for Standardization (ISO) 13485:2016, clauses 7.3.7 and 7.3.9; ISO 20916:2019, clauses 5.5.3 and 8.2; and Clinical and Laboratory Standards Institute (CLSI) EP49, §§3.3 and 3.6, support the previously taught intended-use, change, clinical-study and evidence-accountability distinctions within their own scopes. No licensed procedure or table is reproduced.
- FDA's August 2026 human-factors guidance, §5 supports distinguishing intended users, use environments and interfaces. Qualified specialists and Book 1A own detailed human-factors methods; screen comprehension is not diagnostic performance.

The fictional adequacy stipulations and exhibits, not those source titles, determine the exercise's answer. No real-product approval follows.

WORKING TERMINOLOGY

Working Terminology and Distinctions

A familiar term can answer different questions in different records. This aid retrieves the distinctions taught in the chapters; the links lead to the explanation, source scope and limits. These are not universal legal or standards definitions. A regulated document's own discipline, authority and jurisdiction govern its terminology. Contents gives the full reading route.

Product and evidence questions

Claim. The bounded assertion the evidence is asked to support. Its result or function, population or material, conditions, user and clinical role make it assessable. An aspiration such as “improves care” does not supply those boundaries. See Chapters 1 and 4.

Product boundary and configuration. The components, interfaces, versions, materials, information dependencies, users and conditions connected to the claimed outcome need an explicit account. A release label may identify only part of the configuration. Intended, studied, proposed and deployed states can differ; the route drawn for understanding does not by itself settle a legal product boundary. See Chapter 2 and the change comparison in Chapter 17.

Verification and validation. The object, question, requirement or intended-use basis, acceptance logic and framework determine which job the word describes. Product-software, process, analytical and whole-product intended-use validation can establish different things. See Chapter 3.

Analytical and clinical performance. Examination behavior and its relationship to a clinical condition, state or process are distinct questions within the applicable framework. Neither label establishes utility or removes limits concerning population, reference or workflow. See Chapters 3, 10 and 12.

Acceptance criterion. A rule specified in advance for judging a defined result for a defined purpose. A numerical rule still needs clinical and statistical justification and an appropriate source of authority. Meeting it does not automatically establish sufficient evidence for the whole product. See Chapters 4 and 12.

Materials, references and measurement

Reference process; comparator. The comparison result or label has an assignment process, a role for the endpoint and limits. Agreement with an imperfect comparator need not establish correctness about a clinical state. Calling the comparator “truth” leaves those questions unanswered. See Chapter 5.

Specimen; derived material; analyte. The collected specimen, material made from it during preparation and the constituent being examined are different objects. In the proposed liquid path, venous whole blood, derived plasma and isolated cell-free DNA are different states. Circulating tumor DNA may be a fraction of the cell-free DNA; it is not a separately collected specimen. See Chapter 6.

Material fitness. Suitability for the particular evidence question. Patient-derived, archived, reference and contrived materials each need justification for the role they play; a material's name does not make it interchangeable with another. See Chapters 5 and 6.

Measurand. The quantity intended to be measured, with the system, component, conditions and detail needed to identify it. A unit alone may leave the quantity ambiguous. A quantitative model is not automatically suitable for categorical or qualitative reporting. See Chapter 7.

Metrological calibration; traceability. Calibration establishes a measurement relationship under stated conditions. Adjusting an instrument using that relationship is a separate action. Metrological traceability concerns an identified result, reference, documented chain and associated uncertainty. A calibrator's label alone establishes neither that chain nor fitness for patient material. See Chapter 7.

Precision; trueness; bias. Measurement precision concerns the dispersion of repeated results under stated conditions. Trueness concerns the relationship of an expectation or average to a reference value; bias estimates a systematic difference in that setting. A difference found on one material does not automatically establish bias for patient samples. See Chapter 7.

Measurement uncertainty. Dispersion associated with the value attributed to a measurand under a stated model. The account needs the result, components, conditions and effects the model does not represent. Measurement uncertainty differs from an error, the standard error of a study estimate and the broader uncertainty about clinical applicability. See Chapter 7.

Commutability. Whether the relationship observed for a specified material between specified measurement procedures represents the relationship for clinical samples. Traceability, similar material composition or a shared unit cannot guarantee this property. See Chapters 5 and 7.

Detection boundary; decision boundary; reportable state. A supported signal, a rule selecting a category and permission to report a result are separate steps. Detection alone does not establish quantitation, clinical meaning or the handling of invalid and indeterminate states. See Chapter 8.

Crossed, nested and confounded factors. Crossing allows specified combinations to be compared. Nesting places levels within another factor. Confounding prevents particular effects from being separated. A list of lots, operators and instruments does not identify which of these relationships the study used. See Chapter 9.

Clinical interpretation and study inference

Interpretation provenance. The identity, version, inputs, rules, evidence and reasoning that connect a finding to an assertion. Whether the finding is technically correct, whether the assertion can be traced and whether that assertion is supported are different questions. See Chapter 11.

Estimand. The quantity a study intends to estimate. Population, unit, endpoint, conditions, handling of result states and comparison must fit the question. A number produced by software cannot supply an unspecified target. See Chapter 12.

Enrolled, examined, evaluable and attempted. These may count different things. One participant can undergo an initial examination and a permitted repeat while contributing one final participant-level outcome. Missing specimens, invalids, indeterminate states, exclusions and unresolved references remain part of the account. See Chapters 10 and 12.

Sensitivity and specificity; predictive values. Each estimate belongs with its reference process, target population, unit, handling of result states and denominator. Predictive-value projections also need stated prevalence and assumptions about transfer. Agreement measures cannot simply be renamed sensitivity and specificity. See Chapter 12.

Confidence interval. An uncertainty statement obtained from a specified statistical procedure and its assumptions. Nominal coverage does not repair bias, represent absent users or justify an unsupported change. The interval alone cannot show that a new setting is represented. See Chapter 12.

Diagnostic, prognostic, predictive, monitoring and treatment-selection claims. Each asks a different clinical question. Disease association, future-outcome prognosis and differential treatment response do not establish one another. Companion-diagnostic terminology also depends on jurisdiction. See Chapter 13.

Standalone, unassisted, assisted and workflow evidence. An algorithm's output, a reader alone, a reader using the product and operation of the work pathway have

different study targets. Evidence for one can contribute to another question without answering it completely. See Chapters 14 and 15.

Multi-reader multi-case. Reader-study design and inference must account for readers and cases. Repeated readings are not automatically independent. Fixed or random inference scope, pairing, assigned cells and missing observations affect what can be concluded. See Chapter 15.

Clinical utility. Whether a specified test-informed strategy improves patient-relevant outcomes compared with an appropriate alternative, considering benefits, harms and burdens. A direct study design without results demonstrates no benefit; a conclusion built from linked evidence needs credible, applicable connections. See Chapter 16.

Change and judgment

Applicability; sufficiency. Applicability asks whether evidence addresses the target object and question. Sufficiency asks whether the relevant body of evidence supports the bounded conclusion. An applicable but weak study may be insufficient; strong evidence about a different object may be inapplicable. See Chapter 17.

Bridge. Evidence and reasoning that justify a bounded extension of applicability across an identified difference. Candidate, planned and accepted bridges are different states. Acceptance does not change the configuration of the original study or establish every downstream claim. See Chapters 13 and 17.

Unchanged, changed and uncertain. The fact determines the level at which sameness or difference is supported. Unchanged model weights do not establish unchanged product behavior. A field absent from an earlier record does not establish that it changed. A possible interaction remains a separate question from an observed cause. See Chapter 17.

Supported, not established and contradicted. Missing responsive evidence can leave a claim unestablished without disproving it. An observed result can contradict a particular proposition. Adequate evidence can support a bounded claim while another stays open. Each judgment needs its own question. See Chapters 4, 16 and 17.

Smallest defensible closure plan. The work needed to resolve decision-critical uncertainty, including dependencies, criteria, owner and remaining limits. Adequacy constrains “smallest.” The plan can retain accepted evidence, but cannot omit an independent obstacle merely for convenience. See Chapter 17.

Cross-book terminology boundaries

Calibration. Metrological calibration concerns a measurement relationship under stated conditions in Book 1C, Chapter 7. Predictive-score calibration concerns the relationship between stated probabilities and observed frequencies under an evaluation design in Book 1D, Chapter 9. The two uses do not share a ready-made adequacy criterion.

Precision and reproducibility. Measurement precision describes repeated-value dispersion in Chapter 7. Classification precision is the fraction of positive predictions that match the reference-positive category; Chapter 12 explains this positive-result denominator through positive predictive value. Book 1D, Chapter 9 develops related classification-evaluation and predictive-calibration questions. Reproducibility can concern changed measurement conditions, repeated execution or repeated model training; the intended meaning and relevant configuration must remain explicit.

Traceability. The relationship may connect requirements to evidence in Chapter 4, material and result or metrological chains in Chapters 5 and 7, or data provenance and lineage in Book 1D, Chapter 5. In Book 1C, CRM means *certified reference material*; the same letters may mean something else in another field.

Configuration and predetermined change control plan status. A candidate state, an accepted evidence baseline, manufacturer release and market authorization are separate states. Chapter 17 teaches the first three distinctions for evidence transfer. A predetermined change control plan can be proposed, submitted or authorized only within its actual jurisdiction and scope; Book 1D, Chapter 15 develops that distinction. A plan title alone authorizes no changed product.

Initially open and reopened evidence. An unanswered initial question does not itself establish a change. A documented difference or new information is needed before an earlier conclusion can be described as reopened by it. See Chapter 17.

ABBREVIATIONS

Abbreviations

This table gives expansions of short forms used in the book. The chapter explains the idea, criterion or source authority; the expansion does not replace that explanation. Fictional study, configuration and exhibit labels remain identifiers rather than abbreviations to decode here.

Short form	Expansion or reading
510(k)	United States premarket-notification pathway designation; not an acronym
AI	artificial intelligence
API	Association for Pathology Informatics, in the Chapter 14 source context
ASCP	American Society for Clinical Pathology
AUC	area under the curve, ordinarily a receiver-operating-characteristic curve where used here
CAP	College of American Pathologists
CAPA	corrective and preventive action, a local grouping whose parts retain different triggers
CDx	companion diagnostic
CDRH	Center for Devices and Radiological Health, within the U.S. FDA
CFR	Code of Federal Regulations (United States)
CI	confidence interval
CLIA	Clinical Laboratory Improvement Amendments (United States)
CLSI	Clinical and Laboratory Standards Institute
CRM	certified reference material, in Book 1C's measurement context
CV	coefficient of variation

Short form	Expansion or reading
cfDNA	cell-free DNA
ctDNA	circulating tumor DNA, a possible tumor-derived fraction within cfDNA
DNA	deoxyribonucleic acid
DP	digital pathology
EU	European Union
FDA	United States Food and Drug Administration
FFPE	formalin-fixed, paraffin-embedded
GHTF	Global Harmonization Task Force, for the chapter's legacy source
IEC	International Electrotechnical Commission
IMDRF	International Medical Device Regulators Forum
ISO	International Organization for Standardization
IVD	in vitro diagnostic
IVDR	In Vitro Diagnostic Medical Devices Regulation (EU Regulation 2017/746)
JCGM	Joint Committee for Guides in Metrology
LoB	limit of blank
LoD	limit of detection
LoQ	limit of quantitation
MDCG	Medical Device Coordination Group (EU)
MDDT	Medical Device Development Tool, in the FDA source context
MDIC	Medical Device Innovation Consortium
MDSW	medical device software, in the cited EU guidance context
MRMC	multi-reader multi-case
NGS	next-generation sequencing

Short form	Expansion or reading
NIH	National Institutes of Health (United States)
NIST	National Institute of Standards and Technology (United States)
NPA	negative percent agreement
NPV	negative predictive value
PCCP	predetermined change control plan; its actual jurisdiction, status and scope matter
PMA	premarket approval (United States pathway)
PPA	positive percent agreement
PPV	positive predictive value
QC	quality control
QMSR	Quality Management System Regulation (United States)
RNA	ribonucleic acid
ROC	receiver operating characteristic
ROI	region of interest
RWD	real-world data
RWE	real-world evidence
SaMD	Software as a Medical Device, in the IMDRF source context
SBOM	software bill of materials
SD	standard deviation
SE	standard error
SOP	standard operating procedure
STARD	Standards for Reporting Diagnostic Accuracy Studies
VIM	International Vocabulary of Metrology, cited by JCGM
WSI	whole-slide image or whole-slide imaging, according to the sentence

The API row shows why context matters: in Chapter 14 it names the Association for Pathology Informatics, rather than a software interface. The CRM row likewise means certified reference material in this book's measurement discussion. A familiar expansion from another field could change the meaning of the sentence.

Working Terminology explains the different senses of calibration, precision, traceability and configuration. The chapter-local source trails identify the authority and scope of cited sources.

SOURCE NAVIGATION

Source and Citation Navigation

Each of Chapters 1–17 has a local source trail. The cited passage and its nearby scope note identify what the source contributes: binding law, a voluntary standard, nonbinding regulator guidance, a method reference and a fictional exhibit have different roles. A citation supporting one proposition does not automatically support another product, population, configuration or jurisdiction. Dates and editions stay with the chapter's recorded checks; this aid does not claim to have verified every source anew.

Chapter-local trails

Part I — Make the question visible: Chapter 1 · Chapter 2 · Chapter 3 · Chapter 4

Part II — Analytical case: Chapter 5 · Chapter 6 · Chapter 7 · Chapter 8 · Chapter 9 · Chapter 10

Part III — Meaning and use: Chapter 11 · Chapter 12 · Chapter 13 · Chapter 14 · Chapter 15 · Chapter 16

Part IV — Change: Chapter 17

Chapter 18 is an independent staged assessment, entered through its neutral opening. Its initial judgment, later exhibit, revised judgment and final judgment precede the worked review. The judgments remain distinct so the effect of new information can be understood. The case-specific source trail follows the final judgment inside the chapter; there is no direct source link here.

Working Terminology retrieves distinctions; Abbreviations gives expansions. Contents supplies the reading order.